



Dynamics of Health Care Services in India



A Compendium of Studies Conducted by the
Population Research Centres 2020-21



**Ministry of Health and Family Welfare
Government of India
New Delhi**

Dynamics of Health Care Services in India

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Population Research Centres 2020-21**



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MESSAGE

It gives me immense pleasure to introduce the seventh edition of annual Population Research Centre (PRC) Compendium for the year 2020-21. This compendium contains selected research studies in the area of Population, Health and Nutrition that would further enhance understanding of Policy Makers, Researchers and Field Functionaries on some critical issues related to health and family welfare sector.

I am sure that the findings of the research papers in the compendium will be extremely useful for the policy makers, programme managers, academicians and other stakeholders to assess the progress in the sector of health & wellness and would assist them to identify specific areas where interventions are needed.

I commend the efforts of faculty and staff of the 18 Population Research Centres (PRCs) which are spread all over the country. I am sure that with perseverance and dedication, the PRCs will reach greater heights of accomplishments. I wish them all success in their future academic endeavours.

Place : New Delhi
Date : 25-02-2022

(Rajesh Bhushan)



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FOREWORD

The Population Research Centres (PRCs) in India have a special place in the broad field of Population Studies. With a long scholarly history dating back to 1957, they have contributed significantly in the understanding of health and demographic dimensions of development issues in an interdisciplinary perspective.

During past six decades, the work of the PRCs have added to the available knowledge base and also have been used widely by the Central, State, and Local Governments besides a number of National and International agencies. With an all India presence, the network of 18 PRCs have been continuously providing inputs for the policy formulation and interventions required for the programs through regular research and evaluation with an active support by the Ministry of Health and Family Welfare (MoHFW), Government of India.

The annual Compendium (2020-21) is a glimpse of work that the PRCs regularly undertake. It has a number of well researched and peer-reviewed articles on a variety of themes that may act as valuable input for policy formulation and impact assessment of programme related to health sector.

I congratulate all the PRCs for their contributions to the compendium and for being part of the National endeavor to invigorate the universal access to equitable, affordable and quality health care .

I am confident that scholars, researchers, health professionals and policy makers will make use of the findings for the research papers of compendium in their field of activities.

(Vikas Sheel)

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Message

It gives me immense pleasure to inform that the annual Compendium of the Population Research Centres is being released. I congratulate and appreciate the endeavour of officials of Ministry, GIPE Pune and PRC CRRID to making this Publication available to all stakeholders.

The volume marks a good tradition to further enhance the knowledge and practices in Health and Family Welfare and other related aspects. Under the guidance and support from the Ministry of Health and Family Welfare (MoHFW), Government of India, the Publication adds significantly to the range of works that the PRCs are engaged in. I am sure that this publication would benefit the scholars, programme managers, policy planners and others stakeholders in their work substantially.

I take this opportunity to congratulate the faculty and staff at different PRCs in various parts of the country for contributing to this volume. I hope that the endeavour to undertake quality research continues with same spirit.


(Sandhya Krishnamurthy)

Healthy Village, Healthy Nation



एड्स - जानकारी ही बचाव है
Talking about AIDS is taking care of each other



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PREFACE

Eighteen Population Research Centres (PRCs), spread over Universities and research Institutions in different States and Union Territories in India, are devoted to research and evaluation in the field of health, population and other related aspects. Set up by the Ministry of Health and Family Welfare (MoHFW), Government of India, these PRCs have regularly provided critical inputs for programme intervention and policy planning at different levels. The research feedback from the PRCs are regularly used by the MoHFW and others.

Publication of the Compendium for the year 2020-21 is a regular annual exercise and the volume "Dynamics of Health Care Services in India" this year is an effort in this direction. This publication contains a set of 17 papers in six thematic sections such as Infant, Child and Adolescent Health, Maternal and Reproductive Health, Family Planning Services, Non-Communicable Diseases and Mental Health, Health Infrastructure, Care Services and Use and Sex Composition, Education, Insurance and Data Quality. This is a useful compilation of contemporary assessment of health care services and related components in various States as well as Union Territories of India. The papers for this volume have been finalized through an academically rigorous peer-review process where a total 47 papers reviewed by the experts in the relevant fields for user content and quality.

I take the opportunity to place on record my heartfelt thanks to the 46 experts from reputed Institutions and Universities in different parts of the Country who readily agreed to be a part of the review process in spite their hectic engagements. I also extend my sincere thanks to the authors from PRCs who contributed to this volume taking time out from their regular engagements. I express my special appreciation to the faculty and staff at PRC in the Centre for Research in Rural and Industrial Development (CRRID), Chandigarh for arduously undertaking the task of bringing out this volume.


(Anjali Rawat)

ACKNOWLEDGEMENTS

This compendium is a successful culmination of selected research on diverse aspects of population and health in India undertaken at the Population Research Centres (PRCs) all over the country during 2020-21. I take this opportunity to mention names of a few organizations and individuals whose efforts at different stages of the publications added value to our work for the volume at the Centre for Research in Rural and Industrial Development (CRRID), Chandigarh.

At the outset, we are grateful to the Ministry of Health and Family Welfare (MoHFW), Government of India for their overall support and guidance in the preparation of the compendium. The direction, guidance and encouragement of Ms. Sandhya Krishnamurthy, DG (Statistics) was always available. She has been source of great inspiration in bringing out this publication. Ms. Anjali Rawat, Deputy Director General (Statistics) took special interest for this publication, through regular monitoring and supervision. She effectively coordinated the task. Her valuable suggestions and advice have been useful in selection of the papers after the peer-review. Her help was always available for smooth and efficient completion of the assignment. So, I gratefully acknowledge her contributions and express my gratitude to her. Special thanks to Shri Kumar Sundaram, Director (Statistics), Mr. Anindya Saha, Deputy Director (Statistics), Ms. Preeti Tiwari, Assistant Director (Statistics), Ms. Deepshikha Arora, Senior Statistical Officer, Shri Rajiv Ahuja, Assistant Computer Programmer, Ms. Meenakshi, JSO, Sh. Tamal Bandyopadhyaya, Consultant and Dr. Rakesh Sharma, Consultant (HMIS), for their continuous cooperation during the preparation of the compendium.

We are extremely grateful to the reviewers of the volume who are from various research Institutions, Universities and other specialized agencies in the field of health in the country. We extend our gratitude to them for sparing their valuable time for the review. In a very short span of time, they offered their critical comments which after incorporation substantially improved the quality of the articles. Help received from Professor Suresh Sharma IEG, Delhi and Professor T. S. Syamala, ISEC, Bengaluru in screening of papers is also thankfully acknowledged.

In the CRRID, I gratefully acknowledge the encouragement that I regularly received from the Vice Chairman Dr S.K. Mangal and the Executive Vice Chairman Shri D.P.S Sandhu during the course of preparation of the compendium. The (Acting) Director Dr Rajesh Kumar Aggarwal was always available for help. My special thanks are due to my young colleagues Ms Babita Chauhan and Ms Prabhjot Kaur Turna, who as Research Fellows, were always available for all basic work of facilitation including communication, office editing, etc. I also express my thanks to other colleagues in PRC and CRRID.

Chandigarh
2 March 2022

Aswini Kumar Nanda
Volume Editor

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INTRODUCTION

The current volume “*Dynamics of Health Care Services in India*” grows out of the selected research at some of the Population Research Centres (PRCs) during 2020-21. With a smattering of fifteen papers spread over six broad yet interlinked themes such as infant, adolescent and child health; maternal and reproductive health; family planning services; health infrastructure, care services and use; sex composition, education and health insurance and data quality. In the backdrop of an ever-increasing effort by the government to reach beyond the frontiers and take good health and wellness to the doorsteps, it presents spatially diverse narratives that construct an unifying discourse. Common concern for progress and yearning to build on the already achieved gains in the sector of population, nutrition and health remains the underlying essence of all the contributions.

Outlining the issues, the chapters provide disparate snapshots that are based on a primary as well as on a variety of secondary sources. The focus of the Section I in the compendium remains infant, child and adolescent health in which substantial progress has been achieved in the country. Adolescence is recognized as a distinct life stage with significant health risks. In Chapter 1, *Syamala* and *Kavitha* peruse over the adolescent marriages in India that have been a major policy and programme concern for long. Using various rounds of NFHS data, they document regional concentration of such marriages in the country and examine variations across different socio-economic sections—particularly in terms of poverty and illiteracy. They echo the view that education of girls, beyond secondary level, significantly reduces the probability of adolescent marriages. Prevalence of anaemia among children under age five at district level in India is examined also in this section by *Ghosh* and *Desai*. Both use specialized spatial analytic software such as ArcGIS and GeoDa to represent the differentials in levels of anaemia indicated by the NFHS-4 and map the vulnerable clusters in Chapter 2.

Maternal and reproductive health brings focus on women and children, and is discussed in the Section II. Hysterectomies provide relief to women in certain conditions but are often laden with psycho-social and health consequences when such procedures are performed unnecessarily. Self-reported hysterectomy prevalence in Punjab increased marginally from 2.6 percent in 2015-16 to 3.2 percent in 2019-21 among women aged 15-49 (NFHS-5). Using data from the NFHS-4, *Nanda* and *Aggarwal* concentrate on the prevalence of hysterectomy in this State and examine selected predictors for the women aged 15-29 on the basis of binary logistic regression in Chapter 3. They indicate that that hysterectomy is linked significantly to age, religion, cohabitation and fertility of women but not to the caste and standard of living of the ever-married women. In Chapter 4, *Anitha Kumari* also studies hysterectomy in southern Indian States, namely Andhra Pradesh, Telengana, Karnataka, Tamil Nadu and Kerala classifies them into high, medium and low prevalence categories on the basis of NFHS-4 data. She examines

the varying role of age, education, place of residence, standard of living etc. of woman as determinants in these states besides the part played by private health facilities.

Elimination of iron deficiency among females has been a top policy as well as programme priority in India for long time. Keeping this in view and using NFHS statistics, *Sharma, Sharma* and *Mukherjee* examine the prevalence of anaemia among women and bring to the fore the vulnerability in terms of spatial patterns and socio-economic groupings in Chapter 5. They recommend, among other things, a diversified diet to overcome the problem of such deficiencies. Caesarian section births are considered medically unnecessary and potentially harmful in many circumstances and rates above 10-15 percent are often viewed with concern. The share of C-section births, as per the NFHS, increased from 17.2 percent in 2015-16 to 21.5 percent in 2019-21 in India. In this backdrop, *Bhatt* examines the prevalence and determinants C-section births in Jammu and Kashmir in Chapter 6, and observes that such procedures are more prevalent in Srinagar region than in Jammu region. He also flags that the planned C-sections exceed the emergency C-section in this Union Territory. Using binary logistic regression model, he finds that the age at marriage is an important predictor of the procedure. He discounts the argument that complications at birth add to the C-sections births and point to a contributing role of the private sector health facilities to increasing C-section births. In Chapter 7, wastage of pregnancy in Karnataka is examined at district-level by *Pundappanavan* and *Noolvi* through HMIS data. They estimate that on an average in thousand pregnancies, 62 are wasted due to both abortions and still births. According to them induced abortion is the leading cause of pregnancy wastage followed by spontaneous and then still births. Their regional analysis shows that the southern districts in Karnataka report a higher level of pregnancy wastage than their northern counterparts. Moreover, they refer to the private sector facilities where most induced abortions are undertaken.

Section III in the volume includes family planning services that complement empowerment of women in wider terms. *Subaiya* and *Bidari* investigate the contraceptive acceptance among young women (15-24 years) in India in Chapter 8 using data from the fourth round of the NFHS and estimate that two in three refrain from using any method. They also record regional disparity not only in acceptance of contraception but also in choice of specific methods. Lack of awareness about methods of family planning and low ability to take decisions on reproductive matters are specified along with others as reasons for non-acceptance.

Non-Communicable diseases and mental health are the focus of papers in the Section IV. National Health Mission (NHM) has been focusing on Mental Health with a renewed vigour after the onset of the COVID-19 pandemic. In Chapter 9 *Mathew* takes the case of school students in southern Kerala during the pandemic and analyses the impact of anxiety as well

as depression among them through a web-based survey. Using standardized protocols, he finds that about one-quarter of such students reported mild to severe anxiety and depression due to a long duration of online classes, lack of physical exercise, and confinement due to quarantine.

Section V of the compendium covers health infrastructure, care services and use. Availability and accessibility of health care services hold key to health outcomes and the NHM has brought in a series of interventions in rural and urban areas for the underserved segments of population. In Chapter 10, *Kayakalp*—a score based external assessment and award scheme on a set of predefined parameters—which has been instrumental in changing the face of the public health institutions in the country—is examined. *Kumari, Sharma and Sharma* evaluate the progress in the State of Rajasthan across four selected districts and credit *Kayakalp* initiative significant improvements in the public health institutions. They observe a scope for improvement in reducing the delay in distribution of prize money, routine training, manpower deficit, etc. *Muniswamy, Vijay Krishna and Kumar* use rural health statistics to discuss the growth in the number of health centres at different levels in the States and Union Territories all over the country during 2005-19 in Chapter 11. *Aggarwal and Mishra* evaluate the facilities for child care at ground level, and select a Urban Primary Health Care Centre (UPHC) in Lucknow City, Uttar Pradesh for effective delivery of health care services in Chapter 12. Their assessment of 120 clients at the facility where the child was less than five years old indicates that the seeking of health care from public sector is largely related to one's socio-economic background. The probabilities of accessing facilities for sick children at UPHC increased with income of the parents but reduced with their education. A male child was more likely to be brought to the urban health facility than the female counterparts revealing a cultural bias disadvantaging the girls.

Section VI covers areas such as sex composition, education, health insurance, data systems and quality. *Muniswamy, Suri babu and Vijay Krishna* in their paper look at the changing sex ratio among the tribals in the state of Andhra Pradesh through census based statistics and point to a fluctuating sex composition of the population during 1961-2011 in Chapter 13. Their district level analysis reveals that the sex ratio rose relatively faster in urban areas than in the rural areas. Currently, health insurance is considered to augment the universal health coverage (UHC) and enhance access to effective health care services. In relation to this, *Kumar S and Nair* examine in Chapter 14 the penetration of the health insurance schemes in urban Kerala through a primary survey. They observe low levels of awareness among the urban poor and recommend wider campaigns to bring them into the fold of insurance coverage offered at both public and private health facilities. Robust data system is essential for delivery of effective health care services and in country like India hold good potential for health

governance. In this context, *Jha* and *Arora* trace the growth Health Management and Information System (HMIS) since 2008 in Chapter 15 and outline the overarching features and also highlight some data quality dimensions. They also layout the mechanisms in which HMIS can be a source of great thrust to the health system in the country.

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Section I
Infant, Child and Adolescent Health

1

Adolescent Marriages in India: Trends and Patterns

T. S. Syamala and N. Kavitha

Abstract

Despite several legislative measures, a significant proportion of marriages in India take place before the age of 18. As curbing child marriage is an important aspect to achieve sustainable development goals 5, this study aims to understand the trends and patterns of child marriages in India. This study uses data from all the four rounds of National Family Health Surveys. This analysis shows that the prevalence of marriage among female children has considerably declined from 1992–1993 (NFHS-1) and 2015–2016 (NFHS-4). There are also wide variations in proportion of child marriages across states. In states of west Bengal, Bihar and Jharkhand, the proportion of child marriages are hovering around 40 percent whereas in states like Kerala and Punjab, this proportion is below 10 percent. Child marriage, on the other hand, persists among illiterate, poor, and rural women. The results of multivariate logistic regression reveal that the educational attainment and wealth status of the household are the most important drivers of female child marriage in India. Women with secondary and higher educational levels are less likely to be married before 18 years compared to the women who are illiterates. Further, females from poorer households have higher incidence of child marriages compared to those from wealthier households. The findings suggest that improving female education and providing financial assistance to the poor families would reduce the proportion of child marriages in India.

Introduction

Child Marriage is a cause for concern in most countries of the world irrespective of cultures, religions and ethnicities. Every year, 12 million girls marry before the age of 18 and 70 million women in the age group 20-24 around the world had been married before the age of 18. Further, as reported by International Centre for Research on Women (ICRW) in the year 2016, one third of the girls in the developing world are married before the age of 18 and the largest number of child brides reside in South Asia. Estimates suggest that 1.5 million girls under age 18 get married in India each year which accounts for one-third of the global total (UNICEF, 2012). Although there is a strong commitment to eliminate child marriages, a considerable proportion of marriages in India still happen below the age 18.

Legislation for eliminating child marriage in pre-independent India was put in place in 1929. Keeping this in mind, several legislative measures were initiated to regulate child marriages in India. Legislation for eliminating child marriage in pre-independent India was put in place in 1929 is an initiative in this direction. The purpose of the Child Marriage Restraint Act of 1929

was to prohibit the solemnization of child marriages. The Union Government has endeavoured to curb the practice in recent years through a more progressive Prohibition of Child Marriage Act, 2006 where in child marriage is a punishable offence that includes punitive measures against those who perform, permit and promote child marriage. The marriage of males under the age of 21 and females under the age of 18 is defined as child marriage under this Act. Keeping this in mind, several legislative measures were initiated to regulate child marriages in India. Legislation for eliminating child marriage in pre-independent India was put in place in 1929 is an initiative in this direction. The purpose of the Child Marriage Restraint Act of 1929 was to prohibit the solemnization of child marriages. The Union Government has endeavoured to curb the practice in recent years through a more progressive Prohibition of Child Marriage Act, 2006 where in child marriage is a punishable offence that includes punitive measures against those who perform, permit and promote child marriage. The marriage of males under the age of 21 and females under the age of 18 is defined as child marriage under this Act.

The incidence of child marriage is generally reported to be higher among the socially, economically and educationally backward sections. Further, the proportion of child marriages also varies across states and the highest proportion being reported from the states of Rajasthan and Bihar. As curbing child marriage is an important aspect to achieve sustainable development goals 5, this study aims to understand the trends and patterns of child marriages in India.

Though it is illegal to marry before the age of 18 for the girls, the practice remains unabated in the country. Factors responsible for the high incidence of child marriage in India are different in different societies. The four rounds of the National Family Health Survey showed a declining trend but the rate at which the declining happens is very slow. Marriage at younger ages not only results in low educational levels and economic dependence among women, it has far reached consequences on their mental as well as physical health. In western societies, it is common that births occur to women outside the marriage but in eastern societies, particularly, in countries like India, childbearing occur within the wedlock. Women are expected to go for childbearing soon after the marriage. Hence, early marriage has direct bearing on their maternal health as they become mothers at early ages which would ultimately affect the health of their off-springs.

Data suggest that though there has been decrease in the proportion of adolescent pregnancy, significant proportion of girls (10 percent) in the age group 15-19 become mothers worldwide. About 12 million girls in the same age group and at least 777000 girls under age 15 give birth every year in less developed countries (Darroch et al., 2016; UNFPA, 2015). Pregnancy and childbirth are the leading causes of death among women in the age group between 15 and 19 in developing countries (Neal et al., 2012). Studies have also brought out that the risk of maternal

mortality is highest among the adolescent girls under 15 years old and complications in pregnancy and childbirth are higher among adolescent girls aged 10-19 compared to women aged 20-24 (Ganchimeg et al., 2014; Althabe et al., 2015). In India, though the legal minimum age at marriage is defined as 18, according to fourth round of National Family Health Survey (NFHS), about eight percent of 15-19 age group women were already mothers or pregnant at the time of survey (IIPS and ICF, 2017).

Consequences of Child Marriage

Early marriage can have detrimental effects on girl's life socially, economically, psychologically and physically. The primary risk factor of child marriage is that gender inequity often characterized by low level of education and less opportunities for employment (Raj, 2010; Klugman et al., 2014; Malhotra et al., 2011; Vogelstein, 2013). Studies consistently brought out that child marriage has detrimental effect on the psychological or mental well-being of girls (UNICEF, 2001; Nour, 2006; Clark, 2004). Many studies have also pointed out that early marriage and childbearing affects the health of the mother and their off-springs negatively. Women who enter into the wedlock early tend to go for early and frequent childbearing, which would ultimately affect their health as well as the health of their children (Trussell and Menken, 1978; Ravi Prakash et al., 2011). Young women are at a higher risk of poor birth outcomes particularly premature birth, low birth weight and infant death than women who postponed their marriage and childbearing (Bacci et al., 1993; Chen et al., 2007; Cooper et al., 1993; LeGrand and Mbacke, 1993; Trivedi and Pasrija, 2007). Risk of maternal death is higher among teenagers than older women (Gupta et al., 2010; Midhet et al., 1998; Neto et al., 2009). Women who are married young are often exposed to various forms of domestic violence (Fig. 1). Lack of negotiating power and effect of gender inequity among young mothers may result in less use of health care services (Reynolds et al., 2006; Obermeyer and Potter, 1991; Sharma et al., 2001, Thato et al., 2007). Further young married girls are often exposed to unplanned pregnancies and termination of pregnancy which are the results of lack of bargaining skills for using the contraception as well as lack of knowledge on contraception (Kamal and Hassan, 2013; Machel et al., 2013; Raj et al., 2009).

A review of earlier studies has also revealed that certain predictors are important in determining the marriage age among girls as the marriage is mostly decided by the family members and not the girl herself. Women who are educated, women belong to higher economic strata, urban women tend to marry at later ages than their counterparts (Wong, 2005; Subramanian, 2008; Pandey, 1984; Chandrasekhar, 2010; Kamal, 2011; Modak, 2019). It is also observed that religion and caste or tribe which the women belong to are also important in determining the marriage age (Bhagat and Unisa, 1991; Das et al., 2011; Desai, 2010; Modak, 2019).

The above discussion clearly brings out that undoubtedly child marriage has detrimental

impact on women's physical, mental, social and economic conditions. It is also a fact that a significant percentage of population of women of India belongs to younger age groups and considerable proportion of them enters into wedlock as well as childbearing at early ages. Though age at marriage of women in India has been in focus for long time and legal marriage age of women and men has been defined by the government, still marrying at early ages is a common phenomenon in the society. Further, in order to improve the maternal and child health situations in the country, along with several other programmes, it is very essential to eliminate adolescent marriages.

In view of this background, it is essential to understand the trends in percentage of girls marrying at ages 18 or below in India and states over a period of time. It is also equally important to understand the impact of background factors on the percentage of women who are married at ages 18 or early. This study aims to understand the prevalence, trends and patterns of child marriages in India, to analyze the state level variations in child marriages and also to analyze the socio-economic drivers of child marriages in India.

Data and Methods

Data for this study is taken from all the four rounds of the National Family Health Survey (NFHS) conducted during 1992-93, 1998-99, 2005-06 and 2015-16 for analyzing the trends in the percentage of women married before the age of 18. For further analysis, data from the recent round is used. NFHS provides information on exact age of women at the time of marriage for women in the age group 15-49 and men in the age group 15-54. The survey provides information on population, health and nutrition for India, each state and union territory. The primary objective of the 2015-16 NFHS is to provide essential data on health and family welfare, as well as data on emerging issues in these areas. The clinical, anthropometric and biochemical (CAB) component of NFHS-4 is designed to provide vital estimates of the prevalence of malnutrition, anemia, hypertension, HIV and high blood glucose levels through a series of biomarker tests and measurements (IIPS and ICF, 2017).

Four types of questionnaires: household questionnaire, woman's questionnaire, man's questionnaire, and biomarker questionnaire) were used in NFHS to collect information on various aspects. In household questionnaire, basic demographic information was collected on the characteristics of each person listed, such as age, sex, marital status, schooling, and relationship to the head of the household. The information on age and sex of household members obtained in the household questionnaire was used to identify women and men who were eligible for individual interviews. Information on characteristics of the household's dwelling unit such as source of water; water treatment; type of toilet facilities; type of cooking fuel; materials used for the floor, roof, and walls of the dwelling unit; and ownership of various durable goods was collected through the household questionnaire.

The woman's questionnaire collected information from all eligible women age 15-49 on background characteristics of woman, reproduction, hysterectomy, menstrual hygiene, family planning, maternal and child health, breastfeeding, and nutrition, fertility preferences, domestic violence etc. the man's questionnaire covered information on man's background characteristics, media exposure, marriage, employment, number of children, presence at antenatal care visits, contraceptive knowledge and use, fertility preferences, nutrition, sexual behaviour, attitudes toward gender roles, HIV/AIDS, tobacco and alcohol use, knowledge of tuberculosis, current morbidity (diabetes, asthma, goitre, heart disease, cancer), and household decision making. The biomarker questionnaire covered measurements of height, weight, and haemoglobin for children, and measurements of height, weight, haemoglobin, blood pressure, and random blood glucose for women age 15-49 and men age 15-54. In addition, eligible women and men were requested to provide a few drops of blood from a finger prick for HIV testing.

In the first part, analysis is having been carried out to understand the prevalence, patterns and trends of child marriages in India and states across urban and rural areas. Subsequently, the association between socio economic factors and child marriages have been analyzed and finally, logistic regression analysis is employed to understand the major drivers of child marriages in India.

Explanatory Variables Used in the Analysis

Studies have shown that place where the women reside is an important factor that determine the age at which the women marry and in general rural women are more likely to be married early compared to their urban counterparts. Therefore, place of residence i.e., whether the woman resides in urban or rural area, is used as an explanatory variable in the analysis. Religion and caste are important social variables that affect the age at marriage of women. Hence, religion (Hindu, Muslim, Christian and Others) and caste (SC/ST, Other Backward Caste and Others) are also used to understand the association between these factors and early marriages. Education of women and economic level of household proved to be significant factors influencing the age at marriage. These two variables are also included in the analysis and are categorized as follows: educational level of women viz., illiterates, primary, secondary, and higher levels; Wealth status of the Households (Poorest, Poorer, Middle, Richer and Richest).

Analysis and Discussion

Levels and trends in child marriages

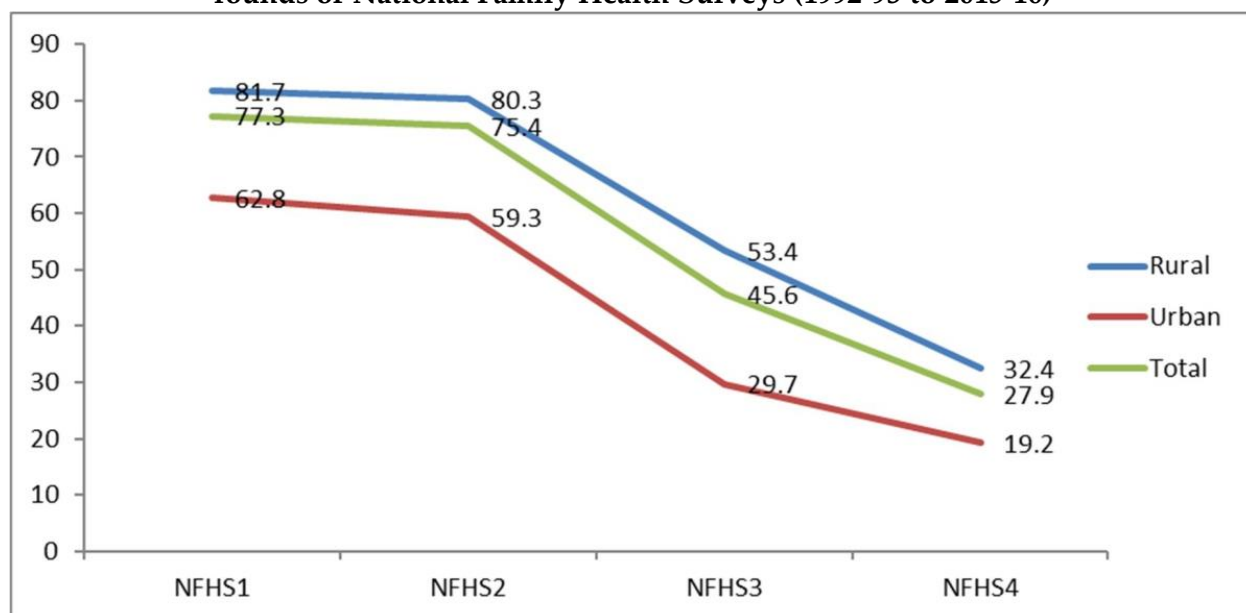
There are different indicators used to measure the child marriages. Using census data, some studies have used percentage of girls married before age 18 among those who married 0-4 years prior to the Census ((Kumar, 2019; Xavier et al., 2019) as an indicator to measure the child marriages. The other measures used in National Family Health Survey are the proportion of 20–

24-year-old women who married before age 18. Each of these measures provides different levels of child marriages. In this study, the indicator used to measure child marriage is the proportion of 18–29-year-old women married below age 18.

The figure 1 shows the trends in child marriages in India over a period of last 25 years. The figure also shows the levels for rural and urban areas separately. The figure clearly shows that child marriages in India have reduced by nearly one third during this two and half decade's time (from 77 percent in 1991-92 to 28 percent in 2015-16). The pace of decline was sharper during NFHS 2 and 3. The rural urban disparities in child marriages are also wide with higher proportion of rural women marry very early as compared to urban women. In the year 2015-16, 32 percent of the rural women married below age 18 whereas this proportion is 19 percent in urban India. In other words one in every 3 women in rural India and one in every 5 women in urban India gets married before age 18, which is a cause of concern.

Table 1 presents the variations in child marriages across selected states by rural and urban areas. It is clear from the table that in India as well as in states, rural women are at the disadvantageous side as they marry early compared to urban women and similar trend is noticed for all the time periods. In 2015-16, the state with highest proportion of child marriages is West Bengal (43.6) and the lowest is Kerala (9.3 percent). In Bihar also the proportion of child marriage is above 40 percent and in Punjab it is below 10 percent.

Figure 1 Percentage of women married at age 18 or before for first four rounds of National Family Health Surveys (1992-93 to 2015-16)



Results also brought out that in the first round of NFHS, all the states, except, Kerala, Punjab and urban areas of Jammu and Kashmir, more than half of the women aged 18-29 were married at below age 18. In many states such as Andhra Pradesh, Assam, Bihar, Haryana, Karnataka,

Madhya Pradesh, Maharashtra, Odisha, Rajasthan, Uttar Pradesh and West Bengal, more than three-fourth of the rural women were married before the age of 18. Though percentage is slightly lower in urban areas, the similar trend could be observed.

The percentage of women marrying before the age of 18 declined gradually in the consequent two rounds of NFHS and a significant decline in the percentage marrying below age 18 between round 3 and round 4. Overall, one could observe a drastic decline in the child marriages over a period of time. However, it is noticed that in spite of the drastic decline, still many states (Andhra Pradesh, Assam, Bihar, Gujarat, Haryana, Jharkhand (bifurcated from Bihar), Karnataka, Madhya Pradesh, Maharashtra, Rajasthan, Telangana (bifurcated from Andhra Pradesh), Uttar Pradesh and West Bengal) have more than one-fourth of women marrying at or below age 18. Percentage of women married before the age of 18 from round 1 of NFHS (1992-93) to round 4 of NFHS (2015-16) has declined significantly, still half of women in rural areas of West Bengal and about 30 percent of women in urban areas are getting married before the age of 18.

**Table 1 Percentage of women age 18-29 who married below age 18 years
across rural and urban areas (1992-93 to 2015-16)**

States	NFHS 1			NFHS 2			NFHS 3			NFHS 4		
	Rural	Urban	Total	Rural	Urban	Total	Rural	Urban	Total	Rural	Urban	Total
India	81.7	62.8	62.8	80.3	59.3	75.4	53.4	29.7	45.6	32.4	19.2	27.9
Andhra Pradesh	93.8	73.9	88.3	90.8	72.4	86.4	62.9	73.4	56.2	38.9	30.4	36.2
Assam	80.9	61.3	78.8	71.9	47.8	70.2	40.9	25.7	38.2	34.3	22.0	32.6
Bihar	84.3	69.3	82.3	84.9	69.2	83.2	68.6	37.8	63.7	43.6	31.0	41.9
Chandigarh	-	-	-	-	-	-	-	-	-	-	12.7	12.7
Delhi	63.9	59.2		69.5	47.2	49.5	53.4	22.6	45.6	10.6	17.2	17.2
Gujarat	69.5	48.7	62.5	73.1	51.4	64.7	40.9	28.2	35.4	29.4	18.2	24.5
Haryana	82.2	59.7	76.7	76.7	54.2	71.1	45.2	32.9	41.4	21.9	21.9	21.9
Jammu and Kashmir	58.3	32.9	54.6	59.1	40.4	55.8	18.6	9.2	16.1	11.1	4.2	9.2
Jharkhand	-	-	-	-	-	-	70.1	32.7	60.2	44.5	24.9	39.2
Karnataka	84.6	67.4	79.2	81.5	59.3	74.4	48.8	29.7	41.0	29.3	19.3	25.0
Kerala	49.1	45.6	48.2	47.4	31.0	44.0	20.1	12.1	17.2	9.9	8.6	9.3
Madhya Pradesh	89.6	70.2	85.7	90.0	68.0	84.8	60.4	33.7	52.6	38.5	20.3	33.0
Maharashtra	87.8	69.1	80.5	85.1	65.9	77.5	51.8	29.2	40.2	32.1	19.6	26.0
Odisha	77.7	69.9	76.1	70.6	60.5	69.6	39.5	27.8	37.5	23.6	18.9	22.8
Punjab	49.7	42.0	47.7	47.1	28.0	42.1	22.9	19.4	21.6	9.5	10.4	9.8
Rajasthan	86.1	72.1	83.7	87.9	67.0	83.2	67.4	36.3	58.4	36.6	21.8	32.9
Tamil Nadu	69.9	52.0	63.6	60.4	46.6	55.7	29.5	20.7	25.2	20.1	15.3	17.7
Telangana	-	-	-	-	-	-	-	-	-	40.1	21.2	30.8
Uttar Pradesh	83.2	59.9	79.0	86.9	62.6	82.7	59.4	31.8	52.2	26.0	13.1	22.5
Uttarakhand	-	-	-	-	-	-	29.5	16.7	26.0	17.8	14.1	16.4
West Bengal	85.6	65.7	80.9	80.5	57.1	75.9	62.3	31.3	53.3	50.0	29.6	43.6

The pace of overall decline in child marriages across states show that the decline was more than 50 percentage points in states like Haryana, Maharashtra, Orissa, Karnataka and Andhra Pradesh and the pace of decline was least in West Bengal. Further, the pace of decline was higher in rural areas than in urban areas. The pace decline is highest in rural Maharashtra (56 percentage points) and lowest in rural West Bengal. Similarly, in the urban areas the pace of decline is highest in Rajasthan (50.3 percentage points) and lowest in Gujarat (30.5 percentage points).

Geographic Variations in Child Marriages

The geographic variations in child marriages across states of rural, urban and total India is presented in figure 2, 3 and 4

Figure 2 Percentage of women married below age 18, India, 2015-16

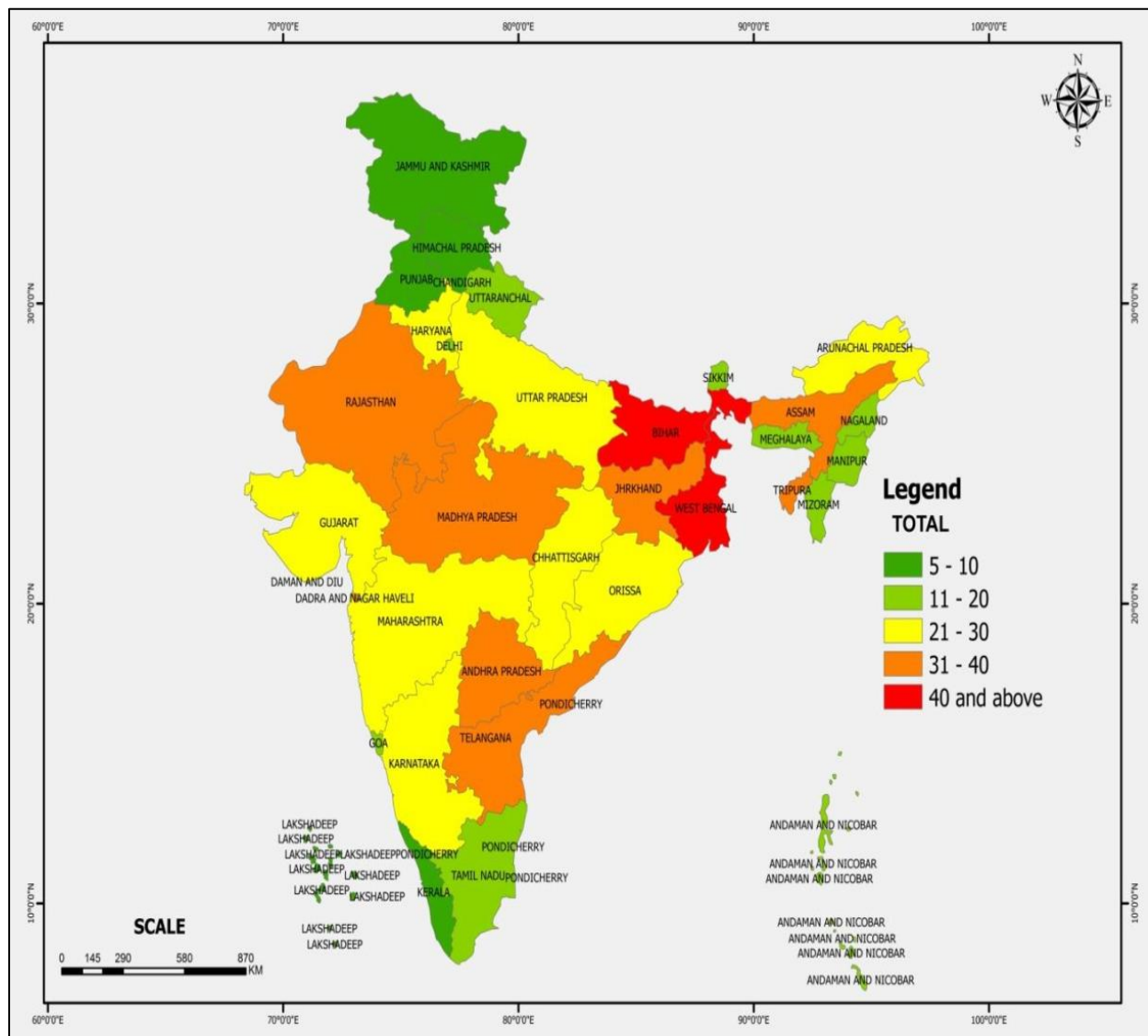


Figure 2 shows that the proportion of women marrying below age 18 is highest in eastern India particularly in states of Bihar, West Bengal and Jharkhand and lowest in Kerala, towards extreme south and Jammu and Kashmir, Himachal Pradesh and Punjab towards extreme north.

The map of Urban India shows that only in Jammu and Kashmir, the proportion of child marriages is less than 5 percent. Child marriage is more than 21 percent in urban areas of Rajasthan and Haryana towards north, Bihar, Jharkhand and west Bengal towards east, Assam, Arunachal Pradesh and Tripura towards north east and Andhra Pradesh, Telangana and Pondicherry towards south.

The map of rural India shows that the child marriage proportions are lowest in Kerala, Goa, Punjab and Himachal Pradesh and highest again in the eastern part of India in Jharkhand, Bihar and West Bengal.

Figure 3 Percentage of women married below age 18, urban India, 2015-16

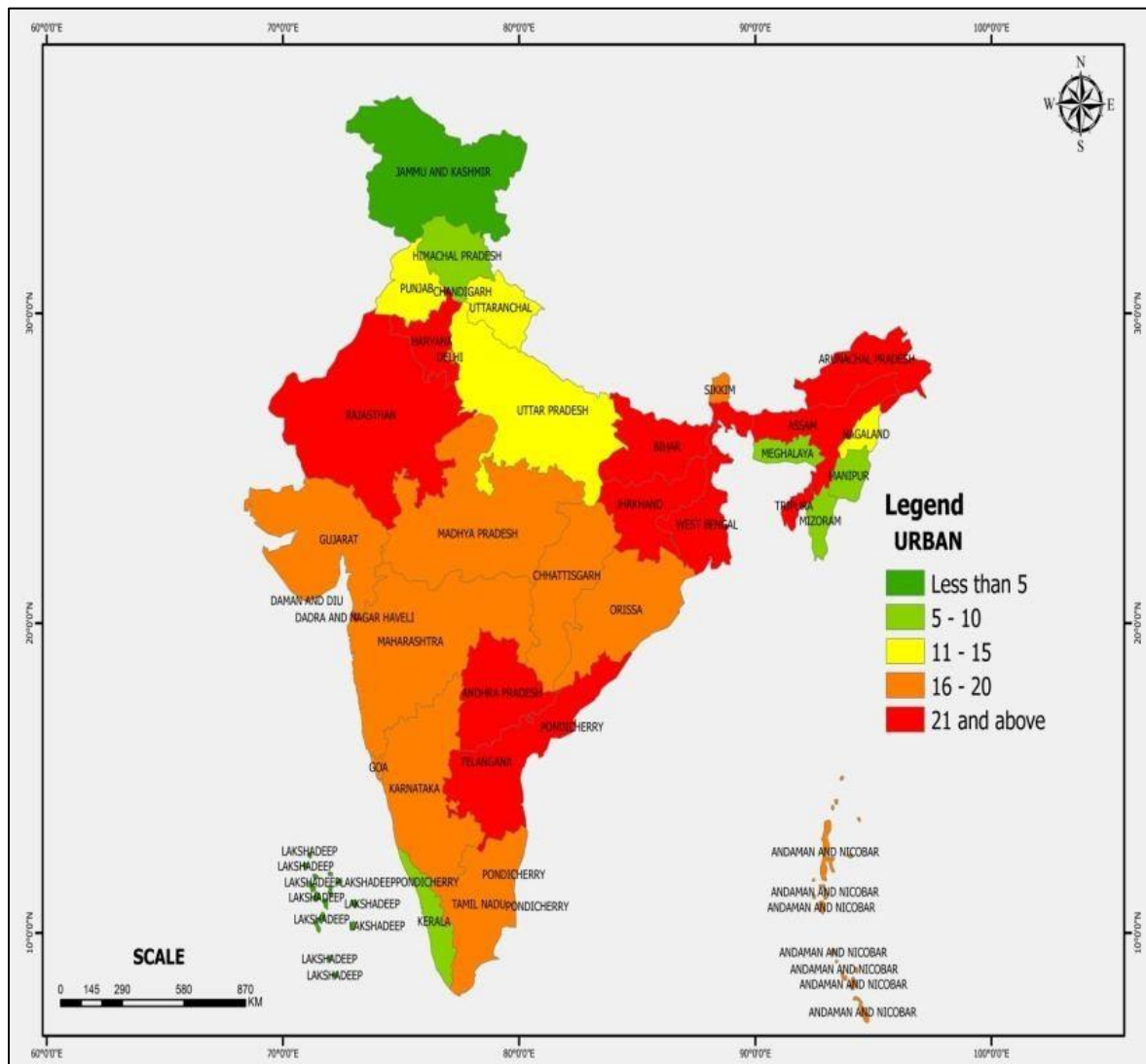
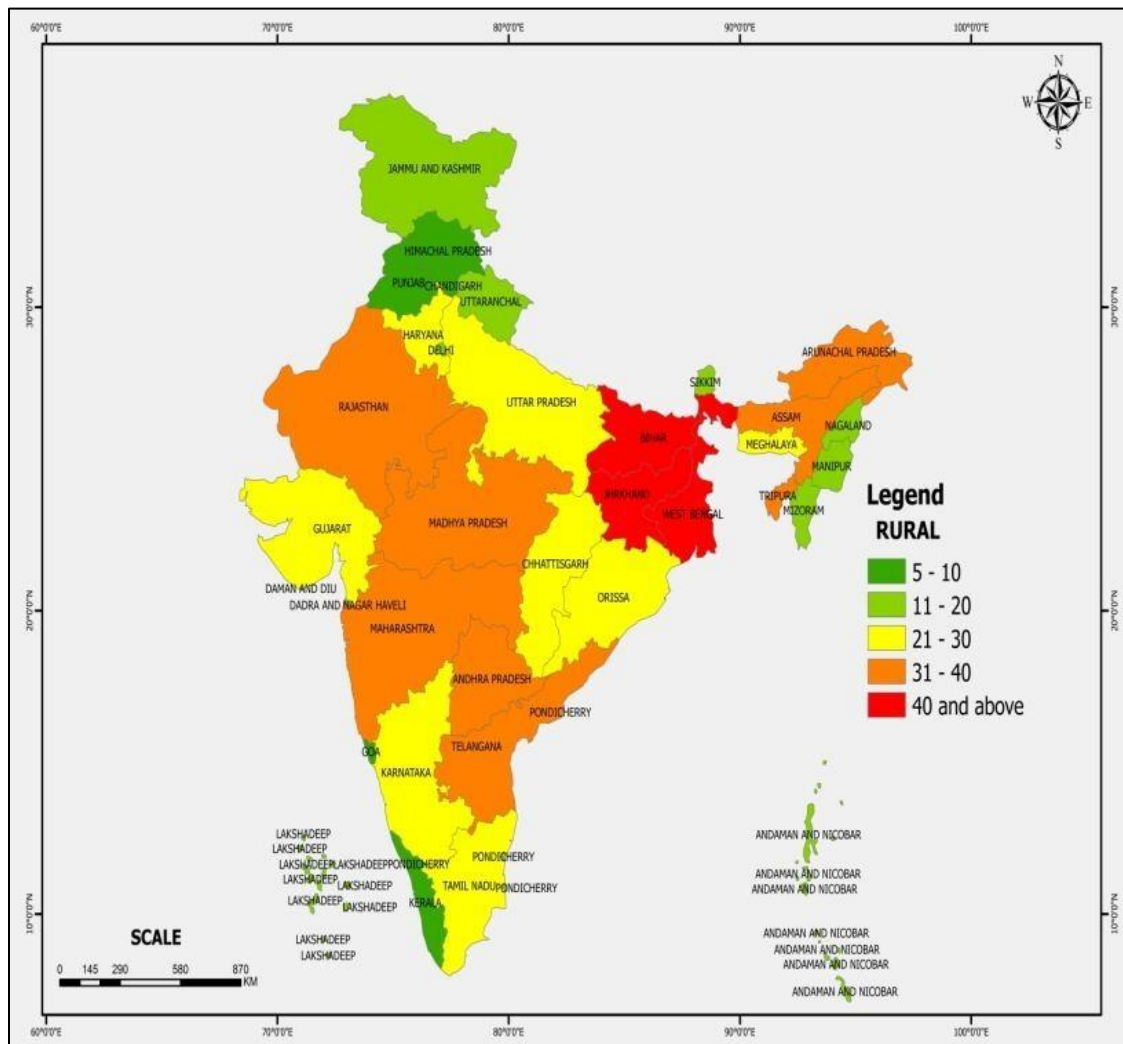


Figure 4 Percentage of women married below age 18, rural India, 2015-16



Scio-Economic Factors and Child Marriages

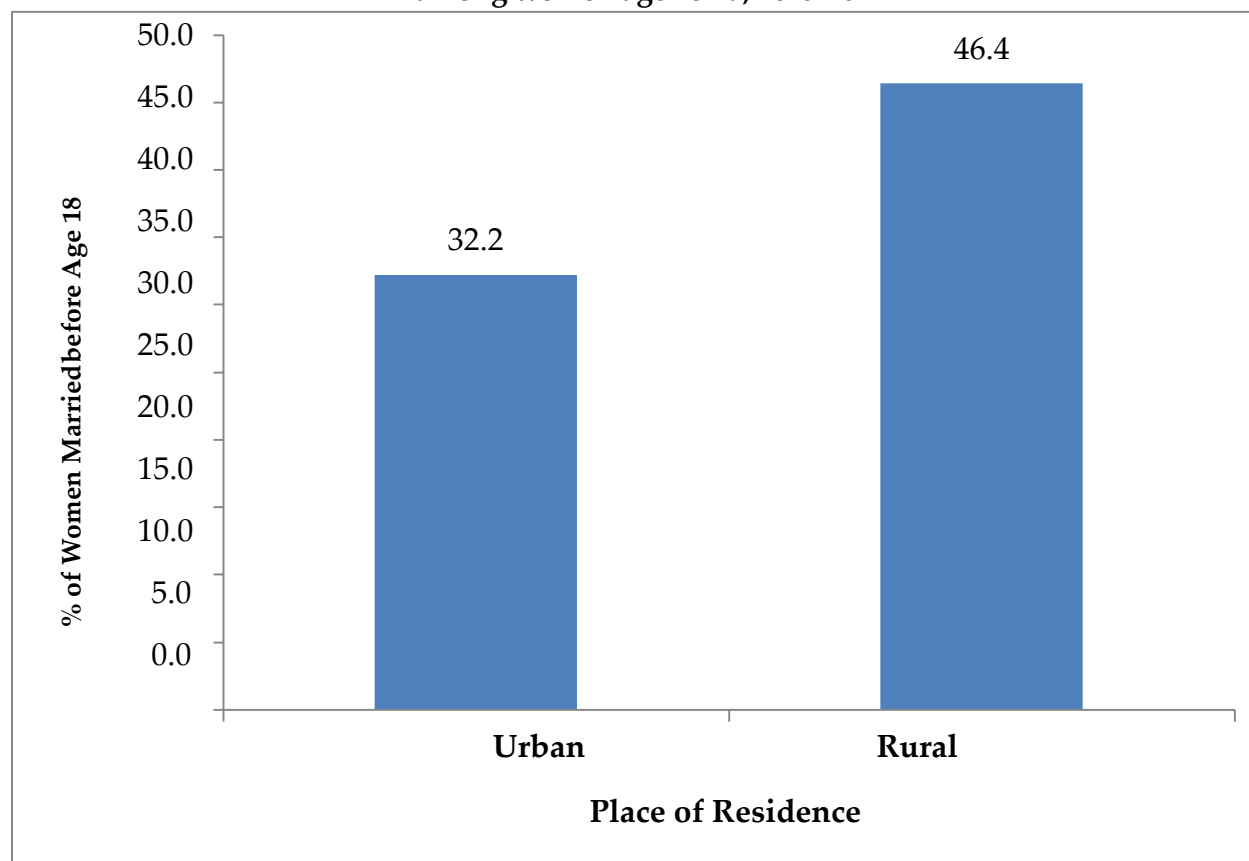
Marriages are nearly universal in India and the institution of marriage is still very strong despite rapid levels of modernization. Only 12.1 percent of women remain unmarried by age 29 and this percentage of 1.8 when they reach the age of 35 years. This pattern is more or less same with different states with minor variations. However, the timing of marriage varies widely across states and also across different socio-economic groups. Child marriages are often associated with socio economic background of the women. It is generally observed that women belonging to poorer households, those who have the least education and those from rural areas marry at an early age than their counterparts. Since child bearing in India occurs within marriage, an adolescent who gets married soon gets pregnant and becomes mothers. Data from the Indian Census 2011 shows that around 30 percent of women aged 15-19 years who are ever married are already mothers and 10 percent have had at least two children. Hence, child marriages have serious consequences on the health of the women.

In this section, an attempt has been made to understand the influence of socio-economic factors on child marriages. The major socio-economic factors considered are place of residence, religion, caste, household economic status, women's educational attainment and current age of the women.

Rural Urban Differentials in Child Marriages

Rural Urban Differentials in Child Marriages are presented in Figure 5.

Figure 5 Rural urban differentials in child marriages among women age 18-29, 2015-16



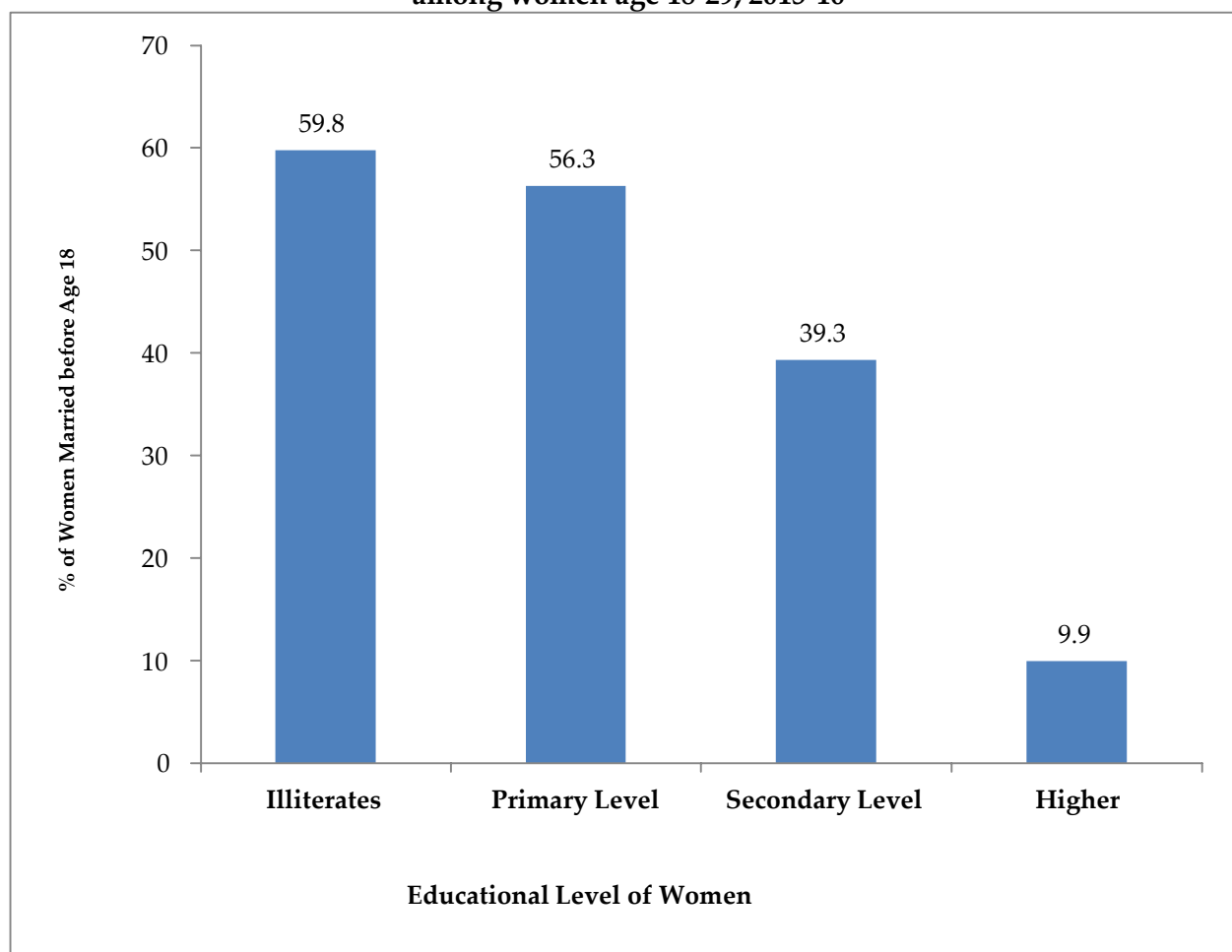
As indicated in the previous section, the incidence of child marriages is much higher among rural areas than that of urban areas. What is surprising here is that even in urban India, one-third of the women age 18-29 got married before attaining the legal age of 18 years. Hence child marriage is a concern both in rural and urban India.

Women's Education and Child Marriages

The relationship between education and early marriage is bidirectional. Generally, girls who are not in schools are likely to get married early. At the same time, marriage itself may be hindrance to girls' education. Many community-based studies have shown that marriage is one of the reasons for school dropout among adolescent girls. Therefore, policies to promote girls'

education are likely to reduce the incidence of early marriages. Figure 6 shows that education and early marriages are negatively correlated. The proportion of child marriages among the women who are illiterate is as high as 60 percent. This proportion reduces to 39 percent when they are educated up to secondary and 10 percent if they have higher level of education. This clearly shows that promotion girl's education can bring down the incidence of child marriages. Therefore, the country needs to invest in promoting girls' education and also in making education accessible and affordable to girls.

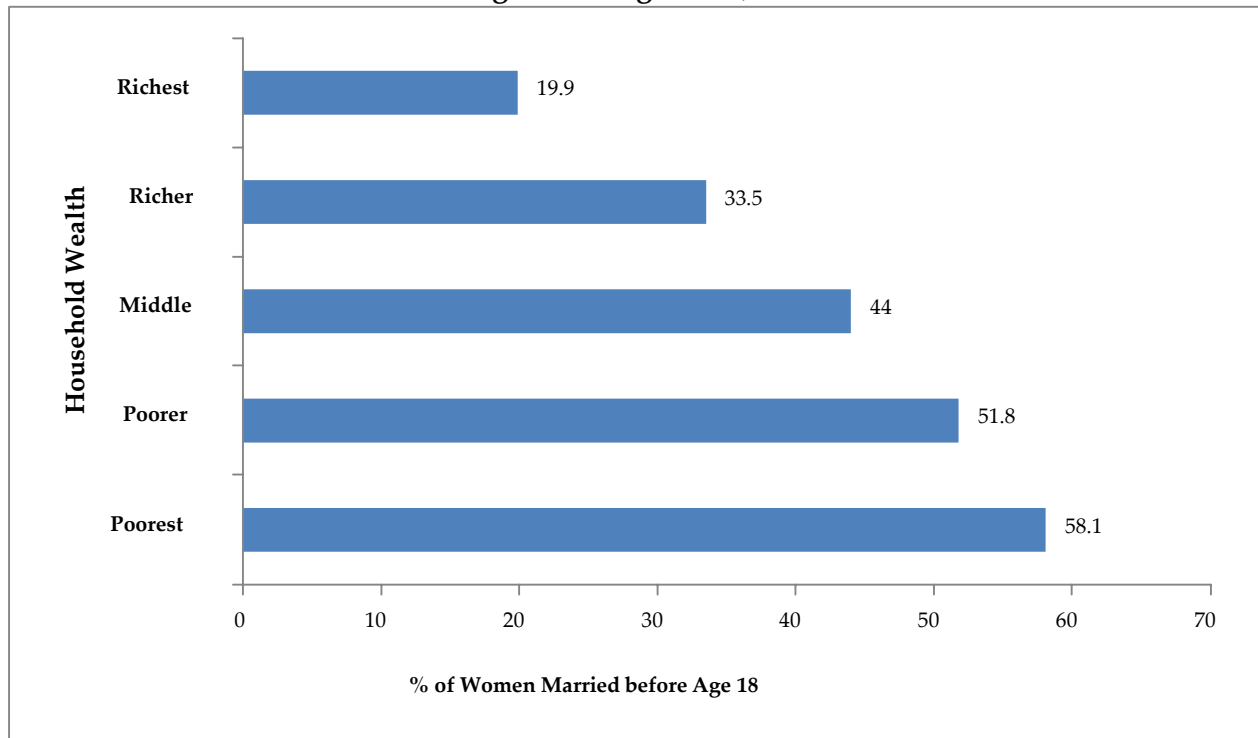
**Figure 6 Educational differentials in child marriages,
among women age 18-29, 2015-16**



Household Economic Status and Child marriages

Poverty is yet another reason for early marriages. Although, the practice of dowry is prohibited since 1961 it continues to exist in Indian societies. Dowry perpetuates child marriage as it encourages parents to marry off their girls early to avoid an increase in the dowry amount. Parents of poor household generally considers girl as economic burden and they tend to reduce their burden by marrying them at an early age. Figure 7 provides the proportion of child marriages across household wealth quintile.

Figure 7 Household economic status and child marriages among women age 18-29, 2015-16



This figure clearly shows a negative correlation between household economic status and early marriages. The proportion of women married before age 18 is nearly triple among women belonging to poorest household than women from richest households. Although the proportion of child marriages are declining across higher wealth quintiles, even among the richest quintile 20 percent of girls gets married before age 18 is a serious concern.

Child Marriages Across Religious and Caste Groups

Marriages in India are governed by a host of customs and traditions and these customs vary across religious and caste groups. The marriage age also varies across religion and caste. It is often mentioned that religion and caste are one of the determinants of marriage age, when it comes to child marriage, it is not only confined to one particular religion, and it cuts across all religious groups.

Figure 8 shows that proportion girls married below age 18 is highest among Muslims (43.1 percent) followed by Hindus (42.6 percent). Similarly, the child marriages are highest among girls belonging to Scheduled Caste or Scheduled Tribes (46.3 percent) than the other caste groups (Figure 9).

Figure 8 Child marriages among women age 18-29 across religious groups, 2015-16

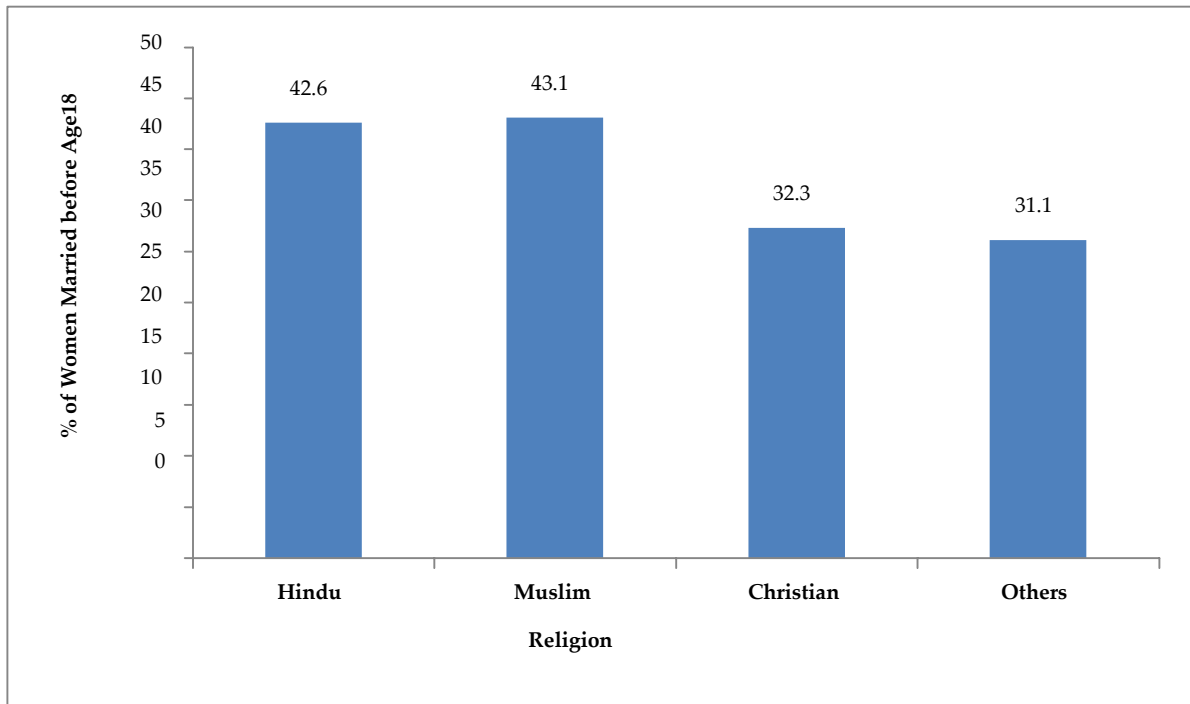
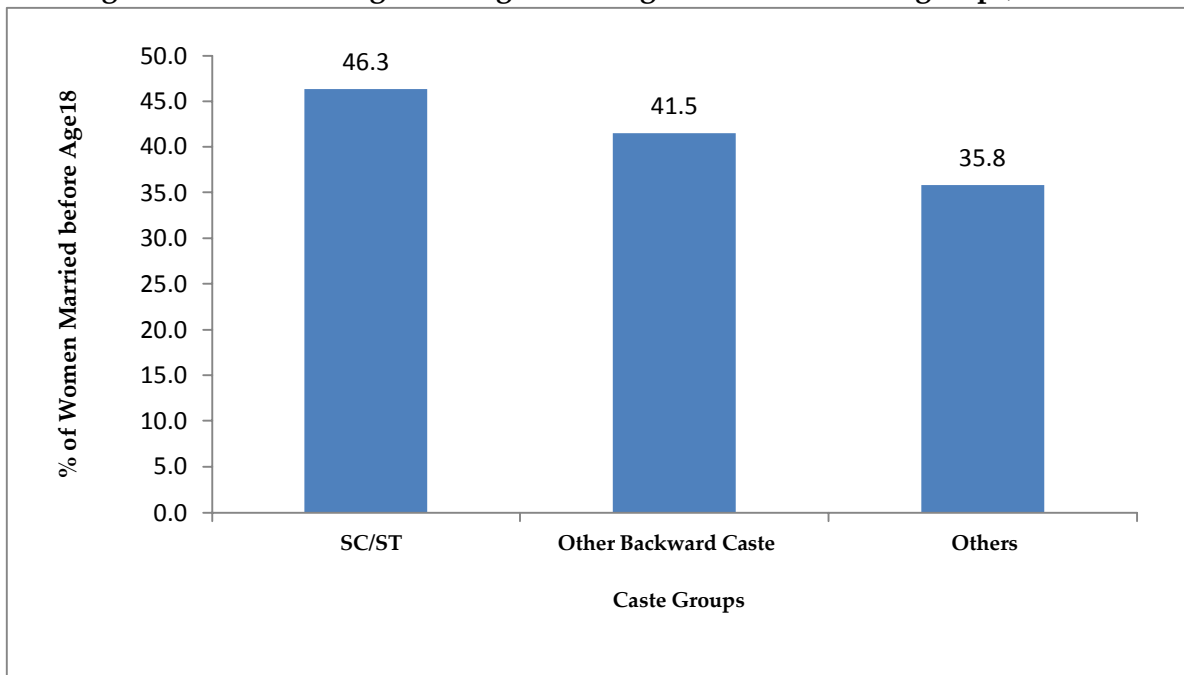


Figure 9: Child marriages among women age 18-29 across caste groups, 2015-16



Drivers of Child Marriage in India: A Logistic Regression Analysis.

The previous section has clearly shown that the child marriages are higher among women who belong to rural areas, those with least education and also among poorer households. It also

varies across different religious and caste groups. However, without controlling the effect of background variables, these are only gross differences. In order to quantify the net effect of the socio-economic factors on child marriages, a logistic regression has been carried out. Since the dependent variable in the present analysis is dichotomous (whether a woman age 18-29 married before age 18 or not), logistic regression is an appropriate technique to assess the influence of explanatory variables on the dependent variables. The explanatory variables included are place of residence (with rural as the reference category), Religion (with Hindu as the reference category), caste with SC/ST as the reference category), education of women (with illiterate as the reference category) and household wealth index (with poorest as the reference category). The results of logistic regression analysis is presented in Table 2.

Table 2 Logistic regression analysis of child marriages among women age 18-29, 2015-16

Independent Variables	Place of Residence	Coeff	Odds Ratios
Urban (Ref.)			
Rural		0.008	1.007
Religion			
Hindu (Ref.)			
Muslim		-0.175	0.840***
Christian		-0.536	0.585***
Others		-0.618	0.539***
Caste			
SC/ST (Ref.)			
Other Backward Caste		0.101	1.106***
Others		-0.060	0.942***
Educational Level of Women			
Illiterates (Ref.)			
Primary Level		-0.205	0.814***
Secondary Level		-0.818	0.442***
Higher		-1.967	0.140***
Wealth Index			
Poorest (Ref.)			
Poorer		0.001	1.001
Middle		-0.111	0.895***
Richer		-0.243	0.784***
Richest		-0.459	0.632***
Coefficient		0.540	1.716***
Loglikelihood		588495.944	
Chi-square Value		49090.732	
significant Level		0.000	
Number of Women		499627	

The results clearly show that after controlling the effect of all the socio-economic factors, all variables except place of residence are statistically significant in determining child marriages. Hindu women are more likely get married before age 18 than Muslims or Christians. The most important drivers of child marriages appear to be women's education and household economic status. Women who have higher levels of education are significantly less likely to marry below 18 compared to their illiterate counterparts. Similarly, women belonging to economically better off households are less likely to marry before age 18 than the poorer households.

Conclusion

Marriage is one of the important social institutions. This social process is expressed in the form of rituals and practices followed in the society. In India, marriage related decisions are usually taken by the parents, particularly on issues related to whom to marry and when to marry. In many communities' girl child has not been considered as an asset but they have been viewed as a liability. Therefore, girls are married off early due to variety of reasons. Dominant notions of morality and honour are important factors encouraging the practice of child marriages. Although the proportion of child marriages has declined over the years, it is still a common practice in many societies. Poverty and lack of education are the two main drivers of child marriages. Child marriage is widely recognized as a violation of human rights and it continues to deprive girls of their basic and equal rights to health, education and development.

There have been concerted efforts from the government to curb the child marriage in India and the age at marriage was raised from 15 to 18 years for the girls and from 18 to 21 for the boys under The Child Marriage Restraint Act, 1978. Despite continuous efforts from the government, still a significant proportion of girls marry at lower ages or below the minimum legal age of marriage specified by the government. Analysis of the present study clearly shows that in India, although child marriages have declined over time, it is still common and one out of every three women age 18-29 are being married off before they attain the age 18. Although the risk of child marriage is much higher in certain sections of the society, it cuts across all sections of the society. The incidence of child marriages also varies widely across states. In states of west Bengal, Bihar and Jharkhand, the proportion of child marriages are hovering around 40 percent whereas in states like Kerala and Punjab, this proportion is below 10 percent. Child marriage is also persistent among uneducated, poor and rural women. Girl's education beyond secondary levels brings considerable reduction in child marriages. The incidence of child marriage is almost double for girls who are from the poorest households as compared to the richest households. Similarly, considerable reduction in child marriages is also seen among girls who have secondary and higher levels of education. Religious and caste differentials are also pronounced with lower incidence of child marriages among Christians and also among other caste groups.

Hence, there is a need to effectively tackle the issue of child marriage with a stricter implementation of laws. Further, increasing the literacy levels and making education more accessible to girls could also be yet another strategy to eliminate child marriages. There is need to educate people regarding the ill effects of marriages and the repercussion on their overall development.

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2

Prevalence of Anemia among Children and Its Covariates in Districts of India: A Geospatial Analysis

Koustav Ghosh and Gayatri S Desai

Abstract

Prevalence of anaemia among children has been a major health problem in developing countries. It is one of the major causes of concern for the health programmers in India. In India the prevalence of anaemia among under five children has been reduced by only 11 % from 2005-06 to 2015-16. The present study has utilised NFHS-4 (2015-16) data to show the spatial prevalence and clustering of anaemia in the State/UTs and districts of India, and to show Spatial Correlation between childhood anaemia and its Co-variates, among 5-59 months' children. ArcGIS 10.8 and GeoDa software with Moran's I technique was utilised. In India, around 58.65% of children < 5 years are living with anaemia. Anaemia prevalence is the highest in the UTs of Dadra & Nagar Haveli and Daman & Diu. It is lowest in Meghalaya followed by Maharashtra. Around 52% of districts mostly from central, south-west, and northern and eastern parts of India, are having more than the national level of anaemia. There is a medium to high level of clustering of anaemia in the districts (Moran's $I=0.31$). A total 110 districts have been identified as a hot-spot area from MP, Gujarat, UP, Bihar, Jharkhand and Jammu & Kashmir and 105 districts shows < 0.05% level of significance. Through the Bivariate Moran's I value, it was found that spatial autocorrelation exists between childhood anaemia with certain other Co-variates. It was found to be the highest with mother anaemia (Moran's $I=0.39$), followed by underweight condition of children (Moran's $I=0.38$), illiteracy of mother (Moran's $I=0.31$), childhood diarrhoea (Moran's $I=0.15$), and poor household condition (Moran's $I=0.04$). The study shows remarkable geographical variations in childhood anaemia and its correlates in the country. It recommends that the public health programmes need to target the hot spot districts with high anaemia prevalence as well as co-variates of childhood anaemia. The risk of anaemia can be reduced by increasing mother's education and anaemic status along with improving children's condition. Programmatic measures to improve nutritional status and occurrence of diarrhoea among children can reduce the risk of anaemia among children.

Introduction

Prevalence of anaemia among children has been a major health problem in developing countries. It is one of the major causes of concern for the health programmers in India. It is the result of lack of nutritional intake of iron, folates, vitamin A, B12 and other micro-nutrients (Sharma et.al. 2020). Among pre-school children it is also caused by other factors such as vitamin A and vitamin B12 deficiencies, and hookworm and malaria infections (Pasricha et al., 2010; Dey et al., 2013). Iron deficiency anaemia has been reported to have many harmful effects among pre-school children as it reduces learning capacity, decreases attentiveness and causes low intelligence (Agaoglu et al., 2007;

Zhao et al., 2012). Moreover, WHO region-wise data (2008) shows that anaemia among children has also been indirectly linked to the developmental stage of a region. Around 90,000 maternal and neonatal deaths were due to iron deficiency anaemia in developing regions in 2013 (WHO, 2014). According to the World Health Organization, in 2015 more than 50% of people were leaving with anaemia in the South East Asia region, and about half of cases were caused by iron deficiency (WHO, 2015). Among the developing countries, during the last decade, India was the largest contributor to child anaemia (WHO, 2008; Pasricha et al., 2010).

Childhood anaemia was found to be a severe public health problem irrespective of urban residence, better wealth status and educational levels (Habte et al., 2013). The most recent rounds of National Family Health Survey (NFHS-4, 2015-16) estimated that 58% of children age 6–59 months were anaemic in India (haemoglobin levels below 11.0 g/dl), an improvement over the NFHS-3 estimate of 69%.

Under the economic development and the national anaemia control program there was no major reduction in the rate of anaemia. In India the prevalence of anaemia among under five children has been reduced by only 11% from 2005-06 to 2015-16. In view of the gradual decline of this morbid condition among children a district level intervention study need to identify the high prevalence area with particular Co-variates which may help to reduce the risk of anaemia in India.

Objectives

The overall aim of the study is to examine the distribution of anaemia and contextual risk factors among 5-59 months' children. The specific objectives are as follows-

- To show the spatial prevalence and clustering of anaemia (5-59 months of children) in the State/UTs and districts of India.
- To show Spatial Correlation between childhood anaemia and its Co-variates.

Materials and Methods

Data Source

The study considers National Family Health Survey (NFHS-4) conducted in the year 2015-16 in India. It has covered 35 states/UTs including 640 districts in India (IIPS & ICF, 2017). This national level large-scale survey conducted under the supervision of the Ministry of Health & Family Welfare (MoHFW), Government of India and Institute for Population Sciences (IIPS), Mumbai, was the nodal agency. Multistage stratified sampling design has been used and it provides the estimation on clinical, anthropometric and biochemical (CAB) components along with the prevalence of diabetes, hypertension, malnutrition, and HIV through the biometric measurement. Total 601,509 households were interviewed along with a response rate of 98% (over 90% in the case of every state and union territory), and 97% for eligible women (aged 15-49) and 92% for men aged 15–54 years.

In NFHS, 2015-16 haemoglobin testing was performed on all children age 6–59 months in the sample households using capillary blood. As per the standards of NFHS, children were considered as

anaemic when their haemoglobin level was below 11.0 g/ dL; as mildly anaemic with 10.0–10.9g/ dL, moderately anaemic with 7.0–9.9 g/ dL and less as 7.0g/ dL as severely anaemic.

Methodology

To understand the geographical variation and spatial clustering of child anaemia in the districts of India, the study uses spatial analysis software ArcGIS 10.8 and GeoDa. To determine the existence of clustering, spatial heterogeneity and spatial auto-correlation in prevalence of anaemia, Exploratory Spatial Data Analysis (ESDA) has been attempted. Geographical Information System (GIS) and spatial statistical technique based on ESDA are useful in describing and representing spatial distribution, detecting hot-spot and cold-spot areas and suggesting spatial regimes or other forms of spatial heterogeneity (Anselin, 1988). For hot-spot analysis in GIS or Geo-Da a scale is important to specify the distance value (Pravin et.al, 2020). The computation of spatial autocorrelation requires the construction on the base of matrix known as spatial weight matrix (W). We use Moran's I (spatial autocorrelation) and Local Indicators of Spatial Autocorrelation (LISA) to measure the extent of autocorrelation among the neighbourhood districts. Moran's I helps to measure the intensity of clustering (Z-score) with increasing distance. LISA statistics has been used to detect the hot-spot areas across a district considering the percentage of children suffering from anaemia (Kamath et al., in 2018). The LISA values consent the computation of its similarity with its neighbouring districts and also test the level of significance (p values) for each location. LISA map portrays results in five scenarios- cluster with high values (high-high or hot spot), cluster with low values (low-low or cold spot), outlier in which a high value is surrounded by a low value (high-low), outlier in which a low value is surrounded by a high value (low-high) and no significant local autocorrelation.

Bivariate Moran's I: examines the relationship between the exposure and the outcome variables for the geographic region of India.

Univariate Moran's I: It measures the correlation of neighbourhood values around a specific spatial location. It determines the extent of spatial non-stationery and clustering present in the data. STATA-14 has been used for analysis the data.

Outcome variable: Prevalence of anaemia among children is used as the dependent variable in the study. The dependent variable is dichotomous having 0 as "Not Anaemic" and 1 as "Anaemic".

Co-variates: A set of background characteristics has been used as co-variates based on existing literature and availability of data. The Co-variates which include % of children with experience of Diarrhoea (in last 2 weeks), % of children Underweight, % of Illiterate mother, % of poor household (household belongs from very poor and poor quantile considered as poor household) and % of anaemic mother.

Discussion

Prevalence of Anaemia Among Children (6-59 Months) with geographical variation: Table 1 represents the Geographical Variation of anaemia among children under-five years of age at the

state/UTs level in India (2015-16). At the national level, the prevalence of anaemia has been reported to be 58.65%. Fourteen states/UTs showed more than the national average of anaemia prevalence. The prevalence of anaemia is high in Dadra & Nagar Haveli (84.35%) followed by Daman & Diu (74.46%), Chandigarh (72.18%), Gujarat (71.84%), Jammu & Kashmir (70.1%). On the contrary, Meghalaya (19.63%), Maharashtra (23.96%), Mizoram (26.59%), Assam (23.88%), and Karnataka (35.95%) reported a low level of anaemia among children in India.

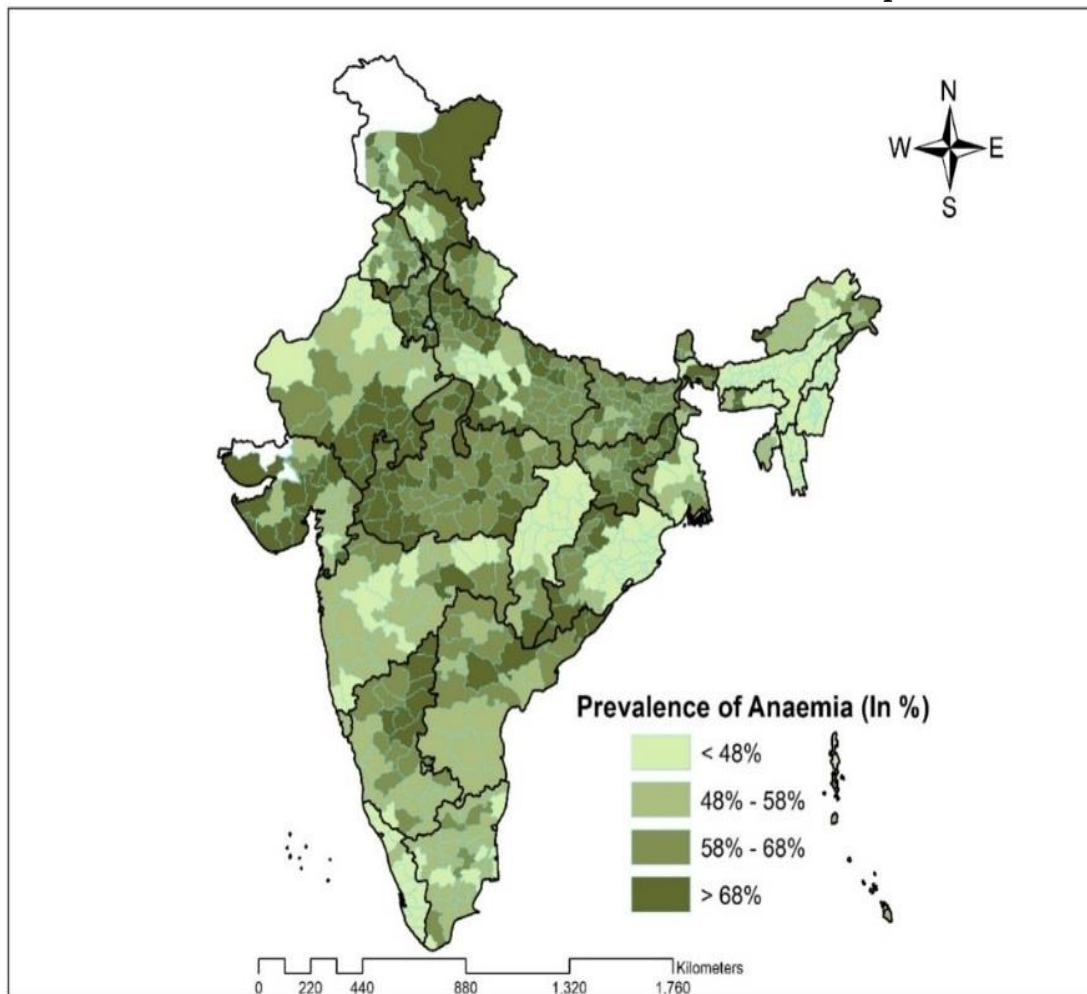
Table 1 Prevalence of children anaemia (6-59 months) by gender in the State/UTs in India, NFHS 2015-16

State/ UTs	Anaemia Prevalence (in percentage)	
	Total	Total Sample Size
Andaman & Nicobar Islands	50.27	44
Andhra Pradesh	58.5	6413
Arunachal Pradesh	54.38	162
Assam	35.88	4954
Bihar	63.54	26786
Chandigarh	72.18	122
Chhattisgarh	41.74	4994
Dadra & Nagar Haveli	84.35	63
Daman & Diu	74.46	20
NCT Of Delhi	48.35	1972
Goa	62.4	193
Gujarat	71.84	8621
Haryana	53.7	4923
Himachal Pradesh	54.16	923
Jammu & Kashmir	70.1	1878
Jharkhand	61.25	6337
Karnataka	35.95	8548
Kerala	52.61	3762
Lakshadweep	69.02	10
Madhya Pradesh	53.97	14514
Maharashtra	23.96	17669
Manipur	48.14	429
Meghalaya	19.63	626
Mizoram	26.59	183
Nagaland	59.86	264
Odisha	44.61	6957
Puducherry	44.9	192
Punjab	56.69	3668
Rajasthan	60.44	12535
Sikkim	55.54	51
Tamil Nadu	50.77	12460
Telangana	48.23	5260
Tripura	63.35	481
Uttar Pradesh	60.21	37428
Uttarakhand	54.47	1700
West Bengal	60.63	14354
India	58.65	209495

Figure 1 illustrates the spatial prevalence of anaemia in the districts (total 640) for India. 52% of districts (333 out of 640) in India have reported more than 58.65% (national

average) of prevalence of childhood anaemia. Spatial map shows that 26% (164) districts have reported <48% level of anaemia, and 49% of districts (316 districts) reported the range between 48% to 68% level of anaemia and the remaining 25% districts (160 districts) reported >68% prevalence of anaemia in the year 2015-16. Leh (95.8%) from Jammu & Kashmir and lahul and spiti (94.85%) from Himachal Pradesh show the highest prevalence of anaemia whereas, the Champhai (7.87%) and Aizwal (11.05%) in Mizoram and mokokchung (12.96%) in Nagaland represent a low level of anaemia prevalence. The high prevalence districts come from central, south-west, Northern and eastern part of India, mostly from Uttar Pradesh, Madhya Pradesh, Jharkhand, Bihar, Uttarakhand, Gujarat, Karnataka, Haryana, Panjab, and Jammu& Kashmir.

Figure 1 Spatial prevalence of anaemia among children (6-59 months) in India (2015-16): A district level map



Spatial clustering of anaemia in districts of India: Moran's Index statistics of spatial autocorrelation shows the prevalence of anaemia in the districts of India as a cluster

(Moran's I 0.307 in Figure 2-C) and the results are also highly significant (p value <0.001). Based on these results, the LISA map has identified the probable hot-spot area of childhood anaemia prevalence at the district level in India (depicted in Figure-2). In Figure 2, map A presents the clustering levels in the prevalence of anaemia, and map B depicts the significant levels of clustering. The hot spots (high to high) are shown in red colour indicating that districts with a higher prevalence of anaemia are surrounded by neighbouring districts with greater levels of anaemia. Contrarily, cold spots (low to low), given in blue colour, are the districts with lower levels of anaemia prevalence which also have similar neighbourhoods.

Table 2 Hot-spot districts based on prevalence of childhood anaemia from India, 2015-16

States	Hot-spot Districts
Madhya Pradesh	Ashoknagar; Jabalpur; Rajgarh; Shivpuri; Katni; Sehore; Mandsaur; Dewas; Chhatarpur; Sagar; Guna; Raisen; Panna; Gwalior; Ujjain; Narsimhapur; Vidisha; Satna; Neemuch; Bhind; Datia; Jhabua; Dhar; Khargone (West Nimar); Shajapur; Ratlam; Khandwa (East Nimar) ; Barwani
Uttar Pradesh	Budaun; Ghaziabad; Siddharthnagar; Bulandshahr; Shrawasti; Gonda; Bijnor; Meerut; Bareilly; Moradabad; Lalitpur; Saharanpur; Jhansi; Baghpat; Muzaffarnagar; Jalaun
Rajasthan	Chittaurgarh; Bhilwara; Kota; Pratapgarh; Rajsamand; Dungarpur; Baran; Jhalawar; Udaipur; Bundi; Banswara
Gujarat	Rajkot, Bhavnagar, Sabar Kantha, Gandhinagar, Surendranagar, Mahesana, Ahmedabad
Haryana	Yamunanagar; Sonipat; Panipat; Hisar; Panchkula; Jhajjar; Faridabad; Karnal; PalwalJind; Rohtak; Rewari
Jharkhand	Khunti; Deoghar; Hazaribagh; Ranchi; Bokaro; Giridih; Dumka; Godda
Bihar	Madhepura, Madhubani, Banka, Bhagalpur
Himachal Pradesh	Sirmaur; Kinnaur; Lahul & Spiti
Jammu & Kashmir	Bandipore; KargilLeh(Ladakh)
Karnataka	Bagalkot; Raichur; Yadgir
Maharashtra	Nandurbar
Odisha	Koraput, Malkangiri
Punjab	Sahibzada; Ajit Singh Nagar
Andhra Pradesh	Visakhapatnam
Uttarakhaand	Udham Singh Nagar

A total of 110 districts (Table 2) belong to the hot-spot areas which are mostly from Madhya Pradesh, Gujarat, Uttar Pradesh, Bihar, Jharkhand and Jammu & Kashmir. Whereas 89 districts are form cold-spot areas from the south and north-east and some

are from central parts of India. Level of Significance map represents that around 105 districts are reported in cluster with less than 5% significance level.

Figure 2 LISA cluster and LISA significance map of anaemia disease in districts of India, 2015-16

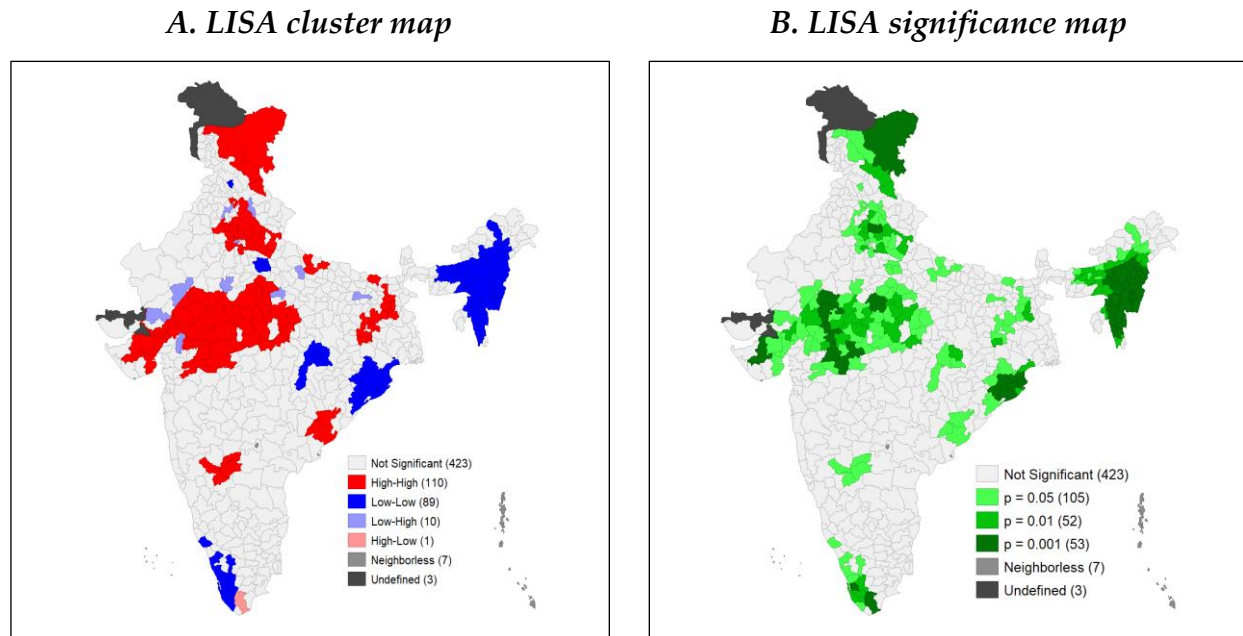
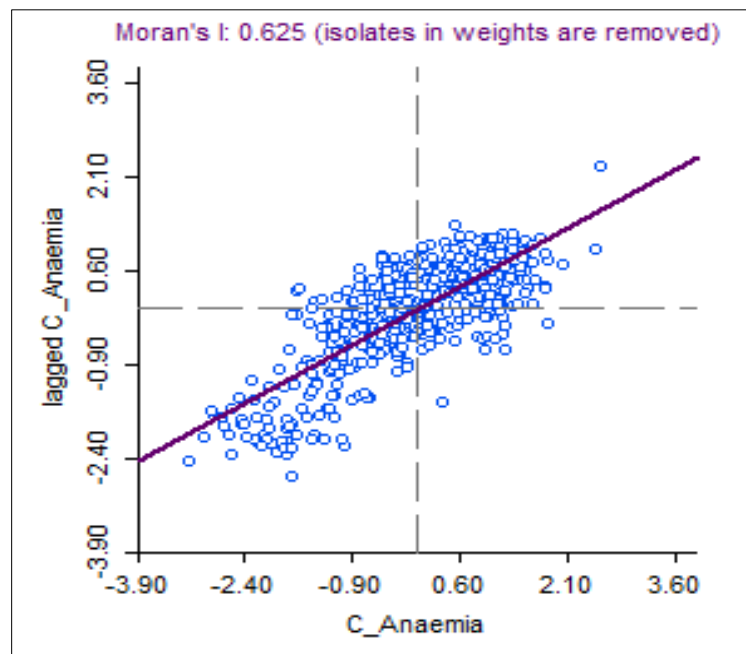


Figure 2(C) Moran's Index



Spatial Correlation between child anaemia and its Co-variables: Table 2 depicts the Bivariate Moran's I statistics for anaemia among children with some covariates in India

across 640 districts of India. With regard to Bivariate Moran's I value, it was found that spatial autocorrelation was highest with Childhood anaemia and Mother Anaemia (Moran's I: 0.39), followed by Underweight children (Moran's I: 0.38), Illiterate mother (Moran's I: 0.31), childhood diarrhoea (Moran's I: 0.15) and poor household (Moran's I: 0.04).

Table 3 Moran's I statistics showing spatial clustering of child anaemia with some co-variables across districts of India, NFHS 2015–16

Co-variables	B-Moran's Index	P-value	Hot-spot Districts (N)	Cold-Spot Districts (N)	Significance districts (p<0.05)
% Poor Quantile Household	0.04	<0.001	120	89	133
% of Illiterate Mother	0.31	<0.001	115	110	120
% of Anaemic Mother	0.39	<0.001	94	75	114
% of Child having Diarrhoea	0.15	<0.001	55	55	87
% of Underweight child	0.38	<0.001	145	104	154

Fig. 3 shows Bivariate LISA (Local Indicators of Spatial Association) maps that depict the spatial association between independent variables and dependent variable. The hot spots (high to high) are shown in red colour indicating that districts with a higher prevalence of anaemia are surrounded by neighbouring districts with greater levels of anaemia and others **Co-variables** (i.e., % of poor household; % of illiterate mother; % child having diarrhoea). Contrarily, cold spots (low to low), given in blue colour, are the districts with lower levels of anaemia prevalence which also have similar neighbourhoods.

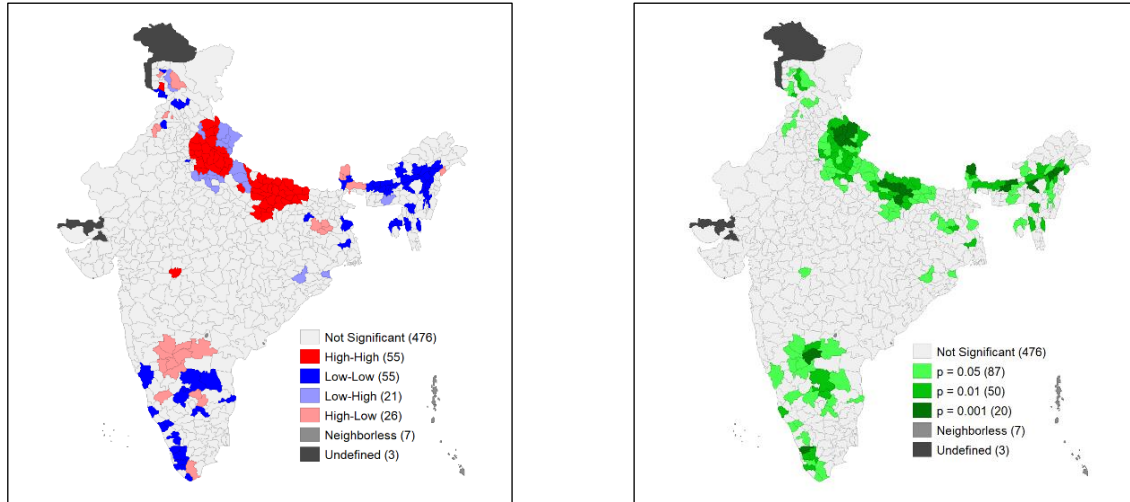
Figure 3 (A) & Figure 4 (A) shows the clustering between childhood anaemia and children having Diarrhoea. Result of Moran's I value (0.15) shows that auto-correlation exists between these variables. A total of 55 districts belong to the hot-spot areas which are mostly from central and northern (UP, Bihar, Himachal Pradesh & Uttarakhand) parts in India. Level of Significance map represents that around 87 districts are reported in cluster with less than 5% significance level.

Similarly Figure 3 (B) & Figure 4 (B) represents that spatial clustering exists between childhood anaemia and underweight children and identifies 145 districts as hot-spots areas which are mostly from central, eastern, and western, and some of the southern part of India. Level of Significance map represents that around 154 districts are reported in cluster with less than 5% significance level.

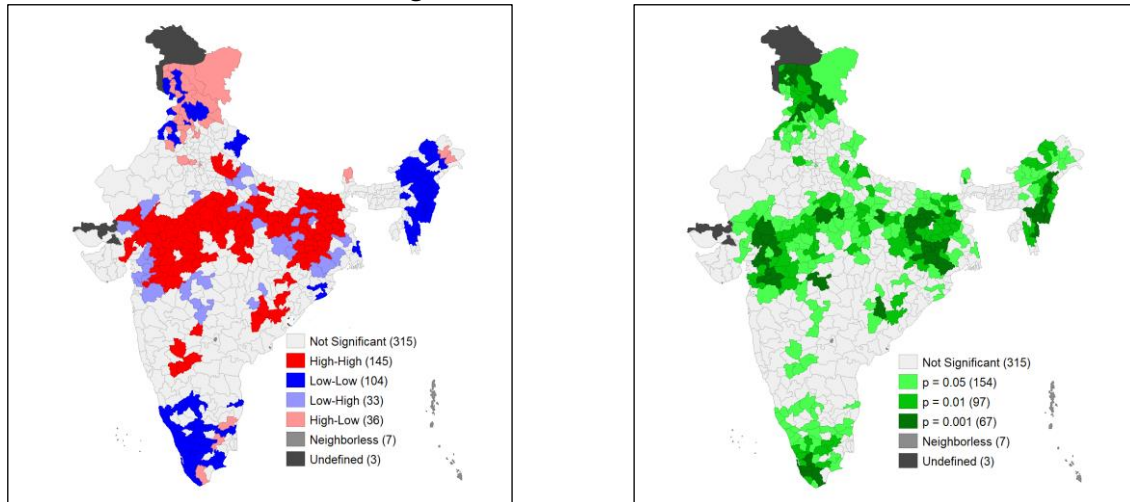
Figure 3 (C) & Figure 4 (C) shows that the auto-correlation exists with 120 hot-spots (mostly from central, eastern and western parts from India) and 89 cold spots between Childhood Anaemia & Poor Household. Level of Significance map represents that around 133 districts are reported in cluster with less than 5% significance level.

Figure 3 Bivariate LISA maps showing spatial correlation between child anaemia and its co-variates.

A. Children Anaemia Vs Children having Diarrhoea



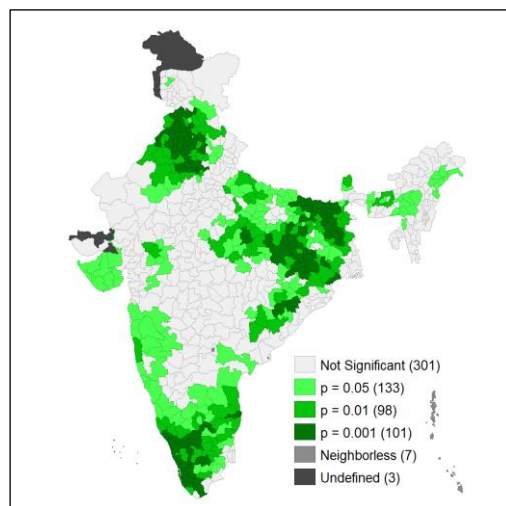
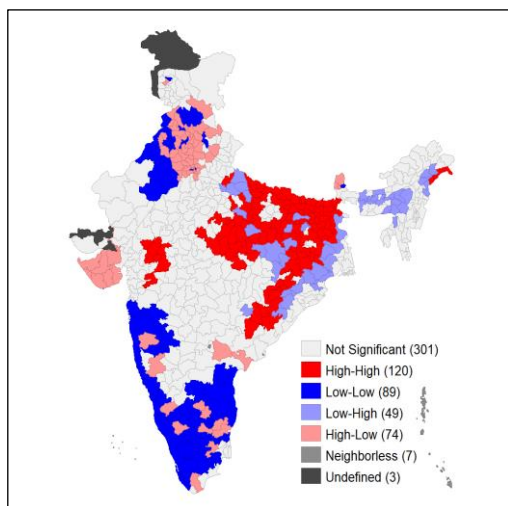
B. Children Anaemia Vs Underweight children



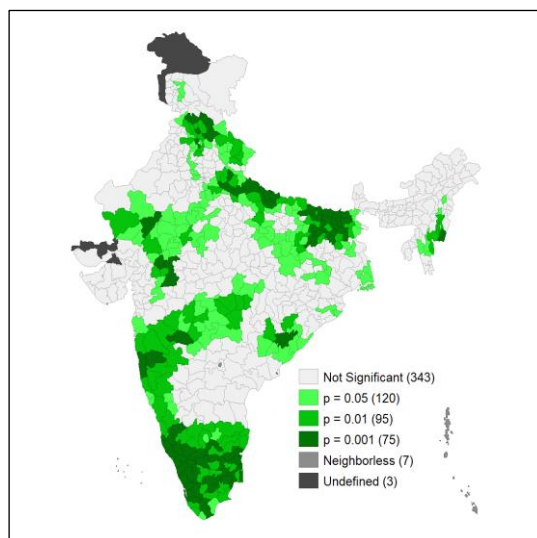
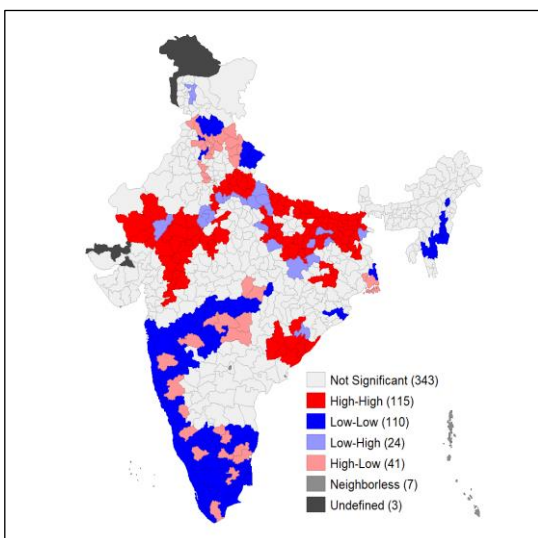
The auto correlation between childhood anaemia & mothers who are illiterate identify 115 hot-spots (mostly from east, west, central and southern parts of India) and 110 Cold-spot districts in the figure 3 (D) & Figure 4 (D). On the other hand, level of Significance map represents that around 120 districts are reported in cluster with less than 5% significance level.

The auto correlation is found between Childhood Anaemia & Mother Anaemia in 94 hot-spots (mostly from North, eastern, Southern and western part of India) and 75 Cold-spot areas in the Figure 3 (E) & Figure 4 (E).

C. Childhood Anaemia Vs Poor Household



D. Childhood Anaemia Vs Mothers who are Illiterate



E. Childhood Anaemia & Mother Anaemia

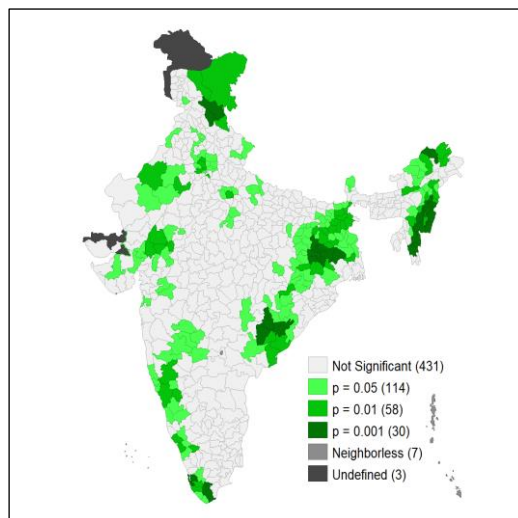
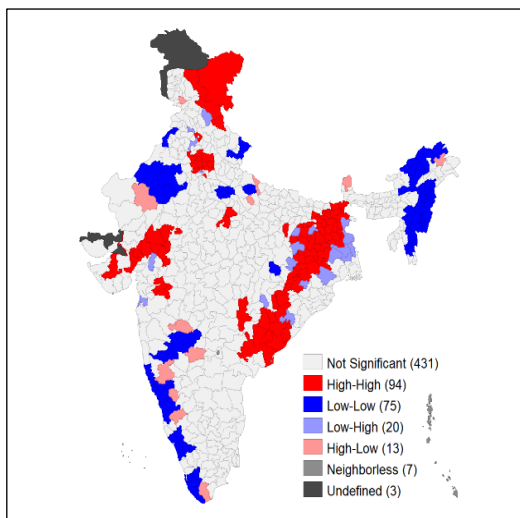
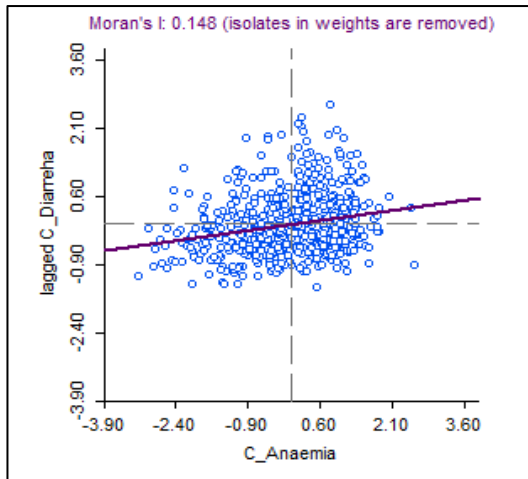
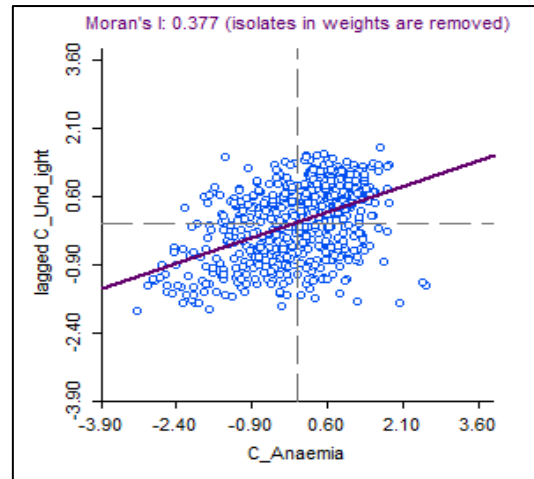


Figure 4 Bivariate Moran's Index for childhood anaemia with others co-variates

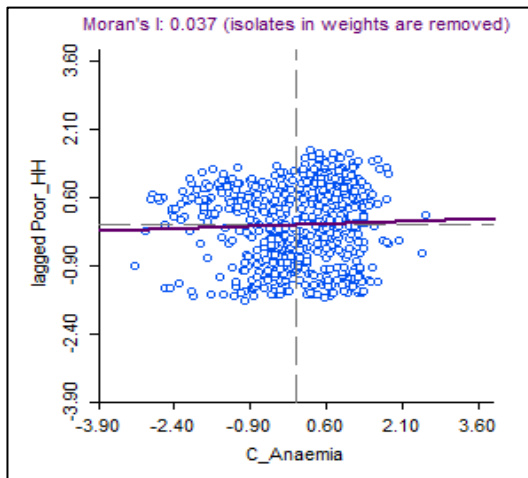
A. Children Anaemia Vs Children having Diarrhoea



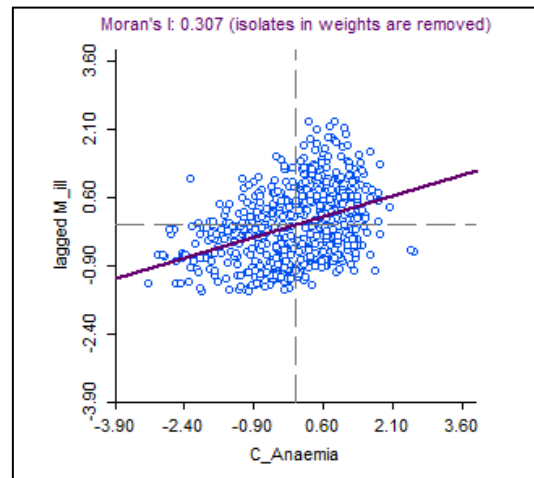
B. Children Anaemia Vs Underweight children



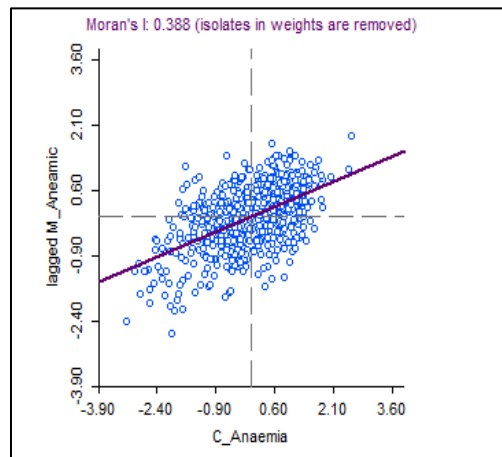
C. Childhood Anaemia Vs Poor Household



D. Childhood Anaemia Vs Mothers who are Illiterate



E. Childhood Anaemia & Mother Anaemia



Conclusion

In India around 58.65% of children less than 5 years are living with anaemia in 2015-16. The prevalence is highest in the Union territories of Dadra & Nagar Haveli and Daman & Diu. The reason for this is not clear, but it might be due to an improper diet (Zhang et al., 2018), childhood illness such as diarrhoea (Tariq et al., 2017), worm infestations (Seidelman et al., 2016) or lack of appropriate iron supplementation (Okam et al., 2017). On the other hand, Meghalaya (19.63%), and Maharashtra (23.69%) show a low level of anaemia among 6-59 months of children.

The present study reveals a remarkable geographical variation in the risk of childhood anaemia in the districts of India. The study finds that the prevalence of anaemia is the highest in central, south-west, and northern and eastern parts of India. Around 52% of districts are having more than the national level of anaemia. The present study also finds the medium to high level of clustering of anaemia in the districts with 0.31 Moran's Index, which means autocorrelation exists between neighbourhood districts in India. Total 110 districts have been identified as a hot-spot area from Madhya Pradesh, Gujarat, Uttar Pradesh, Bihar, Jharkhand and Jammu & Kashmir and 105 districts shows < 0.05% level of significance.

Through the Bivariate Moran's I value, it was found that spatial autocorrelation was highest with children anaemia and Mother Anaemia, followed by Underweight, Illiterate mother, childhood diarrhoea, and poor household. Among the maternal characteristics increase of mother's age and education reduce the risk of childhood anaemia. Whereas, anaemic mother increases the risk of anaemia. Underweight children increasing the risk of childhood anaemia (Bharati, et. al., 2020; Sharma et. al., 2020).

The present study does not only contribute in providing remarkable geographical variations in childhood anaemia in India but also examines the contextual co-variables factors affecting its occurrence. Based on these findings, the study recommends the public health programmers to target hot spot districts with high prevalence of anaemia. The risk of childhood anaemia can also be reduced by focusing on mother education and anaemic status as also on household condition. Furthermore, measures to improve nutritional status, underweight and diarrhoeal condition among children may help in reduction of anaemia. Moreover, districts specific measures are equally important to achieve the desired outcome.

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Section II
Maternal and Reproductive Health

3

Levels and Patterns of Hysterectomy in Punjab: An Empirical Analysis

Aswini Kumar Nanda and Rajesh Kumar Aggarwal

Abstract

The paper focuses on the State of Punjab in the north-west India and is based on the data from the fourth round of the National Family Health Survey (NFHS) conducted during 2015-16. It covers ever-married women aged 15-49 years on the basis of a few direct questions on hysterectomy. The general goal of the study is to unfold the dynamics of acceptance of hysterectomy in the State of Punjab in India. Using binomial logistic regression, the study uncovers those socio-economic factors significantly push women towards hysterectomy in the State of Punjab. Such factors include woman's current age, age at cohabitation, number of living children, etc. Findings of the study are expected to add to overall understanding of hysterectomy prevalence in the State of Punjab besides being of substantial academic and policy interest to the stakeholders.

Introduction

Since the time large scale data were made available for the first time in India in 2012-13 through the District Level Household and Facility Survey (DLHS-4), hysterectomies are seen to be rising in the country. Prevalence of hysterectomies, nationwide, increased from 1.7 percent during 2012- 13 to 3.2 percent during 2015-16. However, this consolidated national statistics tend to masks sharp regional variations that are vivid when the figures at the state-level are examined (Chander Sekar et al., 2019). Southern states of Andhra Pradesh and Telengana distinctively stand out as leaders—with much higher levels than the national average followed by Bihar in the east and Gujarat in the west. Acceptance of hysterectomy is comparatively less in the northern states namely Uttar Pradesh, Rajasthan and Himachal Pradesh.

Though the developing world in general has lower prevalence of hysterectomy as compared to its developed counterpart, yet, due to conservative approach, it is witnessing a slow but consistent decline in these surgeries (Domingo and Pellicer, 2009). Hysterectomies have been a source of controversy in many parts of the world, and have been hit by the issues of non-exploration of alternatives while before recommending hysterectomy, low-value care, sidelining of ethical concerns, surgery-for-profit by private health facilities, etc. Reports of increasing hysterectomy among the women workers in sugarcane farms, textile manufacturing for the fear of losing working hours, wage, and productivity and absence or distance of washroom facility

from place of work have raised concerns of basic reproductive rights (Narayana, 2020). Hysterectomies in India and elsewhere are viewed as totally unwarranted (Gaber 2019, Davis 2019). Additionally, hysterectomies in India need to be viewed in the perspective of the huge burden that they impose on already stressed health system in terms of availability of hospitals beds, skilled manpower, equipment's, drugs, consumables and other scarce resources.

Hysterectomy is a major intervention either through open surgery or through laparoscopy to remove uterus (womb) from the female body. It can be of different type depending on its procedures and method, such as: i) total abdominal hysterectomy, ii) vaginal hysterectomy, and iii) laparoscopy-assisted vaginal hysterectomy. Hysterectomy for non-cancerous reasons is usually considered only after all other treatment approaches have been exhausted without success. A commonly held view is that hysterectomy substantially improves quality of life and makes living easier for woman and her family (Thakar, 2004). However, the counter opinion is that healing is never easy as physical, emotional and psychological conditions continue to ravage woman for a long period after hysterectomy (Vomvolaki et al., 2006). But it is clear that hysterectomies are surely not a one-way process and may bring significant side effects—temporary or long term—to the body of woman.

In the backdrop of a rising global uterine fibroids market over time, growing hysterectomies in India are considered as totally unwarranted and a cause for concern. Expanding privatization of health care services in the country has given a big push to hysterectomy among women in the country. Hospitals, nursing homes, clinics etc. in the private sector offer the facility of hysterectomy at a fee which is widely advertised the public domain in the hoardings, pamphlets, wall writings, print media, and also on the internet to attract possible clients. By boasting about “high” quality services, experience of medical professionals, latest technology, insurance support, fixed price packages, etc., attempts are made to build confidence. Large scale hysterectomies in India have been related corruption practices among doctor who are anxious to earn quick money (Gadre and Shukla, 2020). It has been found that the hysterectomies have been financially burdensome for the households (Singh and Arora, 2008). A recent study by the National Health Authority (NHA), India on the *Ayusman Bharat Prime Minister Jan Arogya Yojna* (ABPMJAY)—a government funded health insurance programme for the poor and vulnerable—which offers 1,393 treatment packages for men, women and children in 24 States and UTs (excluding the state of Punjab) red-flagged hysterectomy. About two percent of all claims by women in India related to hysterectomy and 69 percent of these claims were from the private sector during the period of nine months from the start of this Scheme (Kaur et al., 2019).

Table 1 Percentage of women age 15-49 with hysterectomy in selected States, India 2012-13 and 2015-16

S. No.	State/Country	Prevalence		National share (2015-16)
		2012-13	2015-16	
1	Andhra Pradesh	6.3	8.9	4.2
2	Telengana	5.6	7.7	2.7
3	Bihar	--	5.4	11.5
4	Gujarat	--	4.2	4.4
5	Tamil Nadu	0.6	3.4	4.5
6	Karnataka	2.9	3.0	3.6
7	Madhya Pradesh	--	3.0	8.6
8	Maharashtra	1.4	2.6	3.4
9	Punjab	2.3	2.6	2.3
10	Jammu & Kashmir	--	2.6	3.1
11	Goa	0.9	2.6	0.2
12	INDIA ^{\$}	1.7	3.2	100.0

Source: 1. District Level household and Facility Survey (DLHS-4), 2012-13. IIPS, Mumbai, 2. National Family Health Survey (NFHS-4), 2015-16, IIPS, Mumbai. 2018. Note: '\$' indicates that the total cases for India do not add to 100 as other states and UTs are excluded from the calculation.

Punjab is one among the eleven states in Indian union with high prevalence of hysterectomy in 2015-16 (NFHS-4, 2018). It is also at the top in the north-west India in terms of hysterectomy acceptance along with adjacent Jammu and Kashmir—each with a prevalence rate of 2.6 in 2015-16. Comparable data record that the prevalence was marginally lower in Punjab in 2012-13 at 2.3 percent (table 1).

Objectives

The general goal of the study is to unfold the dynamics of acceptance of hysterectomy in the State of Punjab in India while the specific objectives of the exercise are:

- To examine the prevalence of hysterectomy in the State of Punjab,
- To study the inter-district variations in the prevalence of hysterectomy.
- To relate the patterns of hysterectomy to social-cultural, economic, and other correlates of women, and
- To highlight the possible consequences of hysterectomy on women and her family changes,

Figure 1 Administrative district, Punjab, 2017-18



Source: Maps of India, 2021

Investigations of these objectives is expected to add to overall understanding of hysterectomy prevalence in the State of Punjab besides being of substantial academic and policy interest.

Data and Methods

The study uses self-reported data on hysterectomy from the fourth round of the National Family Health Survey (NFHS) undertaken during 2015-16. NFHS is an internationally recognized data source in India since 1992-93 that offers periodic information on fertility, contraception, infant, child and adult mortality, child health, reproductive health, women empowerment, anthropometric measures, domestic violence, and so on. For the first time, NFHS, in its fourth round, collected nationwide data on hysterectomy. Prior to NFHS-4, DLHS also captured prevalence of hysterectomy for the non-empowered action group (EAG) states only in its fourth round in 2012-13. Out of three questionnaires—for the household, for the women and for the men—canvassed in the survey, the Woman’s Questionnaire contained a few direct questions to the ever-married woman aged 15-49 years to explore the dynamics of hysterectomy. The questions asked were “Some women undergo an operation to remove the uterus. Have you undergone such an operation?”, “How many years ago it was performed?” Where was this operation performed?”, and “Why was this operation performed?” in sequence. Answers by women to these questions, mostly pre-coded, are the basis of this paper. Two-step

sampling design was followed in both rural as well as urban areas to target the sample. NFHS also collects data on consumer goods and housing characteristics that are used for computation of the health index divided into lowest, second, middle, fourth and heist quintiles computed the paper uses multivariate analysis to undertake predictive analysis between independent and dependent variables.

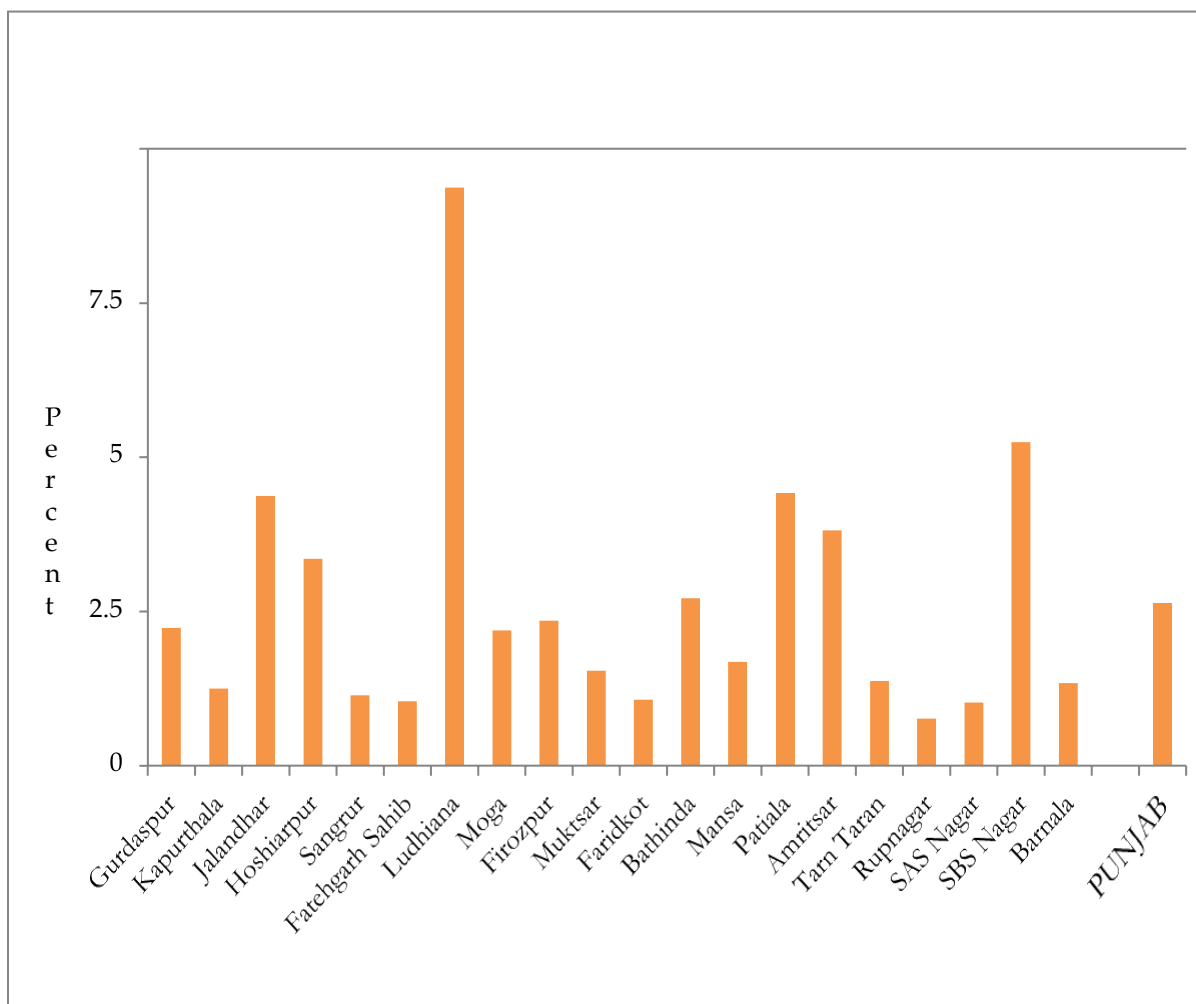
Regional Spread

Spread of hysterectomy among the 20 districts in the State indicate that prevalence is substantially higher in the districts of Ludhiana and Patiala in *Malwa* region, districts of SBS Nagar, Jalandhar and Hoshiarpur in *Doaba* region, and Amritsar in *Majha* region (figure 1 & 2, table 2). Except for SBS Nagar and Hoshiarpur, all these districts are highly urbanized by state standard—with the share of urban population, in total, being 59.2 percent in Ludhiana district, 40.3 percent in Patiala district, 52.9 percent Jalandhar, and 53.6 percent in Amritsar. These districts are also home to a wide network and presence of private sector health facilities. Conversely, SAS Nagar district has a high proportion of population living in urban areas and epicenter of leading private sector health care providers.

Table 2 Prevalence of hysterectomy by districts, Punjab 2015-16

Districts	Number of women (N)	Women with hysterectomy	Prevalence of hysterectomy
Gurdaspur	941	21	2.23
Kapurthala	960	12	1.25
Jalandhar	986	43	4.36
Hoshiarpur	925	31	3.35
Sangrur	881	10	1.14
Fatehgarh Sahib	864	9	1.04
Ludhiana	918	86	9.37
Moga	1,003	22	2.19
Firozpur	1,022	24	2.35
Muktsar	975	15	1.54
Faridkot	1,026	11	1.07
Bathinda	1,033	28	2.71
Mansa	1,013	17	1.68
Patiala	1,042	46	4.41
Amritsar	1,051	40	3.81
Tarn Taran	1,019	14	1.37
Rupnagar	925	7	0.76
SAS Nagar	880	9	1.02
SBS NAGAR	1,050	55	5.24
Barnala	970	13	1.34
PUNJAB	19,484	513	2.63
Source: National Family Health Survey-4, 2015-16, IIPS, Mumbai. 2018.			

Figure 2 Percent of women (15-49 years) with hysterectomy by districts, Punjab 2015-16



Discussion

Hysterectomy in the State reveals some interesting patterns when seen in the context of the social, economic and demographic characteristics of the acceptors (table 3). Its acceptance rises progressively with increasing age and falls with the rise in level of education of women. Among the two major religions which account for about 96 percent of total population in the State, Sikh women (2.9 percent) report a marginally higher level of hysterectomy surgeries than their Hindu counterparts (2.2 percent). Similarly, occurrence of hysterectomy is also higher among the women from the other backward castes (OBC) or general castes than among women from scheduled caste (SC) communities.

Table 3 Acceptance of hysterectomy among women aged 15-49 years classified by socioeconomic background characteristics, Punjab, 2015-16

Background Characteristics	Number of women	Number with hysterectomy	Prevalence of hysterectomy
Age			
15-29	9,317	5	0.1
30-39	5,575	112	2.0
40-49	4,592	396	8.6
Residence			
Rural	7,717	193	2.5
Urban	11,767	320	2.7
Schooling			
No schooling	3,274	152	4.6
<5 years	217	16	6.1
5-9 years	5,153	170	3.3
10-11 years	4,002	105	2.6
12 or more years	6,737	70	1.0
Religion			
Hindu	6,985	152	2.2
Muslim	329	6	2.0
Sikh	11,867	346	2.9
Other	2	9	2.9
Caste			
Scheduled Caste	7,488	170	2.3
Scheduled Tribe	37	1	--
Other Backward Caste	3,829	97	2.5
Others	8,124	245	3.0
Wealth Index			
Poorest	106	3	2.8
Poor	601	9	1.5
Middle	2179	51	2.3
Rich	4489	87	1.9
Richest	12109	363	3.0
ALL	19,484	513	2.6
Source: National Family Health Survey (NFHS-4), 2015-16, IIPS, Mumbai, 2018			
Note: Percentage not calculated because of fewer cases in the sample			

As a medically invasive procedure, hysterectomy is primarily motivated by medical condition(s). Yet, it is inextricably related to a range of social, economic and demographic background characteristics of woman that often act as ultimate predictors. On the basis of binary logistic regression, selected predictors for the women aged 15-29 in the State of Punjab are examined for their roles (table 4). Coefficients indicate that that hysterectomy is linked significantly to age, religion, cohabitation and fertility of women but not to the caste and standard of living of the ever-married woman.

Table 4 Adjusted and unadjusted odds ratios of social, economic and demographic predictors of hysterectomy among women 15-49 years, Punjab, 2015-16

Background characteristics/categories	B	Unadjusted odds	B	Adjusted odds
Current age of woman (in years)				
<30R				
30-39	-3.102	0.045***	-3.054	0.047***
40-49	0.571	1.956***	0.663	1.041***
Place of residence				
UrbanR				
Rural	-0.038	0.963	0.046	1.047
Educational attainments				
Years of education	-0.014	0.986	0.001	1.001
Caste				
Scheduled CasteR				
Other Backward Caste	-0.761	0.467	-2.463	0.085
General Caste	-0.846	0.429	-2.639	0.071
Others	-0.496	0.609	-2.435	0.088
Religion				
HinduR				
Sikh	-0.422	0.655***	-0.332	0.718**
Others	0.534	1.705	0.831	2.296
Number of living children				
0R				
1	-1.494	0.224***	-0.972	0.378*
2	-0.434	0.648*	0.226	1.254
3	0.545	1.724***	0.717	2.049***
4 and more	0.500	1.649***	0.537	1.711***
Age at cohabitation (in completed years)				
Less than 18R				
19-23	0.690	1.994***	0.646	1.908***
24-28	0.533	1.704***	0.541	1.718**
29-40	0.181	1.199	0.106	1.112
Index of wealth				
PoorestR				
Poorer	-0.670	0.512	-0.143	0.867
Middle	-0.954	0.385**	-0.546	0.579
Richer	-0.227	0.797	-0.021	0.979
Richest	-0.332	0.718**	-0.112	0.894
Note: 1. Level of significance, * = $p < .10$, ** = $p < .05$, *** = $p < .01$, 2. Others includes caste not mentioned				

Adjusted odds-ratio indicates that with rising age, the chance of proceeding for the procedure declines among women. For example, women in 30-39 and 40-49 age-groups are 95 percent and 96 percent less likely respectively to accept the hysterectomy than women who are 30 years of

age or younger. Similarly, delayed age at cohabitation among women increased the odds of their hysterectomy. The odds of hysterectomy increased by 90 percent among those who started their first cohabitation during 19-23 years, and increased by 72 percent among those who started first cohabitation during 24-28 years when compared to the ever-married women who were exposed to their first cohabitation prior to age 19 years. Number of living children of a woman also influenced her chances of hysterectomy. For instance, the odds of going for the procedure went substantially higher after the second living child. When classified by two major religions in the State, Sikh women have comparatively 28 percent less likely to have a hysterectomy as against the Hindu women.

Timing of Hysterectomy

Five-yearly data point to strong evidence of rising prevalence of hysterectomies in the State. The procedures among women seem to have taken a quantum leap specifically after 2006, as about four-fifth of the total hysterectomies in the state since early 1990s were performed during the decade of 2007-2016 (table 5). Marginal dip in the share from 2007-2011 to 2012-16 is not because of stagnancy in acceptance of hysterectomy but because the survey in Punjab was conducted in mid- 2016 and did not cover all the twelve calendar months of the year. Such a spurt may be seen in the context of twin developments of rising urbanization and growth in private sector health services in the State.

Table 5 Percent share of hysterectomies by five-year period, Punjab

Period	Number of hysterectomies performed	Percentage share in total number of hysterectomies
1992-1996	9	1.8
1997-2001	28	5.4
2002-2006	69	13.5
2007-2011	205	40.0
2012-2016	202	39.3
1992-2016	513	100.0

Source: National Family Health Survey (NFHS-4), 2015-16, IIPS, Mumbai. 2018

Reasons for Hysterectomy

Hysterectomy is generally performed on medical grounds for a variety of conditions, such as: pain or bleeding or related problems (uterine fibroids), sliding of the uterus from its normal position into the vaginal canal (uterine prolapse), chronic pelvic pain, abnormal vaginal bleeding (endometriosis), thickening of the uterus (adenomyosis), and cancer of the uterus, cervix, or ovaries. Women (15-49) in the state of Punjab were asked to list the health condition(s) that prompted them to undergo the procedure. Responses reveal that Uterus Prolapse (56 percent) followed by Fibroids/Cysts (48 percent) and Excessive Bleeding and or Pain (44 percent) are the common reasons for women opting for hysterectomy (table 6).

Table 6 Acceptors of hysterectomy classified by reasons for hysterectomy, Punjab, 2015-16

Reasons	Number of women with hysterectomy	Percent
Excessive menstrual bleeding and or pain	225	43.86
Fibroids/Cysts	248	48.34
Uterine disorder (Rupture)	73	14.23
Cancer	90	17.54
Uterus prolapsed	288	56.14
Severe postpartum hemorrhage	62	12.09
Others	46	8.97
ALL	513	100.00

Source: National Family Health Survey-4, 2015-16, IIPS, Mumbai. 2018.

Assessment of the reported reasons of hysterectomy in Punjab points not only to the prevailing reproductive health condition of women but also to the inadequate role that the health system currently plays in dealing with the reproductive health needs of women. Lack of appropriate health care facilities—particularly in remote and rural areas, nature of first line of treatment to women coming with health complaints, absence or tardy alternative care through oral and hormonal treatment, inadequate provisions of effective counseling of women, proliferation of private facilities to make profit on health front etc. need attention to improve the state of affairs concerning hysterectomy.

Place of Hysterectomy

Source-specific data on hysterectomies of women aged 15-49 years indicate that little less than two-third of the procedures were performed in the private sector health facilities in Punjab in 2015-16 (table 7). Such a lead role by the private sector health facilities in undertaking the surgical procedures is often attributed to its commitment to profit from health interventions.

Table 7 Hysterectomy among women 15-49 classified by the source of procedure, Punjab 2015-16

Type of facility	Percent Share	Number of women (N)
Government Facility		
Government Hospitals	35.2	181
Non-Government facility		
Private Hospitals	59.6	306
Private Doctors/Clinics	3.6	18
NGOs/Charitable Trust Hospitals/ Clinics	1.6	8
All	100.0	513

Source: National Family Health Survey (NFHS-4), 2015-16, IIPS, Mumbai. 2018.

Data on hysterectomy acceptance, when disaggregated by social and economic characteristics, show that private sector is relatively more active in the urban areas, among the women who have education up to 12 years or more, Hindu women, and women belonging to OBC as well as general category castes.

Greater preference for the private hospitals in India accessing hysterectomy needs to be seen in the context of a wide range of factors. Private interest has grown substantially in health sector and is well entrenched (Baru, 2005). Primary, secondary and tertiary cares are firmly in the grip of the private sector. Move towards public and private partnership in health care provisioning at government facilities that are relatively more crowded has consolidated such a hold. such as physical distance of the facility from place of residence, availability of medical professionals, quality of care expected, waiting period for the procedure at the hospital, capacity to for the procedure if needed, level of familiarity with the hospital and its staff, etc. beside the status that the women enjoy in household decision making. Statistics indicate that for general day-to-day health care household in state of Punjab rely more on the private sector health facilities than on the public sector health facilities as members from 72 percent of the households visited the former as against 27 percent to the latter during 2015-16 (NFHS, 2015-16). In spite of this dependence on private sector, most women in the State are dependent on the public sector (64 percent) providers for their need of modern methods of contraception than on the private sector (36 percent) counterparts (NFHS, 2015-16). Even a majority share of child births (52 percent) takes place in public sector facilities compared to private health facilities (39 percent) during 2011-16. However, research shows that countries with better health outcomes have lower commercialization of health care (Macintosh and Koivusalo, 2005, p.14).

Conclusion

Data on hysterectomy suggests that the socio-economic factors which significantly push women towards hysterectomy in the State of Punjab are her current age, age at cohabitation, number of living children and religion. Other related characteristics that seem to influence the access are the urbanization level of the district along with availability of and access to private sector health services. Attempt to weed out unwanted hysterectomies in the state of Punjab need a holistic strategy involving management of problems and symptoms before surgical intervention. It would be of interest to programme and policy to explore in depth what attracts women to private sector health facilities, what makes them loath the public sector hospitals in spite of its cost advantage? Is it quality of care, or something else associated? Why women readily agree to undergo procedure in spite of its direct and indirect costs to them and their family. Are the alternative treatments explored by the medical professionals in appropriate cases before zeroing on the hysterectomy? Do women repent undergoing the procedure in view of the fact that many consider uterus an inalienable part of women identity and women-hood. Further research is needed to explore the scope of interventions at the non-medical level to stem the tide of rising hysterectomy.

Notes

- NFHS was conducted in the State of Punjab during January-June 2016 and covered 19,484 women (15-59 years) and 3,250 men (15-54 years) in 16,449 households with a response rate of 93 percent for the men and 99 percent for the women. International Institute of Population Sciences (IIPS), Mumbai was the main organization that spearheaded the survey under the overall guidance of the Ministry of Health and Family Welfare (MoHFW), Government of India (GoI) in partnership with several funding as well as field agencies (IIPS 2018).
- Even if Punjab is a highly urbanized state in India, yet the districts in the state differ widely from each other in terms of their share of the urban population in the total population. For instance, Ludhiana (59 percent), Amritsar (54 percent), Patiala (40 percent), Jalandhar (53 percent), and SAS Nagar (55 percent) are highly urbanized districts and record much higher level of urbanization as compared to the State average (37 percent) in 2011.

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4

Hysterectomy in Southern States of India: Findings from National Family Health Survey (NFHS-4) Data

Anitha Kumari K R

Abstract

Number of women undergoing hysterectomy is on the rise in India and the variations in geographical pattern of hysterectomy is wide, more so in the southern states. Based on the data from NFHS-4 the southern states are classified as high, medium and low prevalent states. The objectives are to find out the variations in rates of hysterectomy along with their predictors in these states. Menorrhagia or excessive bleeding is the most common cause for hysterectomy in all southern states except for Kerala where uterine fibroids and cysts are the leading reasons. Mean age at hysterectomy is 31 years in Andhra Pradesh and 32 years in Telangana whereas in Kerala it is 40 years. Overweight and obesity in women in reproductive age group has been found to be a common risk factor in southern states. Hence good health through life style modification, adequate physical activity and a balanced diet invite attention and need to be emphasised at the grassroots. Primary health centres and sub centres should undertake regular awareness programmes through their village health and nutrition day programmes. PHCs may also promote wellness programme like physical exercise and yoga classes arranged with the help of LSGDs. It is pertinent to mention that unlike other southern states, religion and education are not a contributory factor to the hysterectomy rates in Kerala.

Introduction

Hysterectomy is a surgical operation whereby the uterus of a woman is removed. It is the second most frequently performed non-obstetric surgery after caesarean section in many parts of the world (Byles et al., 2000; Hammer et al, 2015; Whiteman et al, 2008; Stankiewicz et al., 2014). It is done for different gynecological conditions that include uterine fibroids that cause uterine prolapse; cancer of the uterus, cervix or ovaries; endometriosis; abnormal vaginal bleeding; chronic pelvic pain; or adenomyosis. One-third of all gynaecological consultations are because of abnormal uterine bleeding (AUB). This proportion rises to more than 2/3rd when peri and post-menopausal women are considered (Guin et al., 2011).

Even though hysterectomy is an option for many gynaecological problems, it is one of the risk factors of cardiovascular diseases (Howard et al. 2005; Ingelsson et al, 2011) and osteoporosis (Choi et al., 2019). Women who undergo hysterectomy face a multitude of physical and psychosocial problems both before and after the hysterectomy (Judith,2003). Studies indicate that urinary incontinence (Brownet al., 2000), sexual dysfunction (Dragisic, 2004; Wilmoth,

2000), late medical problems such as backache and weakness (Sing and Arora, 2008) and earlier onset of menopause (Farquhar, 2005) are adverse effects of this surgery.

Prevalence of hysterectomy varies from country to country. An estimated 5.1 women per 1000 women above age 15 underwent hysterectomy in 2004 in the United States, compared to 3.1 per 1000 women in Australia (Whiteman et al. 2008; Hill et al. 2010). In 2017, China had 3.3 prevalence among all women and 7.2 percent among women 45-49 years (Liu et al, 2017). Within Germany, incidence varies across states, ranged from 2.1 to 3.6 per 1000 women (Stang et al., 2011).

In India, only small community studies provide prevalence rates (Desai et al., 2017) in Gujarat estimated an incidence of 20.7 per 1000 women. (Pandey et al., 2014) and (Judith, 2003) analysed psychosocial problems among women who had undergone hysterectomy in SAT hospital, Thiruvananthapuram. However, DLHS and NFHS 4 data provides nationwide picture of hysterectomy. Studies by Prusty (et al., 2018) (a per prevalence of 17 per 1000 among ever married women) and (Shekhar et al., 2019) (a prevalence of 6% among women aged 30-49 years) using these data are providing insights into hysterectomy prevalence and its determinants at the national level. According to DLHS-4, India as a whole, 6 % of women aged 30-49 years had undergone a hysterectomy. A higher prevalence of 9.3 % among the age group 40-49 was found in NFHS4. Media reports indicate a surge in the number of young women undergoing hysterectomy in the past few years in some areas especially in AP. These reports highlight the place where hysterectomy was conducted and brought the attention of Government about the misuse of health insurance. Andhra Pradesh has imposed restrictions on private hospitals to perform hysterectomy under public insurance schemes (Desai, 2016) because of the rising hysterectomies in private hospitals. Studies or media reports in AP, Telangana and Gujarat had awaken research authorities and government authorities on the misuse of procedure especially in private sector, and intense debates on its potential ill health-effects on young women.

In India, the reported prevalence of AUB is around 17.9% (Sharma and Dogra 2013). The highest incidence of AUB was noted in the 41-50-year group (Nair et al, 2015) and in Agra it is 40 to 49 years age group and this is often due to ovulatory cycles in 80% cases (Agarwal et al, 2017). In Kasturba Medical College, 9% of all gynecological admissions ended up in hysterectomies and around 60% of major gynecological surgeries are hysterectomy (Pandey et al, 2014). Bleeding is said to be abnormal when the pattern is irregular, abnormal duration (>7 days), or menorrhagia or abnormal amount (>80 ml/menses) (Sper off and Fritz, 2005). It is usually due to hormonal disturbances: reduced levels of progesterone cause low levels of prostaglandin F2alpha and causes menorrhagia; increased levels of tissue plasminogen activator (TPA) (a fibrinolytic enzyme) lead to more fibrinolysis. Decision to undergo a hysterectomy was also associated with the importance of sexual relationships and attitudes, e.g., whether the uterus is “useless” to women who have completed childbearing (Kupperman et al, 2010).

There are many studies and documentation regarding reproductive and child health issues. This subject continues to be in the focus of attention of most health care providers, researchers and policy formulators but issues with respect to menopausal and postmenopausal women are often neglected. State specific studies are required in order to make different policies in each state.

Scenario of hysterectomy in India shows that southern states have different prevalence rates. Karnataka (3.0) and Tamil Nadu (3.4) have hysterectomy percent equal to that of India (3.2). It is low in Kerala (1.8 percent). But it is 8.9 percent in Andhra Pradesh and 7.7 percent in Telangana which is the highest and second highest percent of women who had undergone hysterectomy in the country. Among the southern states Andhra Pradesh and Telangana are in the high prevalent status, Tamil Nadu and Karnataka are in medium prevalence level and Kerala is in the lowest prevalence level. Hence this study aims to classify these states as high, medium and low and find out what factors contribute to high, medium and low prevalence of this surgery in these states.

Objectives

- To examine the variations in prevalence of hysterectomy among southern states
- To find out the predictors of hysterectomy in low, medium and high prevalence states in southern India.

Data and Methodology

The National Family Health Survey (NFHS) fourth round data were used for this study. It allowed separation of natural menopausal women from women who had undergone hysterectomy. As part of the ever-married women's questionnaire, all respondents in the age group of 15-49 year were asked the following questions.

- *Some women undergo an operation to remove the uterus. Have you undergone such operation?*
- *How many years ago this operation (Hysterectomy) was performed?*
- *Where was this operation performed?*
- *Why was this operation performed? Any other reason?*

Hysterectomy was analysed as a dichotomous variable (yes/no).

Descriptive statistics were used to describe demographic and clinical data on subjects. Socio-economic and demographic variables like age of women, religion, age at first birth, place of residence, working status, household wealth, household with health insurance, women sterilization and education of women were used to find out its association with hysterectomy. Morbidity status of various non-communicable diseases are also analysed with hysterectomy status. Age at hysterectomy was calculated from the current age and duration of hysterectomy.

Logistic regression was performed to estimate the odds ratio between hysterectomy status and socioeconomic and clinical variables. All independent variables explored were categorical (except age) and included education level, religion, place of residence, wealth index, insurance status, sterilization status, age at first birth and BMI. Since the different states have different predictors, variables those showed significance in bivariate analysis are included for each state. Out of the non-communicable diseases, only thyroid status is related to hysterectomy status and hence it is included in the logistic regression. Five models are formed in order to find out predictors in high, medium and low prevalence states.

Limitations

In the data set, it was not possible to identify whether hysterectomy was done before or after menopause.

Findings

Table 1 reveals that three percent of women had a hysterectomy in India. There is wide regional variation exists in India. In the northern and central parts of India it is below 3 percent. In North eastern states less than 2 percent women had undergone hysterectomy.

Table 1 Percentage of women age 15-49 who have had a hysterectomy, by State/Union Territory, India, 2015-16

State/union territory	Percentage of women who have had a hysterectomy	Number of women
India	3.2	699,686
North		
Chandigarh	1.5	573
Delhi	1.1	10,540
Haryana	1.9	15,587
Himachal Pradesh	2.2	3,842
Jammu & Kashmir	2.6	6,811
Punjab	2.6	15,215
Rajasthan	2.3	36,540
Uttarakhand	2.0	5,930
Central		
Chhattisgarh	1.9	16,505
Madhya Pradesh	3.0	43,747
Uttar Pradesh	2.2	105,115
East		
Bihar	5.4	56,309
Jharkhand	2.3	17,602
Odisha	2.1	24,939
West Bengal	2.0	55,734
Northeast		
Arunachal Pradesh	1.8	603
Assam	0.9	17,309
Manipur	1.6	1,222

State/union territory	Percentage of women who have had a hysterectomy	Number of women
Meghalaya	1.1	1,590
Mizoram	1.0	584
Nagaland	1.6	798
Sikkim	1.3	319
Tripura	1.3	2,172
West		
Dadra & Nagar Haveli	3.6	171
Daman & Diu	3.0	91
Goa	2.6	861
Gujarat	4.2	32,682
Maharashtra	2.6	66,655
South		
Andaman & Nicobar Islands	1.8	231
Andhra Pradesh	8.9	30,417
Karnataka	3.0	34,897
Kerala	1.8	19,270
Lakshadweep	0.9	43
Puducherry	1.7	795
Tamil Nadu	3.4	51,601
Telangana	7.7	22,385

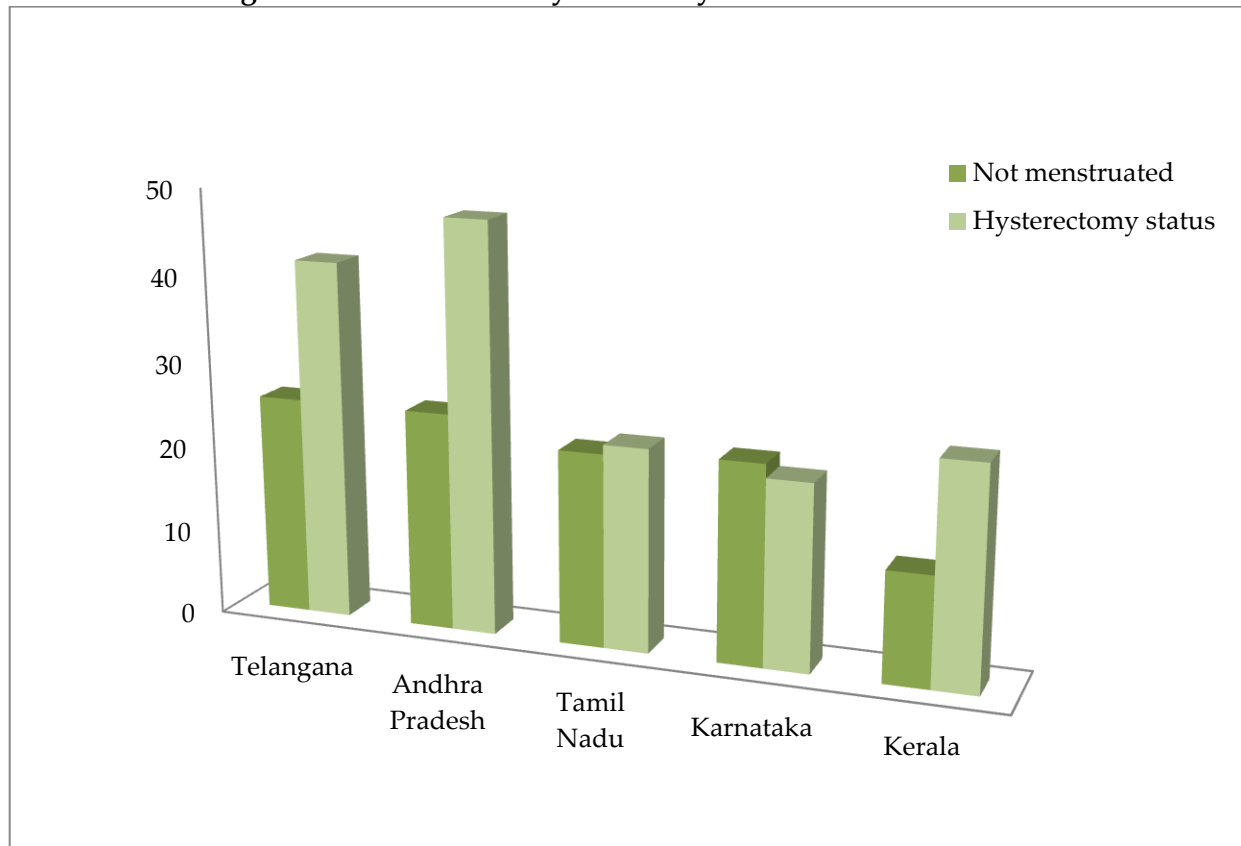
In all eastern states except Bihar hysterectomy is low among women. In West it is higher in all states and UTs except Goa and Maharashtra. Southern states show a different picture. Karnataka and Tamil Nadu have hysterectomy similar to that of India. It is very low in Kerala (1.8 percent) but it is 8.9 percent in Andhra Pradesh and 7.7 percent in Telangana. These states had reported the highest and second highest percent of women who had undergone hysterectomy in the country. Hence among the southern states Andhra Pradesh and Telangana are in the highest status, Tamil Nadu and Karnataka are in medium prevalence level and Kerala is in the lowest prevalence level. Hence these states are compared as high, medium and low and find out what factors contributing high, medium and low prevalence of this surgery in these states.

Table 2 Percent of women who have not had a menstrual period in the last six weeks and their hysterectomy status in southern states, 2015-16

	High		Medium		Low
	Telangana	Andhra Pradesh	Tamil Nadu	Karnataka	Kerala
Menstruation status during last 6 weeks before survey*					
Menstruated	74.5	74.6	77.5	76.8	87.2
Not menstruated	25.5	25.4	22.5	23.2	12.8
Total	22384	30417	51601	34897	19270
Undergone Hysterectomy *					
Yes	41.8	47.7	23.6	21.6	25.6
No	58.2	52.3	74.4	78.4	74.4
Total	4112	5681	7536	4850	1375

*Significant $P < 0.000$

Figure 1 Menstrual and hysterectomy status in southern states



Hysterectomy status was asked to those women who have not had a menstrual period in the six weeks before the survey. It excluded women who are either pregnant or postpartum amenorrhoeic. These five states constitute 23,554 women who had not menstruated in last six weeks. In Kerala 12.8 percent women had not menstruated and this percent is about 23 percent in medium level states and 25.5 in high status states.

Those not menstruated were asked about their hysterectomy status. Table 2 shows hysterectomy status of these states. Out of these 23,554 women 7,594 had undergone hysterectomy. In Telangana out of the 4,112 women who are not menstruated, 1,720 had removed their uterus. Out of the 5,681 women (not menstruated) 2709 are removed their uterus in Andhra Pradesh. Out of the 7536 women in Tamil Nadu, 1767 reported they had removed their uterus. In Karnataka 1046 women had hysterectomy. Similarly in Kerala out of the 1375 women without menses, 352 are removed their uterus. Hence these women are analysed to find out hysterectomy status in different groups. Out of the women who had not menstruated in the last month about 42 percent of women in Telangana are moved their uterus whereas this percentage is 48 percent in Andhra Pradesh, 24 percent in Tamil Nadu and 22 percent in Karnataka and 26 percent in Kerala.

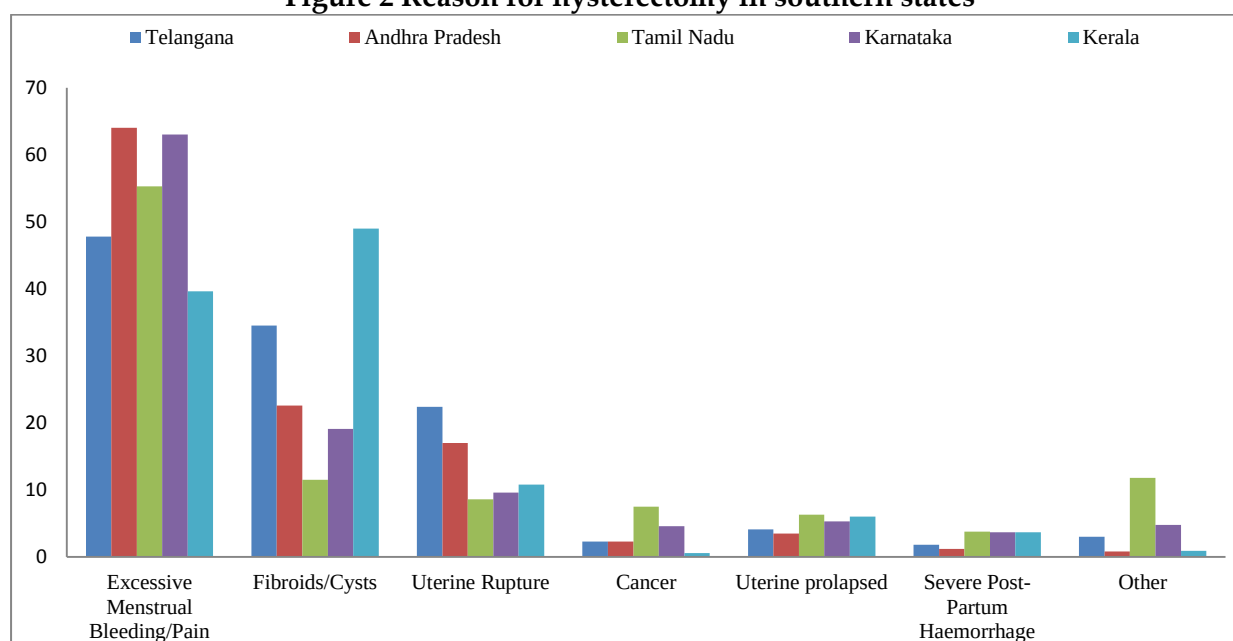
Table 3 Reasons for hysterectomy, place of hysterectomy, age and duration of the surgery

Hysterectomy variables	High		Medium		Low
	Telangana	Andhra Pradesh	Tamil Nadu	Karnataka	Kerala
Reason for hysterectomy					
Excessive Menstrual Bleeding/Pain	47.8	64.0	55.3	63.0	39.6
Fibroids/Cysts	34.5	22.6	11.5	19.1	49.0
Uterine Rupture	22.4	17.0	8.6	9.6	10.8
Cancer	2.3	2.3	7.5	4.6	0.6
Uterine prolapsed	4.1	3.5	6.3	5.3	6.0
Severe Post-Partum Haemorrhage	1.8	1.2	3.8	3.7	3.7
Other	3.0	0.8	11.8	4.8	0.9
Place of hysterectomy					
Public: Govt./Municipal Hospital	10.4	9.9	41.4	32.9	38.6
Public: Govt. Dispensary/UPHC	3.3	3.0	8.6	10.2	0
Public: CHC/Rural Hospital/Block PHC/PHC/SC	4.4	8.6	2.3	9.0	1.1
Other Public Sector Health Facility/Camp	0.75	0.2	0.2	0.8	1.4
NGO or Trust Hospital/Clinic	0.2	0.5	0.62	0.3	0.3
Private: Hospital	76.0	77.3	44.0	40.7	57.2
Private: Doctor/Clinic	3.5	5.1	2.0	3.7	0.8
Private: Mobile Clinic	0.5	0.3	0.6	2.6	0.1
Total Public	18.8	16.7	52.5	52.7	41.6
Total private	81.2	83.3	46.6	47.0	58.4
Age at hysterectomy*					
<25	18.0	17.5	22.0	20.5	2.3
25-29	23.9	27.4	20.9	20.7	6.0
30-34	25.8	24.3	15.6	17.9	6.5
35-39	19.8	15.9	15.8	23.2	25.0
40-44	10.9	11.3	19.7	14.8	38.6
45-49	1.6	3.5	6.0	3.0	21.6
Mean age at hysterectomy	30.9 (30.4-31.3) SD 9.1	30.9 (30.6-31.2) SD 7.9	31.3 (30.7-31.9) SD12.5	30.9 (30.1-31.6) SD12.2	39.3 (38.4-40.2) SD9.0
Hysterectomy variables					
	High		Medium		Low
	Telangana	Andhra Pradesh	Tamil Nadu	Karnataka	Kerala
Median age at hysterectomy	31.0	30.0	32	32	41
Duration of hysterectomy*					
0-4	23.3	26.8	37.4	38.7	65.1
5-9	32.5	24.8	22.9	29.5	20.9
10-14	26.3	21.2	17.2	14.5	7.7
15 and more	17.9	27.3	22.5	17.3	6.3
Total	1712	2704	1751	1034	350

* Significant at $P < 0.001$

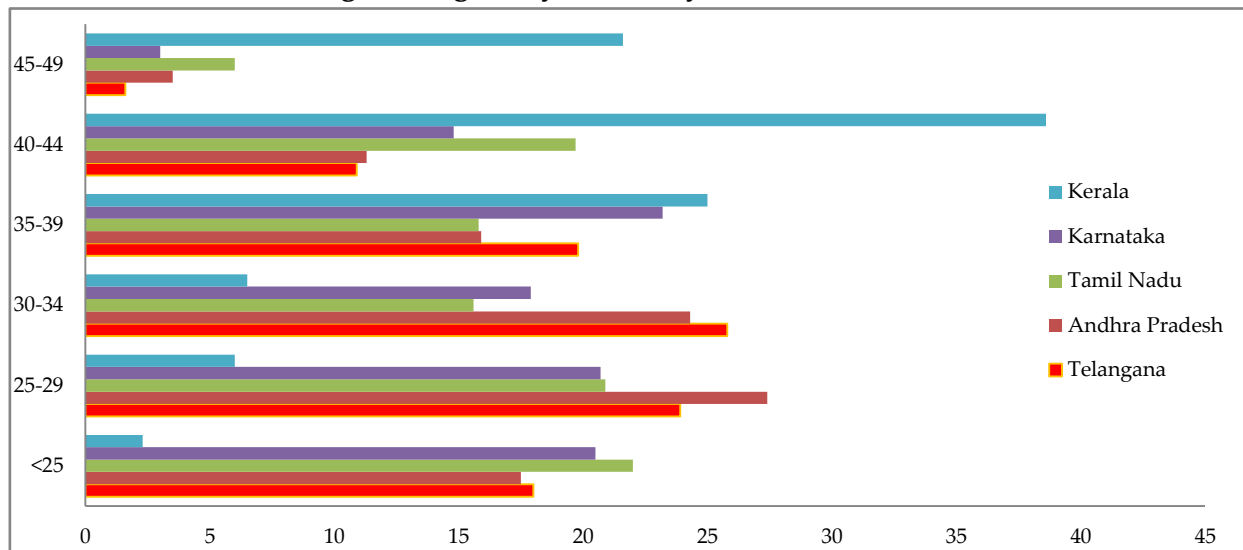
Excessive bleeding is the most reported reason for hysterectomy in Telangana (48 percent), Andhra Pradesh (64 percent), Tamil Nadu (55.3percent) and Karnataka (63percent) whereas fibroids and cysts in the uterus constitute the most important reason for hysterectomy among women in Kerala (49 percent). About 40 percent women in Kerala reported of menstrual excessive bleeding. More than one third of women in Telangana had the problems of fibroids and cysts. Uterine rupture is the third most indication for hysterectomy in these states. Cancer is reported as a reason among 7.5 percent of women in Tamil Nadu whereas this reason is reported by only 0.6 percent of women in Kerala.

Figure 2 Reason for hysterectomy in southern states



Age at which women had removed their uterus reveals that only 2 percent of the women in Kerala removed it in less than 25 years. This percent is more than one fifth in Tamil Nadu and Karnataka. Table 2 reveals that mean age at hysterectomy is 31 years in Andhra Pradesh and Telangana whereas it is high in Kerala with 40 years. Kerala shows a unique picture among the southern states. In the state, after 34 years of age, percent of women who had undergone hysterectomy increases and reaches highest in 40-44 age group. In all other states a higher percent of women had removed their uterus before 34 years. Even below 25 year of age, 18 percent of the women in the two states had undergone hysterectomy. Median and mean age at hysterectomy of women is respectively 40 and 41 years in Kerala. About 68 percent women in Telangana and 69 percent of women in Andhra Pradesh had their surgery before 35 years. Only less than 15 percent of women in Kerala reported that they had undergone the surgery before 35. Duration of hysterectomy shows that it is recent (0-4 years) surgery among women in Kerala (65 percent). About 18 percent women in Andhra Pradesh, 27 percent women in Telangana 22.5 percent in Tamil Nadu and 17 percent in Karnataka did the surgery 15 or more years ago.

Figure 3 Age at hysterectomy in southern states



In India more than two-thirds (67%) of hysterectomies were performed in the private sector (NFHS-4). In Kerala 58 percent of the hysterectomies were performed in private sector especially in private hospitals. In Tamil Nadu and Karnataka about 47 percent of these surgeries are conducting in private facilities. But in Telangana and Andhra Pradesh more than 80 percent surgeries were conducted in private firm. About 41 percent of the women in Tamil Nadu, about 33 percent in Karnataka and 39 percent in Kerala perform hysterectomy in public hospitals where specialists are available. This percent is around 10 percent in other two states. Kerala women utilize secondary or tertiary care hospital facilities (whether it is public or private) more than PHC, CHC and private clinics. But in other states use of primary health institutions are higher.

Figure 4 Place of hysterectomy in southern states

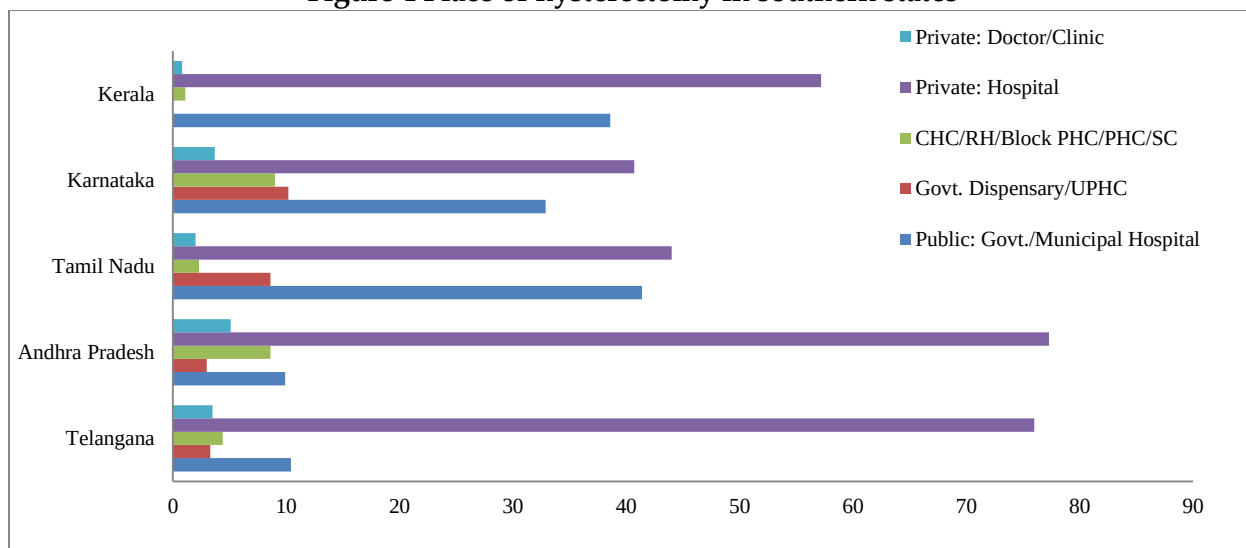


Table 4 Characteristics of women who had undergone hysterectomy

<i>Characteristics</i>	<i>Telangana</i>	<i>AP</i>	<i>Tamil Nadu</i>	<i>Karnataka</i>	<i>Kerala</i>
Age					
15-19	1.6*	0*	3.4*	0.4*	0*
20-24	3.0	2.9	3.8	2.6	0
25-29	27.5	32.3	11.8	10.8	2.7
30-39	66.3	77.4	34.4	33.9	22.6
40-49	44.5	49.4	27.1	26.6	31.3
Mean	40.2 (39.9-40.5) SD6.4	40.8 (40.5-41.0) SD6.2	40.7 (40.4-41.1) SD6.9	39.9 (39.5-40.3) SD6.4	44.5 (44.1-44.9) SD3.8
Median age	41.0	42.0	42.0	41.0	45.0
Highest educational level					
No education	48.0*	50.5*	25.8*	25.5*	20.0NS
Primary	59.4	53.6	22.6	28.3	23.1
Secondary	25.2	41.7	24.0	17.6	26.6
Higher	23.1	24.9	15.5	11.5	24.3
Religion					
Hindu	43.2*	48.6*	23.7**	23.2*	29.4NS
Muslim	28.6	35.2	17.2	12.6	17.7
Christian	38.5	48.4	24.1	6.2	24.5
Place of residence					
Urban	36.8*	46.6NS	26.0*	17.8*	25.6NS
Rural	44.7	48.1	21.4	23.8	25.4
Wealth Index					
Poor	36.3*	36.1*	17.4*	20.0NS	20.0*
Middle	41.0	46.8	23.2	23.2	10.8
Rich	45.7	53.8	26.2	21.3	28.8
Working status					
No	33.0*	42.9*	25.2NS	23.4NS	23.5*
Yes	58.3	53.7	27.9	23.8	51.0
CEB					
No child	13.6*	23.6*	11.5*	6.8*	12.2*
1-2 children	39.8	49.1	24.9	23.4	28.8
3 or more children	47.3	48.1	25.5	24.6	17.4
BMI					
<18.49	30.0*	29.7*	16.1*	17.9*	16.3*
18.50-24.99	41.0	44.6	20.9	19.9	19.4
25.0-29.99	56.0	54.7	27.7	29.7	31.4
>30.00	36.8	56.8	30.7	22.0	34.5
Covered by insurance					
No	30.7*	39.2*	20.8*	19.7*	23.0**
Yes	47.2	50.5	27.0	29.5	28.6
Currently sterilized					
No	33.2*	23.8*	22.5**	17.9*	24.8 NS
Sterilized	50.8	57.3	24.9	26.8	26.6
Age at sterilization					
<25	55.2*	63.0*	24.7NS	30.7*	29.4**
25-29	47.3	48.6	26.1	20.9	27.0
30-34	40.3	38.2	24.3	25.7	26.5
35-39	39.0	42.1	14.8	25.0	12.0

<i>Characteristics</i>	<i>Telangana</i>	<i>AP</i>	<i>Tamil Nadu</i>	<i>Karnataka</i>	<i>Kerala</i>
40-44	20.0	43.5	28.6	13.0	0.0
45-49	28.6	52.9	33.3	33.3	25.0
Age at first birth					
18 years and below	49.3*	54.6*	25.8**	29.6*	29.0NS
19 years and above	37.2	38.3	23.2	21.7	27.1
Hypertension					
No	42.5***	47.6**	22.8*	21.3**	25.8NS
Yes	46.3	53.1	30.3	25.3	23.8
Thyroid					
No	41.1*	47.5NS	22.9*	20.9*	24.3**
Yes	53.1	49.0	39.0	44.1	33.9
Diabetes					
No	42.0NS	47.1*	22.3*	20.7*	23.3*
Yes	38.5	57.7	36.7	46.4	38.1
Cancer					
No	41.8NS	47.6**	23.2*	21.2*	25.7NS
Yes	63.6	80.0	43.8	95.5	20.0
Asthma					
No	41.5**	47.3*	23.0*	21.1*	25.0***
Yes	48.0	55.9	30.6	47.4	34.7
Heart Disease					
No	41.3*	47.6 NS	23.2*	21.4**	25.1**
Yes	61.1	52.9	31.0	34.9	42.2
Morbidity status					
No	41.2**	46.3*	21.7*	19.6*	21.7*
Yes	45.0	54.4	33.2	43.4	34.8

* $p < 0.000$, ** $p < 0.05$, NS- Not significant

Current age of the women shows that only 2.7 percent women below 30 years in Kerala had undergone hysterectomy. But the percent is 27.5 in Telangana and 32 in AP, 12 in Tamil Nadu and 11 in Karnataka. Highest age group of women reporting hysterectomy status is 30-39 age group in southern states other than Kerala but it is 40-49 age group in Kerala. Mean and median age of women in Kerala is highest among the states 44.5 and 45 respectively. Median age of women in Telangana and Tamil Nadu is 42 years each and Andhra Pradesh and Karnataka is 41 year each. When status of education and hysterectomy is analysed, in all state's education is significantly related to hysterectomy except in Kerala where it is found that percentage is almost same in different educational group. But in other southern states, there is differential by educational groups as the education increases hysterectomy status decreases. Hence education of women is an important factor in high and medium level states in taking the decision of a surgery.

There is significant difference in hysterectomy status among religious groups in southern states except in Kerala. Women belonging to Hindu are reporting more than Muslim and Christian in Telangana and Karnataka. About equal percent of Hindu and Christian women in Andhra Pradesh and Tamil Nadu had undergone hysterectomy. There is no rural-urban difference in

performing surgeries in Kerala. Among those performing surgeries rural women are more in high prevalent states and in Karnataka in contrast to Tamil Nadu where urban women are performing more surgery. Wealth Index shows women belonging to middle income (11 percent) reported less hysterectomy status than rich (29 percent) and poor (20 percent) in Kerala. In Karnataka difference in wealth is not significantly related to hysterectomy status. But in other states as wealth status increases hysterectomy status also increases. With regard to working status, there is not much difference in status in Tamil Nadu and Karnataka and in other states working women had performed surgeries more than non-working women. Children ever born and hysterectomy status reveals that about 24 percent women in AP had opted hysterectomy before they had a child. Childless women who had opted hysterectomy before having a child is 12.2 in Kerala and 6.8 percent in Karnataka, 11.5 percent in Tamil Nadu and 13.6 in Telangana. Women who had three or more number of children had undergone surgery more (47 to 48 percent) in Telangana and AP (high prevalent states) and one fourth in Tamil Nadu and Karnataka.

BMI is an important biological factor in the hysterectomy. Hysterectomy status increases as weight of women increases that is from thin to obesity, in states like AP, Tamil Nadu and in Kerala. In Telangana and Karnataka highest percent of women who report hysterectomy status is those who are overweight. Hence hysterectomy is more among women who are overweight and obese than those who are normal and thin. In all southern states more women who had undergone hysterectomy had health insurance coverage than who did not. But the difference of percent of women in the two groups is very less in Kerala than other four states. Hysterectomy is found to be high among women who were sterilized (more than half) in high prevalent states whereas the proportion is one fourth women in Tamil Nadu and more than one fourth of the women in Karnataka. But low prevalent state Kerala is different in this aspect. There is no significant difference among women in their hysterectomy status and sterilization status. The prevalence of hysterectomy is different according to their sterilization status in these states. In Tamil Nadu, 25 percent women who have sterilized before 25 years had undergone hysterectomy whereas this percent is 63 and 55 percent respectively in Andhra Pradesh and Telangana and 31 percent in Karnataka and about 30 percent in Kerala. Table 5 shows that sterilization and hysterectomy both are more in younger ages in high prevalent states. Women whose age at first birth is below 18 years at the time of hysterectomy are more in percent than women whose first birth above 18 years or more. It is significant in all states except in Kerala.

Morbidity status of women who had undergone hysterectomy is analysed in the Table 5. Except Kerala hypertension is significantly high among women with hysterectomy. Thyroid is associated with hysterectomy as hormone imbalance is the reason for abnormal bleeding for many women. In all states women who had undergone hysterectomy had thyroid problems more than women without thyroid problems. Except in Telangana all women with hysterectomy were diabetic. Except in Kerala in all other states hysterectomy status is more

among women with cancer. Asthma and heart disease are also related with hysterectomy. We do not know whether these diseases developed after hysterectomy. But women with hysterectomy are suffering all these morbidities.

Table 5 Determinants of hysterectomy among women in high prevalence states

Déterminants	Telangana			Andhra Pradesh		
	Exp(B)	CI	Sig	Exp(B)	CI	Sig
Age	1.02	1.01-1.03	0.000	1.02	1.01-1.03	0.000
Education						
Illiterate and Primary®	-	-	-	-	-	-
Secondary & higher	0.392	0.33-0.47	0.000	0.700	0.60-0.81	0.000
Place of Résidence						
Urban®	-	-	-	-	-	-
Rural	1.805	1.5-2.2	.000	-	-	-
Religion						
Hindu ®						
Muslim	0.666	0.67-0.51	0.004	0.610	0.48-0.78	0.000
Christian	0.978	0.65-1.5	0.914	1.100	0.90-1.3	0.353
Children Ever Born						
No child®	-	-	-	-	-	-
Have children	0.801	0.68-0.94	0.005	0.576	0.51-0.66	0.000
BMI						
Under & Normal weight ®	-	-	-	-	-	-
Over weight & obese	1.414	1.2-1.7	0.000	1.446	1.3-1.6	0.000
Thyroid						
No ®	-	-	-	-	-	-
Yes	1.70	1.3-2.3	0.000	-	-	-
Wealth Index						
Poor®	-	-	-	-	-	-
Middle	1.223	1.2-1.02	0.030	1.491	1.3-1.8	0.000
Rich	2.173	1.8-2.7	0.000	1.920	1.8-2.3	0.000
Age at First Birth						
Below19 years®	-	-	-	-	-	-
19 years and above	0.719	0.62-.083	0.000	0.61	0.54-0.69	0.000
Stérilisation Status						
Yes®	-	-	-	-	-	-
No	0.696	0.60-0.80	0.000	0.288	0.25-0.33	0.000
Covered by Insurance						
No ®	-	-	-	-	-	-
Yes	1.437	1.2-1.7	0.000	1.371	1.2-1.6	0.000
Constant	0.223	-	0.000	0.491	-	0.000

Table 5 explains predictors or risk factors of hysterectomy in high prevalence states of Andhra Pradesh and Telangana. It is known fact that prevalence is more among higher aged women than younger women. In both states increase in age of one year increases 1.02 times chance of hysterectomy. It is highly significant. In both states, education is an important significant factor in the removal of uterus. There is 60 percent and 30 percent less chance of removal of the uterus for those women with secondary and higher educated than women with less education or

illiterate in Telangana and Andhra Pradesh respectively. With respect to urban women, rural women have 1.8 times more chance of undergoing hysterectomy in Telangana and place of residence is not a significant factor in Andhra Pradesh. With respect to Hindu women Muslim women have less probability for hysterectomy and are not significant compared to Christian women in these states. With respect to women with no child, probability of hysterectomy is 20 percent less than those with children in Telangana and 42 percent less in Andhra Pradesh. The model suggests that BMI status of the women is significantly related with hysterectomy. Overweight and obese women have 1.4 times chance in Telangana and Andhra Pradesh to have hysterectomy. Model shows that thyroid dysfunction status is independently associated with hysterectomy in Telangana (There is 1.7 times chance to have hysterectomy for women with thyroid problem). As it is not significant in AP in bivariate analysis, it is omitted in the logistic regression model. Household Wealth in the states shows its relation with hysterectomy. Women of middle and rich households are at more risk of hysterectomy than women belonging to poor families. Age at first birth is another important variable. Those with age 19 years and above as age at first birth have less chance of hysterectomy (28 percent less chance in Telangana and 40 percent less chance in AP) than those with age at first birth 18 years and below. There is 30 percent less chance of hysterectomy in AP and 70 percent less chance in AP for those women who have not been sterilized compared to sterilized women. Hysterectomy is more probable for those women with health insurance than without insurance in both states.

Table 6 Determinants of hysterectomy status among women in medium prevalence of hysterectomy

Déterminants	Tamil Nadu			Karnataka		
	Exp(B)	CI	Sig	Exp(B)	CI	Sig
Age	1.032	1.02-1.04	0.000	1.039	1.03-1.05	0.000
Education						
Illiterate and Primary®	-	-	-	-	-	-
Secondary&higher	1.08	0.94-1.23	.271	0.863	0.72-1.02	0.099
Place of residence						
Urban ®	-	-	-	-	-	-
Rural	1.03	1.03-1.04	0.000	1.469	1.23-1.74	0.000
Religion						
Hindu®	-	-	-	-	-	-
Muslim	0.446	0.32-0.63	0.000	0.477	0.36-0.62	0.000
Christian	0.982	0.75-1.3	0.896	0.281	0.11-0.70	0.007
Children Ever Born						
No child®	-	-	-	-	-	-
Havechildren	0.984	0.86-1.12	0.815	0.706	0.60-0.84	0.000
BMI						
Under and Normal weight®	-	-	-	-	-	-
Over weight& obese	1.332	1.18-1.5	0.000	1.465	1.24-1.73	0.000

Déterminants	Tamil Nadu			Karnataka		
	Exp(B)	CI	Sig	Exp(B)	CI	Sig
Thyroid						
No ®	-	-	-	-	-	-
Yes	2.246	1.68-3.0	0.000	2.530	1.72-3.74	0.000
Wealth Index						
Poor®	-	-	-	-	-	-
Middle	1.344	1.13-1.6	0.001	1.268	-	0.017
Rich	1.381	1.15-1.65	0.000	1.396	-	0.003
Age at first birth						
18 years & below®	-	-	-	-	-	-
19 years and above	0.858	0.76-0.98	0.019	0.638	0.54-0.75	0.000
SterilizationStatus						
No ®	-	-	-	-	-	-
Yes	1.227	1.09-1.4	0.001	1.00	0.85-1.2	0.980
Covered by insurance						
No ®	-	-	-	-	-	-
Yes	1.250	1.12-1.4	0.000	1.462	1.2-1.7	0.000
Constant	0.062	-	0.000	0.076	-	0.000

Table 6 shows predictors of hysterectomy in medium prevalence states in south. Current age of women is significantly related to their hysterectomy status in medium prevalence states. One year of age in women increases the chance of hysterectomy as 1.03 times in both states. Education shows different scenario in two states. In Tamil Nadu women with higher education have higher probability for the surgery whereas in Karnataka women with higher education level have less chance of hysterectomy status. In Tamil Nadu education is not a significant factor and in Karnataka it is significant at $p < 0.1$ level. In both states rural women have more probability to have hysterectomy. With respect to Hindu women, Muslim and Christian women have less chance of hysterectomy in both states; it is not significant in case of Christian women in Tamil Nadu. Probability of hysterectomy is lesser for women who have children than women with no child. But relationship is not significant in Tamil Nadu but highly significant in Karnataka. Thin and normal weight women have significantly less chance of hysterectomy in both states. Women who reported thyroid have more chance for hysterectomy than those without. Chance of hysterectomy is 2.2 time higher in Tamil Nadu and 2.5 times higher in Karnataka for those with thyroid problems. With respect to poor women, middle and rich class women have more chance to have uterus removed. In both states it is significant. Age at first birth is related to hysterectomy status in both states. Women who have age at first birth above 18 years have less chance for surgery in both states. There is more chance of hysterectomy for women who are sterilized in Tamil Nadu. It is not significant in Karnataka. Women who have health insurance are most probable for hysterectomy in both states. It is highly significant.

Table 7 Determinants of hysterectomy in low prevalence state Kerala

Déterminants	Exp (B)	CI	Sig
Age	1.067	1.05-1.09	0.000
ChildrenEver Born			
No child®			
1-2 children	3.057	1.55-6.02	0.001
3 or more	1.345	0.64-2.83	0.434
Wealth Index			
Poor®			
Middle	0.360	0.14-0.91	0.031
Rich	1.251	0.54-2.87	0.596
BMI			
Under and Normal weight®			
Over weight& obese	1.884	1.45-2.45	0.000
Thyroid			
No			
Yes	1.321	0.92-1.89	0.128
Covered by insurance			
No®			
Yes	1.25	0.96-1.62	0.092
Constant	0.005		0.000

Age of the women is a risk factor for the hysterectomy. As age of women increases by one year hysterectomy status increases at 1.06 times. Women who have one or two children have 3 times more probability to have uterus removal and it is 1.3times chance in case of three or more children (not significant) with respect to childless women. With respect to women belonging to poor family, women in middle class household have 64 percent less chance for hysterectomy and women in rich families have 1.3 times more chance (not significant) for hysterectomy. BMI is an important factor regarding hysterectomy in the Kerala. Overweight and obese women are more likely (1.9 times) to undergo hysterectomy. There is 1.3 times more chance for the women with thyroid than women who have not reported thyroid. At $p < 0.1$ level women with health insurance are related to the hysterectomy status.

Discussion

Excessive bleeding is the most reported indication for hysterectomy in all states except in Kerala (where fibroids and cysts in the uterus is the indication for the hysterectomy. Many studies reported that menorrhagia affects 10-30% of menstruating women at any time, and may occur at some time during the perimenopause in up to 50% of women (Mirza et al,2012). Indication in Kerala is consistent with the hospital study in south India which confirmed fibroids as the most common indication for hysterectomy (about 40 percent) followed by uterovaginal prolapse (16.3%) (Pandey et al, 2014). A study in Trivandrum Medical College also confirms this. In 2001, 58% of the hysterectomies and in 2002, 65% of the hysterectomies are due to fibroids of the uterus (Judith, 2003).

In Kerala 58 percent of the hysterectomies were performed in private sector especially in private 'hospitals. But in Telangana and Andhra Pradesh (high prevalent states) more than 80 percent surgeries were conducted in private firm and this percent is 47 in Tamil Nadu and Karnataka (states with medium prevalence). Here we can notice that Kerala women chose a hospital (whether it is private or public) more than PHC, CHC and private clinics whereas in other states a major surgery like hysterectomy is done even in private clinics. It may be because of the good hospital facilities in Kerala. Role of private sector in surgery is consistent with other studies in AP which show that 95% hysterectomies are carried out in private sector (Kameswari and Vinjamuri, 2013). Duration of hysterectomy shows that it is recent (0-4 year) surgery among women in Kerala (65 percent). About 18 percent women in Andhra Pradesh, 27 percent women in Telangana, 22.5 percent in Tamil Nadu and 17 percent in Karnataka did the surgery 15 or more years ago. Median age at hysterectomy in medium and high prevalent states is almost same and 31 years and it is 41 years in Kerala. That is a woman undergone surgery in other southern states is 10 year earlier than in Kerala. In Gujarat 36 years is found to be age at hysterectomy (Desai et al, 2017). Comparison of states in the southern region regarding characteristics of women and hysterectomy status reveals that socioeconomic, demographic, BMI and insurance status are significantly related. Bivariate analysis shows that women aged below 30 years in Kerala are very less in reported hysterectomy status compared to women in high prevalent states and medium prevalent states. While only 23 percent women in Kerala undergone hysterectomy in the current age group 30-39, about 66 percent in Telangana, 77 percent in AP, 34 percent each in Tamil Nadu and Karnataka had performed this surgery. Median age of respondents who have undergone hysterectomy in Telangana and Karnataka is 41 years and it is 42 years in Andhra Pradesh and Tamil Nadu and it is comparable with DLHS. The median age of women who had undergone hysterectomy in India according to the DLHS is 42 years (Prusty et al, 2018). Median age and mean age in Kerala are higher than other southern states. As education increases percent of women reporting hysterectomy status decreases in all states except Kerala where education is not a significant factor. Education is one of the predictors in Telangana, Andhra Pradesh and Karnataka while it is not significant in Tamil Nadu and it does not enter in the model in Kerala. It is consistent with DLHS study which showed 53% less likely chance of women who had education levels of matriculation and above in India (Prusty et al, 2018). Wilson and Mishra (2016) conferred low education levels related with higher odds of hysterectomy in the lowest versus highest level meta-analysis.

Hysterectomy status is more among Hindus followed by Christians and Muslims in all states and is significant except in Kerala. In all other southern states, being Muslim is a protective factor in hysterectomy. Hysterectomy status is high among rural women in Telangana, Andhra Pradesh and Karnataka and but is slightly high in urban areas in Tamil Nadu and Kerala. Rural residence is a risk factor in Telangana, AP and in Karnataka. Place of residence and religion are

not appeared as a significant predictive factor in Kerala. It may be because there is no rural urban divide in Kerala and religious backwardness or other difference is not much wide like other states in India. Education of women is almost same in all groups of women.

Women reporting hysterectomy status increases as wealth quintile increases in high prevalent states and in Tamil Nadu. Karnataka does not observe a significant relationship between household wealth and hysterectomy but in Kerala women with rich quintile have highly reporting and women with middle level wealth have the lowest reporting status. In Kerala middle income women are more protective than women belonging to poor families whereas in all other southern states women belonging to rich households are more likely to have hysterectomies than women from the poor households. It is in contrast to studies where low-income status is related to hysterectomy by Desai et al (2017) and Kjerulff et al, (1993).

Working status of women has not a significant relation with uterus removal in medium prevalence districts. In all states more percent of women who are currently working have undergone surgery than their nonworking counterparts. Working status is not considered for the logistic regression as number of women is less.

Bivariate analysis reveals that percent of women with no child is lesser in performing hysterectomy than women with children in all states. In Kerala, with reference to women without children, women with one or two children have 3 times higher chance for hysterectomy. In Tamil Nadu, number of children and hysterectomy status are not significantly related. In all other three states women with children have less probability in hysterectomy. In India women with two or more children had higher chance of hysterectomy is found (Prusty et al, 2018). It is consistent with the result of Kerala but is in contrast to the findings in other southern states. Parity did not remain a significant determinant of hysterectomy is found in Hawaii (Sievert et al, 2013) controlling for other demographic and lifestyle factors.

As BMI of women increases from thin to obese, hysterectomy status also increases in Andhra Pradesh, Tamil Nadu, and in Kerala. In Telangana and Karnataka overweight women are reporting more hysterectomy status than other women. In Logistic regression BMI is divided into overweight and obese as one category and thin and normal as other category. It shows that in all states thin and normal category women has less chance to have surgery. BMI is significantly related to hysterectomy is found in other studies. For example, women who were overweight at the age of 30 were more likely to have undergone a hysterectomy (Sievert et al, 2013). A study among 80,204 California teachers, the risk of surgery for fibroids was found to be high among overweight women and obese women compared to women of normal body size, after controlling for numerous demographic and lifestyle factors (Templeman et al, 2009).

The study found that in all states women with insurance is reporting more hysterectomy status and is highly significant. Women covered by insurance are more likely to have hysterectomy in all states. It is consistent with studies in other states (Desai et al, 2017).

Bivariate analysis shows that sterilized women are more in percent in having hysterectomy than non-sterilized counterparts. But the difference is very low in Kerala and hence not significant. Age at sterilization in high prevalent states is very low. Multivariate analysis shows that women who are not sterilized are less likely to have hysterectomy in Telangana and in AP whereas in Tamil Nadu they are more likely to have hysterectomy. In Karnataka it is not significant. In Gujarat, sterilization status is not a predictor of hysterectomy (Desai et al, 2017)

More than half of the women whose age at first birth 18 and below has hysterectomy status in high prevalent states (Telangana and Andhra Pradesh) whereas this percent is 26 and 30 in medium states (Tamil Nadu and Karnataka respectively). Very few numbers of women before age 19 years in Kerala gave birth to a child and hence it could not be considered in the logistic regression. Higher age at first birth is a protective factor in all other states.

Deficiency in thyroid hormone level is related to a decrease in progesterone secretion. Deficiency in progesterone causes abnormal uterine bleeding. Abnormal bleeding is the most important reason in hysterectomy in all southern states except Kerala.

Descriptive analysis shows that women with thyroid dysfunction have reported hysterectomy status more than those without thyroid issues and it is significant in all states except AP. It may be because testing is less in number in AP. Women with thyroid has more chance in hysterectomy in medium prevalence states, and in Telangana. In the study done by Komathi et al (2016) reported that thyroid dysfunction is the indicator for about 30% of abnormal uterine bleeding. Most of the women with abnormal uterine bleeding were euthyroid was also reported (Thakur et al, 2020).

Morbidity status of the women who have undergone surgery shows that self-reported hypertension is high for women with hysterectomy in all states except in Kerala. Diabetic status is also related to hysterectomy in all southern states except in Telangana.

Except in Telangana and Kerala cancer is related to hysterectomy. Asthma status is also related to hysterectomy status in all states. All southern states hysterectomy status is related to heart disease except in Andhra Pradesh. A significant morbidity status is related to hysterectomy status in all states. We do not know whether morbidity occurs before hysterectomy or morbidity is followed by hysterectomy. From the life style diseases, only thyroid status is related with hysterectomy as per medical studies.

Conclusion

In conclusion it is observed that different characteristics are related to the hysterectomy status in each state. The common factors are age of the women, BMI, wealth index and insurance status. Hence strict protocol for the observance of gynaecological problems leading to

hysterectomy should be impart as very young women in south India especially in Telangana and AP are victims of unnecessary surgery.

Overweight and obese of women are found to be one of the common risk factors in southern states. Hence health through exercise should be emphasised at the grass root level and primary health centres and sub centres should take necessary awareness classes through VHNDs. PHCs should take necessary steps for the regular exercise or yoga through arranging rooms and equipment's and space by utilising LSGDs. Pre-Menopausal and post-menopausal problems should be addressed at the primary level and referred if it needs the service of specialist and provide continuum of care at PHC.

Low education, socioeconomically poor background, low age at sterilization and low age at first birth are indicating high prevalence of hysterectomy in high prevalent states. Improving education of women is the only way to rise age at first birth, age at sterilization and age at hysterectomy and prevent women from exploiting them for money through insurance coverage.

In high and medium prevalent states, abnormal uterine bleeding is the major reason for hysterectomy. Women with AUB should initially be treated with medical therapy and when medical therapy fails, the physician should explain to the patient the necessity of the surgery. In Kerala majority are seeking care at secondary and tertiary level hospitals and hence women get the service of Gynaecologist and Obstetrician. Moreover, most women are educated and they are eager to know about merits and demerits of surgery or seek second opinion before taking a decision of a major surgery. These may be the reason for low prevalence of hysterectomy. Surgical management is the last choice for these women after medical management. Hence thyroid testing and monitoring is necessary to avoid bleeding and hence hysterectomy. In contrast to other states in Kerala, religion and education are not related to hysterectomy status in Kerala.

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5

Determinants and Patterns of Anaemia Among Women in India

Vandana Sharma, Suresh Sharma and Ritwika Mukherjee

Abstract

This study examines the relative importance of individual, socio-economic, nutritional and empowerment variables affecting different levels of anaemia using nationally representative cross-sectional data of NFHS-III & IV. Even though there has been improvement in overall anemic levels, it appears that the problem of iron-deficiency anaemia continues to remain a major public health issue in India. Econometric modeling through multinomial logistic regression depicts that caste, class, occupation, place of residence, marital status more or less depicts similar relationship across different levels of anaemia. However, certain variables like maternal education, women's empowerment variables and diet intake assume a significant effect over moderate/severe anaemia relative to no anaemia in comparison to mild anaemia relative to no anaemia. With the temporal analysis carried out, it seems there has been improvements in the poorer households, but some of the richest communities from the richer states have seen a rise of severe anemic women which calls for important policy discussions in the near future. This study can be useful to policymakers to understand the mechanism of anemic intensity among women and to focus on which variables to address any particular anemic level through a low-cost approach.

Introduction

Anaemia or low concentration of hemoglobin (Hb) is a condition in which the number of red blood cells in the body are insufficient to meet the physiological needs of it. Iron deficiency is thought to be the most common cause of anaemia worldwide (DeMaeyer, et al., 1989). Children, women of reproductive age, and pregnant women are at-high risk of developing anaemia. Maternal anaemia is associated with maternal and child, morbidity, mortality, increased risk of miscarriage, stillbirth, prematurity and low birth weight of the baby. About 20% of perinatal mortality and 10% of maternal mortality in developing countries is attributed to iron deficiency. Numerous factors including age, sex, residential elevation (altitude), smoking behavior, social infrastructure, and pregnancy status influence hemoglobin concentration (Kavitha, 2014; Balarajan, 2013; Jose and Navaneetham, 2010). In addition to being a medical condition, anaemia is also an important socio-economic issue (Ghosh, 2009) given its association with decreased physical and cognitive productivity. A complex interplay of political, ecological, social, and biological factors determines the prevalence and distribution of anaemia in a population. Past studies have highlighted women's empowerment in improving the nutritional status of the children (Cunningham, 2015; Holland, 2019) but comparatively fewer studies have

been done to check role of women's empowerment in reducing anaemia recently (Gupta et al., 2019). Dietary diversity among Indian households is low but it might vary with household wealth and maternal education (Agrawal, 2019). Besides excessive vegetarian diet can also induce anaemia among mother and the neonates (Kaur et al., 2006). Analyzing all these factors and keeping in mind the levels of hemoglobin deficiency can bring new insights about the role of the determinants influencing different levels of anaemia. This will help give the policymakers an outlook to address the issue of anaemia among women in the light of current programs.

Objectives

The study pertains to three aims:

- To explore the levels of hemoglobin deficiency across regions and socio-demographic attributes.
- To examine the effect of rising wealth across states and accordingly, illustrate the shift in severe anaemia.
- To account for the relative importance of individual, socio-economic, nutritional and empowerment variables affecting different levels of anaemia.

Data and Methods

Source and target population: Data for the study has been extracted from nationally representative cross-sectional survey, NFHS-III & IV covering a sample size of 124,385, ever married women in 2005-06 and 699,686, ever married women in 2015-16 with detailed information relating to population, health and nutrition. The surveys used a stratified cluster multistage sampling design, consisting of two stages in rural regions and three stages in urban localities. In the first stage, in rural areas, primary sampling units (PSU) were chosen with a probability proportional to the population size. In the second selection stage, households were selected systematically for the survey. The wards constitute the first PSU in urban areas in the first stage after which census enumeration block was randomly selected from every ward; and in the final stage, households were randomly selected from every census enumeration block.

Dependent variable: The main response variable is Hb level measured in gm/dl, categorized as a 'categorical variable' that has predefined cut-off criteria for mild, moderate and severe anaemia as by the WHO recommendations for over 15 years of age. For non-pregnant sample, anaemia is defined as Hb < 12 gram/dl, and for pregnant sample it is < 11 g/dl. Mild anaemia is defined as 10–11.9 gram/dl for non-pregnant women and 10–10.9 gram/dl for pregnant women. Moderate/severe anaemia is defined as Hb < 10 gram/dl. Both pregnant women & non-pregnant are included in the analysis. Hb levels are adjusted to altitude and smoking status as by the NFHS.

Independent variables: The study has used a host of socio-demographic, nutritional and women's empowerment variables. Predictors used in the analysis were caste, wealth index, age, marital

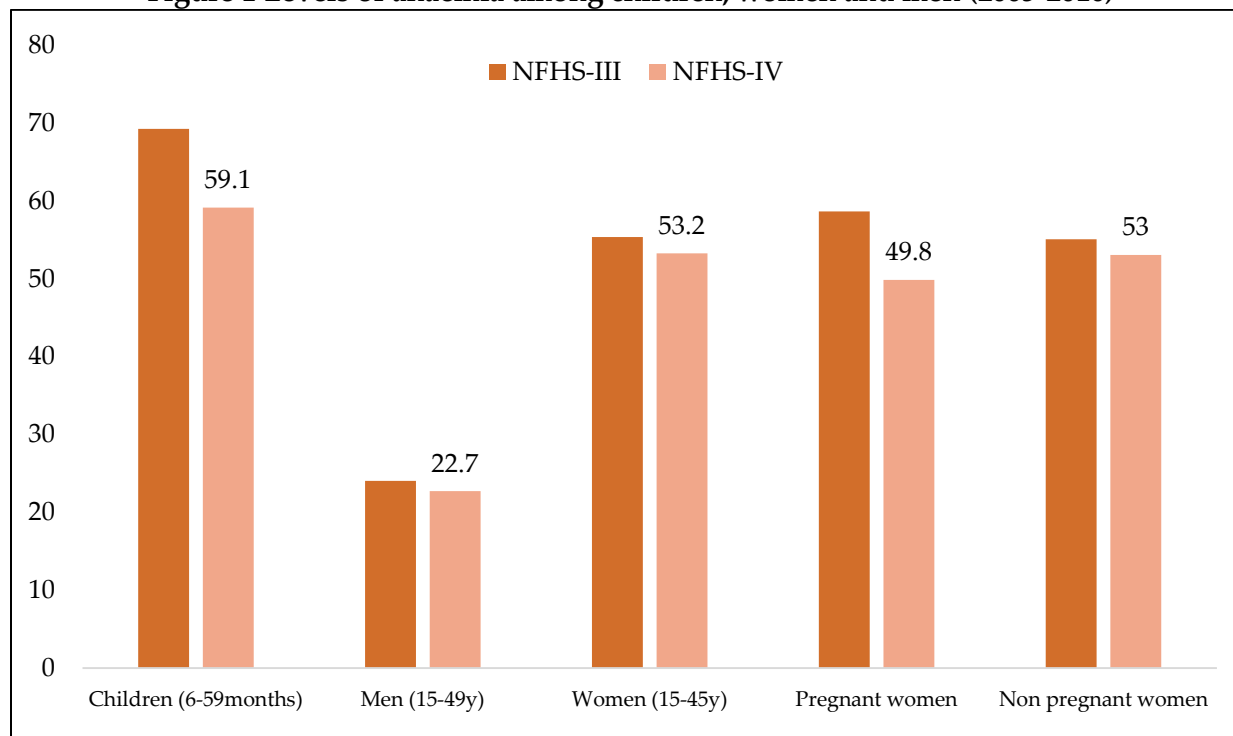
status, religion, number of living children, place of residence, maternal education, occupation, dietary diversity and consumption of vegetarian/non-vegetarian foods.

Statistical analysis: Descriptive and multinomial logistic regression models with the dependent variable as different levels of anaemia (no anaemia, mild anaemia, moderate/severe anaemia) had been used. A detailed explanation of the model is given in the analysis section. Econometric modeling has been done through STATA.

Trends and patterns of anaemia prevalence

Prevalence of anaemia is highest among children aged between 6-59 months. Anaemia due to iron deficiency is more common among children mostly arising due to lack of iron in the diet. Infants generally need more iron in their diet than the older population. So, a steady amount of iron is required to fuel a child's rapid growth and development. Premature babies, babies who are fed only cow milk, breastfed but without complimentary food, etc. are the ones at highest risk of developing iron-deficiency anaemia. Although, there was a decline of about 10 percentage points in childhood anaemia in 2016, still the prevalence is quite high. On the contrary, the incidence of anaemia is very less among men since men need far less iron than women as they don't menstruate.

Figure 1 Levels of anaemia among children, women and men (2005-2016)



The worrying population to consider here are the females in reproductive age group, as it is estimated that in India, one out of every two women suffer from some kind of anaemia. The reasons are biological and social. In the following analysis, we will discuss this issue in depth. However, the decline of anaemia among pregnant women is a good sign, as across the world

anaemia tends to be higher among pregnant than non-pregnant women. This points to the efficient functioning of the nutritional and maternal programs targeted for adequate supplies of IFA tablets to pregnant women. Among non-pregnant women the intensity was less to start with but in the latest years the incidence has been higher compared to pregnant women which means that even though some decline has happened, the decline is less than that of pregnant women.

Figure 2 Varying intensity of anaemia among women (15-45years)

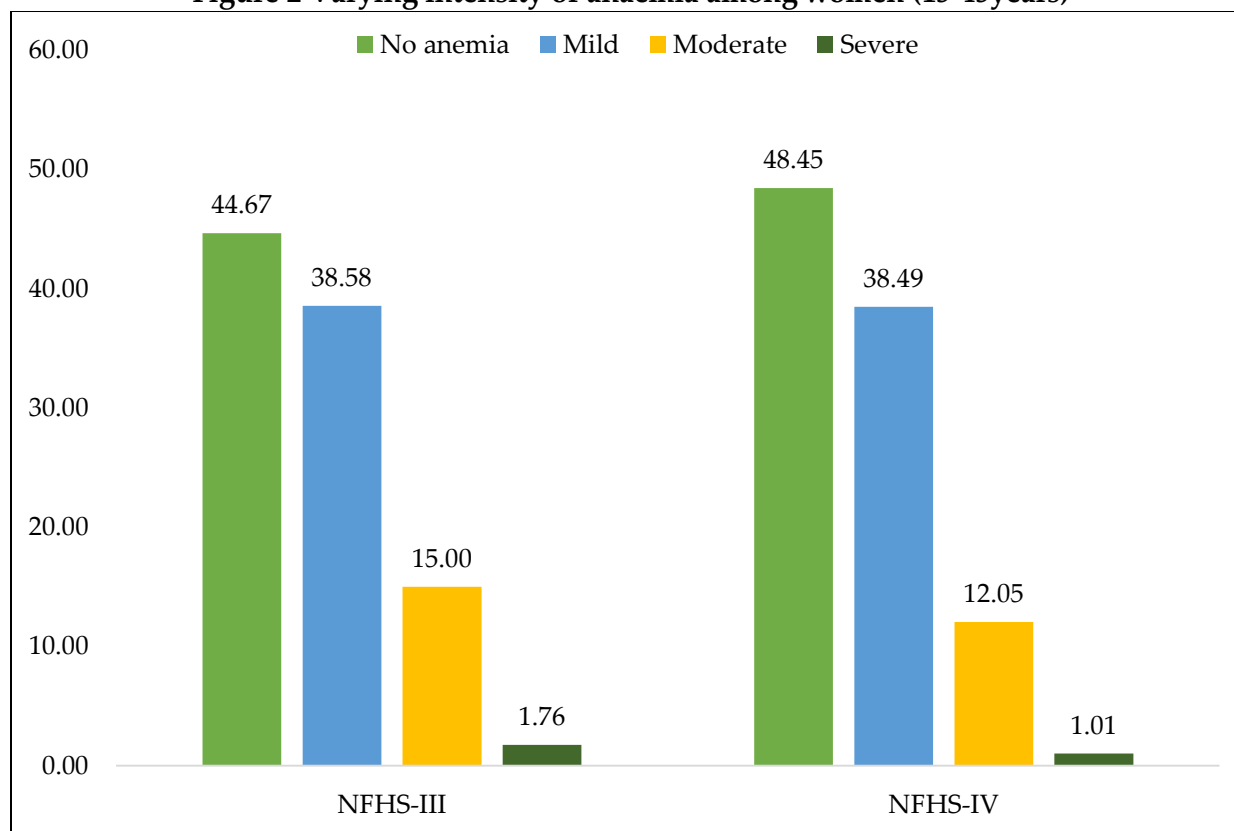


Figure 2 shows the different types of anaemia according to intensity among all women in the 15-45 years age group. As seen in the above figure, this graph also tallies the fact that there has been a decline in anaemia which is found across all categories, mild, moderate and severe. The proportion of women having mild anaemia has only partially declined from 38.58% to 38.49%, while proportion of moderate anaemia has shown a higher decline from 15% to 12.05%. Severe anemic women have also come down marginally from 1.76% to 1.01%. Moderate and severe anaemia are the major limiting factors as they can lead to severe fatigue and inability to carry out daily routine.

State level variations in anaemia among women in the reproductive age group

The spatial variation in anaemia shows a mixed picture. The states that have anemic share greater than 50% are Bihar, Jharkhand, West Bengal, Orissa, Telangana, Gujarat, Madhya Pradesh, Haryana, etc. These states have performed equally bad in both the rounds.

Figure 3 Women having any kind of anaemia (2005)

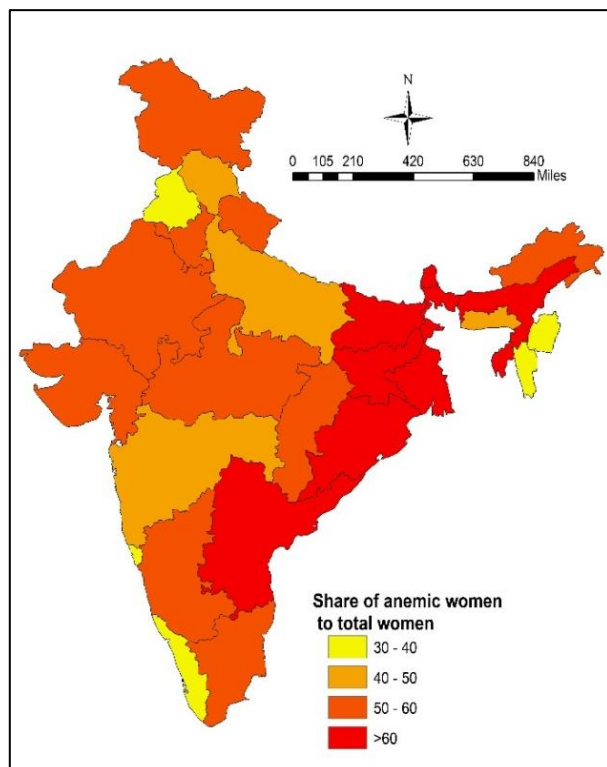
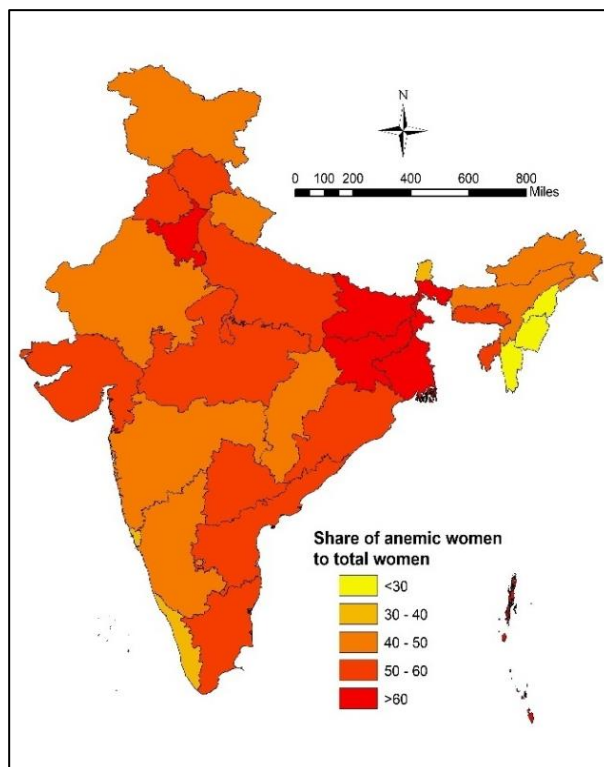


Figure 4 Women having any kind of anaemia (2016)



Although iron-deficiency is considered as an important cause of anaemia, it can also be caused due to deficiencies in vitamins such as Vitamin C, Vitamin B9, Vitamin B12, etc. For example, a large part of Andhra Pradesh and Telengana's anemic deficiency is due to Vitamin B-12 deficiency. The states where the situation has worsened are the states of Punjab, Haryana, Himachal Pradesh, Delhi and Uttar Pradesh. In Punjab, the share of anemic women has gone up from 38% to 54%, a rise of 16%. This is followed by Himachal Pradesh which saw a rise by 10% between both the periods. Delhi has registered an upward jump by 10% from 44% to 54% between 2005 and 2016 respectively. Haryana has registered a rise from 56% to 63% from 2005 to 2016. Interestingly the rise in anaemia has been in some of the most prosperous states. Although the link between nutrition and household wealth is positive, in this aspect of women's health, the link is not very clear. The relation between prosperity and women's health will be discussed in depth in later sections to follow. Some of the poorer states like Orissa, Jharkhand, Chattisgarh, etc. have reportedly shown improvement in anaemia. Maharashtra, Gujarat, West Bengal showed marginal deterioration in anaemia.

Some of the states which, as discussed earlier, showed worsening of anemic conditions, did not show as worse situation in moderate and severe anaemia. For example, Delhi which registered a rise by 10% in overall anaemia, have shown a rise of 5% in moderate and severe anaemia.

Figure 5 Women having any moderate/severe anaemia (2005)

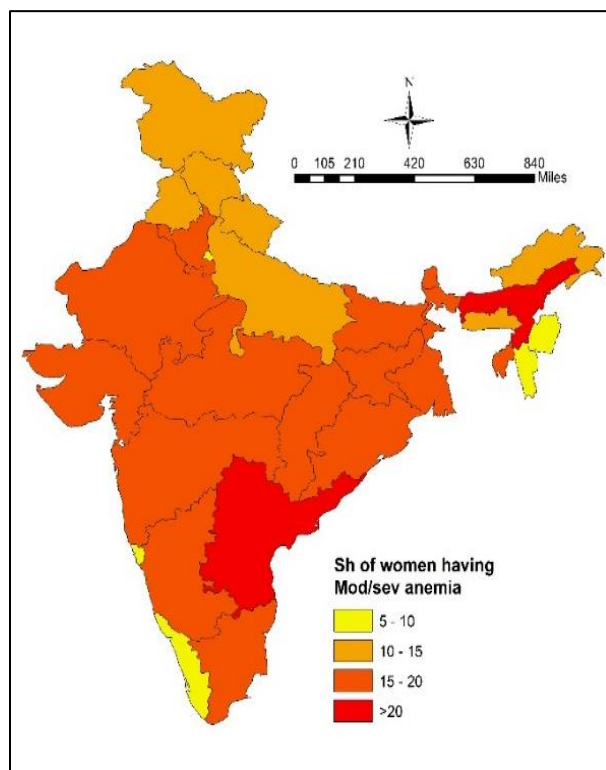
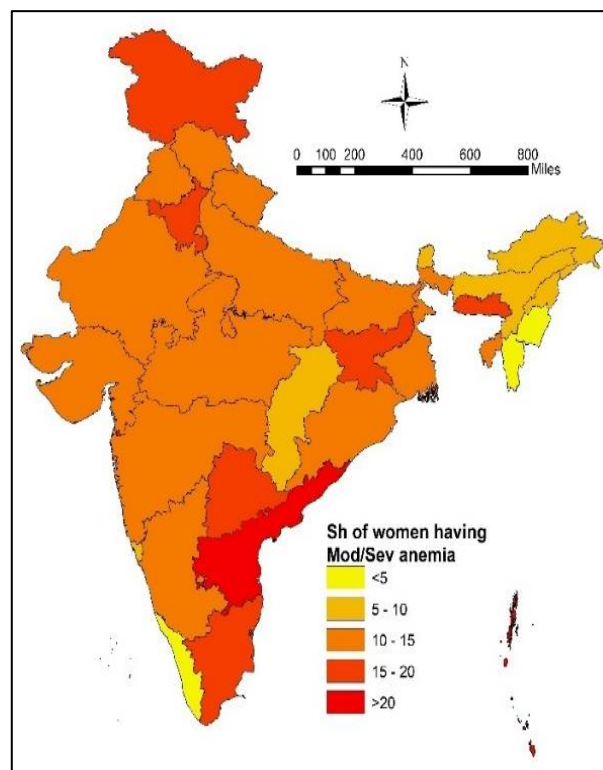


Figure 6 Women having moderate/severe anaemia (2016)



Same applies to Himachal Pradesh which registered a rise in 10% of overall anaemia, has shown only a 2% jump in severe anaemia. Punjab, on the other hand has registered a decline in moderate/severe anaemia while its overall anaemia has gone up which goes to say that majority of worsening impact of anaemia is due to the rise in mild anaemia rather than moderate/severe anaemia.

Intra-state variation in anaemia

A spatial continuity can be observed across the share of anemic women, that is the hotspots of high anaemia are surrounded by the peripheries next to its lowest category. For example, majority of the entire eastern region have high shares of anemic women, West Bengal, east Jharkhand, east Bihar adjoining West Bengal have shares of anaemia immediately surrounded by the orange zone, the next highest category and thereafter it slowly disperses to the softer categories.

Within the state, a lot of internal heterogeneity is observed. For example, the western part of Orissa has a high number of anemic endemicity areas, whereas the coastal part shows a stark difference with as much as 30-40% difference between western and coastal Orissa. The overall improvement in anaemia as seen in the above figures (Figure 5 & 6) is attributed to the improvements coming from the coastal part of the state. Some of the high endemic areas are

tribal populated areas where women undergo many nutritional deficiencies like Orissa. A lot of internal heterogeneity can also be observed in the state of Uttar Pradesh. Even though the central regions have a low prevalence of anaemia, the western and eastern districts show high anemic prevalence, which has pulled up the overall value of anemic women in Uttar Pradesh between the two periods as observed in Figure 7 & 8. Severe/moderate anaemia do not have much internal heterogeneity. It almost disappears from the eastern part which otherwise shows high proportion of any-anemic women (refer Figure 8). Moderate/severe anaemia constitutes the largest part out of any anemic women in the southern states of Telangana, Andhra Pradesh, Karnataka and Tamil Nadu and the northern states of Haryana, Punjab and West Uttar Pradesh, and western states of Gujarat and Rajasthan. The severity of anaemia is high in the districts belonging from the above-mentioned states. On the other hand, severity is less in the eastern region, where although high overall anaemia exists, the contribution of moderate/severe anaemia is lower.

Figure 7 Women suffering from any anaemia 2015-16

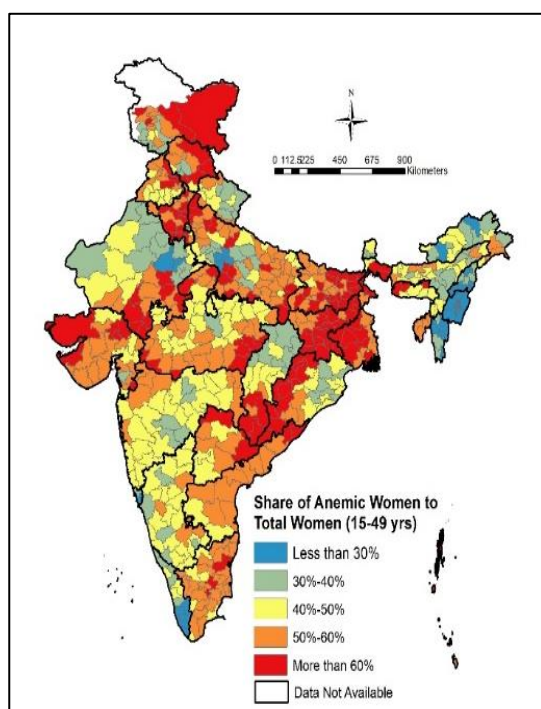
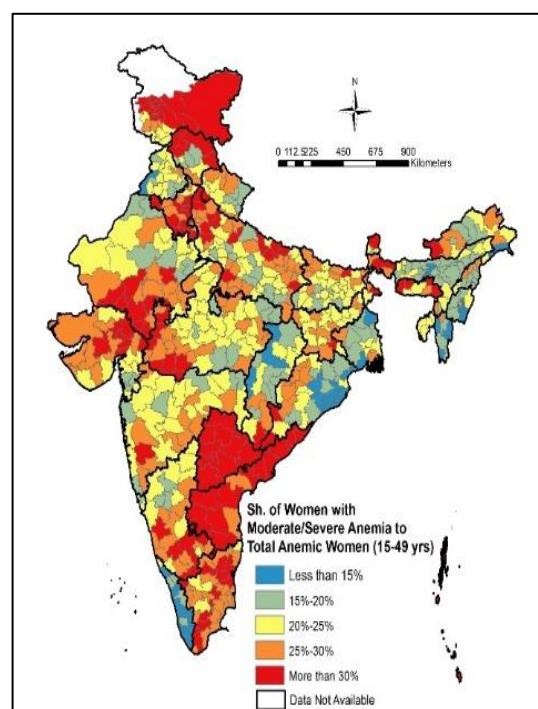


Figure 8 Women suffering from moderate/severe anaemia 2015-16



Anaemia prevalence across social and economic groups

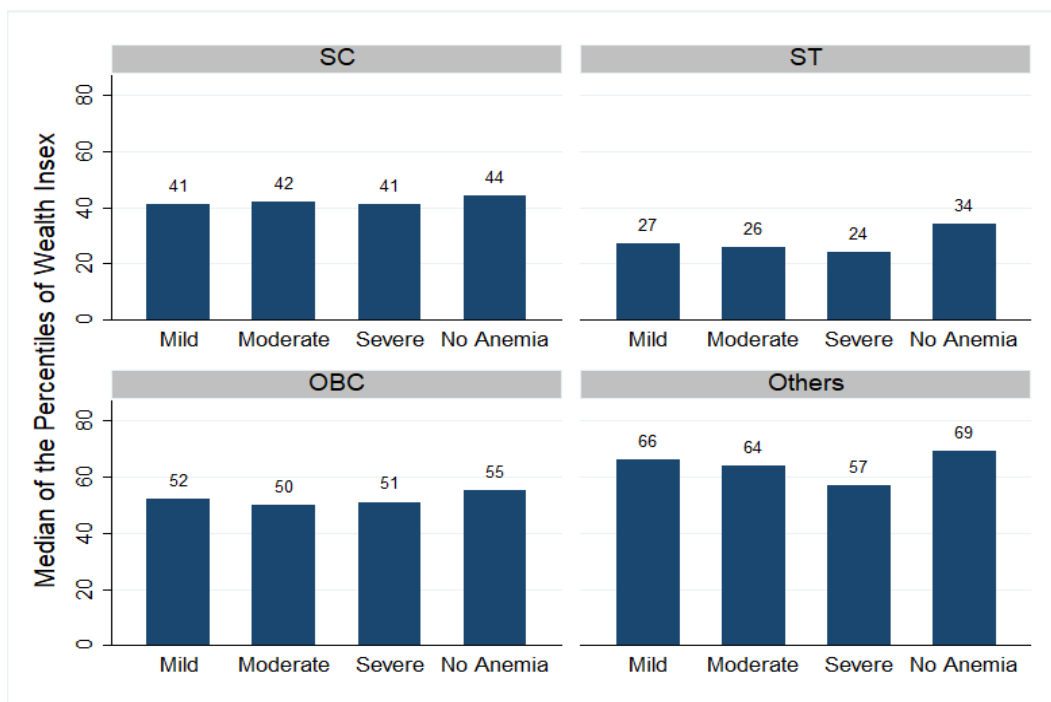
There exists a difference in anemic levels among the different castes. Anaemia prevalence is highest among the STs followed by SCs and OBCs. About 59.9% of STs suffer from any anaemia while 49.7% of General community have any form of anaemia. There is an inverse link between anaemia prevalence and economic prosperity. Among the different social groups, the richer communities have low iron deficiency anaemia. The proportion of moderate and severe anemic

women is lowest among the richest General category (9.1%). The poorest STs have the highest anemic level (17.9%). Belonging from a particular social group entails some advantages of becoming less anemic as observed across the general category where the poorest group have better anemic levels than the richest SCs. The following figure (Fig 11) shows the median percentile of wealth index among the different social groups. Irrespective of the social groups the wealth index score is highest for no anemic women implying wealthier households have less anemic prevalence. Those who suffer from different forms of anaemia don't show much variation with the economic profile in case of SCs and OBCs. However severe anemic women for STs and General community have a relatively lower economic profile, although the wealth profile of the general community is much higher than any communities suffering from severe anaemia. This shows that although there is a variation of wealth profile among the different communities (lowest among the STs and highest for the General), within the communities the prevalence of severely anemic women is highest for the lowest wealth percentile.

Table 1 Class-caste wise variation of anemic prevalence

Social Groups	Wealth Quintiles	Mild	Moderate/Severe	No Anaemia
SC	poorest	42.5%	15.2%	42.3%
	poorer	40.6%	14.9%	44.5%
	middle	38.4%	15.6%	46.1%
	richer	38.3%	14.2%	47.5%
	richest	38.4%	12.4%	49.2%
ST	poorest	46.5%	17.9%	35.6%
	poorer	42.5%	16.3%	41.2%
	middle	40.2%	14.2%	45.6%
	richer	37.1%	12.6%	50.3%
	richest	34.1%	10.5%	55.3%
OBC	poorest	40.8%	14.2%	44.9%
	poorer	39.3%	13.9%	46.8%
	middle	37.8%	13.8%	48.4%
	richer	36.7%	12.6%	50.7%
	richest	36.1%	10.7%	53.3%
Others	poorest	41.9%	13.0%	45.1%
	poorer	40.0%	11.6%	48.3%
	middle	38.2%	12.0%	49.8%
	richer	36.0%	11.5%	52.5%
	richest	34.5%	9.1%	56.4%

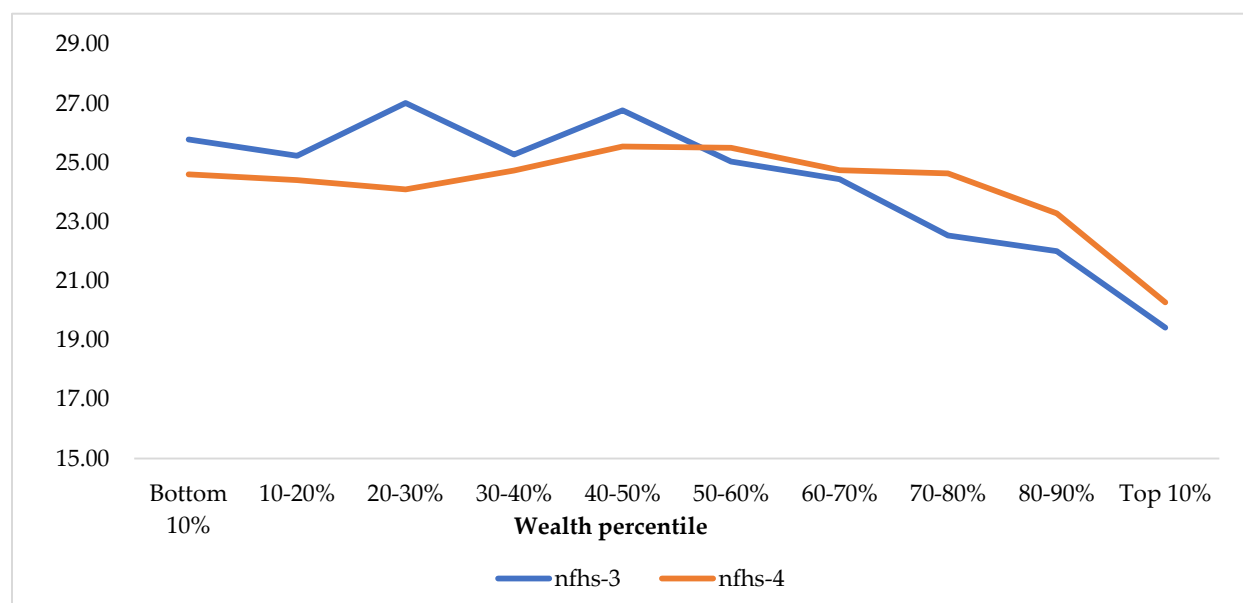
Figure 9 Anaemia prevalence among social groups and median wealth index



Shifts in severe anaemia and wealth status of states: a temporal overview

From the above analysis the wealth status of the household comes out as an important variable. Women belonging from poorer households tend to be more anemic. In the following section, this link will be discussed at greater detail. Specifically, we use two NFHS rounds to see the distribution of women having anaemia across the different wealth quintiles.

Figure 10 Share of women in the highest hemoglobin deviation from normal



Generally, women from richer households have lower hemoglobin deviation in both the rounds, i.e. they have higher hemoglobin levels than poorer households. According to NFHS-III, 26% of women from the bottom 10% of the wealth index had the highest Hb deviation compared to 19% from the top 10% households. However, this gap has reduced between the two rounds. In NFHS-IV, the respective values come to 25% and 20% from the bottom and top 10% respectively. Two things can be observed here: a) between the rounds, the women from poorer households have seen a negative change in contributing to the share of largest Hb deviation, b) between the two rounds, women from richer households have seen a positive shift in contributing to the share of largest Hb deviation. Put simply, this means that the share of severe anaemia has improved among the poorest categories than the richest category. The point of deflection between the rounds is seen at 50-60% wealth quintile, which means that the richest 40-50% of households are found to have women with higher anaemia than the bottom 50-60% of households in NFHS-IV as compared to NFHS-3.

Table 2 Shifts in severe anemic women and wealth class from 2005 to 2016

State		Poorest	Poorer	Middle	Richer	Richest
Andhra Pradesh	Shift in class	-5.15***	-1.88**	+0.92	+4.41***	+1.7*
	Shift in severe anaemia	-5.45**	-2.91	+3.04	+2.93	+2.39
Bihar	Shift in class	+22.43***	-5.11**	-4.16***	-6.23***	-6.93***
	Shift in severe anaemia	+22.37***	-4.81*	-8.6**	-3.08	-5.88**
Chhattisgarh	Shift in class	-9.26***	-1.2	+1.68	+4.6*	+4.18*
	Shift in severe anaemia	-11.49 **	-1.53	+1.68	+5.61	+5.73
Gujarat	shift in class	+1.67	+3.58**	+2.57**	-2.74**	-5.09***
	Shift in severe anaemia	+3.83	+0.63	+3.04	-1.74	-5.76**
Haryana	shift in class	-1.85**	-4.5**	-7.14***	-0.39	+13.88***
	Shift in severe anaemia	-2.86	-4.37	-3.2	-3.9	+14.34**
Himachal Pradesh	shift in class	+0.16	+0.73	-0.15	+4.14	-4.88
	Shift in severe anaemia	-1.24	+4.13	+3.03	+1.18	-7.1***
Jammu and Kashmir	shift in class	+4.33	+9.17**	-4.12	-6.35*	-3.04
	Shift in severe anaemia	+2.53	+9.17	-2.31	-9.76	+0.37
Jharkhand	shift in class	-4.22***	+6.24**	+3.3	-1.63	-3.69*
	Shift in severe anaemia	-9.02**	+8.99	+2.39	-2.1	-0.26
Karnataka	shift in class	-3.35***	-2.27*	+2.84**	+3.56***	-0.77
	Shift in severe anaemia	-4.64	-4.63	+4.92	+2.94	+1.41
Kerala	shift in class	-0.54	-2.01	+1.24	-1.23	+2.53
	Shift in severe anaemia	-2.79	-2.32	+6.3	-1.92	+0.73
Madhya Pradesh	shift in class	-6***	-1.5*	+3.03**	+2.27**	+2.24
	Shift in severe anaemia	-9.49***	0.83	+2.02	+4.55	+2.09
Maharashtra	shift in class	-1.89**	+1.56**	+5.21***	+3.0***	-7.87***
	Shift in severe anaemia	-1.94	-0.21	+4.22	+0.25	-4.91**

State		Poorest	Poorer	Middle	Richer	Richest
Odisha	shift in class	-0.85	+6.7***	+0.85	-2.32**	-4.41***
	Shift in severe anaemia	+1.71	+8.53*	-3.27	-4.06	-2.91
Punjab	shift in class	-0.52	-2.44*	-2.6*	-6.77***	+12.34***
	Shift in severe anaemia	-0.46	-3.56	-1.97	-10.72**	+16.70***
Rajasthan	shift in class	-6.12***	+4.95***	-0.06	+1.96	-0.72
	Shift in severe anaemia	0.93	+7.61	+0.86	-6.64**	-2.76
Tamil Nadu	shift in class	-6.36***	-0.84	-2.65*	+7.03***	+2.83*
	Shift in severe anaemia	-10.18***	-1.33	-2.12	+7.13**	+6.50*
Uttar Pradesh	shift in class	+4.06***	-0.86	-1.84**	-2.38**	+1.02
	Shift in severe anaemia	+3.77	-3.48	-1.85	+0.49	+1.08
Uttarakhand	shift in class	-1.21	+2.14	+4.42	-0.28	-5.1*
	Shift in severe anaemia	-1.7	+3.46	+2.27	-2	-2.03
West Bengal	shift in class	-1.28	+5.04***	+2.67**	-0.48	-5.95***
	Shift in severe anaemia	-2.94	+1.14	+1.71	+1.62	-1.52
Total	shift in class	+0.26***	+0.58***	+0.39***	0.17	-1.39***
	Shift in severe anaemia	-0.75	-0.88	+0.12	+0.67	0.85
*** sig at 1% level, ** sig at 5% * sig at 1% level Source: Computed from NFHS-III & IV						

The above table shows two things: the shift in class referring to the point difference in the proportion of households falling under different wealth quintiles between NFHS-III & IV; and the shift in severe anaemia referring to the point difference in the proportion of women suffering from severe anaemia in the respective wealth quintiles between two NFHS periods. The analysis will give a hint about the general condition of severely anemic women of any particular state which has moved towards poorer or richer categories. It can be gleaned that the states of Andhra Pradesh, Chhattisgarh, Haryana, Kerala, Punjab, Tamil Nadu, Uttar Pradesh among others have shown an addition to the proportion of households in the richest category. In the majority of these mentioned states there has been an increase in the proportion of severely anemic women. The most significant among them being the states of Haryana and Punjab, contributing to an increase by 14% and 17% of severely anemic women in the richest category. On the other hand, for all these states, there has been a decline in households in the poorest category and severe anemic levels have also dropped accordingly. The states where the share of households in the richest category has declined are Bihar, Himachal Pradesh, Gujarat, Karnataka, Maharashtra, Orissa, Rajasthan, Uttarakhand and West Bengal. Barring Karnataka, all the states have also shown a statistically significant decline in severely anemic women. Although at the same time it is also true that the decline in households in the poorest category has also resulted in a decline in shares of severely anemic women. This refers to the fact that as states move away from the poorest class, severe anaemia reduces, at the same time, as they become richer with addition of households in the richest class, they also fall back on

severely anemic women, i.e., anemic levels increase among the richest class. This is a paradox and requires more emphasis. In the multivariate framework we will check further the probabilities of anemic women in different wealth quintiles.

Role of women empowerment

Existing literature (Alaofe, 2017) has most often relied on indicators of empowerment like women's education and control over income to study the effect of women's agency on nutritional outcomes. For instance, investments in human capital resulted in improved iron levels and a significant reduction in anaemia in Peruvian women (Romana, et al. 2006), and improved intake of Vitamin A in preschool children in Thailand via women's leadership abilities (Underwood et al 1996). Women's control over income can have positive implications for child anthropometric outcomes and women's dietary intake (Sheila, C 2016). Women can play a crucial role in the alteration of intra-household dynamics that can result in improved health and nutrition outcomes especially for children. This is driven, globally, by a change in their control of income, decision making power, access to health services and nutrition education while recognizing constraints of time and resources faced by them.

Table 3 Exposure to mass media and levels of anaemia among women 15-49 years

	Reads newspaper/magazine			Listens to radio			Watches Television		
	Mild	Moderate	Severe	Mild	Moderate	Severe	Mild	Moderate	Severe
Not at all	39.62	13.02	1.17	38.73	12.17	1.03	40.7	13.37	1.12
Less than once a week	38.46	11.9	0.88	38.21	11.52	0.88	39.52	12.5	1.05
At least once a week	36.91	10.52	0.85	36.54	11.47	0.95	39.27	12.08	1.01
Almost everyday	35.12	9.43	0.62	36.89	11.16	0.86	37.44	11.51	0.96

According to Table 4 anemic levels tends to decline as more women are exposed to mass media like reading newspaper/magazine, listening to the radio and watching television. 39.62% women who never read newspaper suffer from mild anaemia vis-à-vis 35.12% women have anaemia who read newspaper/magazines almost every day. In case of moderate anaemia, the proportion decline from 13.02% to 9.43% among those women who never read newspaper/magazines to those who read almost every day. A similar decline in anaemia is also noticed among women who listens to the radio and watches television, but the decline is prominent in case of reading newspaper/magazines as it is tied to literacy levels. Educated women read newspapers more than uneducated women and henceforth the level of awareness of these women is higher and hence, they take prompt actions with respect to health care needs.

Figure 11 State wise anemic prevalence who reads newspaper/magazines daily



The above figure shows the state-wise picture of women who reads newspapers/magazines daily and the anemic prevalence among women aged 15-49 years. Except for Kerala and Chandigarh, all the major states are clustered around a patch where the variation of reading newspapers is from 5 to 22%. Share of anemic women for these states vary from about 45 to 65%. Uttarakhand, Karnataka, Rajasthan, Chhattisgarh and Assam have similar shares of anaemia but share of women who read a newspaper daily varies among them. More women from the state of Karnataka read newspaper daily while women from the state of Assam reads the least among these groups of states. Although the share of anemic level is widely different between Chhattisgarh and Jharkhand, about equal proportion of women read newspapers/magazines every day. Kerala is situated as an outlier with a low proportion of anaemia as well as the highest proportion of women reading newspapers regularly. Chandigarh has a lower share of women reading newspapers than Kerala but much anemic prevalence rate compared to Kerala.

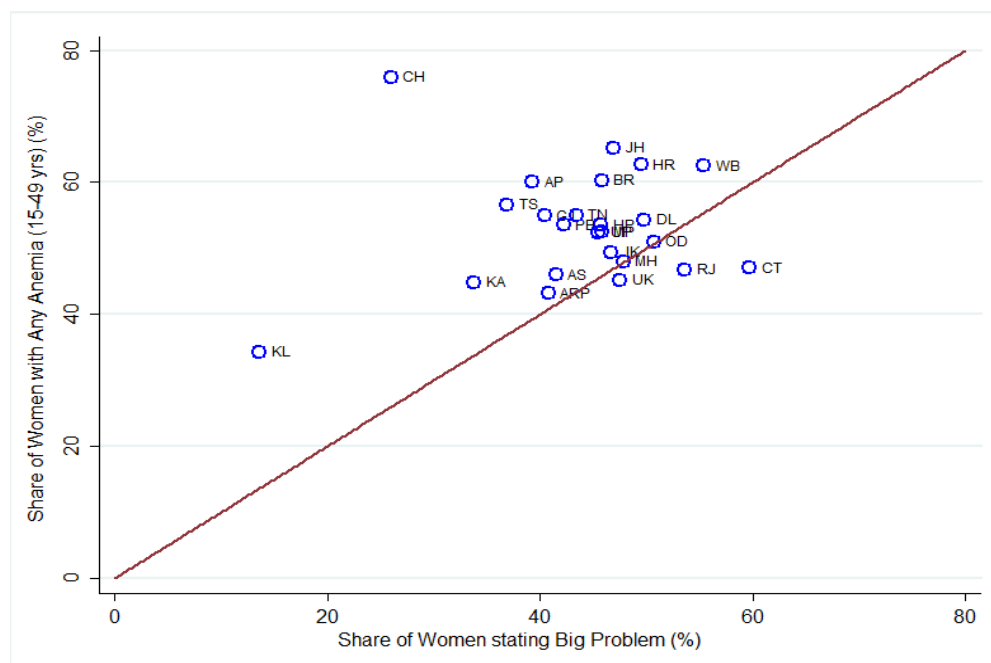
Numerous factors prevent women from getting medical advice or treatment for themselves. NFHS IV asks a set of questions to women as factors that prevent women from seeking medical treatment when she is sick and want to get advice or treatment. The set of questions are:

- a) Getting permission to go
- b) Getting money needed for treatment

- c) Distance to the health facility
- d) Having to take transport
- e) Finding someone to accompany
- f) Concern there may not be a health care provider
- g) Concern there may not be a female health care provider
- h) Concern there may not be any drugs available

Each of these indicators are categorized into three categories of responses: big problem, small problem and no problem. For the analysis purpose, the response big problem has been taken under consideration to build a composite score using the additive method. The newly constructed variable is known as the medical treatment score. A positive relationship can be observed among the two variables. As medical treatment score increases, anemic prevalence goes up. Thus, if women face constraints to access to medical treatment and otherwise states it to be a big problem, they are prone to be more anemic. Classic example is the case of Kerala. The share of anemic women is lowest in this state and also the least proportion of women have stated access to medical treatment as a big problem. Karnataka, Arunachal Pradesh and Assam find access to medical treatment as somewhat of a big problem, less than other states. This is reflected in the anaemia levels too. However, there are some states like Chhattisgarh and Rajasthan where medical treatment score is quite high, but the share of anemic women is not as like other states having the same medical treatment score.

Figure 12 State wise anemic prevalence and medical treatment score



Determinants of levels of anaemia: a multilevel analysis

A multinomial logistic regression model has been used here to determine the relative risk ratios to fall into mild and moderate/severe anaemia in comparison to non-anemic people. Since the set of factors and degree of impacts of a different kind of anaemia is different across currently married women (*the incidence of anaemia varies across marital and pregnancy status*), this exercise has been run twice, one for all women and the other for the currently married woman subsample. Three kinds of factors have been analyzed to explain the variation in anemic level across these population: individual level factors (age, education, wealth, socio-religious sections, occupation, household size, number of living children and pregnancy status etc., controlled by state-specific fixed effects), dietary preferences and diversity (controlled by all the individual level factors and state-specific fixed effects), factors related to women empowerment (mobility, decision making, and mass-media exposure, controlled by individual socio-demographic factors and state-effects). The specifications of these three models is described below.

When there are more than two groups in the dependent variable, as in the present study, each of the odds derived from one group of the dependent variable will be compared to the odds of the reference category, to obtain a relative odds ratio. In the present case, the baseline category is the people with no anaemia, on the basis of which the relative odds ratios of mild anaemia and moderate/severe anaemia are measured. As mentioned above, the core model specification includes three steps for each kind of sample (all women and currently married women subsample): first only using the individual socio-demographic factors in response to the multi-category response variable, and then adding up the dietary diversity and preferences variables and women empowerment variables sequentially. Hence, there are three models for each kind (all women and currently married subsample): These models can be specified like the following

For all women sample:

$$P(\text{Level Anem}_i) = \alpha + \beta_i x_i + \dots + \mu \dots \dots \dots (1)$$

$$P(\text{Level Anem}_i) = \alpha + \beta_i x_i + \Gamma_i DIET_i + \dots + \mu \dots \dots \dots (2)$$

$$P(\text{Level Anem}_i) = \alpha + \beta_i x_i + \Pi_i EMPW_i + \dots + \mu \dots \dots \dots (3)$$

For Currently married subsample:

$$P(\text{Level Anem}_i) = \alpha + \beta_{1i} x_i + \beta_{2i} k_i + \dots + \mu \dots \dots \dots (1)$$

$$P(\text{Level Anem}_i) = \alpha + \beta_{1i} x_i + \beta_{2i} k_i + \Gamma_i DIET_i + \dots + \mu \dots \dots \dots (2)$$

$$P(\text{Level Anem}_i) = \alpha + \beta_{1i} x_i + \beta_{2i} k_i + \Pi_i EMPW_i + \dots + \mu \dots \dots \dots (3)$$

Where x_i is a set of individual characteristics, k_i is a set of specific characteristics applicable for currently married women, and $DIET_i$ is a bunch of dietary diversity and preferences measure, and $EMPW_i$ are the women empowerment variables.

Results and Discussions

Table 5 shows the distribution of anaemia (mild & moderate/severe) among women in the reproductive age group in India for the year 2015-16. Across communities, the prevalence of mild and moderate/severe anaemia is highest among the Hindus, OBCs, married women and women not engaged in any work. Empowerment variables are showing a strong connection where women who are least empowered and do not read newspaper and magazines are equally likely to be mild and moderate/severe anemic. The relationship will be explored with greater precision in the multivariate model as exhibited in Models 1.

Table 4 Descriptive statistics of mild and moderate/severe anaemia prevalence among ever married women in 15-49 years age group, NFHS-IV (2015-16)

Background		Mild anaemia		Moderate/Severe anaemia	
		N	%	N	%
Social Group	Non-SC/ST	50,679	20.04	15,291	17.88
	SC	49,190	19.45	18,013	21.06
	ST	47,241	18.68	17,145	20.05
	OBC	104,626	41.37	34,611	40.47
	Others	1,149	0.45	471	0.55
Religion	Hindu	201999	76.77	68454	76.54
	Muslim	34113	12.96	12097	13.53
	Christian	14636	5.56	4595	5.14
	Others	12384	4.7	4294	4.79
Wealth Index	Poorest	56,364	21.42	19,988	22.35
	Poorer	57,383	21.81	20,078	22.45
	Middle	53,817	20.45	19,001	21.24
	Richer	49,341	18.75	16,758	18.74
	Richest	46,227	17.57	13,615	15.22
Marital Status	Never Married	63,352	24.08	19,350	21.63
	Currently Married	188,350	71.58	65,847	73.62
	Widowed	7,827	2.97	2,862	3.2
	Others	3,603	1	1381	1.54
Age	Less than 20	58,795	22.34	19,140	21.4
	20-25	47,223	17.95	17,117	19.14
	25-30	43,071	16.37	14,724	16.46
	30-35	35,010	13.31	11,706	13.09
	35-40	33,094	12.58	11,311	12.65
	40 & above	45,939	17.46	15,442	17.27
No of HH Members (mean)		5.85		5.83	
Maternal Education	No Education	78,129	29.69	29,308	32.77
	Primary	33,445	12.71	11,938	13.35
	Secondary	123,718	47.02	40,456	45.23

Background		Mild anaemia		Moderate/Severe anaemia	
		N	%	N	%
	Higher	27,840	10.58	7,738	8.65
Place of Residence	Urban	72,501	27.55	23,543	26.32
	Rural	190,631	72.45	65,897	73.68
Occupation Type	Not in Work force	31,411	69.27	11,644	68.02
	Professional/Technical Managerial	1,180	2.6	360	2.1
	Clerical	161	0.36	54	0.32
	Sales	613	1.35	229	1.34
	Agricultural	7,150	15.77	2,988	17.46
	HH and Domestic Services	1,474	3.25	566	3.31
	Manual Labour	2,838	6.26	1,102	6.44
No of Living Children	No Children	82,533	31.37	27,378	30.61
	1	37,400	14.21	13,190	14.75
	2	63,501	24.13	21,616	24.17
	3	42,103	16	14,432	16.14
	4+	37,595	14.29	12,824	14.34
Currently Pregnant	Yes	7,534	2.86	8,272	9.25
	No	255,598	97.14	81,168	90.75
Diet Type	Veg only	71,945	27.34	24,824	27.75
	Veg and Occasional Non-Veg	134,948	51.29	46,059	51.5
	Non-veg Frequently	56,239	21.37	18,557	20.75
Whether eats Fast Food	Daily/Weekly	137,066	52.09	44,735	50.02
	No/Rare	126,066	47.91	44,705	49.98
Decision Making	Low (None)	235,550	89.52	78,945	88.27
	Medium (Score 1-3)	7,356	2.8	2,751	3.08
	High (Score>3)	20,226	7.69	7,744	8.66
Mobility	Low (None)	220,022	83.62	73,313	81.97
	Medium (Score 1-3)	16,578	6.3	6,202	6.93
	High (Score>3)	26,532	10.08	9,925	11.1
Reads Newspaper/ Magazines	No	165,828	63.02	59,475	66.5
	Once a week or less	68,360	25.98	21,781	24.35
	Everyday	28,944	11	8,184	9.15

Results from model 1 (all sample women) rendered quite strong effects for all the socio-demographic variables. It can be seen that the relative odds of falling into mild anaemia compared to any anaemia is higher for the SCs, STs and the OBCs which also corroborates our analysis carried out earlier. Comparing religion groups, Hindus are greater disadvantaged to having higher mild anaemia than Muslims, Christians and other religions. Among all, the Christian have the least relative odds of having mild anaemia over no anaemia. For Christians relative to Hindus, the relative risk for mild anaemia compared to no anaemia would be expected to increase by a factor of 0.564 given other variables in the model are kept constant.

Similarly, for Muslims relative to Hindus, the relative risk for mild anaemia compared to no anaemia would be expected to increase by a factor of 0.846 given other variables in the model are kept constant. It is well known that the Christians and Muslims are known adhering for non-vegetarian dietary habits and have less inhibitory cultural practices related to food distribution and allocation within the family. A class-wise distinction can also be observed such that the richest class were found to have least relative odds for mild anaemia relative to no anaemia compared to the poorest class. Nutritional outcomes of women are often mediated by wealth structure of the family and richer families can better afford to take care of the health needs of the family including women's health. Married and widowed women have higher odds of falling to mild anaemia relative to no anaemia than never married women. Child bearing imposes reproductive burden among women and upon entering into marital homes, women lose autonomy and eating order in the family of the women is often last from usually the left-over food. Across age groups, older women stand at higher risk of developing mild anaemia relative to no anaemia. The health of the women is dispensed to other members of the family and women's health needs are sought after the child and other members health needs are addressed. Maternal education does not vary much in terms of developing mild anaemia compared to no anaemia, however women with some form of education tend to have lower odds of mild anaemia. Women engaged in agricultural, household, domestic and manual labor have higher odds of having mild anaemia in comparison to no anaemia and the similar pattern is also seen for developing moderate or severe anaemia (Model 1b). A major difference between the two models (mild and moderate/severe anaemia models) lies to the reversal of results in age group. The relative odds are higher for younger women in case of having moderate/severe anaemia versus no anaemia. On the other hand, in case of mild anaemia, older women stand at a higher risk. The relationship with education and the economic class becomes stronger in case of moderate and severe anaemia model. The relative risk for moderate/severe anaemia relative to no anaemia is expected to increase by a factor of 0.689 for richest relative poorest whereas for mild anaemia, the relative odds stand at 0.733 for the richest compared to poorest.

For currently married model (Model 1) the relationship among the covariates remains the same only that the younger age groups become more vulnerable to become mildly anemic due to early pregnancy. Women having any children have higher relative risks of being anemic than childless women. The higher chances of being anemic in childbearing years gets carried forward beyond the pregnancy term which might negatively impact the health of the mother.

**Table 5 Model 1 Socio-demographic model (moderate/severe vs no anaemia)
relative risk ratios (base: no anaemia)**

Background	Variables	All Sample					Currently Married Sample				
		Odds Ratio	Standard Error	P	95% CI-	95% CI+	Odds Ratio	Standard Error	P	95% CI-	95% CI+
Social Group	Non-SC/ST										
	SC	1.265	0.038	0.000	1.192	1.342	1.244	0.044	0.000	1.160	1.334
	ST	1.457	0.048	0.000	1.367	1.554	1.440	0.056	0.000	1.335	1.553
	OBC	1.115	0.028	0.000	1.061	1.172	1.095	0.033	0.003	1.032	1.161
	Others	1.058	0.160	0.711	0.786	1.423	1.260	0.205	0.156	0.916	1.733
Religion	Hindu										
	Muslim	0.879	0.026	0.000	0.829	0.932	0.888	0.032	0.001	0.827	0.952
	Christian	0.518	0.022	0.000	0.478	0.563	0.547	0.028	0.000	0.495	0.604
	Others	1.000	0.043	0.998	0.920	1.087	1.007	0.051	0.885	0.912	1.113
Wealth Index	Poorest										
	Poorer	0.888	0.025	0.000	0.839	0.939	0.868	0.029	0.000	0.813	0.926
	Middle	0.884	0.026	0.000	0.834	0.937	0.864	0.030	0.000	0.806	0.925
	Richer	0.856	0.028	0.000	0.803	0.912	0.835	0.032	0.000	0.774	0.901
	Richest	0.689	0.026	0.000	0.639	0.743	0.670	0.031	0.000	0.612	0.733
Marital Status	Never Married										
	Currently Married	1.245	0.039	0.000	1.171	1.323					
	Widowed	1.358	0.082	0.000	1.206	1.530					
	Others	1.412	0.119	0.000	1.197	1.665					
Age	Less than 20										
	20-25	0.970	0.033	0.356	0.908	1.035	0.872	0.043	0.006	0.791	0.961
	25-30	0.793	0.030	0.000	0.738	0.854	0.724	0.038	0.000	0.653	0.803
	30-35	0.737	0.029	0.000	0.682	0.796	0.677	0.038	0.000	0.606	0.755
	35-40	0.746	0.031	0.000	0.689	0.808	0.705	0.041	0.000	0.629	0.790
	40 & above	0.704	0.028	0.000	0.651	0.761	0.676	0.039	0.000	0.603	0.757
No of HH Members		1.010	0.004	0.006	1.003	1.017	1.013	0.004	0.003	1.004	1.021
Education	No Education										
	Primary	0.879	0.027	0.000	0.828	0.933	0.862	0.029	0.000	0.807	0.921
	Secondary	0.792	0.020	0.000	0.753	0.833	0.811	0.023	0.000	0.767	0.859
	Higher	0.637	0.026	0.000	0.588	0.691	0.667	0.035	0.000	0.602	0.738
Place of Residence	Urban										
	Rural	1.021	0.024	0.360	0.976	1.069	1.041	0.029	0.146	0.986	1.099
Occupation Type	Not in Work force										
	Professional/ Technical Managerial	0.980	0.062	0.749	0.866	1.109	1.007	0.079	0.926	0.864	1.175
	Clerical	0.941	0.146	0.694	0.694	1.276	0.846	0.179	0.430	0.559	1.281
	Sales	0.924	0.070	0.298	0.796	1.072	0.856	0.080	0.097	0.712	1.028
	Agricultural	1.086	0.028	0.002	1.032	1.142	1.075	0.032	0.014	1.015	1.139
	HH and Domestic Services	1.076	0.055	0.149	0.974	1.190	1.065	0.065	0.303	0.945	1.200
	Manual Labour	0.999	0.038	0.971	0.927	1.076	1.000	0.046	0.995	0.914	1.095
No of Living Children	No Children										
	1						1.245	0.053	0.000	1.145	1.354
	2						1.433	0.063	0.000	1.314	1.561
	3						1.412	0.069	0.000	1.284	1.553
	4+						1.372	0.072	0.000	1.237	1.521
Currently Pregnant	Yes										
	No						1.990	0.076	0.000	1.847	2.145
	Constant	0.335	0.017	-21.57	0.303	0.370	0.313	0.022	0.000	0.273	0.359
	Likelihood Ratio X2	2202.7					2500.9				
	log likelihood	-114335.6					-81696.35				

In model 2, when the individual factors are controlled with empowerment variables, it can be seen that women enjoying higher control over decision making in the household have a lower relative risk for anaemia than who have no say in decision making. The relationship is insignificant in the mild anaemia model but becomes statistically significant in moderate/severe anaemia model which points to the larger role of empowerment to prevent women from falling into adverse health situations. The same relationship holds for women who read newspapers/magazine have lower relative odds of anaemia than who do not. The situation on the other hand becomes inverse concerning mobility where women with low mobility have lower odds of anaemia. It should be kept into consideration how the family structure operates in India. Generally, the poor and less educated are engaged into outside work which makes them more mobile. However, mobility may not help to cater to the benefits of empowerment because even if more women take part in decision making in household matters, many are poor and belongs to the marginalized communities. For both these attributes, as it had been checked earlier as well, there are higher chances of developing any forms of anaemia.

Table 6 Model 2 Women's empowerment adjusted to individual factors and state effects (moderate/severe vs no anaemia) relative risk ratios (base: no anaemia)

Background	Variables	All Sample					Currently Married Sample				
		Odds Ratio	Standard Error	P	95% CI-	95% CI+	Odds Ratio	Standard Error	P	95% CI-	95% CI+
Social Group	Non-SC/ST										
	SC	1.258	0.038	0.000	1.186	1.335	1.239	0.044	0.000	1.155	1.329
	ST	1.452	0.047	0.000	1.362	1.548	1.437	0.056	0.000	1.332	1.550
	OBC	1.108	0.028	0.000	1.054	1.165	1.088	0.033	0.005	1.026	1.154
	Others	1.051	0.159	0.744	0.781	1.413	1.255	0.204	0.163	0.912	1.727
Religion	Hindu										
	Muslim	0.876	0.026	0.000	0.826	0.929	0.885	0.032	0.001	0.825	0.949
	Christian	0.521	0.022	0.000	0.480	0.566	0.550	0.028	0.000	0.497	0.607
	Others	0.996	0.043	0.926	0.916	1.083	1.004	0.051	0.931	0.909	1.110
Wealth Index	Poorest										
	Poorer	0.890	0.026	0.000	0.841	0.941	0.868	0.029	0.000	0.813	0.927
	Middle	0.892	0.027	0.000	0.841	0.946	0.868	0.030	0.000	0.811	0.930
	Richer	0.874	0.029	0.000	0.819	0.932	0.846	0.033	0.000	0.784	0.914
	Richest	0.723	0.028	0.000	0.670	0.781	0.696	0.033	0.000	0.635	0.763
Marital Status	Never Married										
	Currently Married	1.322	0.052	0.000	1.224	1.427					
	Widowed	1.326	0.081	0.000	1.177	1.494					
	Others	1.387	0.117	0.000	1.176	1.636					
Age	Less than 20										
	20-25	0.972	0.033	0.394	0.910	1.038	0.880	0.044	0.011	0.799	0.971
	25-30	0.801	0.030	0.000	0.744	0.862	0.736	0.039	0.000	0.663	0.817
	30-35	0.746	0.030	0.000	0.690	0.807	0.690	0.039	0.000	0.618	0.771
	35-40	0.756	0.031	0.000	0.697	0.820	0.720	0.042	0.000	0.642	0.808
	40 & above	0.714	0.029	0.000	0.660	0.773	0.692	0.040	0.000	0.617	0.776
No of HH Members		1.009	0.004	0.010	1.002	1.016	1.012	0.004	0.006	1.003	1.021
Education	No Education										
	Primary	0.892	0.027	0.000	0.840	0.947	0.873	0.029	0.000	0.817	0.933
	Secondary	0.836	0.023	0.000	0.791	0.882	0.849	0.027	0.000	0.797	0.903
	Higher	0.701	0.031	0.000	0.643	0.766	0.727	0.041	0.000	0.651	0.811
Place of Residence	Urban										
	Rural	1.015	0.024	0.535	0.969	1.062	1.036	0.029	0.209	0.981	1.094

Background	Variables	All Sample					Currently Married Sample				
		Odds Ratio	Standard Error	P	95% CI-	95% CI+	Odds Ratio	Standard Error	P	95% CI-	95% CI+
Social Group	Non-SC/ST										
	SC	1.258	0.038	0.000	1.186	1.335	1.239	0.044	0.000	1.155	1.329
Occupation Type	Not in Work force										
	Professional/Technical Managerial	1.005	0.064	0.936	0.888	1.138	1.034	0.081	0.676	0.886	1.206
	Clerical	0.958	0.149	0.783	0.706	1.300	0.861	0.182	0.481	0.569	1.304
	Sales	0.932	0.071	0.351	0.803	1.081	0.867	0.081	0.127	0.721	1.041
	Agricultural	1.086	0.028	0.001	1.032	1.143	1.078	0.032	0.010	1.018	1.143
	HH and Domestic Services	1.086	0.055	0.107	0.982	1.200	1.077	0.066	0.223	0.956	1.214
	Manual Labour	1.001	0.038	0.978	0.929	1.079	1.007	0.046	0.877	0.920	1.102
Decision Making	Low (None)										
	Medium (Score 1-3)	0.879	0.031	0.000	0.821	0.942	0.869	0.031	0.000	0.811	0.932
	High (Score>3)	0.925	0.028	0.012	0.871	0.983	0.919	0.029	0.007	0.864	0.977
Mobility	Low (None)										
	Medium (Score 1-3)	0.863	0.035	0.000	0.797	0.933	0.862	0.042	0.002	0.785	0.948
	High (Score>3)	0.888	0.036	0.003	0.821	0.960	0.880	0.042	0.008	0.801	0.966
Reads Newspaper/Magazines	No										
	Once a week or less	0.914	0.023	0.000	0.870	0.960	0.932	0.029	0.022	0.877	0.990
	Everyday	0.808	0.030	0.000	0.752	0.870	0.836	0.039	0.000	0.762	0.916
No of Living Children	No Children										
	1						1.252	0.054	0.000	1.151	1.362
	2						1.441	0.063	0.000	1.322	1.571
	3						1.417	0.069	0.000	1.288	1.559
	4+						1.376	0.073	0.000	1.240	1.526
Currently Pregnant	Yes										
	No						1.996	0.076	0.000	1.851	2.151
	Constant	0.383	0.024	0.000	0.339	0.432	0.380	0.032	0.000	0.323	0.447
	Likelihood Ratio X2	2279.8					2551.4				
	log likelihood	-114297.1					-81671.1				

Conclusion

Iron deficiency anaemia is related to the nutritional intake of women more so given the fact that Indian women do not consume adequate and the right kind of food needed to function properly. Although nutrition is the direct outcome for anaemia prevalence, a bunch of other factors also determine the lack of nutritional intake. The study seeks to throw light on the host of various socio-demographic, dietary and empowerment variables for an in-depth understanding of the variations and relationships between different types of anaemia. In this aspect the study is unique as it envisages to connect the underlying mechanisms between different levels of haemoglobin deficiency.

In India, the gradient of mild and moderate/ severe anaemia is not very clear. Mild anaemia can arise even with a slight departure in haemoglobin from normal levels. This leads to the condition of almost half of the Indian women to be mildly anemic. Eastern states of India fall into this category but the picture fades out in case of moderate/ severe anaemia. Some of the southern states like Telangana, Andhra Pradesh, Tamil Nadu as well as states of Punjab, Haryana and Delhi contribute a larger share of moderate/ severe anemic women out of any anemic women. Some of these states also display an increase in severe anaemia as states tended

to become richer, notably the states of Punjab, Haryana and Delhi. Even though anemic condition is still better in richer households. The poorer households have shown an improvement in anemic levels compared to richer whose proportion of anemic women has gone up between two NFHS rounds.

Among each of the social groups, the richest class has the least anemic prevalence. Richer families are better equipped to meet the nutritional requirements that prevents anaemia among women. Never married women have a better anemic condition than women who are divorced or separated. The age of marriage also displays to be an important factor as the non-anemic women have a higher mean age at marriage. Maternal education does not vary much in terms of developing mild anaemia compared to no anaemia, however, women with some form of education tend to have lower chances of mild anaemia. Women engaged in the agricultural, household, domestic and manual labor have higher chances of having mild anaemia in comparison to no anaemia and the similar pattern is also seen for developing moderate or severe anaemia.

Less diversified diet leads to higher relative risks of falling into anaemia (both mild and moderate/ severe). In the case of mild anaemia, dietary diversity assumes more prominence whereas the intake of veg/ non-veg food becomes more important for moderate/ severe anaemia. Women who have higher decision making in the household have a lower relative risk for anaemia than who have no decision making. The relationship is less important in the mild anaemia model but explains a large part of the variation in the moderate/ severe anaemia model. Thus, this study concludes by saying that although certain parameters help in explaining any anemic conditions such as wealth, religious and social groups, age and education, the moderate and severe prevalence of anaemia is attributed to iron-rich nutritional intake and higher empowerment of women.

Implications for Policy

The problem of iron-deficiency anaemia remains a major concern problem in India. The high levels of anemic prevalence in the reproductive age group have invited widespread implications among academia, health professionals and program managers. Even though anaemia was recognized as a worldwide problem, the Indian Government spearheaded the problem and it appeared to be one of the first few nations to develop an anaemia policy framework. Ministry of Health and Family Welfare first launched National Nutritional Anaemia Prophylaxis Program (NNAPP) in the 1970s to meet the iron deficiency of high-risk population, namely pregnant and lactating women and children below 5 years. With time more population apart from high-risk has been identified constituting both anemic/non-anemic women, adolescents and children below 6 months. Further, integration of programs has been realized and Anaemia Mukht Bharat was launched in 2018 with an ambitious target of reducing anaemia prevalence among reproductive age women to 35%. The current level stands at 53%.

The analysis above can be drawn as an essential input on what aspects state capacity can be strengthened and what measures states can be taken to make the program a success. Apart from the IFA tablets distribution and deworming measures taken by the respective states, addressing some of the women's empowerment variables can lead a sustaining goal-oriented solution. Increasing the age of marriage can be a useful policy factor as women tend to be anemic during early pregnancy and seem to carry that forward during the later years of their lives. Although early marriage has gone down in India, if more can be achieved on that ground, it can be far more effective. One way to increase the age of marriage is through higher education, as education postpones the age of marriage. Indeed, in our econometric modeling exercise, higher maternal education has corresponded to significantly lower chances of developing moderate/severe anaemia. Education can open many gateways as uneducated women often bear the brunt of severe anaemia, having less decision making and low autonomy in the household which hinders their overall attention to health. Poverty acts as an endogenous variable, as many uneducated women fall in this category. Another natural way to increase Hb levels is to incorporate a diverse diet with adequate quantity of non-vegetarian foods (eggs at least who adhere to vegetarianism) as it can act positively to reduce anaemia. With the analysis carried out, it seems there have been improvements in the poorer households, but many richer states have seen a rise of severe anemic women which calls for important policy discussions in the near future. Efforts need to be made to understand the role of women's empowerment in some of these richer states where severe anaemia has gone up. Much of it can be related to patriarchal culture and reduced autonomy of women. If the proportion of anemic women in the richest class in some of the richest states continue to rise, it might reflect a health and developmental paradox. An important case study in this case can be the states of Punjab, Haryana & Delhi where severe anaemia has gone up in the richest class. These states portray many harsh gender relations, one classic example is the huge number of sex-selective abortions done in these states by the upper class and castes that have resulted in thousands of missing girls, although the states progress in several other spheres.

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6

A Study of Trends and Correlates of Caesarean Section Births in Jammu and Kashmir

Bashir Ahmad Bhat

Abstract

Cesarean section surgery, when medically indicated and performed by trained staff with the necessary equipment and supplies can be a life-saving procedure for the mother and baby, but this procedure can also lead to short-term and long-term health effects for women and children. World Health Organization has estimated the ideal population-level cesarean rate at 10-15%, but, unfortunately, the high and increasing levels of cesarean delivery rates around the world illustrate that the procedure is not always medically indicated. Like other parts of the world, C-section rates are also increasing in India but there are few States like Telangana and Jammu and Kashmir where more than 40% of births are delivered through C-section. Based on the National Family Health Survey Data of various rounds, the aim of this study is to study the trends and patterns of caesarean section in Jammu and Kashmir. The study has examined the role of various socio-economic and demographic factors in explaining the variation in caesarean section rates among various groups. We use binary logistic regression model to estimate the increase in the likelihood of cesarean delivery by various background variables. The study found that higher age at marriage is one of the reasons for increasing C-section. C-section rate is very high in Kashmir than in Jammu and very high in private facilities than in public facilities. There is not much difference in C-section rates by complications at the time of delivery and planned C-sections are very high as compared to emergency C-sections.

Introduction

The proportion of CS at population level is a measure of the level of access to, and use of, an obstetric intervention proven to be effective in saving lives. It has served as a proxy measure for governments, policy-makers and public health professionals for assessing progress in maternal and infant health, and for monitoring emergency obstetric and resource use (Zhang J, Liu Y, 2008). Cesarean section surgery, when medically indicated and performed by trained staff with the necessary equipment and supplies can be a life-saving procedure for the mother and baby, but this procedure can also lead to short-term and long-term health effects for women and children. The C-section complications that have adverse consequences for the mortality and morbidity of both the mother and the newborn are well documented in the literature (Baskett TF, 1981; Sachs BP, 1983; Hall MH. 1994; Hopkins K. 2000;). Some of the negative health outcomes in infants born via cesarean delivery include childhood

obesity, respiratory disorders, type 1 diabetes, acute lymphoblastic leukemia, impaired cognitive development, higher rates of autism, and an increased risk of neuro developmental disorders (Zhang T, 2019; Kuhle S, 2015; Keag OE, 2018; Polidano C, 2017).

Cesarean delivery has been reported to be associated with an approximately 4-fold increase in the risk of maternal death (Caughey AB, 2014). Studies have shown that serious risks associated with C-section are deadly bleeding, a lengthy recovery, organ damage and permanent injury. Compared with women having a vaginal birth, those having a C-section for the first time have 3.1 times greater risk of blood transfusions, a 5.7 times greater risk of an unplanned hysterectomy and a six times greater risk of being admitted into intensive care. Studies have also observed that children born via cesarean are more likely to develop respiratory problems, diabetes and obesity later in life. Short-term risks of CS include altered immune development, an increased likelihood of allergy, atopy and asthma, and reduced intestinal gut microbiome diversity.

Recent evidence also suggests that rates of cesarean delivery beyond a 15% threshold may lead to increased maternal and perinatal morbidity (Bogg L, 2011). Although the initial cesarean delivery is associated with some increases in morbidity and mortality, the downstream effects are even greater because of the risks from repeat cesareans in future pregnancies (Han W, Song J, 2011). In addition, unnecessary cesarean deliveries may be associated with higher health care costs in many low-income settings with limited resources (Gilbert SA, 2013). Therefore, cesareans should be considered a major surgical intervention and only be performed when clinically necessary. Furthermore, to minimize the risks associated with cesarean section, the surgery should only be performed by skilled health workers in high quality facilities.

For many years, researchers have debated the optimal cesarean rate for maximizing maternal and infant health outcomes. Since 1985, the World Health Organization has estimated the ideal population-level cesarean rate at 10-15%, although some scientists have suggested a higher figure. But, unfortunately, the high and increasing levels of cesarean delivery rates around the world illustrate that the procedure is not always medically indicated.

Literature Review

A lot of research has been done on various issues of C-section. Though, it is not possible to review all the published material on C-section across the world however, some important publications would be reviewed. O'Leary and Steer (1964) investigated 165 cases of cesarean section with tubal ligation cared during 1952-1961.

The main purpose of the study was to compare the morbidity figures in cesarean hysterectomy with that of cesarean section. About 116 of the 165 patients needed blood transfusion. The study stressed that total cesarean hysterectomy should be limited to the patients who have a definite abnormality. Mishra and Ramanathan (2002) used National Family Health Surveys, 1992-93, concluded that the C-section rates in various States are rising and this cannot be attributed entirely to the rise in institutional deliveries alone because of the strong association between caesarean sections and private sector institutions. They concluded that it is possible that this extremely useful surgical procedure is being misused for profit purposes in the private sector in several states. Guilmoto (2019) has analyzed the trends, regional variations and socio-economic disparities in C-section births in India. He points out that the present level of C-section is already higher than that observed in some industrialized countries in Europe such as the Netherlands or Finland. He also has concluded that there are a significant regional and sex disparities, with a substantial deficit of cesarean deliveries among underprivileged groups and almost 2 million excess cesarean births every year among more advanced sections of the population. Further, India is characterized at the same time by a deficit of access to cesarean deliveries in poorer communities and the emerging overuse of cesarean delivery in more affluent regions and social groups. Bhatia et al (2020) have analysed the C-sections in private sector in India using various rounds of NFHS data. Their findings suggest that the rising trend in C-section rates in the private sector cannot be explained by medical reasons alone. Despite, controlling for medical complications, women's characteristics and preferences, the public-private gap has doubled over the years and that the difference cannot be explained by known determinants of C-section. Over supply of avoidable C-section to the extent of 21%, as a result of physician induced demand and perverse financial incentives was observed in the private sector. Another paper by the same author (2020b) almost on the same topic suggests that the odds ratio for cesarean deliveries in private facilities compared with public facilities increased from 1.62 in the NFHS-3 to 4.17 in the NFHS-4. .

Mitul Surana and Ambrish Dongre (2020) have also analysed the role of private sector in high c-section deliveries in India. Their findings indicate that the probability of C-sections in private medical facilities remains significantly higher than the public sector after controlling for various background characteristics, risk factors and demand factors. Mukherjee et al. (1993) assessed obstetricians compared cesarean section rates between 1990-1991 and 1976-1978 to examine cesarean section-associated perinatal mortality in India. Cesarean section rates increased for almost all indications between 1976-1978 and 1990-1991, except repeat cesarean sections antepartum hemorrhage,

and malpresentation. Between the 2 periods, perinatal mortality increased despite the increase in cesarean sections. Malhotra, Gopalan and Narang (1994) examined if cesarean section offers any advantage over vaginal delivery for a preterm breech fetus in a developing country. A retrospective analysis from hospital records was done at a tertiary care center in North India. In women with a poor obstetric history, the neonatal survival was better in the cesarean group. Even with optimum neonatal care facilities, cesarean section does not offer any advantage over vaginal delivery for a preterm breech fetus in a developing country. Mehta et al. (2001) reported that the steady rise in cesarean section rates is an emerging area of concern in mother-child healthcare and a matter of international attention, since the trend is no longer confined to western industrialized countries. Crude and cesarean section-related perinatal mortality and case-fatality rates may well serve as public-health indicators. No improvement in perinatal outcome was observed beyond a cesarean section rate of 10%, but the perinatal mortality rate in cesarean births increased significantly due to a more liberal use of cesarean sections in preterm deliveries and those that yielded low-birth weight babies.

Objectives

The aim of this study was to study the trends and patterns of caesarean section in Jammu and Kashmir. The study will also study the role of various socio-economic and demographic factors in explaining the variation in caesarean section rates among various groups.

Data and Methods

The disaggregated data used in this study are derived from reanalysis of publicly available micro data from National Family Health Survey-4 and the data from the Health Management Information System (HMIS) web portal of the Ministry of Health and family Welfare. NFHS is a large scale, nationally representative household health surveys that is routinely conducted in India. Till date 5 rounds of NFHS have been completed in India Data and reports for the first four rounds are available on the website of Demographic and Health Surveys (DHS). State and District Fact sheets are also available in the public domain. As the raw data for NFHS-5 is not yet available, therefore for detailed analysis of Jammu and Kashmir, we will use NFHS-4 data. The fourth round of the National Family and Health Survey (NFHS-4) in J&K collected information about 7693 live births born during the five-year period before the survey. More recently, the Health Management Information System has provided annual rates of cesarean births, but the representativeness of these estimates remained untested as private health facilities under report deliveries and c-section deliveries (Krishnan, et al, 2010; Kouser A et al. 2016).

In this study we defined caesarean section rate as the percentage of births achieved by caesarean among all live births in the period before the survey, with data from the NFHS referring to births in the five years before the survey. Data on caesarean sections are based on women's self-report, answering to the question: "Was (NAME) delivered by caesarean, that is, did they cut your belly open to take the baby out?" These data have been shown to be valid and reliable (Li H-T et al, 2016; Liang J et al, 2018). The survey provided additional demographic and health information as well as household socioeconomic status computed by a principal component analysis based on housing type and household amenities (Roustein and Johnson, 2018)

A binary logistic regression model was constructed to estimate the increase in the likelihood of cesarean delivery by various background variables. Cesarean delivery was a binary variable coded as 0 for a vaginal delivery and 1 for a cesarean delivery; thus, an odds ratio (OR) greater than 1 signified that the OR of a cesarean delivery for that particular explanatory variable was higher than that of the reference category. We considered several relevant background characteristics and assessed their associations with the likelihood of having a cesarean delivery. The explanatory variables considered were the size of the child at birth (small, average, or large, as reported by the mother), birth order of the child (1, 2 or 3), maternal age at the child's birth (<19 years, 20-29 years, or 30 years), maternal body mass index (BMI; calculated as weight in kilograms divided by height in meters squared; underweight (BMI, <18.49), normal (BMI, 18.50-24.9), and overweight or obese (BMI, ≥25), maternal educational attainment (no formal education, primary school, secondary school, or higher education (12 years)), household wealth quintile (defined by the NFHS wealth index as the relative index of household wealth based on the standard set of assets owned by the household, including ownership of consumer items and dwelling characteristics; 5 categories of wealth quintiles (poorest, poorer, middle, richer, and richest) from the NFHS wealth index were used), household caste (scheduled caste or scheduled tribe, other backward class, or other caste), household religion (Muslims or non-Muslims), area of residence (urban or rural), and health care facility type (public or private). The adjusted percentages of cesarean delivery rates for selected background variables were calculated from this model. A scenario analysis was performed to estimate the economic burden of avoidable cesarean deliveries in the State by calculating the cesarean deliveries that could have been avoided if WHO recommendation of 15 percent C-section is followed and the potential cost savings that could have been achieved under various scenarios. Data regarding household out-of-pocket expenditures for vaginal and cesarean deliveries, which were available for the first time in the NFHS-4, were used for this analysis. All analyses were performed using SPSS software, version 18. Data were analyzed from January to March, 2021.

Trends and Patterns in Cesarean in India

The proportion of all births delivered by caesarean sections in India increased marginally from 2.5 percent in 1992-93 to 7 percent in 1998-99 and 8.5 percent in 2005-06 (Fig. 1). A rapid increase took place during 2005-06 and 2015-16 when the country witnessed an increase of 102 percent in caesarean deliveries from 8.5 percent in 2005-06 to 17.2 percent in 2015-16. As per NFHS-5, C-section 21.5 percent of births in India are delivered through C-section. Although, C-section rates in India have increased in both rural and urban areas but urban areas have experienced a sharp increase in the C-section rates. During 1998-99, the percentage of births delivered through C-section in rural areas was only 2 percent and it has increased to 5.6 percent in 2005-06 and further increased by 130 percent during the next decade and by 2015-16 around 13 percent of births were delivered through C-section. C-section births have increased much faster during the last five years in rural areas and as per NFHS-5, almost 19 percent in rural areas are delivered through C-section. Urban areas also experienced sharp increase in the percentage of C-section births from 6 percent in 1992-93 to 17 percent in 2005-2006 and to about 29 percent in 2015-16 and further increased to 32 percent in 2019-20. Between 2005-06 and 2019-20, C-section births in urban areas and rural areas have witnessed a threefold increase in C-section births.

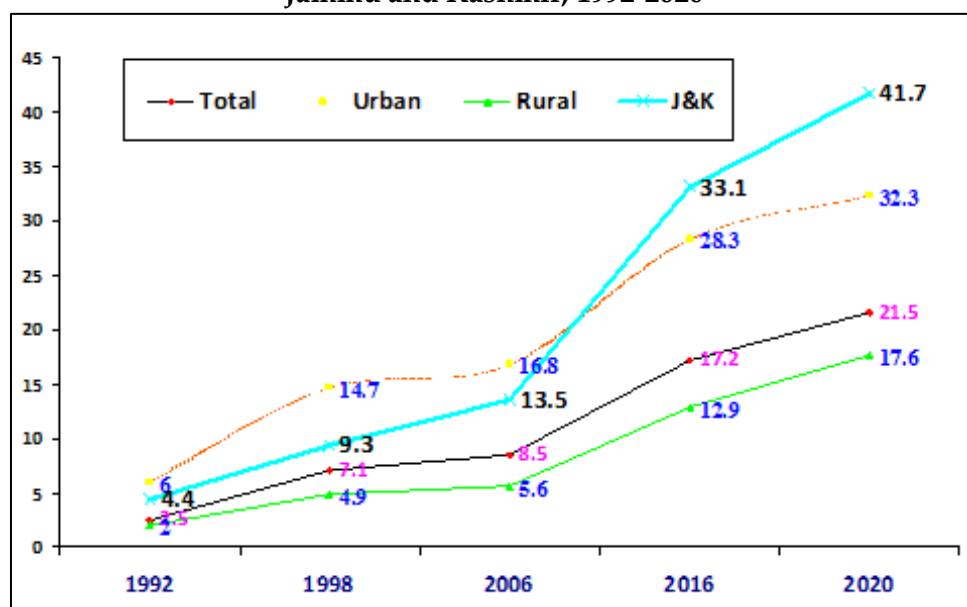
Cesarean Section in Jammu and Kashmir

Jammu and Kashmir have registered an unprecedented increase in the C-section births during the last two decades. (Fig 1) As per NFHS-1, less than 5 percent of the births were delivered by C-section during 1992-93 and it more than doubled to 9.3 percent in just 5-6 years. During the next 6 years, C-section rates increased to 13.5 percent and consequently there was an increase of 45 percent between 1998-99 and 2005-06. The State witnessed an alarming increase in the C-section between 2005-16 and the C-section rate increase to 33.1 percent thus registering an alarmingly growth of 145 percent which was one of the highest in the country. C-section rate continued to increase unabated during 2015-2020 and as per NFHS-5, 42 percent of the births were delivered through C-section as compared to 21 percent at the national level. Thus there was an increase of 26 percent in the C-section rate between NFHS-4 and NFHS-5. If we compare the States that had a C-section rate of about 25 percent or more in 2015-16, J&K has registered one of the highest increases in C-section between NFHS-4 and NFHS-5 period.

Rural-Urban Trends

A sharp urban–rural divide in C-section rates is evident from Figure 2. C-section rates in rural areas have increased from 3 percent in 1992-93 to 9 percent in 2005-06 but it sharply increased to 27 percent within 10 years and currently around 38 percent of births in rural areas are delivered by C-section. In urban areas there has been a steady increase in C-section rates and as per NFHS-1 only 12 percent of deliveries in the State were delivered through C-section, which increased to 27 percent in 198-99.

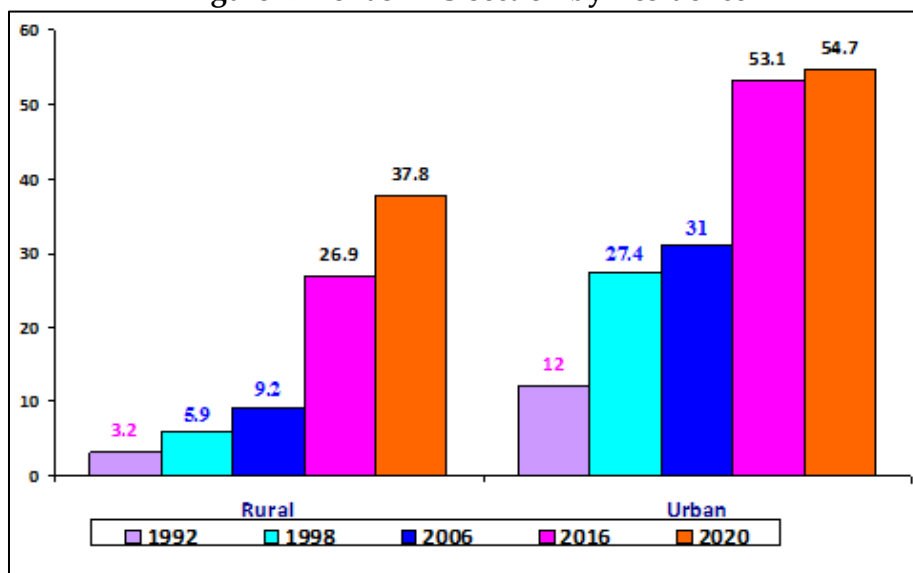
Figure 1 Trends in C-section in India and Jammu and Kashmir, 1992-2020



The C-section rates marginally increased to 31 percent during 2005-2006 but there was an alarming increase in C-section delivery rates between 2005 and 2016 and as per NFHS-4 more than half of deliveries in urban areas in 2015-16 were through C-section.

The C-section rates increased by 1.5 percentage points during 2015-16 and 2019-20 and currently C-section births account for 55 percent of all births in urban Jammu and Kashmir. Thus currently C-section deliveries are increasing more in rural areas as compared to urban areas. While the C-section rate has almost increased four-fold between 2005-06 and 2019-20 in rural areas it has increased 1.7 times in urban areas. The gap in the rural urban C-section rates during 1998-99 and 2005-2006 was about 21 percent and this increased to 26 percent during 2015-16 but as the C-section rate has increased more in rural areas than in urban areas, consequently the rural urban gap has declined to 17 percent in 2019-20. The rural urban ratio of C-section has also declined from 4.6 in 1998-99 to 2.0 in 2015-16 and it currently stands at 1.4. This indicates that while the rising trend in C-sections has slowed down in urban areas but it is increasing in rural areas at an alarming rate.

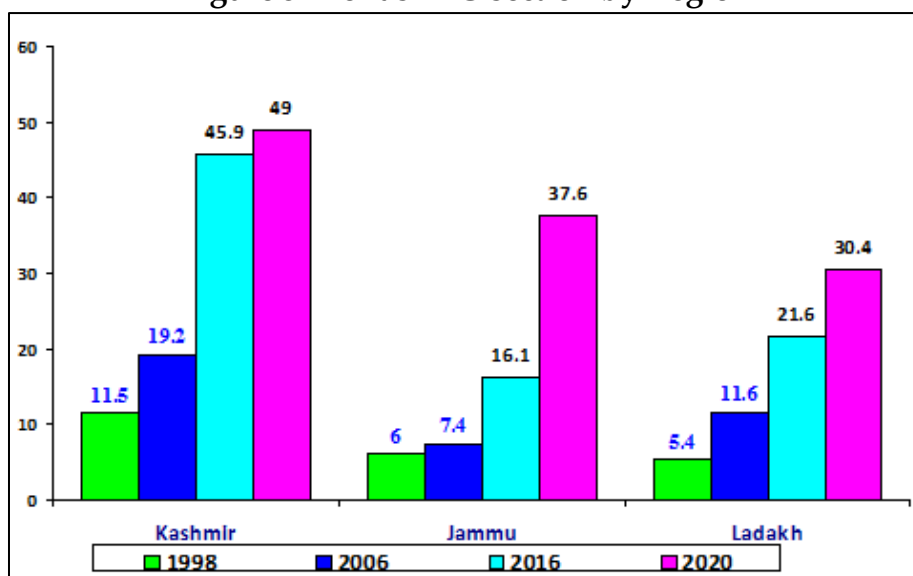
Figure 2 Trends in C-section by Residence



Regional pattern of C-section in J&K

There are sharp regional variations in C-section rates across three geographic regions in J&K (Fig.3). Kashmir Valley has the highest C-section rates and according to NFHS-5 almost half of births in Kashmir Valley are delivered through C-section while as in Jammu division about 38 percent are C-section births and in Ladakh has a C-section delivery rate of 30. Kashmir region has experienced very high increase in C-section between NFHS-3 and NFHS-4 and Jammu division has witnessed highest increase between NFHS-4 and NFHS-5 and Ladakh has also recorded a sharp increase in C-section between NFHS-4 and NFHS-5.

Figure 3 Trends in C-section by Region



District Level Pattern

Table 1 presents district-wise C-section delivery rates in Jammu and Kashmir between 2015-16 and 2019-20. There is a lot of variation in the C-section rates in different districts. As per NFHS-5, C-section rate varied from a high of 60 percent in Pulwama to a low of 20 percent in Ramban. There are 6 districts, all located in Kashmir Valley, having a C-section rate of 50 percent or more. These are Pulwama, Srinagar, Anantnag, Bandipora, Kupwara and Baramulla. Six more districts have C-section rate between 40 and 50 percent. Three of them namely Ganderbal, Kulgam and Budgam and two (Jammu and Udhampur) are from Jammu region. Leh district from Ladakh also has a C-section rate of 43 percent. Six more districts have a C-section rate varying between 30 and 40 percent and five of them are from Jammu division. Kargil with a C-section rate of 32 percent falls in this category. Four districts namely Ramban, Doda, Kathua and Rajouri have a C-section rate of 20-30 percent. During 2015-16, all the 10 districts located in Kashmir region had a C-section rate of 30 and above. In Srinagar district, three out of four children are delivered through C-section. In fact, Srinagar district with a C-section rate of 75 percent had the second highest C-section rate in the country after Karimnagar (80 percent) in Telengana in NFHS-4. Four districts all from Kashmir region had a C-section rate of 30-40 percent. Thus, all the districts of Kashmir region had a C-section rate of above 30. In Jammu division 6 districts had a C-section rate of 20-30 percent and four had a C-section rate of less than 20. Overall, there were only 3 districts in the State which had a recommended C-section rate of 15 percent in 2015-16 and by 2019-20, not even a single district in J&K had C-section rate of less than 20.

During 2015-16 and 2019-20, although C-section rate in the State has increased by 8.6 percentage points but there are four districts namely Srinagar, Shupian, Budgam and Ganderbal, which have experienced a decline in the C-section and highest decline of 18.6 percentage points is observed in Srinagar district. Almost all districts with a C-section rate of 40 or more except Pulwama and Baramulla have registered a decline in C-Section. Shopian also has recorded 12.6 percentage points decline in the C-section rate. Budgam and Ganderbal district also registered 6.8 and 3.3 percentage points decline in C-section rate. There are 8 districts which have registered an increase of less than 10 percentage points in C-Section rate. Pulwama and Baramulla also fall in this category. Four districts namely Jammu, Bandipora, Kulgam and Samba have recorded 10-15 percentage points increase in C-Section delivery rate. The highest increase of 20 percentage points in C-section can be observed in the districts of Leh, Kargil, Kupwara, Anantnag, Udhampur and Kishtwar.

Table 1 District wise rural and urban differences in C-section rates during 2015-16 and 2019-20 in J&K

Districts	2015-16			2019-20	% Change 2015-2020 (Total)	Rural Urban Difference
	Urban	Rural	Total	Total		
Kupwara	46.9%	29.8%	31.6	50.7	19.1	17.1%
Baramulla	66.8%	34.8%	41.5	49.6	8.1	32.0%
Bandipora	50.6%	29.7%	34.6	52.2	17.6	21.0%
Gander	62.5%	46.6%	49.9	46.6	-3.3	15.9%
Srinagar	74.5%	84.4%	74.7	56.1	-18.6	-9.8%
Budgam	62.3%	46.7%	49.8	43	-6.8	15.6%
Pulwama	50.7%	56.3%	56.1	59.7	3.6	-5.6%
Shupian	80.2%	39.3%	42.3	29.7	-12.6	40.9%
Kulgam	30.5%	32.7%	32.3	43.7	11.4	-2.2%
Anantnag	31.7%	32.3%	32.1	55.2	23.1	-.6%
Ramban	27.3%	11.2%	11.5	20	8.5	16.1%
Kishtwar	25.7%	14.7%	15.3	36.9	21.6	11.0%
Doda	38.0%	10.8%	12.0	23.5	11.5	27.2%
Reasi	13.4%	17.9%	17.5	33.1	15.6	-4.5%
Punch	40.3%	23.7%	24.8	34.6	9.8	16.6%
Rajouri	31.1%	22.2%	22.7	30.7	8.0	8.9%
Udhampur	37.2%	16.2%	18.1	41.6	23.5	21.0%
Jammu	33.5%	24.6%	28.4	41.8	13.4	9.0%
Samba	34.3%	27.1%	27.5	35.9	8.4	7.1%
Kathua	50.9%	18.9%	21.8	28.9	7.1	32.0%
Kargil	18.1%	9.5%	11.1	34.2	23.1	8.6%
Leh	22.6%	21.9%	22.1	44.3	22.2	.7%
J&K	53.1%	26.9%	33.1	41.7	8.6	26.2%

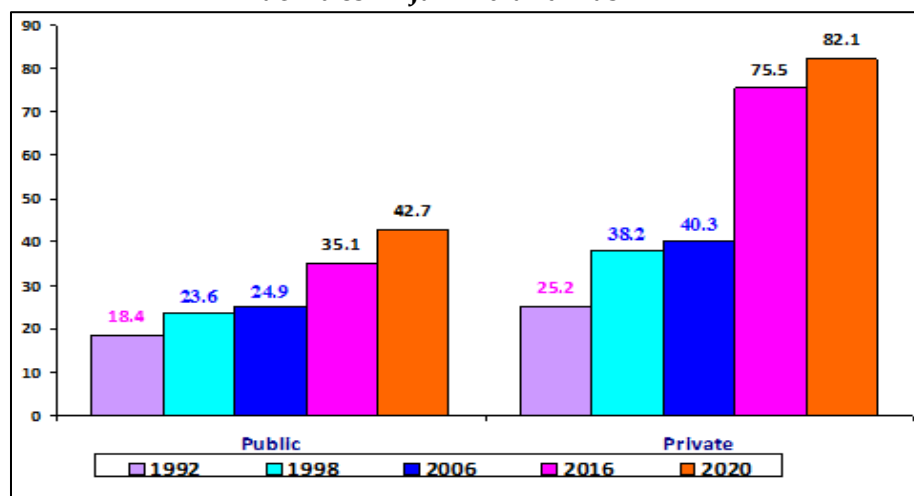
Overall, the urban C-section rate in the State is 26.2 percentage points more than the rural C-section rate. But this pattern is not uniform across all districts. Some of the districts like Srinagar, Pulwama and Kulgam, have very high C-section rates in rural areas as compared to urban areas. There are four districts where the urban C-section rate is more than 25 percent than rural C-Section rate. In many districts, this difference is in the range of 10-15 percent but these are districts which also have high C-section rates. Thus, the C-section rates are increasing more in rural areas and with a span of few years it appears the epidemic of C-section will be more serious in rural areas.

Place of C-section Delivery

Fig. 4 provides the trend in C-section rates in public and private health care facilities in J&K from 1992 to 2020 (NFHS-1 to NFHS-5). The difference in C-section rates between the public and the private sector has widened over the last three decades. From a public-private difference of around 7 percent in NFHS-1, the gap has jumped to 40 percent in NFHS-4 and NFHS-5. C-section rate by public private place also varies

greatly by region of residence. In Kashmir division 45 percent of births at public health facilities are delivered through C-section as compared to 76.5 percent in Private health institutions. In Jammu, 55 percent of births at Private health care facilities are C-section births as compared to 25 percent in Government health facilities. Within the Kashmir region, more than 80 percent of births in Private institutions are C-section births and in North Kashmir this proportion stands at 57 percent.

Figure 4 trends in C-section by in public and private health facilities in Jammu and Kashmir



The percentage of births delivered through C-section in Government health facilities during 2015-16 and 2019-20 in various districts of Jammu and Kashmir is presented in Table 5. During 2019-20, more than 50-60 percent of births at public health facilities in 6 districts of Kashmir Valley are C-section births. In other 6 districts this proportion at government health facilities is 40-50 percent. In most districts of Jammu region, 30-40 percent of births at public health facilities are through C-section. Although the State has witnessed an increase in C-section births both in public as well as private health institutions but 4 districts namely Srinagar, Shupian, Budgam and Ganderbal have witnessed a decline in proportion of C-Section at public health care facilities during 2015-16 and 2019-20. This decline is 14 percentage points in Srinagar, followed by Shupian (8.4 Percentage points) and Budgam (7 points). On the other extreme we have 7 districts which have experienced an increase ranging from 15 percentage points to 22 percentage points in C-Section rates at public health care facilities. These are Anantnag, Kargil, Kupwara, Leh and Udhampur. The district hospitals in these districts were provided manpower and other infrastructure and the labour rooms in the district hospitals were upgraded which has probably helped their work load and substantial number of women who earlier used to have a C-section in private health facility seem to have utilized the services from the public health facilities in these districts.

Differentials in C-Section by Background Characteristics

Table 2 presents the percentage of C-section births by some background characteristics which are known to be associated with the C-section as per NFHS-4 data. As discussed above C-section deliveries as per NFHS-4 are twice more in urban areas than the rural areas. As the age of the mother at birth increases the likelihood of a birth by C-section also increases.

Table 2 Percent distribution of C-section deliveries by background characteristics in J&K (2015-16)

Background Characteristics		No of births	% Normal	% Cesarean
Type of place of residence	Urban	1832	46.9%	53.1%
	Rural	5865	73.1%	26.9%
Mothers age at birth	<20	436	76.4%	23.6%
	20-34	6730	66.6%	33.4%
	35-49	531	63.3%	36.7%
Highest educational level	No education	2543	78.5%	21.5%
	Primary	485	76.0%	24.0%
	Secondary	3887	63.1%	36.9%
	Higher	782	42.3%	57.7%
Religion	Muslims	5528	64.0%	36.0%
	Non-Muslim	2169	74.2%	25.8%
Caste	SC	689	78.0%	22.0%
	ST	1079	86.4%	13.6%
	OBC	353	72.1%	27.9%
	Other	5576	61.4%	38.6%
Wealth index	Poorest	786	91.4%	8.6%
	Poorer	1748	82.4%	17.6%
	Middle	1879	67.0%	33.0%
	Richer	1656	55.3%	44.7%
	Richest	1628	50.0%	50.0%
Mother Tongue	Hindi/Punjabi	1103	74.0%	26.0%
	Kashmiri	3584	54.6%	45.4%
	Urdu	1540	79.6%	20.4%
	Other	1471	78.2%	21.8%
	Total	7697	66.9%	33.1%

The C-section rate among women age less than twenty at the time of delivery is 23.6 percent as compared to 36.7 percent for women age 35 or more. Education is also directly correlated with C-section. The C-section rate is less than 25 percent among women with no education or who have only completed less than 5 years of schooling and it increases to 37 percent among women 10-12 years of schooling and further rises to 58 percent among women who have completed more than 12 years of schooling. C-section rate among Muslim women in the State is almost 10 percentage points higher than women from other religious groups. Women from Scheduled Tribes are likely to

have low C-section rate as compared to women from a Scheduled Castes/OBC category or general category. Almost 39 percent of births to women belonging to a general category are C-section births. Wealth/Economic status is also highly correlated with the likelihood of a C-section delivery. Less than 10 percent of women from poorest families have delivered through a C-section. This proportion increases to 33 percent among women from Middle class families and among women who are from richest families one in two is delivered through a C-section. If we look at the mother tongue, Kashmiri speaking women have the highest C-section rate of 45 percent, followed by women who speak Dogri/Punjabi/Hindi (26 percent) and lowest among women who speak other languages like Pahari/Urdu/Gojri/Budhi.

Differentials by Delivery/Birth Characteristics

Some important characteristics which are associated with C-section like sex of child, weight, of the child, birth order, place of delivery and nutritional status of mother etc, are discussed and information is presented in Table 3. There are not any differences in C-section rates by sex of the child. Similarly, the C-section rate does not vary much by size of the child, however, births whose size is large or very large have higher propensity to deliver through C-section than the births of average or small size at the time of birth. Order of birth is inversely correlated with C-section. The C-section rate is 39-37 percent for the first and second order births and it declines to 27 percent for third order and 12 percent for order 4 and higher order births. Single births have 6 percent lower C-section rate than Twins. The C-section rate among twin births is 39 percent as compared to 33 percent among singleton births. C-section rate is lowest (20 percent) among births who were unwanted, followed by mistimed births (28 percent) and it is highest (34 percent) among planned. Body Mass Index (BMI) is positively correlated with C-section rate. Sixteen percent of births are delivered by a C-section among women who are thin and this proportion increases to 30 percent in case the women have a normal BMI and 46 percent in case the women is overweight. However, 57 percent of the births among obese women are delivered through C-section. There is a perception that if previous birth is delivered through C-section; the subsequent birth is also delivered through C-section. This holds true in the present study also. Almost 97 percent of births among women whose last births was delivered through a C-section were delivered by C-section. C-section rate is 37 percent for births whose mothers had undergone USG during pregnancy as compared to 7 percent who had not undergone USG.

Table 3 Percent distribution of C-Section deliveries in J&K by birth characteristics (2015-16)

Birth-Characteristics		No of Births	% Normal	% Cesarean
Sex of child	Male	4020	66.8%	33.2%
	Female	3677	67.0%	33.0%
Size of child at birth	Very large	189	68.0%	32.0%
	Larger than average	532	57.4%	42.6%
	Average	5856	66.4%	33.6%
	Smaller than average	542	65.4%	34.6%
	Very small	258	69.8%	30.2%
	Don't know	315	90.8%	9.2%
Birth order	First	2968	60.5%	39.5%
	Second	2459	63.3%	36.7%
	Third	1337	73.0%	27.0%
	Fourth or higher	933	88.0%	12.0%
Child is Twin	Single	7551	67.0%	33.0%
	Multiple	146	61.0%	39.0%
Wanted pregnancy when became pregnant	Then	7067	66.0%	34.0%
	Later	294	72.0%	28.0%
	No more	331	80.2%	19.8%
Place of Delivery	Public Facility	6009	64.9%	35.1%
	Private facility	626	30.2%	69.8%
Nutritional Status	Thin	905	84.3%	15.7%
	Normal	4643	70.6%	29.4%
	Overweight	1557	53.6%	46.4%
	Obese	500	43.2%	56.8%
Last birth a caesarean section	No	5104	99.4%	.6%
	Yes	2593	2.9%	97.1%
Ultrasound test	No	721	93.3%	6.7%
	Yes	6970	64.1%	35.9%
	Total	7691	66.9%	33.1%

Differentials by Pregnancy/Delivery Complications

NFHS-4 also collected information related to problems experienced during delivery (breech presentation, prolonged labour, excessive bleeding) but this information was collected for the most recent births. Additionally, information on whether the previous delivery was through C-section—which is an important determinant of whether the current delivery will be through C-section—is available only for those births where the previous birth was also within the reference period. Information about ANC visits is also available for the most recent birth. Almost half of the births are delivered through C-section, if there was a breech presentation during delivery (Table 4). The proportion of births delivered through C-section is less (31 percent) if the women experienced prolonged labour at the time delivery compared to women who did not experience prolonged labour (37 percent). One reason for this may be that

the women who do not wait to undergo labor pain might have planned for a C-section before on set of labour.

Table 4 Distribution of C-section deliveries by ANC/delivery complications

ANC/Delivery Complications		No of Births	% Normal	% Cesarean
Experience a breech presentation	No	4942	67.5%	32.5%
	Yes	918	48.4%	51.6%
	Don't know	36	58.4%	41.6%
Experience prolonged labour	No	4073	62.0%	38.0%
	Yes	1814	69.9%	30.1%
	Don't know	9	90.3%	9.7%
Experienced excessive bleeding	No	4313	63.4%	36.6%
	Yes	1521	68.2%	31.8%
	Don't know	61	46.5%	53.5%
No. of ANC Visits	0-3	1014	81.5%	18.5%
	4-10	3301	65.1%	34.9%
	11-15	1583	52.2%	47.8%
	Total	5899	64.5%	35.5%

Experience of excessive bleeding is also not associated with higher likelihood of C-section. The study shows there is a direct relationship between No. of ANC checkups and the likelihood of C-section. Women who have no or less than 3 ANC checkups are least likely to have delivered through C-section. Probability of C-section increases with increase in the number of ANC checkups. Thirty five percent of births to women who have had 4-10 ANC visits are delivered through a C-section and this proportion steadily increases to 48 percent in case the women have >10 ANC checkups.

Table 5 Logistic regression of various background characteristics on C-section delivery in J & K

Background		B	S.E.	Wald	df	Sig.	Exp(B)
Region	Jammu*						
	Kashmir	.740	.085	75.803	1	.000	2.096
	Ladakh	-.073	.259	.080	1	.778	.929
Age of Mother at Child Birth	<20*						
	20-34	-.081	.130	.391	1	.532	.922
	35-49	.169	.174	.942	1	.332	1.184
Religion	Non-Muslim*						
	Muslim	.271	.094	8.307	1	.004	1.312
Caste	No Caste/tribe*						
	SC	-.035	.114	.093	1	.760	.966
	ST	-.500	.112	19.846	1	.000	.607
	OBC	-.218	.130	2.797	1	.094	.804

Background		B	S.E.	Wald	df	Sig.	Exp(B)
Type of Facility	Public*						
	Private	.937	.101	86.117	1	.000	2.553
BMI	Average*						
	Thin	-.489	.104	22.165	1	.000	.614
	Over Weight	.305	.068	20.445	1	.000	1.357
	Obese	.705	.106	44.087	1	.000	2.024
Birth Order	1*						
	2	-.155	.065	5.689	1	.017	.856
	3 and above	-.734	.079	86.967	1	.000	.480
No. of ANC Visits	0-3*						
	4-10	.408	.066	37.775	1	.000	1.504
	15+	.412	.079	27.102	1	.000	1.510
Size of Child	Large*						
	Average	-.061	.085	.519	1	.471	.941
	Small	-.007	.115	.003	1	.953	.993
Residence	Rural*						
	Urban	.461	.066	49.308	1	.000	1.586
Education	No Schooling						
	Primary*	-.056	.129	.191	1	.662	.945
	Second	.293	.072	16.709	1	.000	1.341
	Higher	.599	.112	28.394	1	.000	1.820
Wealth Quintile	Poorest*						
	Poor	.359	.148	5.850	1	.016	1.431
	Middle	.822	.147	31.377	1	.000	2.276
	Rich	1.054	.152	48.405	1	.000	2.869
	Richest	1.113	.158	49.794	1	.000	3.045
	Constant	-2.441	.205	142.396	1	.000	.087

Logistic Regression Analysis

Table 5 provides the adjusted odds ratios of C-section for various socio-economic demographic and birth characteristics. The odd of C-section among women from Kashmir region was 2.0 times higher than women from Jammu/Ladakh NFHS 4 after controlling all other variables. Urban women are almost 1.6 times more likely to have a C-section birth ac compared to their rural counterparts. Women who had higher educational levels and households in the richest wealth quintile also had a significantly higher likelihood of having a cesarean delivery. Among women with secondary school education, the OR for cesarean delivery was 1.34. In comparison, women with higher education (12 years or above) had an OR of 1.26. For women in the poorer wealth quintile, the OR for cesarean delivery was 1.43. In comparison, women in the rich wealth quintile had an OR of 2.8 and those with the richest wealth quintile it was 3.0. Muslim women are 1.3 times more likely to deliver through C-

section as compared to non-Muslim women. Women from Scheduled tribes are 0.6 times less likely to have a C-section birth as are women from no a general category. Type of facility is the most important determinant of C-section. Regardless of the other factors a woman delivering in private health facility is 2.55 times more likely to have a C-section as are women who delivered in a public health facility. Body Mass Index is also an important predictor of C-section birth. A woman who is obese is 2.0 times more likely to have a C-section birth compared to a woman who had a normal BMI. A woman who is underweight is 0.6 times less likely to have a C-section than a woman with a normal BMI.

Birth order and maternal age at birth are other important determinants-higher the birth order and lower the maternal age at birth, lower is the probability of a C-section. Third and higher order births are 0.5 times less likely to have a c-section birth as compared to first order births. Women who have more than 4 ANC checks up or 1.5 times more likely to have c-section as compared to women with no ANC or fewer than 4 ANCs checkups. Women receiving ANC checks ups from a private doctor/clinic have a higher probability of a C-section birth. Size of the child and pregnancy complications except breech presentation have less impact on the probability of C-section.

Whether the previous delivery was through C-section is the strongest determinant of the current C-section. A woman who delivered through a C-section is 90 percentage points more likely to give current birth through C-section. Except breech presentation, other problems during pregnancy (prolonged labour, excessive bleeding and convulsions) are not statistically significant and have relatively less impact on the probability of C-section.

Planned and Unplanned C-section Deliveries

Planned C-sections can be driven by medical reasons. For example, the American College of Obstetricians and Gynecologists (2014, 2018) recommends a C-section before the onset of labour in first births for: breech or transverse lie, placenta previa, triplets and higher order multiples, uterine rupture, certain rare maternal cardiac or neurologic conditions, or a history of certain uterine surgeries. Planned C-sections may also reflect a demand for C-sections, that is, women who prefer C-sections due to fear of pain in natural births, or preference for the birth of the child on a specific day or time driven by cultural or religious beliefs, or monetary incentives would lead to planned C-sections (Lo 2003; Schulkind and Shapiro 201).

Table 6 Distribution of C-section deliveries by timing of decision to undergo cesarean by various background characteristics

Background Characteristics		No. of Births	Timing of decision to have a C-section			
			Before Onset of Labour		After Onset of Labour	
			No	%	No	%
Division	Kashmir	1678	1199	71.5%	479	28.5%
	Jammu	823	479	58.1%	345	41.9%
	Ladakh	22	11	50.2%	11	49.8%
Type of place of residence	Urban	970	681	70.2%	290	29.8%
	Rural	1553	1008	64.9%	545	35.1%
Mothers age at birth	<20	100	62	61.4%	39	38.6%
	20-34	2229	1476	66.2%	754	33.8%
	35-49	194	152	78.3%	42	21.7%
Highest educational level	No education	535	377	70.4%	158	29.6%
	Primary	116	79	67.9%	37	32.1%
	Secondary	1421	935	65.8%	485	34.2%
	Higher	451	298	66.0%	154	34.0%
Religion	Muslims	1967	1385	70.4%	582	29.6%
	Non-Muslim	556	304	54.6%	252	45.4%
Caste	SC	148	74	50.2%	74	49.8%
	ST	136	82	60.1%	54	39.9%
	OBC	98	62	63.8%	35	36.2%
	Other	2142	1471	68.7%	671	31.3%
Wealth index	Poorest	63	39	62.3%	24	37.7%
	Poorer	300	206	68.7%	94	31.3%
	Middle	613	413	67.4%	200	32.6%
	Richer	735	523	71.1%	212	28.9%
	Richest	812	508	62.5%	305	37.5%
Size of child at birth	Very large	59	42	70.6%	17	29.4%
	Larger than average	223	160	71.5%	64	28.5%
	Average	1954	1292	66.1%	662	33.9%
	Smaller than average	185	125	67.6%	60	32.4%
	Very small	76	51	66.9%	25	33.1%
	Don't know	26	20	74.3%	7	25.7%
Birth order	First	1163	684	58.9%	478	41.1%
	Second	896	653	72.9%	243	27.1%
	Third	355	270	76.0%	85	24.0%
	Fourth or higher	110	82	74.2%	28	25.8%
Wanted pregnancy when became pregnant	Then	2376	1581	66.6%	794	33.4%
	Later	82	60	72.5%	23	27.5%
	No more	65	48	73.4%	17	26.6%
Place of Delivery	Public Facility	2086	1372	65.8%	714	34.2%
	Private facility	437	317	72.4%	121	27.6%

Background Characteristics		No. of Births	Timing of decision to have a C-section			
			Before Onset of Labour		After Onset of Labour	
			No	%	No	%
BMI	Thin	134	85	63.6%	49	36.4%
	Normal	1349	873	64.7%	476	35.3%
	Overweight	720	512	71.1%	209	28.9%
	Obese	284	193	68.1%	91	31.9%
Last birth a caesarean section	No	31	13	42.7%	18	57.3%
	Yes	2493	1676	67.2%	817	32.8%
Ultrasound test	No	45	21	46.1%	24	53.9%
	Yes	2479	1668	67.3%	810	32.7%
	Total	2524	1689	66.9%	834	33.1%
During delivery: experience a breech presentation	No	1591	1067	67.0%	524	33.0%
	Yes	470	321	68.3%	149	31.7%
	Don't know	15	12	82.3%	3	17.7%
During delivery: experience prolonged labour	No	1537	1058	68.8%	479	31.2%
	Yes	538	341	63.3%	197	36.7%
During delivery: experienced excessive bleeding	No	1570	1058	67.4%	512	32.6%
	Yes	472	314	66.5%	158	33.5%
	Don't know	33	27	82.1%	6	17.9%
No. of ANC Visits	0-3	178	111	62.6%	66	37.4%
	4-10	1142	770	67.4%	373	32.6%
	11-15	756	519	68.6%	237	31.4%
	Total	2076	1400	67.4%	676	32.6%

However, in the Indian context, private facilities that provide maternal care predominantly consist of small hospitals or nursing homes, typically owned by a single doctor or a family. These are characterized by limited assistance from other doctors, difficulties in maintaining round the clock monitoring or obtaining a second opinion, inadequate facilities, and lack of skilled human power—including nurses. In these conditions, planned C-sections might be preferred by the provider (Pai 2000).

NFHS-4 added a question about the timing of C-section decision. Women who had delivered their most recent birth through C-section were asked to mention whether the decision to have a C-section was made before or after the onset of labour. This question allows us to identify planned and unplanned (or scheduled and non-scheduled) C-sections. Planned C-sections constitute two-third of all C-sections in J&K (Table 6). Although, there is not much difference between rural and urban areas but Kashmir Valley has a very high proportion of planned C-section (71.5 percent), compared to 58 percent in Jammu region and 50 percent in Ladakh. Planned c-sections are higher among older women, women with no schooling, Muslim women and women who are not from SCST or OBC. Planned C-sections are also very high among

women from richer families and women with a higher birth order. Planned c-sections are also very high among births in a private health sector but even among births in a public health sector 66 percent of c-sections births are planned before on set of labour. Planned c-sections are higher among obese women as compared to women who have thin or have a normal BMI. Further women who had a previous C-section have a higher probability of having a planned C-section for the next birth. This percentage is similar whether a woman has delivery complications or not. As the planned C-sections are not associated with pregnancy or delivery complications, this indicates that the higher proportion of C-sections is not always medically indicated.

Discussion

Jammu and Kashmir have registered an unprecedented increase in the proportion of C-section births during 2005-2016 as the C-section rate has increase from 13.5 in 2005-06 to 33.1 percent in 2015-16. C-section rate continued to increase unabated during 2015-2020 currently, 42 percent of the births in J&K are delivered through C-section. Among the States/UTs, for which NFHS-5 fact sheets are available, J&K has registered the highest increase in C-section during last few years. Though the C-section rates are very high in urban areas of J&K but between 2015-06 and 2019-20, most of the increase in C-section deliveries is being observed in rural areas and consequently, the gap between the rural urban C-section rates which was 26 percent 2015-16 has declined to 17 percent in 2019-20. There is sharp regional variation in C-section rates across three geographic regions in J&K. Kashmir Valley has the highest C-section rates and according to NFHS-5 almost half of births in Kashmir Valley are delivered through C-section while as in Jammu division about 38 percent are C-section births and Ladakh has a C-section delivery rate of 30. During 2015-2020, C-section rates have increased sharply in Jammu and Ladakh division and marginally in Kashmir region. Among the districts, Srinagar and Pulwama have the highest C-section rates.

While looking at the place of C-section in J&K, higher proportion of C-sections are performed at private health facilities. The difference in C-section rates between the public and the private sector has widened over the last three decades. From a public-private difference of around 7 percent in NFHS-1, the gap has jumped to 40 percent in NFHS-5. C-section rate by public private place also varies greatly by region of residence. In Kashmir division 76.5 percent of births in Private health institutions are C-section as compared to 55 percent in Jammu division.

Few socio-economic and birth characteristics are associated with high c-section rates in Jammu and Kashmir. Births of order 1 and older women, women with higher education, Muslim women and women who belong to highest wealth quintile have

very C-section rates as compared to their counterparts. Similarly, births of less than average size, twins and births to obese and overweight women are also more likely to deliver by C-section. There is no significant relationship between various delivery complications and C-section; however, women with a breach presentation are more like to deliver through C-section. Our logistic regression supports our bi-variate results.

Planned C-sections constitute two-third of all C-sections in J&K and this percentage is very high in Kashmir region (71 percent). Planned c-sections are also very high among births in a private health sector but even among births in a public health sector 66 percent c-sections are planned before on set of labour. Planned c-sections are higher among obese women as compared to women who have thin or have a normal BMI. Further women who had a previous C-section have a higher probability of having a planned C-section for the next birth. This percentage is similar whether a woman has delivery complications or not.

The doubling in cesarean deliveries during the last decade is closely related to the simultaneous decline in the share of home deliveries without medical supervision. Their proportion decreased from 50 percent to 7.6 percent during the last 3 surveys in 2005-06 to 2019-20. After the implementation of National Rural Health Mission in 2006, State has witnessed a revolution in the health infrastructure and manpower and emphasis of quality of maternal health care services. J&K was probably the only State which extended the benefit of financial incentives under the Janani Suraksha Yojana (JSY) program to all women irrespective of parity and economic status in 2008. JSY was further supplemented in 2011 with the Janani Shishu Suraksha Karyakram (JSSK) program. The improvement in health infrastructure and manpower and also the financial cash incentives along with zero cost delivery encouraged women to use free prenatal and postnatal care in modern health centers. The programs explain to a large extent the doubling of the proportion of institutional deliveries during the last decade.

The age at marriage in Jammu and Kashmir particularly in Kashmir Valley is increased during the last 20 years. As per NFHS-5, 35 percent of women age 15-49 in the State are unmarried which is one of the highest in India. With delayed marriages, the chances of conception decrease and consequently, even the very first pregnancy becomes precious. The whole family, relatives and the couple see the C-Section to be the sole solutions to ensure that baby is born alive and mother is also safe. This phenomenon is termed as “too posh to push” which is prevalent among educated, rich and urban women who want to avoid labour pain during childbirth is now spreading to middle class families also. Looking at the C-Section by background characteristics, this theory completely befits J&K. It appears that delivery by C-section

in Kashmir has become fashionable and considered “modern” or safer to deliver without labour.

During earlier phase of militancy when the female sterilization facility was not available in Kashmir based public health facilities, women had to go to Jammu and outside State for female sterilization. However, as the years passed a new trend for post-partum sterilization emerged in Kashmir. Now as the ideal family size is less than 2 and Total Fertility Rate has sharply declined women opt for post-partum sterilizations. Although, female sterilizations services were restarted in Kashmir after 2000, but women found postpartum sterilization more convenient due to the social stigma/insecurity associated with female sterilization and it also has helped in increasing the demand for a C-section birth in Kashmir. This is probably a very important reason for the State to have the highest C-section rate in Government hospitals after Telangana.

The Medical Superintendent at one of the district hospitals in South Kashmir mentioned that C-section seems to have become a culture rather than an emergency. He mentioned that it is not only the educated and the wealthier women want a C-section, but even illiterate and poor women also come with the preconceived notion to go for a C-section. Another Gynecologist mentioned that Kashmir is well net society. A woman decides the place of delivery on the basis of whether you have a close contact at the facility that would help her in having a C-section. She mentioned that although the doctors at the public health facility try to convince the to-be-mothers for a normal delivery, but they use one or the other tactics to pressurize us to do the c-section. She also mentioned that whenever, there is any case of a maternal death it is only the doctors who have to face the wrath of the public and in such situations doctors also do not want to take any risk with the vaginal delivery and consequently, the C-section rates even at Sub District Hospitals are also rising.

The role of public facilities in this revolution is crucial because 86 percent of all institutional deliveries occurred in government hospitals during 2015 to 2016, compared with 41 percent 10 years earlier (NFHS-3). However, while in the country the cesarean birth rates remained appropriate in government facilities at 12 percent in 2015 to 2016, but in J&K, C-section rates in public health facilities increased from 18 percent in 2005-06 to 43 percent in 2019-20. In contrast, the cesarean rate in private clinics grew from 30 percent to percent 82 percent during the same period. Therefore, the last 14 years witnessed a complex transition affecting both the places and modes of delivery: many poor women gained access for the first time to safer childbirth in public facilities, but women belonging to higher socio-economic class are opting to deliver through C-section both in public as well as private health facilities. Our

analysis further shows that the diffusion of cesarean deliveries appears closely associated with socioeconomic status, with a variation in cesarean rates from 8.6 percent to 50 percent as women move up the social ladder. While the educated, urban and wealthier women do themselves decide the type of delivery they opt but same is not the case with poor and middleclass women. These women apart from receiving antenatal care from a public health facility also keep on visiting a private clinic/doctor for ANC services. As these women are not aware about the benefits of a normal delivery and they also do not have the power to decide the place and mode of delivery and in such cases, physicians play a crucial role in deciding whether it should be a vaginal delivery or CS. This opportunity gives physicians a window to convert vaginal delivery to CS, as a CS procedure enhance the physicians' income or time spent in patient care. Further, almost all gynecologists are also involved in private practice and pregnant women after initial few ANC visits think it better to visit the same Gynecologist at his/her private clinic who see them at the Government facility doctor and this again gives the Gynecologist to decide the place of C-section at a private hospital. Thus the supplier-induced demand driven by financial incentives especially when the patient has limited information—is probably an important explanation. As per NFHS-4, a natural birth in private facility in J&K costs on an average Rs. 12206, while a C-section costs 24861, more than twice the cost of a natural birth.

Conclusion

Our findings, call for a need to monitor the further progression of cesarean rates in J&K. The drivers of the current enthusiasm for surgical deliveries in J&K are not yet well understood. They may include changes in lifestyles, commercial pressure, and cultural factors, but this emerging situation calls for further investigation. In urban areas and among the middle class, cesarean rates have already reached levels consistently higher than what is considered medically justified. Effective interventions and policies targeted at women and health care professionals will be required to avoid reduce unnecessary cesarean deliveries in the State particularly in Kashmir Valley. The State should follow the WHO guidance on reducing C-section rates which states that comprehensive health education, including tailored information and support about child birth fear, pain relief, and the advantages and disadvantages of caesarean sections, should be provided to all women particularly the young women. The WHO guidance also recommends mandatory second opinions for caesarean section indication, as well as audits and feedback loops within facilities. Financial strategies that remunerate equally for vaginal births and caesarean sections are also recommended. A discussion with the medical officers during field visits to health

facilities reveals that the service providers must become better equipped and confident to have meaningful, evidence-based, and supportive discussions with women about their childbirth options and concerns. Studies have suggested three potential targets for interventions to reduce the rate of C-sections where they are high: the health system, women and families and the culture. The health system must focus on strategies to strengthen the health systems, including increasing the doctor–population ratio, placing highly skilled and well-trained midwives in labour rooms, allowing birth companions during labour and delivery, providing for painless delivery options, improving labour room infrastructure such as bed, electronic fetal monitoring system, neonatal intensive care units and blood transfusion facility. It is necessary to develop a rating system for hospitals, where a hospital promoting normal deliveries gets a high positive score and a hospital conducting caesarean sections without medical indication high negative score. Hospitals and communities should promote health education among women of reproductive age and counsel pregnant women about delivery process. Educating and empowering women and other family members about the risks of c-section and benefits of normal deliveries will make them better advocates for their own care. ASHAs, ANMs and Gynecologists can play critical role in this advocacy during antenatal and delivery care.

Further, the government can consider giving enhanced JSY incentives for normal deliveries only to both the mother and the service provider as J&K has already very high institutional delivery rates and withholding JSY is not going to reduce the institutional deliveries in the State. Another strategy is to have certain guidelines that define the conditions under which the decision for a caesarean surgery can be taken. However, such guidelines are less likely to be effective in practice as providers can always use discretionary interpretation to justify an unnecessary C-section.

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7

Pregnancy Wastage in Karnataka: A District-level Analysis Health Management Information System (HMIS) Data

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Abstract

Pregnancy Wastage (PW) is one of the major health issues in developing countries including India. Poor pregnancy outcome indicates poor Maternal and Child Health (MCH) care. The proportion of abortion and stillbirth are considerably high in India, which has a major share in the PW. The study aims to analyze the levels and trends in pregnancy wastage, district wise variation and to understand the extent of contribution of different level health facilities in the State. Using the HMIS data for Karnataka state for 3 years period, 2017-18, 2018-19 and 2019-20, issues related to pregnancy wastage like reporting on various pregnancy wastages like spontaneous abortion, induced abortion and stillbirth are examined. District wise and year wise as well as the contribution of various facilities in pregnancy wastages are assessed and compared. The pregnancy wastage is found to be 62 per thousand pregnancies in the State. Induced abortion is the most contributing wastage to the total pregnancy wastage and spontaneous abortion is the next major one followed by the stillbirth in the State. Policy needs to address the increase in pregnancy wastage at the private facilities. Further, there should be a proper system at the state and district level to provide appropriate monitoring and supervision of pregnancy wastage.

Introduction

Pregnancy wastage is one of the major health problems in developing countries including India. Poor pregnancy outcome indicates poor Maternal and Child Health (MCH) care. Most of the countries, including India, are expending a huge amount of money on Mother and Child Health programs to improve pregnancy outcomes. Of all the adverse outcomes, the most serious issue is pregnancy wastage (PW) which includes, spontaneous abortion, induced abortion and perinatal mortality, the last variety combines stillbirth and early neonatal death by (Cunningham FG et al., 1993). Found that often, despite good health care, such wastage may occur due to the association of a spectrum of various other factors which, if taken care of, can alleviate the problem and prevent wastage.

Pregnancy wastage such as abortions, be it spontaneous or induced and stillbirth are the most common adverse outcome of pregnancy, yet its prevalence and underlying causes are subjected to continuing investigation and understanding. The reported prevalence of spontaneous abortion is 10 to 15 percent of clinically recognized pregnancies. (García-Enguádanos et al., 2002) However, many spontaneous abortions occur so early in pregnancy that the woman is not even aware that she was pregnant and has aborted. Further (WHO, 2007) reports that globally

approximately 42 million pregnancies are voluntarily terminated each year, 22 million within the legal system and 20 million by unskilled providers or in unhygienic conditions, or both. It is estimated that 19 of every 20 unsafe abortions take place in less developed regions of the world and this is where 98 percent of abortion-related deaths occur. An estimated 6.8 million abortions occur every year in South-Central Asia (including India) at a rate of 17 unsafe abortions per 1000 women.(Shah & Ahman, 2010).

Stillbirth refers to the death of a baby after 28 weeks of pregnancy but before birth (WHO,2013). Unfortunately, each year, 10.6 million children die and this number goes uncounted across the world. (Stanton et al., 2006). Among these 3.2 million annual stillbirths occur in low and middle-income countries(Yakoob et al., 2009). India has the highest number of stillbirths in the world. In India, the stillbirth rates (SBR) varied from 20 to 66 per 1,000 total births in different States. (Bhati, 2014).The magnitude of spontaneous abortion; and induced abortion whether legal or illegal has become an era of serious concern in developing countries. In developing countries, the risk of death following complications of unsafe abortion procedures is several hundred times higher than that of an abortion performed professionally under safe conditions (WHO, 1998).

According to the Consortium on National Consensus for Medical Abortion in India (2008). The Indian government has also repeatedly emphasized that MTP should not be viewed as a method of family planning or of reducing the national birth rate. In India, the incidence of abortion is always under-reported, perhaps because of the guilt or the moral stigma associated with it. Every year an average of about 11 million abortions take place annually and around 20,000 women die every year due to abortion-related complications. Most abortion-related maternal deaths are attributable to illegal abortions. Hence, there is a need to identify the important factors responsible for spontaneous and induced abortion in India and its major States where induced and spontaneous abortion is high according to (Das et al., 2000).

A study conducted by(Khan et al., 1990)and (Chhabra & Nuna, 1994) found that the proportion of illiterate women among abortion seekers is low compared to the literates. A large proportion of abortions are now cited as falling under a special category that was almost non-existent at the time of the framing of the Act. This category is sex determination followed by abortion of the female fetus. In such cases, it is not the pregnancy but its outcome that is unwanted. The first sex-selective abortion was documented in India in the 1970s, with the advent of amnio-centesis and it immediately began to be used in genetic clinics for determining the sex of the fetus, with the sole purpose of circumventing the birth of girls (Oomman & Ganatra, 2002). A study by (Kumari, 2006), portrays that the sex of the child is an important factor for induced abortion as the abortion rate is high with two children or having one son and one daughter. Further (International Institute for Population Sciences (IIPS) and ICF, 2017) data reveals in India, 10 percent is the pregnancy wastage terminated in abortion, miscarriage, or stillbirth. (Miscarriage

6 percent, abortions 3 percent, and 1percent stillbirth. For Karnataka pregnancy wastage, accounted for 6 percent. (Abortion 1.8 percent, miscarriage 3.2 percent, and stillbirth is less than 1 percent) as per the NFHS-4. The occurrence of abortion and stillbirth are always under-reported in the state and nation as well, there are limited studies in the State which highlight the severity of the issue. The proportion of abortion and stillbirth are considerably high in India which has a major share in pregnancy wastage. In the larger perspective pregnancy wastage like, spontaneous abortion, induced abortion and stillbirth are remained underestimated in the Karnataka State and India as well. This issue needs to be addressed through every possible data source; in this context, we are using the Health Management Information System (HMIS) data which gives the facility-based data source on spontaneous abortion, induced abortion (MTP within 12 weeks and more than 12 weeks) stillbirth, these data elements enable the HMIS to capture across districts as well as at different healthcare facilities including Health Sub Centre(HSC) Primary Health Centre (PHC), Community Health Centre (CHC), Sub-Divisional Hospital (SDH), District Hospital (DH) and other public hospitals including medical college and hospital, and even as well as private hospitals. Therefore, the present study is proposed to investigate the district-wise pregnancy wastage in Karnataka State. The findings of this study will highlight the districts and types of health facilities having a high occurrence of pregnancy wastage including spontaneous abortion, induced abortion and stillbirth in Karnataka state and where the district and State officials make necessary policy implications to improve the situation. Accordingly, we have undertaken this study on “Pregnancy Wastage in Karnataka: A district-level analysis based on HMIS data”.

Objectives

- To analyze the levels and trends in pregnancy wastage in Karnataka
- To analyze district wise variation in pregnancy wastage in Karnataka
- To understand the extent of the contribution of different level health facilities in pregnancy wastage.

Methodology

For the study, we have used HMIS data for Karnataka state for 3 years period, 2017-18, 2018-19 and 2019-20. As per the (Cunningham FG et al., 1993) pregnancy wastage (PW) which includes early preclinical loss of conception, spontaneous first-trimester abortion, and perinatal mortality, the last variety combining stillbirth and early neonatal death. All the related HMIS data elements such as the total number of pregnant women registered for ANC, stillbirth, abortion (spontaneous), MTP up to 12 weeks of pregnancy, MTP more than 12 weeks of pregnancy are considered for the analysis. The data has been analyzed to find out the levels and trends in Pregnancy wastage, district wise as well as type of wastages. Further, the contribution of different level health facilities in the Pregnancy wastages is also analyzed for 2019-20.

The Occurrence of all the different types of pregnancy wastages viz. spontaneous abortion, induced abortion and stillbirth at the State and district level is calculated. Further, to understand the occurrence and trend of pregnancy wastage at the district level for the last three years HMIS data. The occurrence of pregnancy wastage has been estimated for India and Karnataka State and type of pregnancy wastage viz stillbirth, spontaneous abortion and induced abortion for Karnataka State using the HMIS data of the year 2019-20. Total pregnancy wastage for India and Karnataka was estimated per 1000 pregnancies and presented in QGIS maps and type of pregnancy wastage at the State and district level was estimated per 1000 pregnancy outcomes.

Results

The occurrence of pregnancy wastage in Karnataka state

The study also examined the levels of reported pregnancy wastages and differentials in wastage like stillbirth, spontaneous abortion and induced abortion in Karnataka State. As presented in fig.1 to 4. fig.1 revealed that the occurrence of overall pregnancy wastage in Karnataka is 62 per thousand pregnancies, in the major districts like Dharwad, Uttar Kannada, Shimoga, Davanagere, Udupi, Dakshina Kannada, Hassan and Bangalore Rural occurrence of pregnancy wastage is more than 100 per thousand pregnancies and these districts have reported above the State average and occurrence of pregnancy wastage in Belgaum, Raichur, Koppal, Bellary, Haveri, Chitradurga, Chikmagalur, Tumkur, Mandya, Kodagu, Mysore, Ramanagar, Chikkaballapur, Kolar and Chamarajnagar ranges from 50 to 100 pregnancy wastage per thousand pregnancies. Districts like Bidar, Yadgir, Gulbarga, Bijapur, Bagalkote, Gadag and Bangalore Urban are showing less than 50 pregnancy wastage per thousand pregnancies.

It is observed from fig.3 that occurrence of stillbirth is 7.7 per thousand pregnancies in the State, districts like Raichur, Koppal, Bellary, Gadag, Dharwad, Davanagere, Shimoga, Dakshina Kannada and Kodagu is more than 10 per thousand pregnancies; and occurrence of stillbirths in Bidar, Gulbarga, Bijapur, Belgaum Uttara Kannada, Haveri, Chitradurga, Hassan, Mandya, Mysore, Chamarajnagar Kolar and Chikkaballapur districts ranges from 5 to 10 stillbirths per thousand pregnancies. Districts like Yadgir, Bagalkot, Udupi, Chikmagalur, Tumkur, Bangalore Urban, Bangalore Rural and Ramanagar are showing less than 5 stillbirths per thousand pregnancies. Hence most of the Northern Karnataka districts were reported high stillbirth compare to the southern Karnataka region.

The occurrence of spontaneous is 26 per thousand pregnancies and induced abortion is 29 per thousand pregnancies in Karnataka shown in fig.4 and 5 most of the districts of the South Karnataka region is reported spontaneous and induced abortion higher than the North Karnataka region districts in the State. However, it is to be noticed that the occurrence of spontaneous abortion in Shimoga, Kodagu and Bangalore Rural showing more than 50 spontaneous abortions per thousand pregnancies, Districts like Bidar, Gulbarga, Yadgir, Bijapur, Bagalkote, Raichur, Koppal, Bellary, Bangalore Urban, Ramanagar and Chamarajanagar are showing less than 25 spontaneous abortions per thousand pregnancies and remaining districts are in the range from 25 to 50 spontaneous abortion per thousand pregnancies.

Figure 1 Proportion of pregnancy wastage to the total pregnancies (per 1,000 pregnancies) by districts, Karnataka, HMIS 2019-20

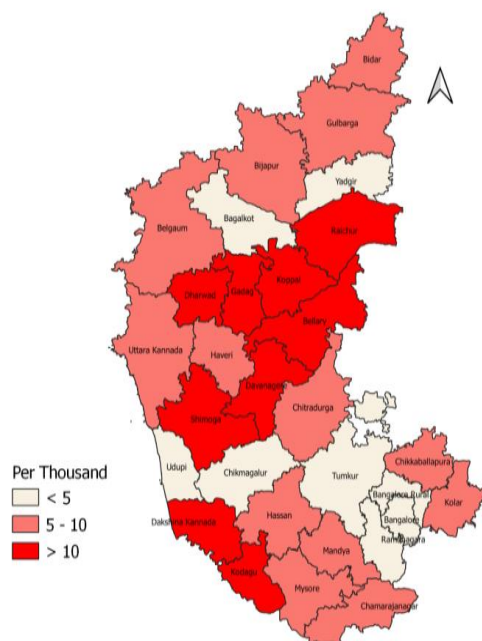


Figure 2 Proportion of stillbirth to the total pregnancies (per 1,000 pregnancies) by districts, Karnataka, HMIS 2019-20

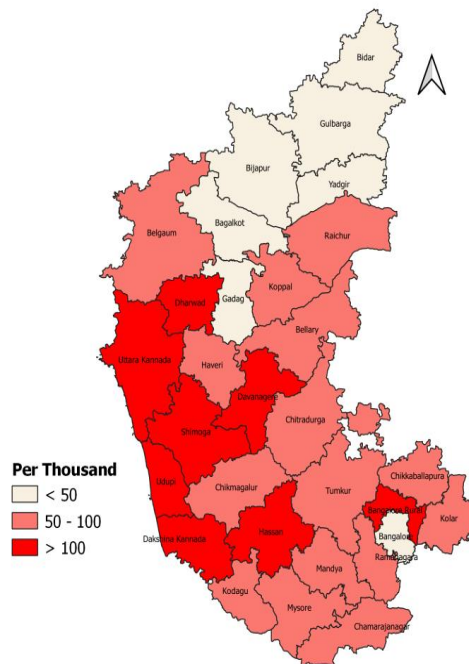


Figure 3 Proportion of spontaneous abortion to the total pregnancies (per 1,000 pregnancies) by districts, Karnataka, HMIS 2019-20

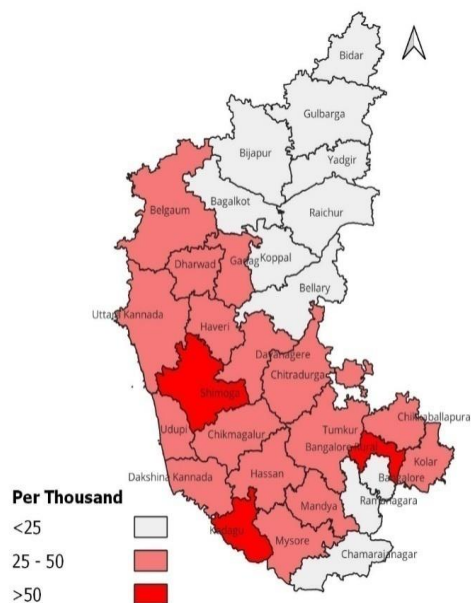
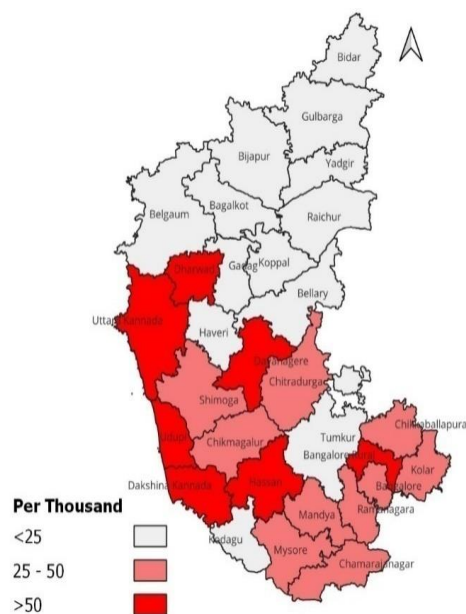


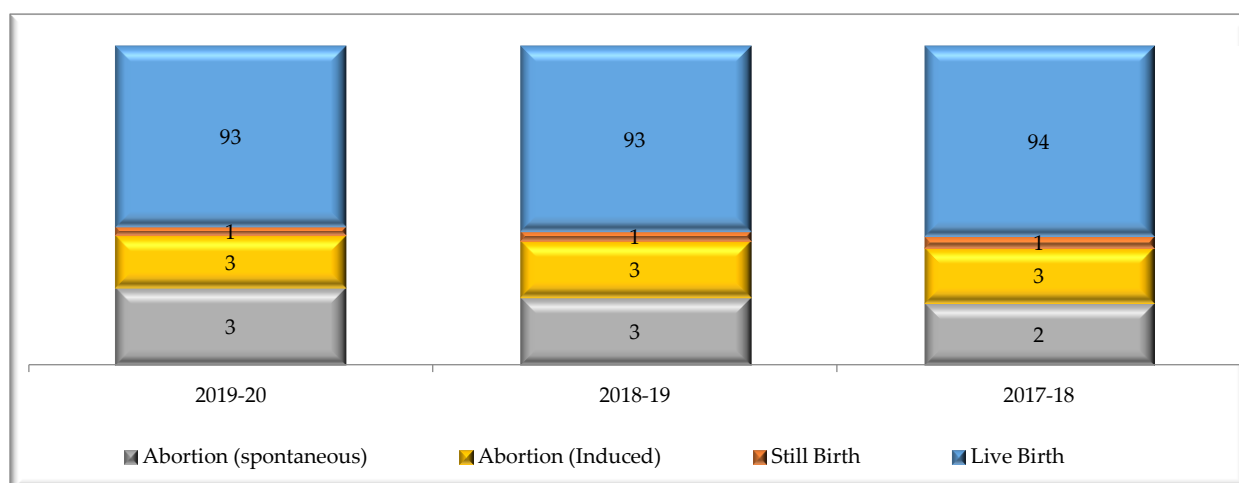
Figure 4 Proportion of induced abortion to the total pregnancies (per 1,000 pregnancies) by districts, Karnataka, HMIS 2019-20



Subsequently, the occurrence of induced abortion in Bangalore Rural, Hassan, Dakshina Kannada, Udupi and Uttar Kannada are showing more than 60 induced abortions per thousand pregnancies and the remaining districts are showing less than 30 induced abortions per thousand pregnancies. It is evident from figures 4 and 5 that, in Karnataka state occurrence of spontaneous and induced abortion is higher in South Karnataka compare to the North Karnataka region.

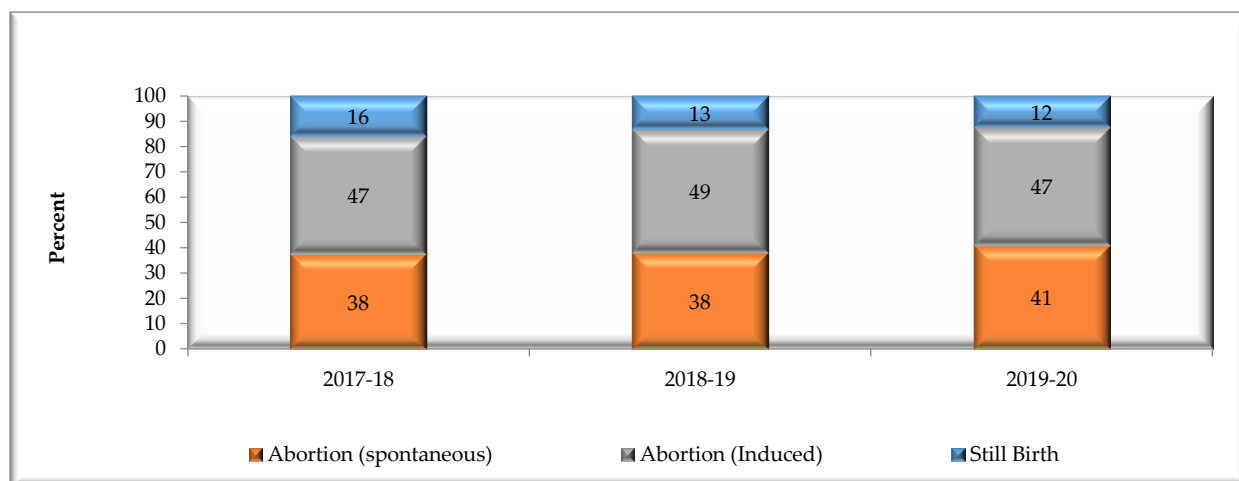
Levels and Trends in Pregnancy Wastage in Karnataka (2017-2020)

Figure 5 Pregnancy outcome in Karnataka, (HMIS 2017 to 2020)



The HMIS data of Karnataka has been analyzed district-wise and year-wise to understand the levels and trends in pregnancy wastage in Karnataka. As presented in Figure 5, among all the pregnancies outcome, the proportion of Pregnancy Wastage viz, spontaneous abortion induced abortion and stillbirth area little low during the last two years compared to the year 2017-18 and live birth reported 93 to 94 percent during the last three years.

Figure 6 Percent distribution of pregnancy wastage, Karnataka HMIS 2017-20



Among the pregnancy wastage as seen in fig. 6, nearly half of the pregnancies are terminated through induced abortion, another 40 percent underwent a spontaneous abortion and 12 to 16 percent had a stillbirth in the State. The trend follows the more or less same pattern for all the three 3 years from 2017-18 to 2019-20 in Karnataka State

Further, when the district wise variation in the levels of pregnancy wastage is analyzed, as shown in figures 7 to 9 below, Bangalore Rural stands at the highest in the outcome of pregnancy wastage during all the years 2017-20, with around 16 to 18 percent of pregnancy wastage. During all years 2017-20, Bagalkote and Bijapur districts have performed a comparatively low proportion of pregnancy wastage (2 to 3 percent) in Karnataka. State and district level officials need to monitor most of the South Karnataka districts like Bangalore Rural, Hassan, Udupi, Mandya, Davanagere, Chikmagalur, Shimoga, Chitradurga, Dakshina Kannada, Bangalore Urban, Kodagu, Ramanagar, Chikkaballapur, Mysore and Kolar including two northern districts (Dharwad and Uttara Kannada).

Figure 7 Percent distribution of pregnancy outcome by districts Karnataka, HMIS-2019-20

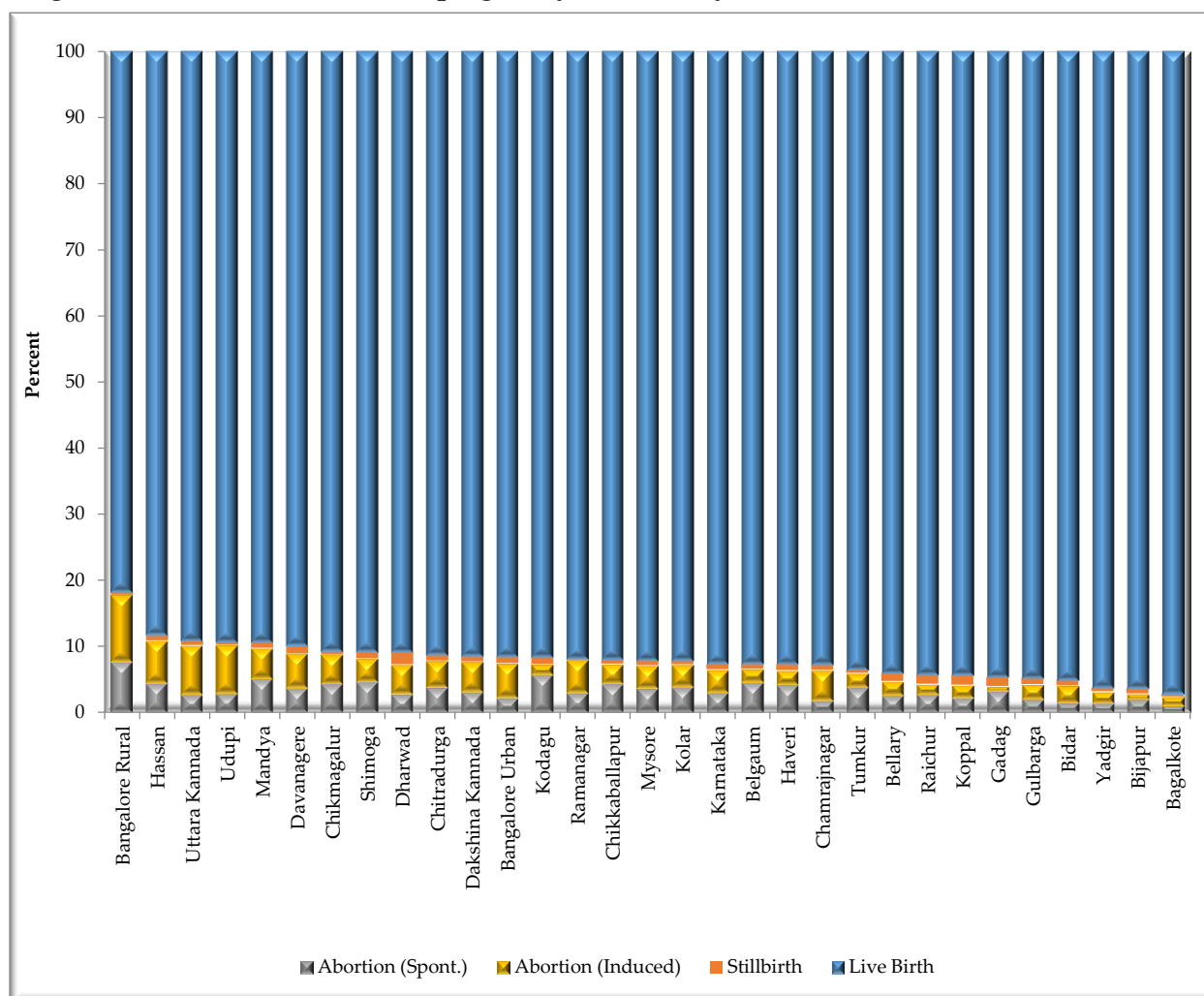


Figure 8 Percent distribution of pregnancy outcome by districts Karnataka, HMIS-2018-19

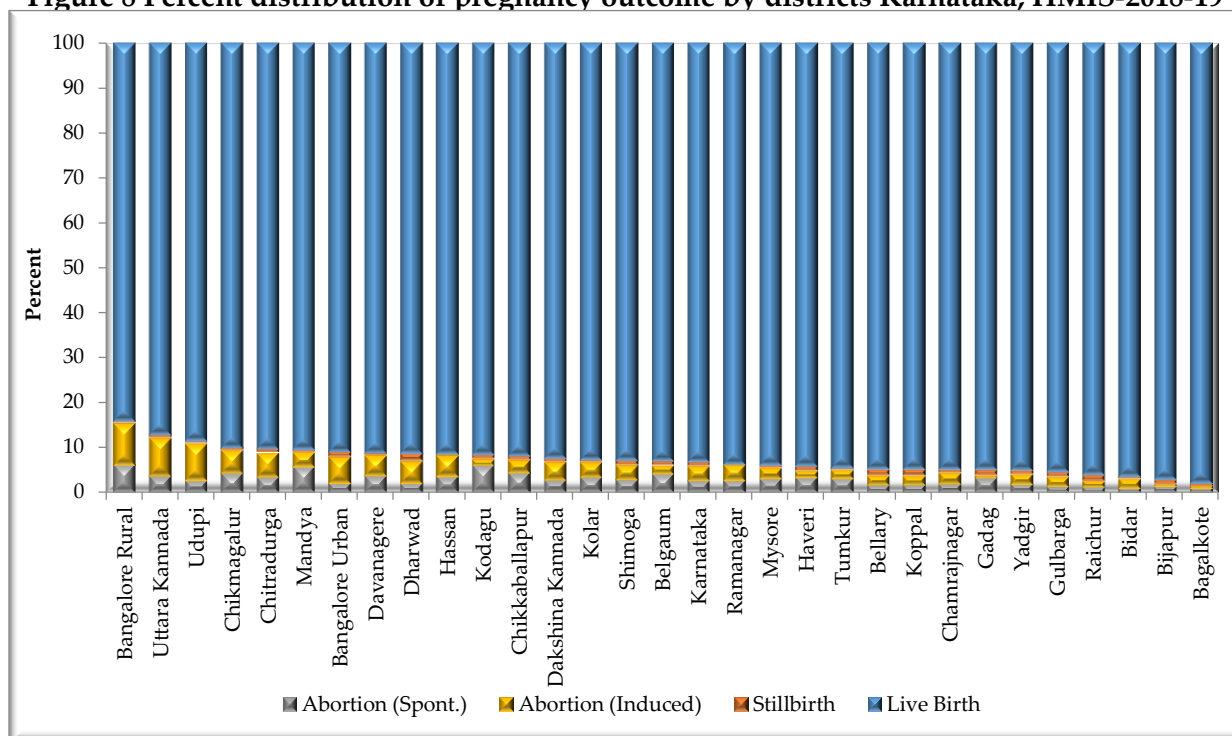
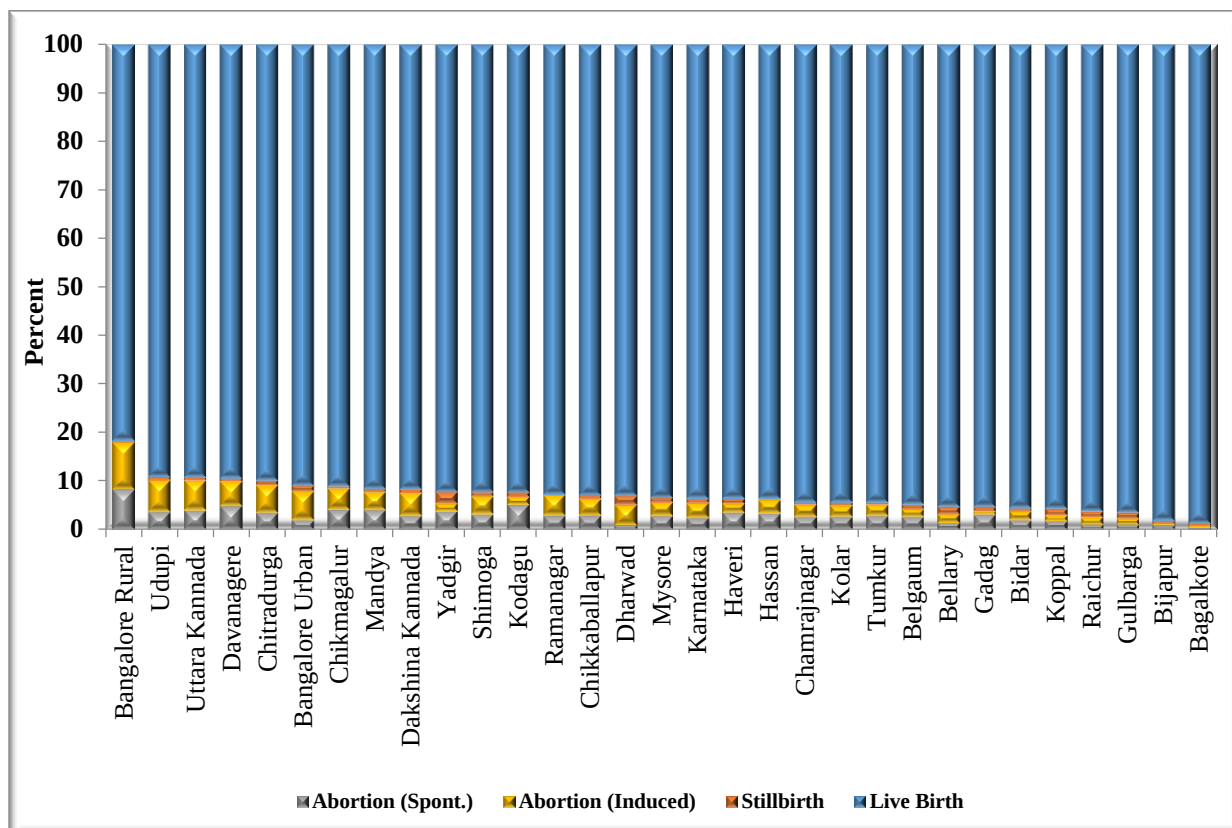


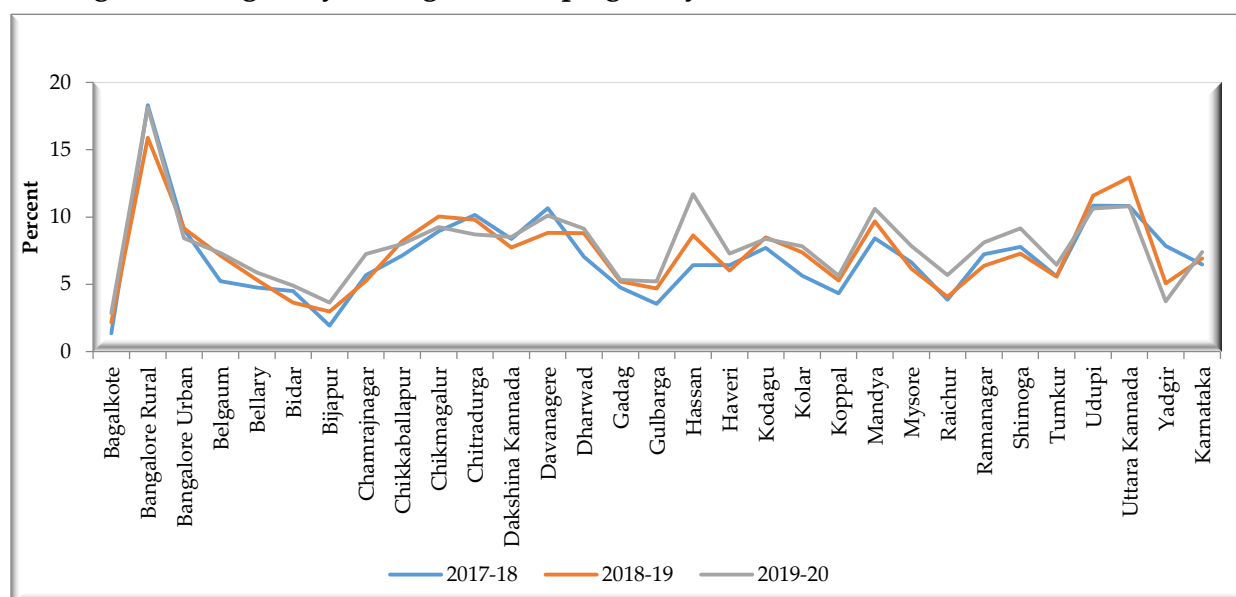
Figure 9 Percent distribution of pregnancy outcome by districts Karnataka, HMIS-2017-18



Trends in Pregnancy Wastage

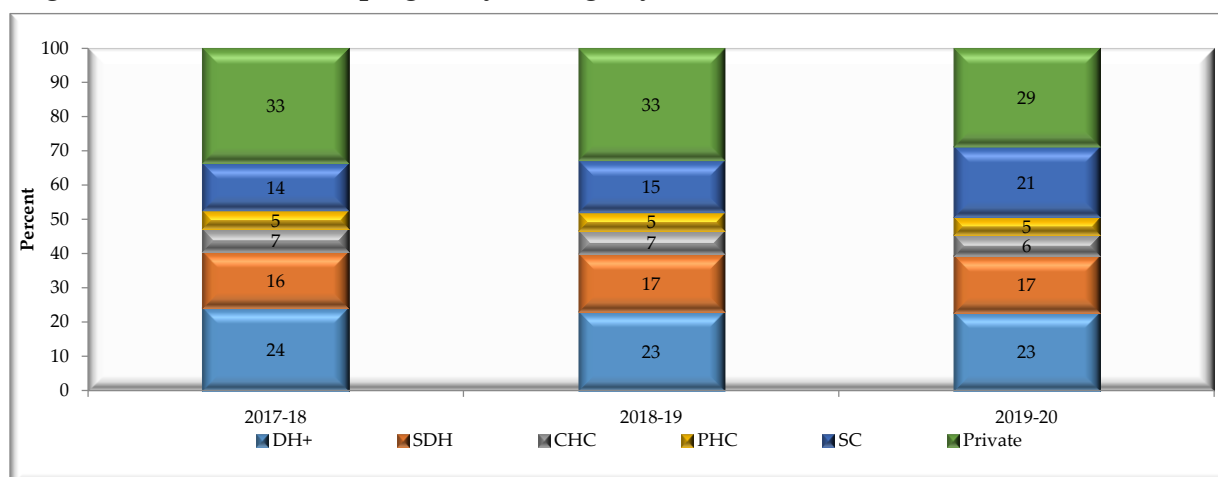
To understand the year-wise variation in reporting of overall pregnancy wastage were analyzed for all the districts of Karnataka for the years, from 2017-18 to 2019-20 and presented in fig. 10 as mentioned below. Regarding reporting of pregnancy wastage, overall, for the State, it has constant at 7 percent during 2017-20, the district wise trend indicates that most of the districts like Bangalore rural, Bijapur, Hassan, Mandya, Raichur, Ramanagar, Mysore and Haveri have reported as high as State average during the year 2019-20 compared to 2018-19. District-wise analysis indicates that Bangalore Rural, Hassan and Yadgir show wide fluctuation among all the districts.

Figure 10 Pregnancy wastage to total pregnancy outcome, Karnataka, HMIS 2017-20



Distribution of Pregnancy Wastage by type of Health Facilities

Figure 11 Distribution of pregnancy wastage by Health facilities, Karnataka, HMIS 2017-20



After analyzing the Levels and trends in pregnancy wastage in Karnataka, the next important issue to be analyzed is the role of different health facilities like HSCs, PHCs, CHCs, SDHs, and DHs and above facilities and private sector contribution in different pregnancy wastages. Hence the distribution of health facilities has been analyzed for different pregnancy wastages, separately for the year 2017-18 to 2019-20 for the State.

Regarding the distribution of pregnancy wastage, one-third of the total wastage that occurred during 2017-18 and 2018-19 were reported through private facilities in the State. Fig.11 presents, around 25 percent of the wastage was reported at DH and above facilities and another 17 percent were performed at SDHs. Reporting of pregnancy wastage from HSCs is 14 to 21 percent and reporting of wastage from CHCs and PHCs is very less during the year 2017-20.

Further, analyzed pregnancy wastages by the health facilities and district wise variation for 2019-20. (Annexure 1 to 4) Regarding the distribution of pregnancy wastage, half of the total pregnancy wastage was reported through private facilities in Udupi, Dakshina Kannada, Dharwad, Uttara Kannada, Gulbarga, Yadgir, Bagalkote, Mysore, Bangalore Urban, Chikmagalur, Hassan, Raichur, Bidar and Bangalore Rural.

Total pregnancy wastage is as high as 81 percent at Udupi and 59 percent at Dakshina Kannada districts and as low as 2 percent in Haveri and 3 percent in Davanagere districts were reported through private facilities. Concerning the reporting of stillbirth, more than 50 percent of the stillbirth was reported through DH and above facilities in Bangalore Urban, Davanagere, Mandya, Kodagu, Chitradurga, Dharwad, Shimoga, Hassan, Chamarajnar, Gulbarga, Mysore and Bellary. Stillbirth is as high as 95 percent in Bangalore Urban and 80 percent in Davanagere districts and as low as 1 percent in Yadgir, 7 percent in Bagalkote, Bangalore Rural and Ramanagar reported nil through DH and above facilities.

Reporting of spontaneous abortion is more than 50 percent were reported through HSCs in Haveri, Bagalkote, Belgaum, Raichur, Bijapur, Bellary, Gadag and Chamarajanagar districts. Spontaneous abortion is as high as 93 percent in Haveri and 81 percent in Bagalkote districts, as low as 3 percent in Udupi, 4 percent in Dakshina Kannada districts reported through DH and above facilities More than half of the induced abortions were reported through the private facilities in Udupi, Dharwad, Gulbarga, Bagalkote, Dakshina Kannada, Raichur, Mysore, Chikmagalur, Uttara Kannada, Bangalore Urban, Hassan Yadgir and Shimoga districts. Induced abortion is a concern as high as 93 percent in Udupi and 69 percent in Dharwad districts, as low as 1 percent in Haveri, 4 percent in Davanagere districts reported through private facilities.

It is indicated that half of the districts were reported high in total pregnancy wastage and induced abortion through private facilities, stillbirths from DH and above facilities and spontaneous abortion at HSCs in the State. The district and state-level officials need to monitor private facilities for induced abortion and stillbirth at DH and above facilities and spontaneous abortion from HSCs.

Discussion

In this study, overall pregnancy wastage was found to be 62 per thousand pregnancies in the State during 2019-20 and Our results are broadly comparable to those in other studies. A study in a district of State, which looked PW is 60.3 per 1000 pregnancies, MTP is 40.1 and miscarriage is 101 per thousand pregnancies (Dhaded et al., 2018). While in our study PW is 62, MTP is 29 and miscarriage is 26 per thousand pregnancies, Induced abortion is the most contributing wastage to the total pregnancy wastage during all three years (2017-20) and spontaneous abortion is the next major one followed by the stillbirth in the State. The majority of the pregnancy wastage is reported through private facilities followed by DHs and above and it is 17 percent at SDHs. Around 5 to 6 percent at CHCs/PHCs. Spontaneous abortion is only reported at HSCs in the State. Most of the stillbirths are reported through DHs and above followed by the private facilities/SDHs and it is 5 to 10 percent at HSCs/PHCs. Around 5 percent at CHCs in the State. Half of the Spontaneous abortion is reported through HSCs followed by DHs and above facilities and it is 14 to 21 percent at private facilities and 14 percent at SDHs. Around 4 percent at CHCs/PHCs in the State.

The majority of the induced abortions were reported through private facilities followed by SDHs/ DHs and above facilities and it is 6-10 percent at CHCs and PHCs in the State. District wise relatively South Karnataka districts are reporting more pregnancy wastage compared to the North Karnataka districts. Pregnancy wastage is found to increase from 2018-19 to 2019-20, which indicates that most of the districts in Karnataka might be reported high figures than that of the earlier year. North Karnataka districts are reporting more stillbirth and spontaneous abortion than the South Karnataka districts, majority of the districts are varied in stillbirths and abortion and it is found to decrease stillbirth from 2019-20 to 2018-19.

Most of the Southern districts are reporting more induced abortion than their counterparts, The widespread use of sex-determination tests and abortion facilities has given an opportunity for parents to achieve the desired family size and the desired gender composition of children (Sekher & Hatti, 2010), which indicates that neither more MTP performed in southern districts nor under-reporting of abortions in North Karnataka districts. The majority of the districts are reported pregnancy wastage through private facilities followed by the DHs and above. Most of the districts are reported stillbirths are from DHs and above. Private hospitals are reporting more induced abortions and spontaneous abortions mainly reported by the SCs.

Conclusion

The study indicated that half of the districts were reported high in total pregnancy wastage and induced abortion through private facilities, stillbirths from DH and above facilities and spontaneous abortion at HSCs in the State. The district and State level officials need to surveillance on private facilities for induced abortion and stillbirth at DH and above facilities

and spontaneous abortion from HSCs in general and particular on facilities of South Karnataka for the reduction of pregnancy wastage in the State.

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Annexure 1 Contribution of public and private health facilities in pregnancy wastage by districts, Karnataka, HMIS 2019-20							
District	DH+	SDH	CHC	PHC	SC	Private	Number
Bagalkote	4.7	15.1	6.2	3.4	29.8	41.0	1,072
Bangalore Rural	NA	48.8	4.8	5.8	9.9	30.7	1,970
Bangalore Urban	33.1	3.7	12.5	9.5	3.6	37.6	11,533
Belgaum	8.3	19.0	4.3	2.0	47.2	19.2	5,423
Bellary	27.7	26.5	8.2	7.5	26.8	3.3	2,716
Bidar	40.5	7.4	3.6	4.6	13.3	30.7	1,377
Bijapur	17.5	5.0	21.2	5.5	35.0	15.8	1,591
Chamrajnagar	39.8	18.3	9.1	9.9	11.1	11.7	656
Chikkaballapur	18.8	28.2	0.4	12.6	22.3	17.7	1,289
Chikmagalur	30.3	20.1	0.3	3.4	12.9	33.0	1,285
Chitradurga	27.6	36.5	9.9	0.1	17.3	8.6	1,758
Dakshina Kannada	21.4	14.3	4.2	0.1	1.5	58.7	3,212
Davanagere	60.1	14.9	4.6	7.2	10.1	3.1	3,265
Dharwad	29.9	6.2	0.0	3.8	9.8	50.4	3,654
Gadag	40.4	10.9	1.0	3.6	34.8	9.3	907
Gulbarga	19.1	13.1	1.7	3.8	16.7	45.7	2,611
Hassan	16.6	26.2	5.5	5.6	14.5	31.6	2,705
Haveri	10.8	20.7	9.4	3.0	53.7	2.3	1,509
Kodagu	39.9	13.9	20.8	2.4	19.1	3.8	581
Kolar	27.3	19.7	2.4	10.1	20.0	20.7	1,743
Koppal	22.8	30.0	13.1	5.1	17.8	11.2	1,617
Mandya	51.1	15.4	2.7	0.3	19.1	11.4	1,982
Mysore	21.9	15.3	7.5	9.2	7.0	39.1	3,326
Raichur	14.0	13.4	1.3	5.2	34.7	31.4	2,465
Ramanagar	9.7	45.5	17.6	2.9	14.6	9.7	732
Shimoga	25.6	30.1	5.6	1.1	12.5	25.0	2,489
Tumkur	14.3	40.2	0.3	7.6	25.5	12.0	2,051
Udupi	5.2	7.6	4.4	0.4	0.9	81.4	1,642
Uttara Kannada	3.6	34.0	5.9	0.1	8.7	47.7	2,347
Yadgir	21.5	6.9	3.9	12.5	10.3	44.9	1,199
Karnataka	24.2	17.6	6.4	5.3	16.9	29.5	70,707
Note: DH+ facilities include District hospital, Govt. Medical college and other public facilities. NA: Not Applicable as DH+ facilities not exists in the district.							

Annexure 2 Contribution of public and private health facilities in stillbirth by districts, Karnataka, HMIS 2019-20							
Districts	DH+	SDH	CHC	PHC	SC	Private	Number
Bagalkote	7.4	18.5	10.1	18.0	0.0	46.0	189
Bangalore Rural	NA	48.8	2.4	4.9	9.8	34.1	41
Bangalore Urban	95.3	1.1	0.8	1.3	0.1	1.3	1,415
Belgaum	45.2	16.7	3.9	6.2	3.4	24.6	533
Bellary	53.6	22.2	10.1	8.2	1.1	4.8	537
Bidar	44.0	17.1	9.4	14.5	6.4	8.5	234
Bijapur	22.4	14.8	6.7	12.2	5.2	38.7	344
Chamrajnagar	57.0	12.7	10.1	10.1	2.5	7.6	79
Chikkaballapur	34.3	52.8	0.9	6.5	0.0	5.6	108
Chikmagalur	32.8	42.2	1.6	4.7	0.0	18.8	64
Chitradurga	62.8	15.2	9.1	0.6	4.3	7.9	164
Dakshina Kannada	36.7	6.0	0.3	0.3	0.0	56.7	319
Davanagere	79.7	8.5	0.8	3.5	1.9	5.6	375
Dharwad	62.5	3.0	0.0	4.1	0.1	30.3	759
Gadag	44.4	19.4	3.9	12.5	1.3	18.5	232
Gulbarga	54.6	10.2	4.7	11.0	1.8	17.6	489
Hassan	61.0	16.9	1.0	3.1	3.6	14.4	195
Haveri	49.7	22.1	5.6	9.7	0.5	12.3	195
Kodagu	67.1	6.8	9.6	6.8	2.7	6.8	73
Kolar	37.8	21.5	0.7	17.8	5.2	17.0	135
Koppal	39.1	28.3	6.0	12.2	4.6	9.8	417
Mandya	68.7	5.5	0.0	2.5	1.8	21.5	163
Mysore	54.4	11.1	2.5	4.7	4.4	22.8	316
Raichur	33.1	17.0	4.4	11.4	5.5	28.5	634
Ramanagar	0.0	43.8	12.5	0.0	31.3	12.5	16

Annexure 2 Contribution of public and private health facilities in stillbirth by districts, Karnataka, HMIS 2019-20							
Districts	DH+	SDH	CHC	PHC	SC	Private	Number
Shimoga	61.8	29.1	1.1	1.8	1.1	5.1	275
Tumkur	15.1	38.0	0.6	9.0	10.8	26.5	166
Udupi	12.1	10.6	0.0	0.0	1.5	75.8	66
Uttara Kannada	9.7	38.9	6.3	0.0	5.6	39.6	144
Yadgir	1.1	5.5	6.6	47.0	27.3	12.6	183
Karnataka	53.4	14.2	3.6	7.4	3.0	18.4	8,860
Note: DH+ facilities include District hospital, Govt. Medical college and other public facilities							
NA: Not Applicable as DH+ facilities not exists in the district.							

Annexure 3: Contribution of public and private health facilities in abortion (spont.) by districts, Karnataka, HMIS 2019-20							
District	DH+	SDH	CHC	PHC	SC	Private	Number
Bagalkote	0.5	6.3	0.8	0.5	80.8	11.1	395
Bangalore Rural	NA	53.8	1.6	0.1	23.0	21.5	829
Bangalore Urban	49.0	7.0	6.1	2.7	13.8	21.4	3,009
Belgaum	2.4	14.1	1.2	0.6	75.5	6.2	3,366
Bellary	14.1	12.7	5.0	8.5	59.8	0.0	1,207
Bidar	33.2	1.0	2.9	1.2	40.1	21.7	419
Bijapur	17.5	1.4	10.1	2.3	61.2	7.6	881
Chamrajnagar	0.0	31.1	6.1	2.4	43.3	17.1	164
Chikkaballapur	20.7	29.3	0.1	8.6	41.2	0.0	699
Chikmagalur	46.9	15.9	0.0	0.0	27.5	9.8	604
Chitradurga	23.8	27.3	3.7	0.1	37.0	8.0	802
Dakshina Kannada	17.7	18.6	5.0	0.1	4.0	54.6	1,173
Davanagere	62.1	6.7	0.7	3.2	26.9	0.3	1,203
Dharwad	25.1	4.4	0.0	4.2	32.0	34.3	1,116
Gadag	35.5	5.6	0.0	0.7	56.2	2.0	557
Gulbarga	13.2	7.8	1.0	0.4	42.5	35.1	1,006
Hassan	20.3	28.9	2.5	4.9	37.3	6.1	1,029
Haveri	0.6	4.3	1.4	0.2	93.0	0.6	870
Kodagu	32.0	14.9	20.9	2.3	27.5	2.5	397
Kolar	24.5	11.7	0.0	9.3	39.5	15.0	864
Koppal	27.5	10.0	8.3	2.0	42.1	10.0	639
Mandya	51.8	8.5	0.1	0.2	39.3	0.1	955
Mysore	29.9	14.7	6.8	12.9	14.2	21.6	1,533
Raichur	5.0	4.8	0.3	3.4	70.1	16.3	1,169
Ramanagar	0.0	46.2	1.5	0.0	38.3	13.9	266
Shimoga	32.6	27.6	4.8	0.4	24.8	9.9	1,248
Tumkur	18.0	29.4	0.0	1.7	41.4	9.5	1,222
Udupi	7.7	16.9	1.1	0.0	3.2	71.1	443
Uttara Kannada	9.8	26.7	4.2	0.3	33.3	25.7	592
Yadgir	18.7	2.9	2.2	9.5	16.1	50.7	454
Karnataka	23.2	14.9	3.1	2.9	40.2	15.6	29,111
Note: DH+ facilities include District hospital, Govt. Medical college and other public facilities.							
NA: Not Applicable as DH+ facilities not exists in the district.							

Annexure 4 Contribution of public and private health facilities in abortion (induced) by districts, Karnataka, HMIS 2019-20						
Districts	DH+	SDH	CHC	PHC	Private	Number
Bagalkote	7.0	20.9	9.0	0.0	63.1	488
Bangalore Rural	NA	45.0	7.4	10.1	37.5	1,100
Bangalore Urban	14.1	2.7	17.5	14.0	51.7	7,109
Belgaum	8.5	30.8	11.2	3.5	46.1	1,524
Bellary	30.2	46.2	11.1	5.9	6.6	972
Bidar	43.5	8.0	2.1	3.3	43.1	724
Bijapur	13.1	4.4	61.5	6.8	14.2	366
Chamrajnagar	52.3	14.3	10.2	12.8	10.4	413
Chikkaballapur	12.4	21.2	0.6	19.7	46.1	482
Chikmagalur	13.8	21.9	0.5	6.6	57.2	617
Chitradurga	24.1	50.1	16.3	0.0	9.5	792
Dakshina Kannada	21.0	12.9	4.3	0.0	61.8	1,720
Davanagere	54.4	22.2	8.2	10.8	4.4	1,687

Annexure 4 Contribution of public and private health facilities in abortion (induced) by districts, Karnataka, HMIS 2019-20						
Districts	DH+	SDH	CHC	PHC	Private	Number
Dharwad	19.0	8.6	0.0	3.4	69.0	1,779
Gadag	55.1	19.5	0.0	0.0	25.4	118
Gulbarga	8.8	19.1	1.0	3.6	67.6	1,116
Hassan	8.1	25.6	8.2	6.5	51.6	1,481
Haveri	13.7	52.5	26.8	5.6	1.4	444
Kodagu	50.5	15.3	27.9	0.0	6.3	111
Kolar	28.5	28.6	5.4	9.7	27.8	744
Koppal	5.3	54.0	23.9	3.2	13.5	561
Mandya	47.0	25.0	6.1	0.0	21.9	864
Mysore	6.7	16.9	9.4	6.3	60.7	1,477
Raichur	11.6	25.1	0.0	2.4	60.9	662
Ramanagar	15.8	45.1	27.3	4.7	7.1	450
Shimoga	6.3	33.6	7.9	1.9	50.3	966
Tumkur	7.4	60.8	0.8	17.9	13.1	663
Udupi	3.8	3.8	6.0	0.6	85.8	1,133
Uttara Kannada	0.8	36.3	6.5	0.0	56.5	1,611
Yadgir	30.4	10.7	4.4	3.7	50.7	562
Karnataka	17.2	20.9	10.2	6.9	44.9	32,736
<i>Note: DH+ facilities include District hospital, Govt. Medical college and other public facilities</i> <i>NA: Not Applicable as DH+ facilities not exists in the district.</i>						

Section III
Family Planning Services

8

Contraceptive Use and Unmet Need Among Young Women in India

Lekha Subaiya and Madhu S. Bidari

Abstract

The knowledge and use of contraception is likely to be of great importance for the health and welfare of young mothers and their children. This study focused on the patterns of contraceptive use among young women aged 15 to 24 across the country utilising data from the National Family Health Survey conducted in 2015-2016. In this study the contraceptive prevalence and unmet need for contraception is presented for young married women in the country, and the protective factors associated with the use of contraception are identified. The findings indicate that, in India, only one in four married women age 15 to 24 years use some form of family planning indicating that there is low contraceptive prevalence in the country. Only, twenty one percent of young married women use modern methods of contraception which include sterilization, pill, IUD, injectables or condom. About eight percent of young married women have undergone sterilization. Adolescents and young women vary in their utilisation of modern methods. Twenty three percent of young women do not use any form of contraception even though they would to delay their next pregnancy or limit their fertility indicating a high level of need for contraception. Education and frequent exposure to media are important correlates of use of modern contraceptive methods and lower unmet need for contraception. There is a significant level of unmet need for contraception among young married women in the North-eastern states, Jharkhand, Kerala, Gujarat and Bihar

Introduction

Youth is a period of rapid growth in an individual's life, wherein the transformation from childhood to adulthood includes many important transitions involving physical, sexual, psychological and social development. The WHO defines adolescents as individuals in the 10-19 years age group, and young people as those falling between the ages of 10 and 24 years. There is an urgent need for researchers and policy makers to focus on this group as the impact of adolescent and youth development is observed throughout one's life. In India, this group is not homogenous in nature, and they have diverse needs. Important challenges for youth in India are poverty and malnutrition as well as the lack of awareness and poor knowledge about physical and psychological changes that affect them (Jeejeebhoy 1998). In particular, for girls, lack of knowledge of proper nutrition, risky behavior with regard to reproductive health and adolescent child bearing have implications for both mother and child health. Young age at childbearing and short birth intervals are risk factors for both maternal and child mortality (Conde-Agudelo et.al. 2006). Hence, the use of contraception by sexually active young women,

particularly the use of spacing methods, is likely to be of great importance for the health and welfare of young mothers and their children.

In India, the average ideal family size among women between the ages of 15 and 49 years was 2.2 in 2015-16 (IIPS and ICF 2017). In other words, the typical woman of reproductive age wants only two children on average. In order to restrict the number of children she bears a woman will have to use contraception throughout her reproductive years, which is typically three decades in length. The primary purpose of contraceptives is for couples to plan their families by limiting, delaying or spacing their fertility. Contraception also has health benefits beyond family planning in itself (Kavanaugh and Anderson, 2013). A review of the literature shows that contraceptive use can reduce abortion and unintended pregnancy among women, which can result in decreases in pregnancy-related morbidity and mortality. The spacing of pregnancies is associated with improved birth outcomes through the reductions in the number of prematurely born babies and babies born with low-birth-weight (Conde-Agudelo et.al. 2006). In India, data from the NFHS 2015-16 shows that 27 percent of births occurred within 24 months of the preceding birth (IIPS and ICF 2017). More young women have short birth intervals compared to older women, and education and wealth are other factors which are associated with longer birth intervals and better pregnancy outcomes. Further, there are high rates of teen pregnancy in India, albeit within marriage. Delaying pregnancy until after the adolescent years is associated with reductions in infant mortality.

The focus of this study is on the contraceptive behaviour of young married women aged 15 to 24 years in India utilising data from the fourth round of the National Family Health Survey conducted in 2015-2016. The study analyses the association between key risk factors and key protective factors with that of contraceptive use and unmet need for contraception among young married women. The aspects of contraceptive behaviour being studied are the use of any modern methods of contraception, female sterilisation and unmet need for contraception among fecund women who do not want any more children or want to postpone childbirth but are not using contraception to limit or space pregnancies. In doing so the study seeks to highlight some of the reproductive issues faced by young women when planning their families.

Data and Methods

From all the states of the country. The data for the study are from the National Family Health Survey (NFHS) which uses a multi stage, stratified design and weights to provide nationally representative estimates of demographic and health variables. This analysis is based on the fourth round of the National Family Health Survey conducted in 2015-16. In general, for the NFHS, unmarried and married women age 15 to 49 are eligible to be interviewed. For the purpose of this study only married women age 15 to 24 are included in the analysis to understand contraceptive behaviour and unmet need among a group which is expected to be in need of contraception. Taking into account marital status and age, the sample employed for this study results in 94,034 married women of age 15 to 24 years. This group forms about 13 percent

of the larger sample of women of reproductive age covered by the survey. Appropriate sample weights are used in the analysis in order to make the sample representative of young married women in the country.

The three indicators of contraceptive behaviour among young married women which form the focus of the study are (1) the utilisation of modern methods of contraception, (2) female sterilization as a permanent method of contraception, and (3) the unmet need for contraception. The analysis will examine the relationship between the social and demographic characteristics of women and their contraceptive behaviour using bivariate and multivariate methods.

Contraceptive prevalence is the percentage of women who are currently using, or whose husband is currently using, at least one method of contraception, regardless of the method used. Modern methods of contraception include male or female sterilization, the use of pill, IUD, injectables and condoms. Traditional methods include the rhythm method, withdrawal or folk methods. This study will analyse the factors associated with two indicators of contraception, the use of any modern methods of contraception and female sterilization, among currently married women age 15 to 24 years.

Unmet need for family planning is defined as the percentage of fecund women, either married or in a union, who are not using contraception but want to stop or delay childbearing. Specifically, women are considered to have unmet need for spacing if they are at risk of becoming pregnant, not using contraception, and either do not want to become pregnant within the next two years, or are unsure if or when they want to become pregnant. Unmet need for family planning is the sum of unmet need for spacing plus unmet need for limiting. Women who are classified as infecund have no unmet need because they are not at risk of becoming pregnant and are not included in this variable. Thus, 3841 women from the larger sample are dropped from the analysis of unmet need for contraception.

Social and Demographic Profile

Information on the social and economic characteristics of women included in the study is presented in Table 1. Out of the sample of 98,767 young married women, 81 percent are in the 20 to 24 age group and three out of four young married women live in rural areas. About 65 percent of young married women have at least one child, and four percent have three or more children. One in four women has no education, 5 percent have a few years of schooling, and a little more than half of young married women in the country have completed primary education. About one in ten women have completed secondary school and the same number have higher levels of education. The primary form of media exposure is the television with about 68 percent of young women reporting that they watch TV at least once a week. On the other hand, one in five young women have no exposure to any form of media, be it newspaper, radio, tv or cinema.

**Table 1 Background characteristics of currently married women
age 15 to 24 years, India 2015-16**

Age	
15-19	18.7
20-24	81.3
Age at marriage	
Below 18 years	38.7
18 and above	61.2
Number of living children	
No children	34.6
One child	39.3
Two children	21.6
Three or more children	4.5
Place of Residence	
Urban	25.9
Rural	74.1
Education	
No education	23.1
Incomplete primary	5.4
Primary	7.7
Incomplete secondary	47.1
Secondary	11.7
Higher	10.4
Media Exposure	
Reads a newspaper/ magazine at least once a week	19.6
Listens to radio at least once a week	8.7
Watches television at least once a week	67.6
Not at all exposed to any media	20.6
Religion	
Hindu	80.6
Muslim	15.4
Christian, Sikh, Buddhist, Jain, other	4.0
Caste/ tribe	
Schedule caste	22.1
Scheduled tribe	10.7
Other backward class	43.4
Other	18.7
Don't know and missing	5.0
Wealth Index	
Lowest	20.8
Second	23.8
Middle	22.7
Fourth	19.7
Highest	13.0
Number of women 15 to 24 years (unweighted)	98,767

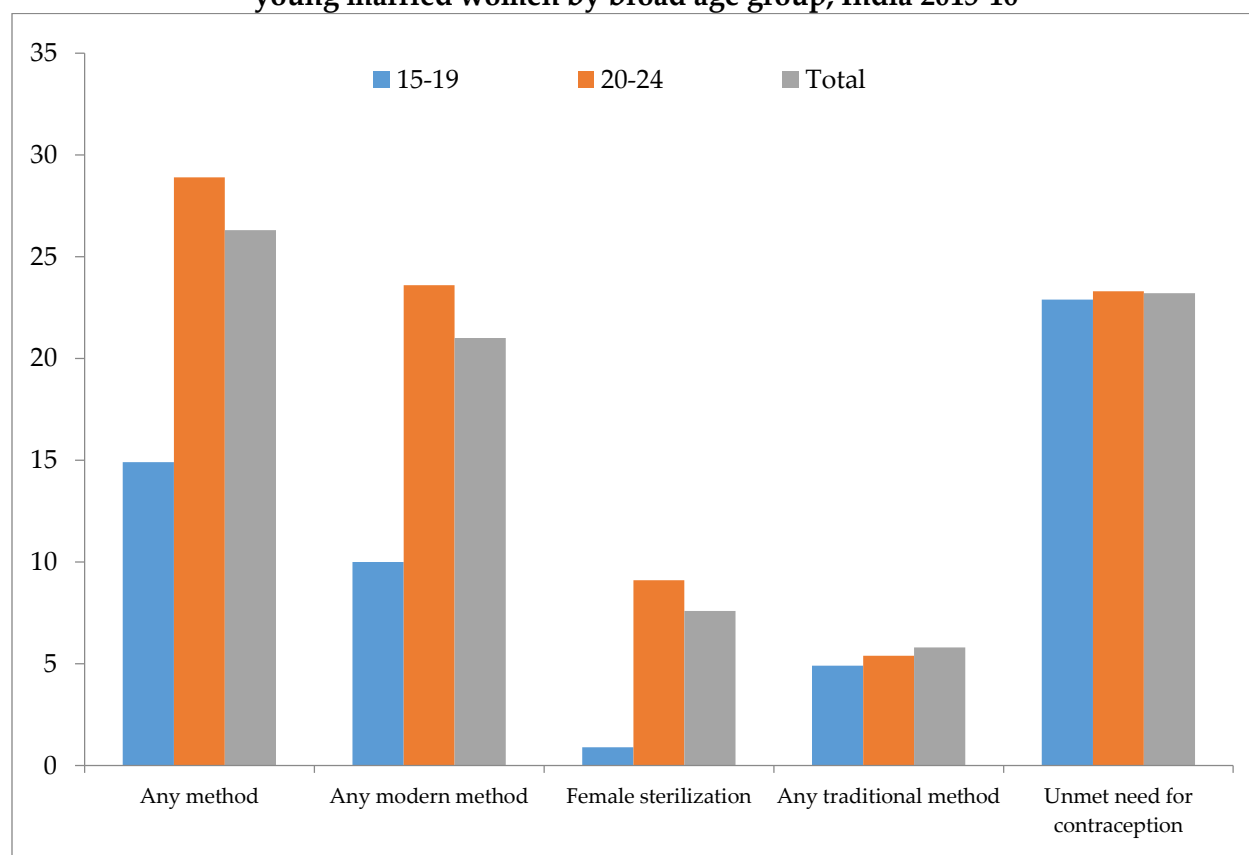
Source: Data are from NFHS 4

Patterns of Contraceptive Use and Unmet Need

In India, significant proportions of women are married during the years of early youth and initiate childbearing. Early age at childbirth and short birth intervals are both risk factors for maternal and child mortality. Hence, the use of contraception by sexually active women age 15-24, particularly the use of spacing methods, is likely to be of great importance for the health and welfare of young mothers and their children.

A weighted analysis of the sample shows that, in 2015 -2016, a little more than one in four married women age 15 to 24 years in the country uses some form of contraception and 21 percent use modern methods which include sterilization, pill, IUD, injectables or condom. About 8 percent of young married women have undergone sterilization. The data by broad age group is presented in the figure below. Double the women in the 20 to 24 age group use modern methods of contraception compared to women in the younger age group. This difference by age is stark in the case of the permanent method of female sterilisation. About one in ten young married women in the age group of 20 to 24 years have undergone sterilisation and finished their childbearing, whereas only about 1 percent of women in the younger age group have done so.

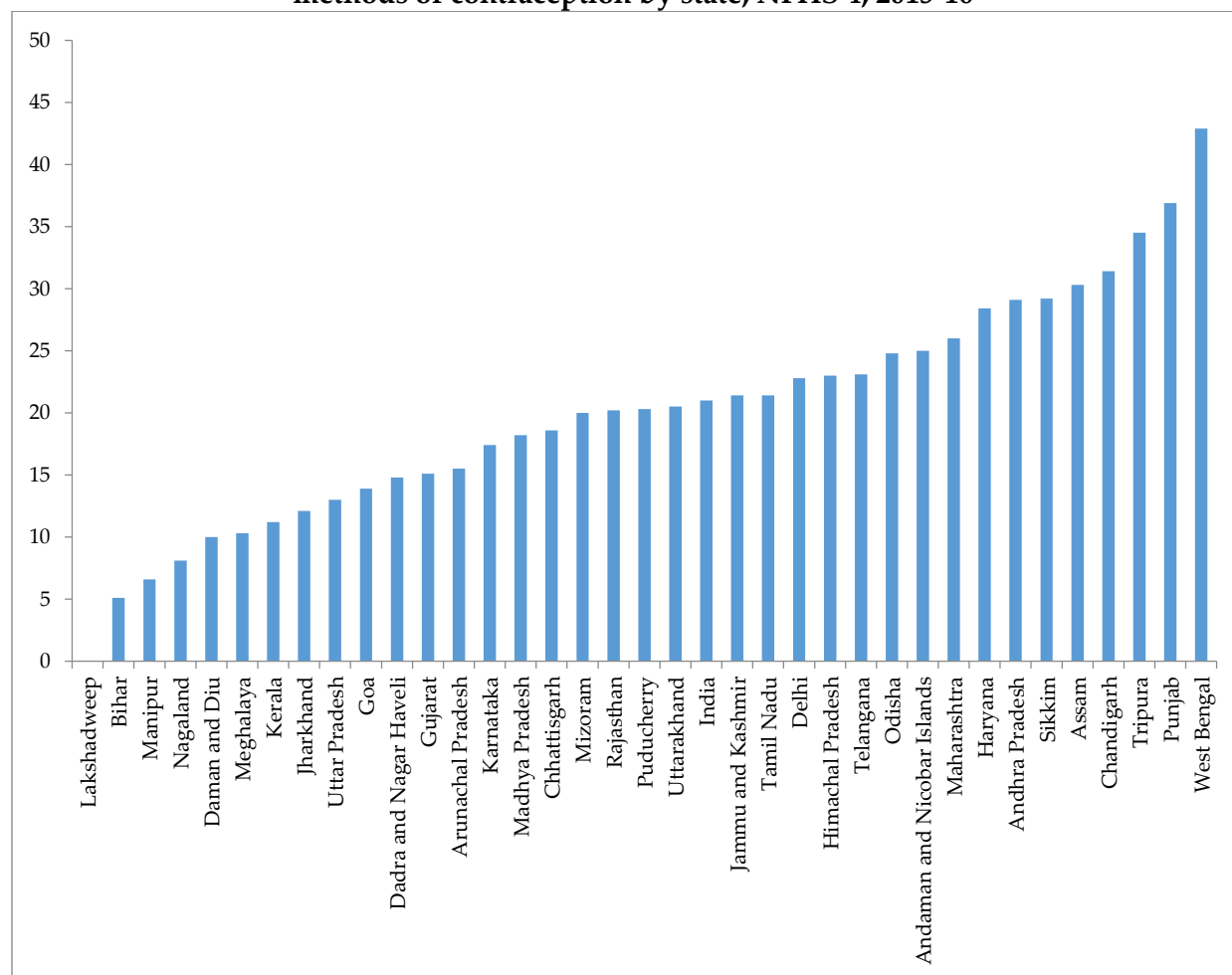
Figure 1 Contraceptive use and unmet need for contraception among young married women by broad age group, India 2015-16



Source: Data are from NFHS 4

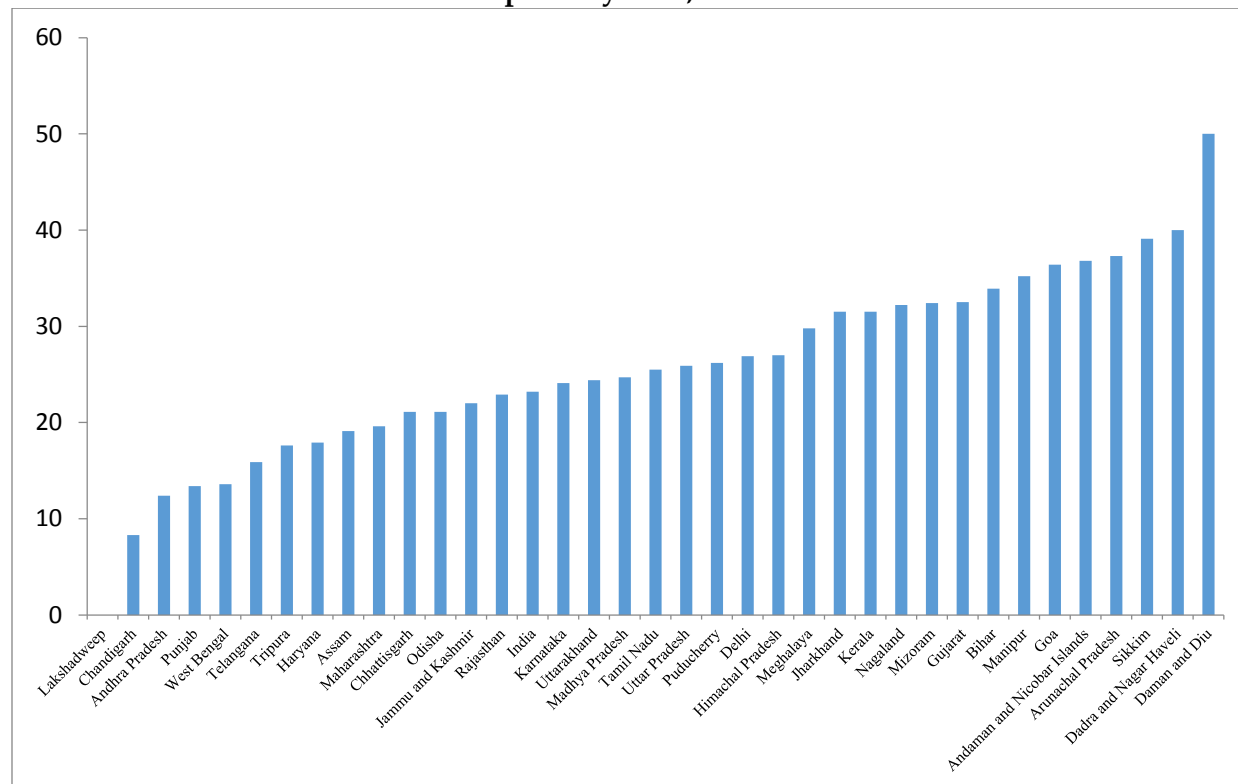
Information on the patterns of young women's use of contraception according to the state they live in is presented in the figure below (Figure 2). The data show contraceptive prevalence is lowest in Bihar with only about 5 percent of young married women using modern methods of contraception, of which female sterilisation is at 3 percent. The proportion of young married women who use modern methods in states including Arunachal Pradesh, Chhattisgarh, Goa, Gujarat, Jharkhand, Karnataka, Kerala, Madhya Pradesh, Meghalaya, Manipur, Nagaland and Uttar Pradesh is less than 20 percent. Among the other states, the proportion of young married women who use modern methods of contraception ranges up to 42.9 percent in West Bengal. In Andhra Pradesh, Goa, Karnataka, Puducherry, Tamil Nadu and Telangana the permanent method of female sterilisation is the main type of modern method utilised. All these states also have relatively high levels of unmet need among young women (see Figure 3). Across the country, the states with the highest proportion of young women with unmet need for contraception are Arunachal Pradesh (37.3%), Bihar (33.9%), Goa (36.4%), Gujarat (32.5%), Jharkhand (31.5%), Manipur (35.2%), Mizoram (32.4%), Nagaland (32.2%) and Sikkim (39.1%).

Figure 2 Percentage of married young women who are using modern methods of contraception by state, NFHS 4, 2015-16



Source: Data are from NFHS 4

Figure 3 Percentage of young married who have unmet need for contraception by state, India 2015-16



Source: Data are from NFHS 4

The distribution of young women's use of contraception by their background characteristics is presented in Table 2. Slightly more young women residing in urban areas use modern methods of contraception compared to their rural counterparts, and there is no variation among the two groups by residence with regard to the proportion which has undergone sterilisation. The use of modern methods of contraception is slightly higher among young women with at least some level of education compared to those with no education. However, women with higher than secondary level of education are much less likely to have undergone sterilization. A similar pattern is observed in the association between the level of wealth of the household the young woman lives in and the choice of contraceptives. Women living in families which are in the highest category of the wealth index are most likely to use modern methods of contraception and least likely to have undergone sterilization. These women are most likely to use condoms as a method of contraception (IIPS and ICF, 2017).

Not surprisingly an important association is the one between the number of children a woman has and her use of modern methods of contraception. While only 5.5 percent of women who have no children use modern methods of contraception to postpone childbearing, that proportion rises to 20.7 percent for women who have one child and 42 percent for women who have 2 or more children. While almost no women who have only one child have undergone sterilisation, about one in three women who have 2 more children have done so.

**Table 2 Contraceptive use and unmet need among young married women
by background characteristics, India 2015-16**

Background characteristics	Any modern method	Female sterilization	Unmet Need for contraception
Age			
15-19	10.0	0.9	22.9
20-24	23.6	9.1	23.3
Age at marriage			
Less than 18 yrs	26.8	13.1	23.6
More than 18 yrs	17.4	4.1	23.0
Number of living children			
No children	5.5	0.0	15.1
One child	20.7	0.6	28.4
Two children	42.0	27.3	25.5
Three or more children	42.4	31.4	27.7
Education			
No education	16.1	8.2	23.2
Incomplete Primary	26.9	11.4	21.3
Complete Primary	20.4	9.7	23.9
Incomplete secondary	23.5	8.5	22.3
Complete secondary	18.6	4.2	26.0
Higher	18.4	2.5	24.9
Residence			
Urban	24.1	7.0	22.7
Rural	19.9	7.8	23.4
Media Exposure			
Reads a newspaper/ magazine at least once a week	21.1	6.0	24.0
Listens to the radio at least once a week	18.9	7.1	27.1
Watches television at least once a week	24.3	8.7	21.9
Visits the cinema / theatre at least once a month	22.7	7.8	23.0
Not at all exposed to any media	13.7	5.3	25.9
Religion			
Hindu	20.7	8.3	23.4
Muslim	20.8	3.6	23.1
Christian	23.3	14.0	21.2
Sikh	37.0	3.9	12.3
Other	27.2	6.8	25.8
Caste/ tribe			
Schedule caste	22.4	8.4	21.7
Scheduled tribe	18.6	8.6	21.8
Other backward class	18.0	7.8	25.1
Other	26.0	6.2	21.6
Don't Know & missing	27.8	4.9	23.2
Wealth index			
Lowest	15.4	6.5	25.5
Second	20.5	7.4	23.3
Middle	22.3	9.8	21.9
Fourth	24.1	8.6	23.0
Highest	24.2	4.2	22.4
Sample size (unweighted)	98,767	98,767	94,558

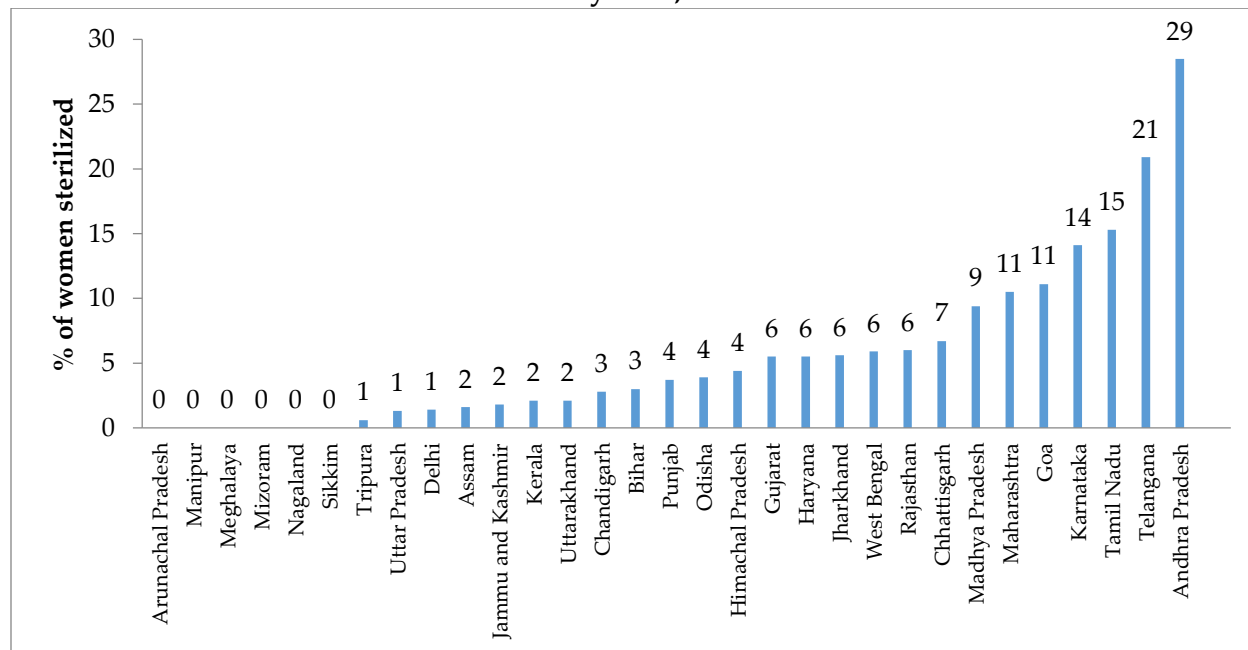
Source: Data are from NFHS 4

The proportion using modern methods also vary by whether women were married before attaining majority or not. It is interesting to note that more women who were married before turning 18 years of age (27%) are utilising modern methods of contraception, compared to women who were married after attaining majority (17.5). However, it is clear that most of this proportion of women who got married young and are using modern methods is made up of women who have undergone sterilisation. From the perspective of intervention, it is notable that women who have no exposure at all to any form of media have the smallest proportion using modern methods of contraception compared to women who watch tv, listen to the radio or go to the cinema at least once a week.

The distribution of young women who have undergone sterilisation according to their background characteristics is presented in Table 2. Among currently married women of age 15 to 24 years, 7.6 percent have undergone sterilisation. The vast majority of young married women who are sterilised are in the 20 to 24 age group, where about 1 in ten women have undergone this permanent method of contraception. Among young women who have two children the proportion who have undergone sterilisation increases to 27.3 percent, and to 31.4 percent among women who have 3 or more children. The proportion of young married women who have undergone sterilisation does not vary substantially by their place of residence. About 7 percent of women in urban areas and 7.8 percent of women in rural areas have used this method of contraception. About 8 percent of women with no education, 11 percent of those with a little education, ten percent of women who have completed primary level of education and 8.5 percent of women with slightly less than secondary education have undergone sterilisation. However, the proportion reduces by half for women with completed secondary levels of education and even more for women with higher education. Among women in various levels of household wealth only the women in the highest level have low levels of sterilisation.

The figure below highlights the proportion of women sterilised according to geographic region. The data show that the states of Andhra Pradesh and Telangana have by far the highest proportions of young women who have undergone this permanent method of contraception. In all the proportions are higher in the southern states compared to the northern, eastern and western states. The states in the North East have zero sterilisation rates among young women in this age group.

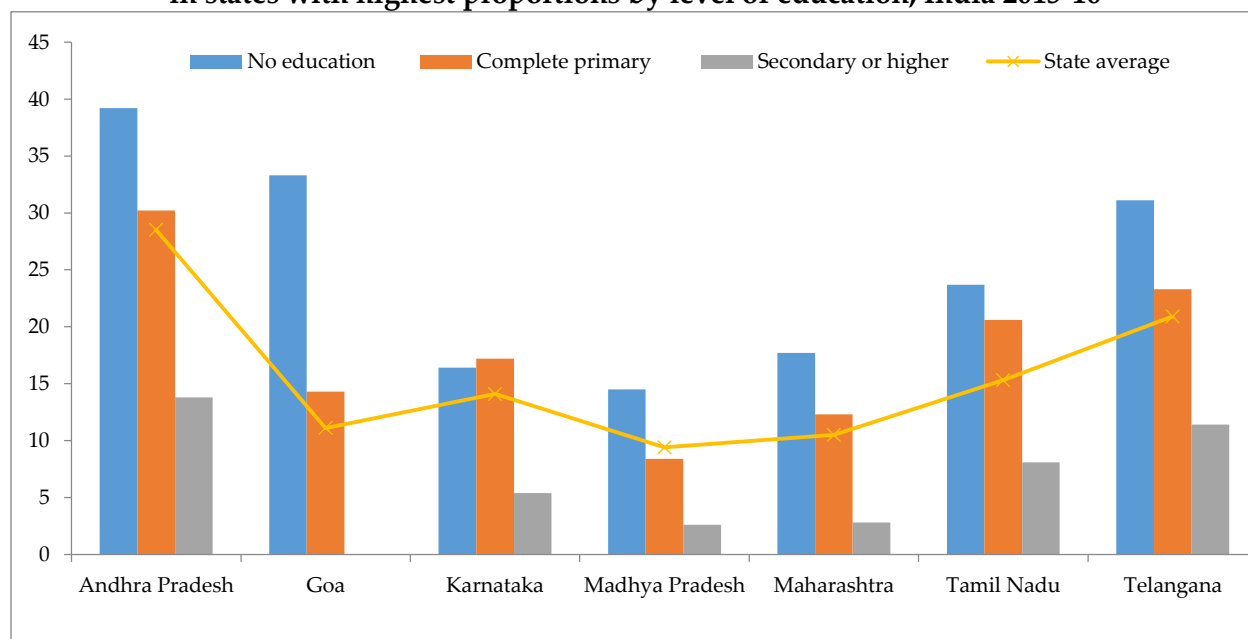
Figure 4 Percentage of young married women who have undergone sterilization by state, India 2015-16



Source: Data are from NFHS 4

A distribution of young women in the select states by levels of education shows the strong relationship between education and sterilisation (Figure 5). The rates of sterilisation are higher among young women with no education or with only primary level of education.

Figure 5 Percentage of young married women who have undergone sterilization in states with highest proportions by level of education, India 2015-16



Source: Data are from NFHS 4

The proportion of young women who have unmet need for methods to space or limit child bearing does not vary significantly by many of their background characteristics. The significant associations are with number of children and education. In particular, a larger proportion of young women who have secondary and higher levels of education have unmet need for contraception compared to women who have no or fewer years of education. Larger proportions of women with children have unmet need for spacing or limiting methods compared to women with no children.

Correlates of Modern Methods of Contraception

The results of a multivariate analysis of the correlates of women's use of modern methods of contraception are presented in Table 3. The analysis indicates that there are strong relationships between women's age, the number of children they have, education, place of residence, exposure to media, wealth of the household and geographical region. Controlling for the effects of number of children, education and wealth, married women age 20 to 24 are significantly more likely to use modern methods of contraception than married women in the younger age group of 15 to 19 years.

It is not surprising that women who have children are more likely to be using contraception to limit their fertility or stop childbearing completely. Women with two children have more than thirteen times the odds of using modern contraceptive methods compared to women who have no children, and women who have three or more have even higher odds of using modern methods to space or limit childbearing. This relationship is by far the strongest among all the factors included in the model. Education and exposure to media are both strongly and positively associated with young women's likelihood of using contraception. Women who have some years of schooling, those who have completed primary education, as well as those who have completed secondary and higher levels of education are more likely than those with no education to be currently using modern methods of contraception. Young women who are exposed to any form of media, be it newspaper, radio, tv or cinema, at least once a week are significantly more likely than those who are not exposed to any media to use modern methods of contraception. These two factors are important indicators of where interventions can take place.

It is notable that the relationship between young women's use of modern methods of contraception with place of residence is strong, with women living in urban areas being significantly more likely than their rural counterparts to use contraception. Further, compared to women living in the southern states, women living in all the other regions but for the central region are more likely to use modern methods of contraception.

Table 3 Odds ratios of logistic regression estimating young married women's use of contraception, India, 2015-16

	Any modern method
Age	
<i>(15-19)</i>	
20-24	1.247***
Age at marriage	
<i>(Less than 18 yrs)</i>	
More than 18 yrs	0.807***
Number of living children	
<i>(No children)</i>	
One child	4.230***
Two children	13.143***
Three or more children	15.668***
Residence	
<i>(Urban)</i>	
Rural	0.908**
Education	
<i>(No education)</i>	
Incomplete Primary	1.592***
Complete Primary	1.207***
Incomplete secondary	1.534***
Complete secondary	1.281***
Higher	1.539***
Any Media Exposure	
<i>(No)</i>	
Yes	1.801***
Religion	
<i>(Hindu)</i>	
Muslim	0.836***
Christian	0.948
Sikh	1.888***
Other	1.213**
Caste/ tribe	
Schedule caste	0.884***
Scheduled tribe	0.765***
Other backward class	0.669***
Don't know & missing	1.079*
<i>(Other)</i>	
Wealth index	
<i>(Lowest)</i>	
Second	1.216***
Middle	1.332***
Fourth	1.546***
Highest	1.890***
Regions	
<i>(South)</i>	
North	1.284***
Central	0.926***
East	1.623***
Northeast	2.166***
West	1.090***

Notes: Data are from NFHS 4; Reference categories are in italics;
 *, **, *** represents odds ratio significant at 10%, 5%, 1% respectively

Corelates of Female Sterilization

The odds ratios calculated from coefficients of the logistic regression for whether a woman has undergone sterilization or not is presented in Table 4.

Table 4 Odds ratios of logistic regression estimating the likelihood of being sterilized among young married women, India 2015-16

	Female Sterilisation
Age	
<i>(15-19)</i>	
20-24	2.258***
Age at marriage	
<i>(Less than 18 yrs)</i>	
More than 18 yrs	0.573***
Number of living children	
<i>(No children)</i>	
One child	10.988***
Two children	606.890***
Three or more children	860.811***
Residence	
<i>(Urban)</i>	
Rural	1.239***
Education	
<i>(No education)</i>	
Incomplete Primary	1.419***
Complete Primary	1.181***
Incomplete secondary	1.106***
Complete secondary	0.673***
Higher	0.738***
Any Media Exposure	
<i>(No)</i>	
Yes	1.446***
Religion	
<i>(Hindu)</i>	
Muslim	0.342***
Christian	1.320**
Sikh	0.983
Other	0.824
Caste/ tribe	
Schedule caste	0.851***
Scheduled tribe	0.952
Other backward class	0.866***
Don't know & missing	0.809**
<i>(Other)</i>	
Wealth index	
<i>(Lowest)</i>	
Second	0.889***
Middle	0.996
Fourth	1.017
Highest	0.882
Regions	
<i>(South)</i>	
North	0.149***
Central	0.127
East	0.143***
Northeast	0.049***
West	0.286***

Notes: Data are from NFHS 4; Reference categories are in italics;

*, **, *** represents odds ratio significant at 10%, 5%, 1% respectively

Each odds ratio gives the increase or decrease in the odds of the women being sterilised for a given value of the independent variable compared to the reference category. The model controls for the independent effects of various social and demographic characteristics including number of children, education, place of residence, social group, wealth of household and geographical region. The results show that women in the 20 to 24 age group have more than twice the odds of being sterilised compared to women in the 15 to 19 age group. By far the strongest relationship is between the number of children a young woman has and her likelihood of being sterilised. While women who have one child have 11 times higher odds of being sterilised than women who have no children, women who have 2 children have 600 times the likelihood of being sterilised. This relationship is significant at the 1 percent level.

Women who have a little education are more likely to have undergone sterilisation compared to women who have no education. However, women who have completed secondary or higher levels of education are less likely to be sterilised than women who have little or no education. This relationship is significant at the 1 percent level. Women who are exposed to some form of media, be it newspaper, radio, tv or cinema, at least once a week are more likely to have undergone sterilisation compared to women who had no exposure to media.

The relationship with place of residence is dissimilar to that found in the bivariate analysis, with young women in rural areas being significantly more likely to have undergone sterilisation compared to young women whose place of residence is urban. There is a strong association with region in the country and the likelihood of young women being sterilised. Compared to women who live in states in the southern region of the country, young women who live in all the other regions are much less likely to have undergone the permanent method of sterilisation.

Corelates of Unmet Need

Among all married women in the 15 to 24 age group, about 23 percent report that they would like to either limit their fertility or delay their next birth but do not use any methods of family planning. The distribution of the prevalence of unmet need for family planning according to background characteristics does not show significant variation. However, in a few cases such as education and media exposure some categories are higher than others. Across levels of education, women who have secondary or higher levels show slightly higher levels of unmet need. Twenty six percent of women who are not exposed to media via newspaper, tv or radio have unmet need for contraception.

The results of the multivariate analysis of the unmet need for spacing or limiting childbearing are presented in Table 5. After controlling for the effects of various social and demographic characteristics, the results show that women in the 20 to 24 age group are less likely to have unmet need than their younger counterparts. While the distribution of the prevalence of unmet need for family planning according to background characteristics did not show much variation, the regression model indicates that age at marriage, education, media exposure, wealth of the household and region of the country have significant association with women's unmet need for

contraception. By far the strongest association of the dependent variable in the model is with number of living children. Women with one child and those with two children have 2.5 times the odds of having unmet need for contraception compared to women with no children. It is notable that women who got married after attaining their majority are more likely to have unmet need for contraception than women who got married earlier.

Table 5 Odds ratios of logistic regression estimating unmet need for contraception among young married women, India 2015-16

	Unmet Need for contraception
Age	
<i>(15-19)</i>	
20-24	0.649***
Age at marriage	
<i>(Less than 18 yrs)</i>	
More than 18 yrs	1.190**
Number of living children	
<i>(No children)</i>	
One child	2.647***
Two children	2.569***
Three or more children	3.038
Residence	
<i>(Urban)</i>	
Rural	0.981
Education	
<i>(No education)</i>	
Incomplete Primary	0.940
Complete Primary	1.133***
Incomplete secondary	1.100***
Complete secondary	1.472***
Higher	1.586***
Any Media Exposure	
<i>(No)</i>	
Yes	0.837***
Religion	
<i>(Hindu)</i>	
Muslim	0.961*
Christian	1.028
Sikh	0.496***
Other	1.148**
Caste/ tribe	
Schedule caste	1.008
Scheduled tribe	0.955
Other backward class	1.230***
Don't know & missing	1.116***
<i>(Other)</i>	
Wealth index	
<i>(Lowest)</i>	
Second	0.905***
Middle	0.853***
Fourth	0.890***
Highest	0.853***
Regions	
<i>(South)</i>	
North	1.186***
Central	1.316***
East	1.228***
Northeast	1.011
West	1.296***

Notes: Data are from NFHS 4; Reference categories are in italics;
*, **, *** represents odds ratio significant at 10%, 5%, 1% respectively

Women's level of education has a strong association with their unmet need for contraception. Young women who have at least completed primary levels of education are significantly more likely to have unmet need for contraception compared to young women who have no education. On the other hand, women who have some media exposure are less likely to have unmet need for contraception. Women living in states which are in the northern, central, eastern and western region of the country are more likely to have unmet need than women living in the southern states.

Conclusion

In this study the contraceptive prevalence and unmet need for contraception is presented for young married women in the country, and the protective factors associated with contraception use are identified. The findings indicate that, in India, only one in four married women age 15 to 24 years use some form of family planning indicating that there is low contraceptive prevalence in the country. Only, twenty one percent of young married women use modern methods of contraception which include sterilization, pill, IUD, injectables or condom. About eight percent of young married women have undergone sterilization. Adolescents and young women vary in their utilisation of modern methods. Twenty three percent of young women do not use any form of contraception even though they would to delay their next pregnancy or limit their fertility indicating a high level of need for contraception. Education and exposure to media are important factors correlated with the use of contraceptives. Women who are exposed to some form of media, be it newspaper, radio, tv or cinema, at least once a week are more likely to use modern methods of contraception than women who are less exposed to media. Women with secondary and higher levels of education are more likely to use modern methods compared to women with primary levels of education or no education, but they are less likely to have undergone sterilisation and are more likely to have unmet need as well. On the other hand, women living in households with more wealth are more likely to use modern methods and less likely to have unmet need for contraception.

Studies have shown that lack of information and fear are important reasons for the low levels of contraceptive prevalence in India. An in-depth village study inquired into the reasons for not accepting contraceptive methods, and the primary reason given by respondents was fear of side effects (Murarkar et.al., 2011). Opposition from family members was another reason for not using contraception (ibid). The literature identifies son preference as another important factor contributing to low contraceptive prevalence. India has a strong preference for sons (Vlassof et al; Rajaretnam and Deshpande 1994, Jejheebhoy and Kulkarni, 1989) which results in families continuing childbearing till the ideal family composition is reached. A study conducted in the 1990s in South India showed that in the absence of son preference, marital fertility would decrease by about 20 percent (Rajaretnam and Deshpande 1994).

The lack of ability to make decisions among young women regarding their own reproduction as well as the lack awareness of contraceptive options available are important correlates of the low prevalence of contraception in the country. However, this is a group for whom contraception is most relevant in order to delay pregnancy for adolescents and to space pregnancies for both adolescents and young women. Both policy and research should intervene so that young married couples can plan their families in a way that is beneficial to themselves as well as society. Some ways that this can be done is to increase information and counselling to meet the needs of youth.

The issue of high levels of female sterilization in the country requires further research. The wide regional variation within the country indicates that the approach of the local health system has a role to play in the type of contraception that is utilised. A significantly higher percentage of sterilisation is conducted in southern states, in particular in Andhra Pradesh and Telangana. Sterilisation is a permanent method, so it is empowering on one hand because women do not have to worry about accessing contraception for the 30 plus years of their reproductive cycle. However, studies have also found there is regret after sterilization, particularly among young women and among those who have lost a child (Singh et. al., 2012). It is recommended that care be taken at the time of the procedure to ensure that full information is provided to young woman about the nature of the procedure and its consequences. Also, personnel at health facilities should take into account the youth and lack of empowerment among young women before suggesting female sterilisation as a method.

Last but not least, it is important to bring men into the picture. Currently, contraception is considered to be a woman's issue and the burden falls solely on women. If men participated in the decision-making, then women and couples could make informed choices about their own reproductive health. This is particularly important in the case of young women for whom spacing and limiting of childbearing is important, and therefore the type of contraception used matters the most among all the age groups while at the same time, they have the least ability to make decisions in their own interest.

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Section IV
Non-Communicable Diseases and Mental Health

9

Depression and Anxiety Among High School and Higher Secondary School Students During COVID-19 Pandemic in South Kerala: A Web-Based Cross-Sectional Survey

Oommen P Mathew

Abstract

Depression and anxiety are the most common and under-recognized psychological disorders in school-aged children and adolescents worldwide. The outbreak of coronavirus diseases (COVID-19) has been substantially influencing the life and living of people across the world. Government of Kerala shut down all colleges and schools on 10th March, 2020 and schools started online classes during the academic year 2020-21, hence students are restricted to live within four walls, and the situation creates psychological problem among students. This study is aimed to assess the depression and anxiety of high school and higher secondary students of south Kerala during the pandemic period and also to find the factors associated with depression and anxiety. Snowball sampling technique was used for the study. An online-based platform was used to collect data. About 24 percent participants showed mild to severe anxiety among these nearly 8 percent were moderate or severe anxious. Socio-demographic factors associated with depression are more time duration of online class, lack of exercise and quarantine during COVID pandemic period. The factors associated with presence of anxiety were students of higher class, higher duration of online class, students from urban area, lack of exercise and quarantined during the pandemic period. Three or more hours of online class daily (OR 2.37), absence of exercise (OR 2.62) and quarantined during the COVID pandemic period (OR 2.46) were found independent predictors of depression. The above three variable namely, three or more hours of online class daily, absence of exercise and quarantined during the COVID pandemic period were found as independent predictors of anxiety.

Introduction

The period of adolescence represents a transitional stage from childhood to adulthood and represents the critical time frame during which an individual undergoes a variety of developmental changes along with an encounter with a number of emotional and psychosocial issues. Depression is one of the most common global mental health problems (Nagendra et.al, 2012). In adolescents, it is one of the under-recognized health problems due to the inability to disclose their feelings and reluctant to seek psychiatric help. In the last century, the medical community did not accept the existence of depressive disorders in children. It was believed that children are lacking the mature psychological and cognitive structure needed to experience symptoms related to depressive disorders. However, a growing body of evidence proved that children not only experience the whole spectrum of mood disorders but also suffer from significant morbidity and mortality associated with them (Nagendra et.al, 2012). Globally, it has

been reported that depressive disorders often start at an early age, with prevalence rates of mental illness among children and adolescents ranging between 1% and 51% (Sandal et al., 2017). More precisely, among adolescents between the age of 14 and 19 years, the prevalence was reported to be from 15% to 20% (Kessler et al., 1998). Depression is a global concern for children, adolescents, and adults even in developed nations (Thapar et al., 2012). Major depression was the fourth most frequent human illness in 1990 and is projected to rank second by the year 2020 in the adolescent age group (Lopez & Murray, 1998). In India, the combined prevalence of depression and anxiety among school-going adolescents has been reported as 57.65%, with 3% cases of extremely severe depression (Nagendra et al., 2012). Depressive disorders are reported to be considerably more common in females, with a lifetime prevalence of 14.1% for females and 8.6% for males (Copeland et al., 1999, Compas et al., 1997). This has been attributed to genetics, increased prevalence of anxiety disorders in females, biological changes associated with puberty, cognitive predisposition, and sociocultural factors (Breslau et al., 1998). From health-seeking and treatment point of view, about half of depressed adolescents are left undiagnosed in primary care settings (Olson et al., 2001).

Depression also has a deep effect on adolescents' psychosocial domain and academic performance. More often, they are more preponderate toward probability for hospitalizations, recurrent depressions, substance abuse, and antisocial behaviours, and with time, the most devastating outcome is suicide, which is the third leading cause of death among older adolescents (Weissman et al., 1999). It may negatively impact peer or family relationship (Bhatia, 2007). Moderate to severe depression can affect the scholastic performance of adolescents (Nair et al., 2004). Above all, there is also a danger of students picking up habits like usage of substances of abuse to get over depression. Major depressive disorder is also found to be a leading cause of youth suicidal behaviour and suicide (Kann et al., 2000).

Anxiety is one of the most common psychological disorders in school-aged children and adolescents worldwide (Costello et al., 2003). The prevalence rates range from 4.0% to 25.0%, with an average rate of 8.0% (Bernstein & Borchardt, 1991). These figures could be underestimated since anxiety among a large number of children and adolescents goes undiagnosed owing to the internalized nature of its symptoms (Tomb & Hunter, 2004). Anxiety is associated with substantial negative effects on children's social, emotional and academic success (Essau et al., 2000). Specific effects include poor social and coping skills, often leading to avoidance of social interactions (Albano et al., 2003, Weeks et al., 2009), loneliness, low self-esteem, perceptions of social rejection, and difficulty forming friendships (Bokhorst et al., 2001, Weeks et al., 2009). Importantly, school avoidance, decreased problem-solving abilities, and lower academic achievement have also been noted as consequences (Donovan & Spence, 2000, McLoone et al., 2006, Rapee et al., 2005). Anxiety is considered to be a universal phenomenon

existing across cultures, although its contexts and manifestations are influenced by cultural beliefs and practices (Good, B. J. & Kleinman, 1985, Guarnaccia, 1997).

In India, the main documented cause of anxiety among school children and adolescents is parents' high educational expectations and pressure for academic achievement (Deb, 2001). In one year alone in India, 2320 children, or more than six children per day, committed suicide because of failure in examinations (National Crime Records Bureau, 2000). This shocking figure underlines the seriousness of this problem and its resounding social costs to communities

Previous research on the socio-demographic correlates of anxiety reveals well-established relationships with gender and socioeconomic status, but little evidence of consideration of school type. Generally, more girls than boys develop anxiety disorders and symptoms. Adolescent girls report a greater number of worries, more separation anxiety, and higher levels of generalized anxiety (Campbell, & Rapee, 1994, Costello et al., 2003, Poulton et al., 2001, Weiss & Last, 2001). While anxiety is known to affect both learning and performance (McDonald, 2001), no empirical research has explored the relationship between adolescent anxiety and school type, school choice, or mode of instruction

The outbreak of coronavirus diseases (COVID-19) has been substantially influencing the life and living of people across the world, especially after the declaration of a global pandemic by the World Health Organization in the second week of March 2020 (World Health Organization, 2020). Hence, many countries implemented a range of anti-epidemic measures, such as restricting travel for foreign nationals, closing down public spaces, and shutting down the entire transit system (Zhai, 2020) to contain the transmission of the highly contagious infections from human-to-human. This unprecedented experience of 'home quarantine' under lockdown with the uncertainty of academic and professional career has multifaceted impacts on the mental health of students (Ahmed MZ et al, 2020). The ongoing COVID-19 pandemic is creating a psycho-emotional chaotic situation as countries have been reporting a sharp rise of mental health problems, including anxiety, depression, stress, sleep disorder as well as fear, among its citizens (Chen L, 2020, The Daily Star, 2020, Bhuiyan et al., 2020, Banik R et al., 2020, Jahid AM, 2020), Researchers in China observed that the greater exposure to 'misinformation' through social media are more likely contributing to the development of anxiety, depression, and other mental health problems among its population of different Socio-economic background (Dong L et al., 2020, Bai Y et al., 2020, Dhaka Tribune, 2020).

The first COVID -19 case of India reported in Kerala on 30th January 2020. The government of Kerala declared high alert from 4 to 8 February. Kerala declared high alert from 8 March 2020 due to coronavirus cases being reported from the state. On 10 March, the government of Kerala shut down all colleges and schools. On 23rd March, announced a state-wide lock-down till 31 March to prevent further spread of the Corona Virus. The lock down continued till 30th May

2020. Usually, Schools and Colleges reopen in June, but due to the spread of Corona virus, schools and colleges continued closed even after June. Schools and higher secondary schools started online classes later. So, children are restricted to live round the clock within the four walls of home. No social or peer group gathering allowed since the shutdown of schools on March 10th. The living conditions created with spread of Corona virus is a factor for the development of depression and anxiety among students. Hence, this study is aimed to assess the depression and anxiety of high school and higher secondary students of south Kerala and also to find the factors associated with depression and anxiety

Objectives

- To Study the depression and anxiety among high school and higher secondary students in South Kerala
- To find the factors associated with depression and anxiety among high school and higher secondary students

Materials and Methods

The survey was conducted during the month of November 2020, from 5th November to 30th November 2020. Students of 8th class to 12th class of different schools of South Kerala were the target population. An easy-to-understand questionnaire was framed to collect basic information along with ‘depression,’ and ‘anxiety’ related information. An online-based platform was used to distribute the e-questionnaire, developed by using the Google Form. For getting clarity of the questions, questions/ statements and options for responses were written in both English and Malayalam. Students studying in State / CBSE and ICSE syllabus from different schools from southern part of Kerala were contacted through different social networks and send the link of e-questionnaire.

Ethical approval for carrying out the study “Depression and anxiety among High School and Higher secondary school students during COVID-19 pandemic in South Kerala: A web-based cross-sectional survey” was obtained from the ethics committee of the department. The informed consent was also taken from all participants to use their data in the study before participation through Google form

Sampling technique The Snowball sampling technique was used for collecting information from students. The participants were asked to share the e-questionnaire with their friends using their personal and institutional Facebook and Messenger. Since the schools were closed, online data collection used, hence there is limitation to get proportional samples from each of the stream of education.

Measures Basic information ‘Basic Information’ contained the personal information of the respondents. The variable included for the section are :- Current age of students, gender (Male, Female) of students, stream of study (State, CBSE, ICSE), duration of online class (No online

class, 1-2 Hours, 3-4 Hours, 5-6 Hours, 7-8 Hours, >8 Hours) during the pandemic period, online tuition class apart from the school online class (No tuition, 1-2 Hours, 3-4 Hours, >4 Hours), place of residence (Panchayath area, Municipality area, corporation area), whether students reside with parents during pandemic period (Yes, No), exercise during the pandemic period (No exercise, <15 minute, 15 – 30 minute, 15-30 minutes, 30 – 60 minutes, >60 minutes), have been Quarantined / isolated in connection with COVID-19 pandemic and Job Loss to father / mother during the COVID pandemic period (Yes, No).

Depression was determined by using the Patient Health Questionnaire (PHQ-9). PHQ-9 is an easy way to use in a questionnaire for screening depression of the responses that are used to predict depression of an individual and what state he/she is in during the survey. Each item of PHQ-9 was scored on a scale of 0–3 (0=not at all; 1=several days; 2=more than a week; 3=nearly every day). Principal components factor analysis showed good construct validity of the PHQ-9. PHQ-9 total score showed a positive correlation with Hamilton Depression Rating Scale score (HAMD-17) ($r=0.610$, $P<0.001$). With HAMD as the standard, PHQ-9 depression scores of 7, 15, and 21 points were used as cut-offs for mild, moderate, and severe depression, respectively. The test-retest correlation coefficient for total score was 0.737 (Kroenke K et al., 2001). The reason for choosing PHQ-9 was that it proved to be a useful tool for detecting depression (Martin A, 2005). The PHQ-9 total score ranges from 0 to 27 (scores of 5–9 is classified as mild depression; 10–14 as moderate depression; 15–19 as moderately severe depression; ≥ 20 as severe depression)

Anxiety was evaluated by using the Generalized Anxiety Disorder (GAD-7). In the questionnaire, the questions were used for screening anxiety state of an individual on a scale ranging from '0 = not at all sure' to '3 = nearly every day'. A 7-item anxiety scale (GAD-7) had good reliability, as well as criterion, construct, factorial, and procedural validity. A cut point was identified that optimized sensitivity (89%) and specificity (82%). The internal consistency of the GAD-7 was excellent (Cronbach=0.92). Test-retest reliability was also good (intraclass correlation=0.83). Comparison of scores derived from the self-report scales with those derived from the MHP-administered versions of the same scales yielded similar results (intraclass correlation=0.83), indicating good procedural validity (Spitzer RL et al., 2006). GAD-7 has been found successful in identifying anxiety among different populations and thus used for its reliability (Hossain S et al., 2019). The levels of anxiety for the study were categorized as 'none-minimal' = <5, 'mild = 5–9, 'moderate = 10–14 and 'severe = ≥ 15 .

Statistical Analysis Frequency tabulation was used to summarize basic information of respondents, as well as their response to depression and anxiety. Chi-square test was used to find association of depression and anxiety with selected background characteristics of students. The outcome variable Depression and anxiety was categorized into two categories, i.e., depressed = 'yes' and 'no' and anxious = 'yes' and 'no'. Multiple logistic regression analysis

was carried out to elicit independent risk factors of depression and anxiety. For the determination of independent predictors of depression / Anxiety, presence of depression / anxiety was taken as dependent variable and variable have $p < 0.20$ in bivariate analysis was taken as independent variables. Backward logistic regression was carried out to derive independent factors. For all statistical interpretations, $p < 0.05$ was considered the threshold for statistical significance. Statistical analyses were performed by using a statistical software package SPSS, version 20.0:

Limitations for the study Snowball sampling technique was used for collecting information from students. Since the sampling procedure was not a probabilistic sample method, the result obtained from the study may vary with that of cross-section of the population.

Result and Discussion

Background characteristics of students Background characteristics of students participated in the study was presented in Table1. Nearly 58 percent of students were females, 52 percent students from 8th or 9th class, 15 percent from 10th class and 32 percent students from higher secondary class. Nearly two third of the sample students are studying in Kerala state syllabus and 25 % students were from CBSE syllabus and 11% from ICSE syllabus. Because of the COVID pandemic the schools closed in March 2020, and not reopened till the date of the survey period. Almost all schools started online classes irrespective of the syllabus. The duration of online classes varies with class, syllabus and type of management of School. Among the sample students, 3 percent reported no online classes to them, 1- 4hours of daily online classes to nearly 40% students, 35 % attending 5-6 hours of online class and 14 percent students have more than 6 hours of online class. In Kerala, a large proportion of students depend special tuition classes apart from the regular class at school. Because of the COVID pandemic the usual special tuition classes were not possible, but online tuition classes started. Among the participants nearly 27 percent students utilizing online tuition classes apart from the general online class of the school. About 22 percent students responded that they are attending 1-2 hours online class daily during the COVID pandemic period. Among the participants, 45.7% percent of students were from rural area. Parental support and presence of parents is a protective factor against depression and anxiety especially during a pandemic period and a situation where high level of social life is restricted. Ninety five percent of students reported that they were living with their parents during this COVID pandemic period. Exercise is a known protective factor against the anxiety and depression, a question was asked to assess the level of exercise of students during the pandemic period. Usually, children spent most of the time in school and study related matters. Because of the pandemic, schools are closed and children are restricted to their home only and have sufficient time for extracurricular activities.

The level of engagement for exercise was categorized as nominal (<15 min per day), Mild level (15-30 minutes per / day) and adequate (>30 minutes per day at least for 5 days per week) patters, About, 30 percent students were not engaged in any exercise activities during the period, 29 percent of participants engaged in nominal level of exercise, 23 percent engaged in mild level of exercise and only 17 percent students engaged in adequate exercise during this pandemic period. Quarantine / isolation is one of the factors responsible for anxiety and depression. During the start of the pandemic 7-14 days of institution quarantine was there for primary contact cases of Corona virus, later the institution quarantine was liberalized and home / room quarantine implemented. Since the quarantine, whether institutional or home/room is a situation of loneliness and a factor for depression / anxiety. Hence, students were asked whether they were isolated or quarantined during the pandemic period.

Table 1 Distribution of the sample according to background characteristics

Background characteristics	Variables	Count	Percent
Gender	Male	146	42.8
	Female	195	57.2
Class of Study	8-9	179	52.5
	10	53	15.5
	11-12	109	32.0
Stream of study	State	219	64.2
	CBSE	86	25.2
	ICSE/ISC	36	10.6
Duration of daily Online Classes during COVID-19	No online class	11	3.2
	1-2 Hrs	97	28.4
	3-4 Hrs	66	19.4
	5-6 Hrs	118	34.6
	7-8 Hrs	39	11.4
	>8 Hrs	10	2.9
Have online tuition class beyond the regular online institutional class	No	250	73.3
	1-2 Hours/ day	76	22.3
	3-4 Hours/ day	11	3.2
	>4 Hours/ day	4	1.2
Place of residence	Panchayath area	156	45.7
	Municipality	95	27.9
	Corporation	90	26.4
Residing with parents	No	17	5.0
	Yes	324	95.0
exercise	No	103	30.2
	<15 min	100	29.3
	15 - 30 minute/day	81	23.8
	30 min - 1 Hr/day	29	8.5
	>1 Hr/day	28	8.2
Have you been Quarantined / isolated during COVID-19 pandemic	No	294	86.2
	Yes	47	13.8
Job Loss to father / mother during the COVID pandemic period	No	267	78.3
	Yes	74	21.7

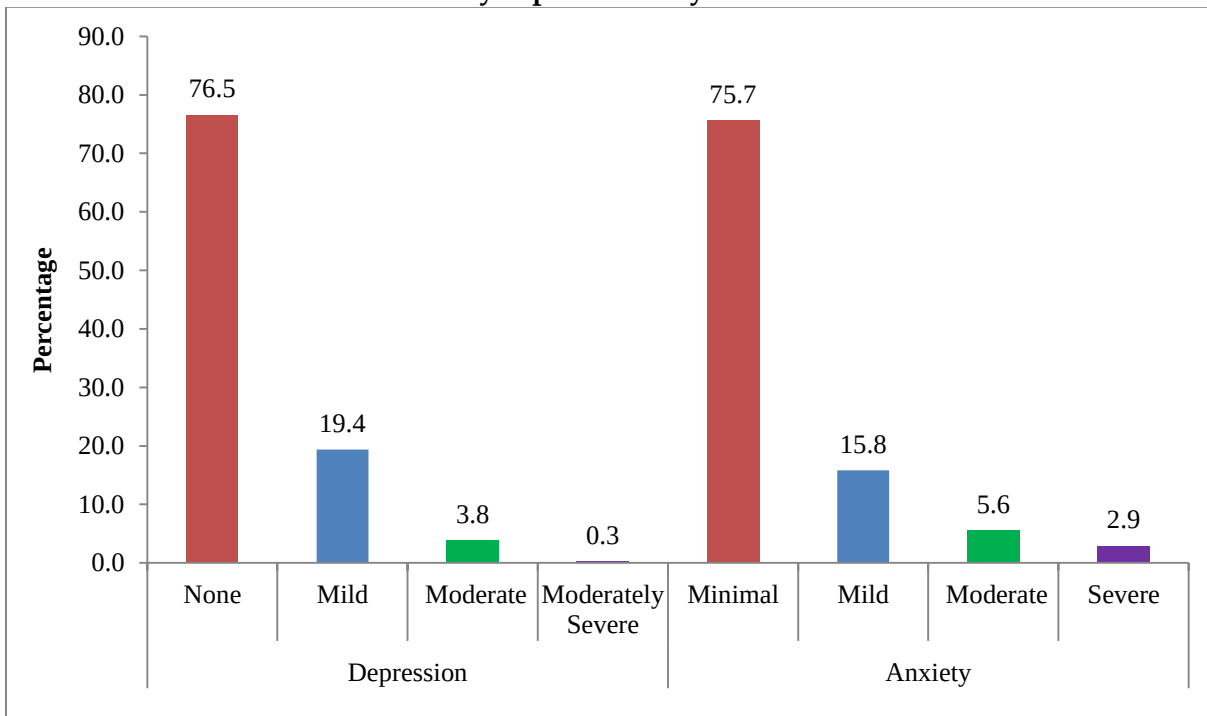
Note: Sample size N=341

Nearly 14 percent of the students were quarantined / isolated during the pandemic period. During the Covid pandemic period, economic crisis was a phenomenon everywhere in the world. A large number of people in private sector and unorganized sector lost their job. The Job loss create financial stability of household and in turn affect the mental health of family members including children. So, one question was included in the questionnaire to assess whether father / mother lost job during pandemic period. From the table, job loss to parents during COVID pandemic period is nearly 22%

Level of Depression and Anxiety Among Students Level of depression and anxiety experienced by students during the COVID pandemic period is presented in Figure 1. Depression was assessed using PHQ-9 scale, 76.5 percent students are absolutely free from depression. Among the participants 19.4 percent shows mild depression, 3.8 percent students showed moderate depression and one student showed moderately severe depression.

Globally, it has been reported that depressive disorders often start at an early age, with prevalence rates of mental illness among children and adolescents ranging between 1% and 51%². Among adolescents between the age of 14 and 19 years, the prevalence was reported to be from 15% to 20%³. In India, the combined prevalence of depression and anxiety among school-going adolescents has been reported as 57.65%, with 3% cases of extremely severe depression¹. The finding of present study reveals nearly 24% of students experiencing mild to severe depressive symptoms.

Figure 1 Distribution according to level of depression and anxiety experienced by students



Anxiety was measured using Generalized Anxiety Disorder (GAD-7) tool. The result showed that 75 percent students are with minimal level anxiety, 15.8 percent students experienced mild level anxiety, 5.6 percent students experienced moderate level anxiety and nearly 3 percent of students live with severe anxiety. Previous, studies showed the prevalence rates range from 4.0% to 25.0%, with an average rate of 8.0%^{15,16}.

Factors Associated with Presence of Depression and Anxiety among Students Association of presence of depression with selected background characteristics of students were presented in Table2. Mild, Moderate and moderate to severe depression were grouped as presence of depression for the analysis. Chi square test was carried to find associated factors of depression. From, the Table, it can be observed that the presence of depression among males is 22.6 whereas its percentage among females is 24.1, Chi square test shows that there presence of depression among students is independent of their gender. Similarly, class of study of students, stream of syllabus, whether attending online tuition class apart from general online class, place of residence, whether residing with parents during COVID pandemic period, job loss of parent during the COVID are found independent of depression of students.

Table 2 Association of depression with selected background characteristics of students

Background Characteristics	Variables	Depression				χ^2	p
		No		Yes			
		Count	Percent	Count	Percent		
Gender	Male	113	77.4	33	22.6	0.1	0.746
	Female	148	75.9	47	24.1		
Class of Study	8-9	145	81.0	34	19.0	4.19	0.123
	10	38	71.7	15	28.3		
	11-12	78	71.6	31	28.4		
Stream of study	State	170	77.6	49	22.4	0.69	0.708
	CBSE	63	73.3	23	26.7		
	ICSE/ISC	28	77.8	8	22.2		
Duration of daily Online Classes during COVI	No class / 1-2 Hours / day	94	87.0	14	13.0	13.47* *	0.001
	3 - 6 Hours / day	137	74.5	47	25.5		
	>6 Hours / day	30	61.2	19	38.8		
Online Tuition Class	No	196	78.4	54	21.6	1.81	0.179
	Yes	65	71.4	26	28.6		
Place of residence	Rural	124	79.5	32	20.5	1.39	0.238
	Urban	137	74.1	48	25.9		
Residing with parents	No	12	70.6	5	29.4	0.35	0.552
	Yes	249	76.9	75	23.1		
Doing any sort of exercise	No Exercise	66	64.1	37	35.9	13.7**	0.001
	Mild Exercise	151	83.4	30	16.6		
	Adequate Exercise	44	77.2	13	22.8		
Quarantined / isolated	No	233	79.3	61	20.7	8.74**	0.003
	Yes	28	59.6	19	40.4		
Job Loss to father / mother	No	205	76.8	62	23.2	0.04	0.843
	Yes	56	75.7	18	24.3		

*: Significant at 0.05 level **: Significant at 0.01 level

Duration of online class is found significantly associated with presence of depression. Thirty nine percent of students have more than 6 hours of online class expressed depression symptom, whereas only 13 percent of students have nominal online class. Exercise is found as a protective factor for depression, nearly 36% student without any exercise showed depressive symptoms, whereas only 17 percent have depression among students with mild exercise and 23% have depression among those have adequate exercise. Quarantine / isolation during COVID period is another risk factor for development of depression among students. Among the students went for quarantine during the COVID period, 40 percent were developed depression when compare with only 21 percent experience depressive symptoms among non-quarantined students

Association of Anxiety of students with selected background variables presented in Table 3. For the purpose of find the risk factors of Anxiety, minimal level of anxiety was categorised as “No anxiety” and other categories such as Mild, Moderate and severe anxiety are grouped as presence of Anxiety.

Table 3 Association of anxiety with selected background characteristics of students

Background Characteristics	Variables	Anxiety				χ^2	p
		No		Yes			
		Count	Percent	Count	Percent		
Gender	Male	109	74.7	37	25.3	0.14	0.709
	Female	149	76.4	46	23.6		
Class of Study	8-9	146	81.6	33	18.4	7.41*	0.025
	10	38	71.7	15	28.3		
	11-12	74	67.9	35	32.1		
Stream of study	State	168	76.7	51	23.3	0.81	0.666
	CBSE	62	72.1	24	27.9		
	ICSE/ISC	28	77.8	8	22.2		
Duration of daily Online Classes during COVI	No class / 1-2 Hours / day	97	89.8	11	10.2	19.29**	p<0.01
	3 - 6 Hours / day	131	71.2	53	28.8		
	>6 Hours / day	30	61.2	19	38.8		
online tuition class	No	192	76.8	58	23.2	0.66	0.416
	Yes	66	72.5	25	27.5		
Place of residence	Rural	126	80.8	30	19.2	4.08*	0.043
	Urban	132	71.4	53	28.6		
Residing with parents	No	10	58.8	7	41.2	2.75	0.097
	Yes	248	76.5	76	23.5		
Doing any sort of exercise	No Exercise	65	63.1	38	36.9	16.04**	p<0.01
	Mild Exercise	152	84.0	29	16.0		
	Adequate Exercise	41	71.9	16	28.1		
Quarantined / isolated	No	233	79.3	61	20.7	14.94**	p<0.01
	Yes	25	53.2	22	46.8		
Job Loss to father / mother	No	203	76.0	64	24.0	0.09	0.762
	Yes	55	74.3	19	25.7		

*: Significant at 0.05 level **: Significant at 0.01 level

From the table it can be found that there are five variables significantly associated with anxiety of students. The variables significantly associated with anxiety are class of study, duration of daily online classes, place of residence, exercise and quarantined during COVID pandemic period.

Anxiety is found less among students of lower classes as compared to higher classes. The percentage of students expressed anxiety among classes 8-9 are 18% only, whereas 28% of students from 10th class and 32% of students from higher secondary level expressed anxiety. Percentage of anxiety cases is found higher among students attending more time for online classes. Among students have online classes more than 6 hours daily, 39% were expressed anxiety, whereas only 10 % were anxious among those have less than 2 hours of online classes. Place of residence of student is found another risk for anxiety. Among the urban students, 28% expressed anxiety whereas only 19% of students from rural area were anxious during the COVID period. Among those students who were not engaged in any physical activities showed anxiety for 36.9 percent, among those engaged in mild exercise only 16 percent expressed anxiety. Quarantine during COVID is another risk for anxiety. Among those quarantined during pandemic period, 47% percent were expressed anxiety.

Independent Predictors of Depression and Anxiety Independent predictors of depression and anxiety were assessed using multiple logistic regression. For the determination of independent predictors of depression / Anxiety, presence of depression / anxiety was taken as dependent variable and variable has $p < 0.20$ in bivariate analysis was taken as independent variables. Backward logistic regression was carried out to elicit independent factors. Independent variables selected for the study are duration of online classes during pandemic period, online tuition, exercise during the period, quarantined or not during the pandemic period. The result of the table is presented in Table 4. The variables found independent predictors of depression are duration of online class, exercise during the period and quarantine during the pandemic. Students spend more than 3 hours for online class are more prone to develop depression with an odd of 2.37, those students without any physical activity is 2.62 odds time more chance to develop depression and those who were quarantined during the pandemic period are 2.46 odds time more chance to develop depression.

Table 4 Independent predictors of depression and anxiety (multiple logistic regression)

Depression	Variables	B	Odds (95 % CI)	p
Duration of daily Online Classes (1-2 Hours / day ®)	>=3 Hours / day	0.863	2.37 (1.24 - 4.53)	0.009
Exercise (Yes ®)	No	0.964	2.62 (1.53 - 4.49)	p<0.01
Quarantined / isolated (No ®)	Yes	0.899	2.46 (1.25 - 4.83)	0.009
Anxiety				
Duration of daily Online Classes (1-2 Hours / day ®)	>=3 Hours / day	1.257	3.52 (1.74 - 7.09)	p<0.01
Exercise (Yes ®)	No	0.982	2.67 (1.55 - 4.61)	p<0.01
Quarantined / isolated (No ®)	Yes	1.145	3.14 (1.6 - 6.16)	0.001

For the determination of independent predictors of Anxiety, presence of anxiety was taken as dependent variable and class of study, duration of online classes during pandemic period, place of residence, whether residing with parents during pandemic period, exercise during the period, quarantined or not during the pandemic period were taken as independent variables. The variables found independent predictors of anxiety are duration of online class, exercise during the period and quarantine during the pandemic. Students spend more than 3 hours for online class are more prone to develop anxiety with an odd of 3.52, those students without any physical activity is 2.67 odds time more chance to develop anxiety and those who were quarantined during the pandemic period are 3.14 odds time more chance to develop anxiety.

Findings

Depression is one of the under-recognized health problems in adolescents, it has a deep effect on adolescents' psychosocial domain and academic performance. Anxiety is one of the most common psychological disorders in school-aged children and adolescents worldwide⁵¹. The outbreak of coronavirus diseases (COVID-19) has been substantially influencing the life and living of people across the world and is creating a psycho-emotional chaotic situation. On 10 March, the government of Kerala shut down all colleges and schools and continued closed even at year ending period. Schools and higher secondary schools started online classes and children are restricted to live round the clock within the four walls of home. This study is aimed to assess the depression and anxiety of high school and higher secondary students of south Kerala during the pandemic period and also to find the factors associated with depression and anxiety. Snowball sampling technique was used for the study. Depression was determined by using the Patient Health Questionnaire (PHQ-9) and anxiety was evaluated by using the Generalized Anxiety Disorder (GAD-7). Both the tools were standardised tools to assess depression and anxiety.

Background characteristics of students shows, 58 percent of students were females, 52 percent students from 8th or 9th class, nearly two third of the sample students are studying in Kerala state syllabus, 50% students have online class for <4 hours daily, no online classes for 73 % students, 45% students were from rural area, 95 % students residing with parents, no physical exercise for 30 % participants, 14% participants quarantined during COVID pandemic period and father / mother of 22 % students lost their job during the pandemic.

Nearly 24% participants showed mild to severe depression and about 4% cases were moderate or moderate to severe depressive disorder. About 24 % participants showed mild to severe anxiety among these nearly 8% were moderate or severe anxious. Socio demographic factors associated with depression are more time duration of online class, lack of exercise and quarantine during covid pandemic period. The factors associated with presence of anxiety were students of higher class, higher duration of online class, students from urban area, lack of

exercise and quarantined during the pandemic period. Multiple logistic regression was carried out to find independent predictors of depression and anxiety. Three or more hours of online class daily (OR 2.37), absence of exercise (OR 2.62) and quarantined during the covid pandemic period (OR 2.46) were found independent predictors of depression. The above three variables namely, three or more hours of online class daily, absence of exercise and quarantined during the covid pandemic period were found as independent predictors of anxiety.

Conclusion

Nearly one in four students showed mild to severe depression as well as mild to severe anxiety in south Kerala during the COVID pandemic period. About 4 percent were moderate / severe depressed and moderate to severe anxiety was for 8 percent students. Independent risks identified for depression and anxiety were three or more hours of daily online classes, absence of exercise and quarantined during the COVID pandemic period.

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Section V
Health Infrastructure, Care Services and Use

10

Assessment of *Kayakalp* Programme in Rajasthan

Bindiya Kumari, Vandana Sharma and Suresh Sharma

Abstract

To address the challenge of the continuing perception of low-quality services in the public healthcare premises, the policy makers in India introduced a set of quality-oriented programmes under the National Health Mission. Quality improvement initiatives i.e., Indian Public Health Standard (IPHS), NQAS, KAYAKALP, MERA ASPATAAL and LaQshya programme, have revolutionized the pathways of public healthcare service delivery in the country. Among these initiatives, Kayakalp programme has brought a remarkable transformation vis-a-vis sanitary standards, infection control and hygiene practices, and facility upkeep by incentivizing & recognizing the facilities where service delivery is complemented with quality care. Healthcare facilities all over India are enthusiastically participating under the said scheme every year, which shows the tremendous interest this programme has garnered. In this context, this study assessed the phase-wise assessment trends and growth of Kayakalp awarded facilities in four districts of Rajasthan – Ajmer, Churu, Hanumangarh, and Jhunjhunu. The study argues that sustained implementation of the scheme is crucial to materialize the objectives of the programme. The study found significant improvements in the Kayakalp scores across districts and facilities, in the backdrop of baseline performance. The major bottlenecks as observed, and as asserted by the healthcare service providers pertain to delayed fund disbursement of the Kayakalp award money, non-compliant distribution of the staff specific Kayakalp fund, paucity of routine trainings, and crunch in manpower availability. While it is most certain that the impact of the programme in improving the nooks and corners of the healthcare facilities are commendable, nevertheless, the existing bottlenecks seriously hinder the extreme potential of this programme.

Introduction

Kayakalp programme, launched in the year 2015 was initiated to promote proper sanitary standard in the public health premises of India. To complement the swachhta guidelines of the public health facilities, the Ministry of Health & Family Welfare inaugurated the *Kayakalp* program to commend those health facilities that had demonstrated excellent sanitation, cleanliness and hygiene promotion. In the year 2015-16, *Kayakalp* programme was implemented in the District Hospitals across all states. In the year that followed, it was implemented at both the CHC as well as PHC Level and at present this programme has been taken up to the SC level (MoHFW, GoI, 2019). Cleanliness and Hygiene promotion practices have been regarded among

the most neglected areas with respect to public health care facilities. Therefore, *Kayakalp* programme is seen as a revolutionary strategy towards up-gradation of public healthcare facilities. Its main objective is to inculcate a culture for promotion of cleanliness, infection control and hygiene practices in the facilities by incentivizing & recognizing the facilities showing exemplary performance while adhering to the *Kayakalp* guidelines & thereafter sustaining such practices as well (MoHFW, GoI, 2015). The assessment process consists of three steps:

- Internal Assessment (by facility staff & scores to be reported to the district: Facility)
- Peer Assessment (by team of other district hospital under the divisional purview: Other District)
- External Assessment (by team of trained external assessor: State)

The assessment involves a stepwise procedure. Once a facility has been nominated in the internal and peer assessment it goes for review for external assessment that is usually constituted at the state level. After reviewing the steps, the facilities are ranked based on scores obtained and the facilities acquiring maximum scores are declared the winners and awarded accordingly.

Criteria for the Application to the Awards Scheme

Following are the prerequisites in order to apply as a recipient of *Kayakalp* award -

- Constituted a Cleanliness and Infection Control Committee.
- Instituted a mechanism of periodic internal/peer assessment based on defined criteria.
- Achieved at least 70 % in the criteria during the peer assessment process.

As per *Kayakalp* revised guidelines 2019, the winner and runner-up facilities in the previous year would have to show an improvement in the score by at least 5 percentages from previous year scores. If the winner and runner-up facility do not meet the said criterion of 5 % increment, then it would only receive the commendation award subject to facilities scoring >70%.

Selection of Facilities

The awards for individual public health facility are given to those that score the highest based on a set of standard defined criteria (MoHFW, GoI, 2015). The table 1 clearly depicts that in Rajasthan (33 Districts), the first- and second-best district hospitals shall receive a cash award of Rs. Fifty and Rs. Twenty lakhs, respectively. For small states only the first ranking facility in this category will be awarded. At CHC level, first and second ranked CHCs/ SDHs will receive cash awards of Rs. Fifteen and Ten lakhs, respectively. In relation to PHC, the best PHC will receive a cash award of Rs. two lakhs. Moreover, in order to motivate, sustain and improve performance in facilities that scored over 70%, but could not make it to the list of top two/one in a particular year, a Certificate of Commendation plus cash award is given as follows:

- District Hospital: Rs. 300,000
- CHC / SDH: Rs. 100,000 & PHC: Rs. 50,000

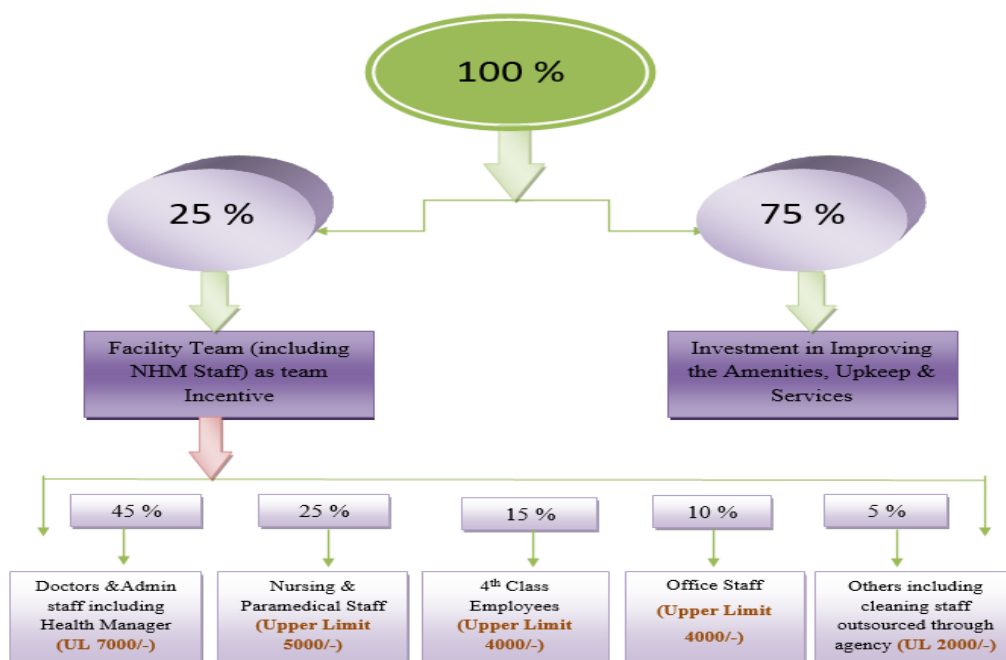
Table 1 Cash awards for States/UTs (DH, CHC and PHC)

Category	Number of Districts	Assessment Score	Amount (In Lakh)
DH level			
1. A	10-25	Highest (Winner)	50
2. B	26-50	Highest (Winner)	50
		Runner up	20
3. C	>50	Highest (Winner)	50
		Runner up-1st	20
		Runner up-2nd	10
SDH/CHC Level			
Small State	<10	Highest (Winner)	15
Large State	≥10	Highest (Winner)	15
		Runner-up	10
PHC Level			
For all States		One in each District	2
HWCs operational in SCs			
1.A	10-25SCs	Highest (Winner)	1
2. B	26-50 SCs	Highest (Winner)	1
		Runner up	0.50
3. C	>50 SCs	Highest (Winner)	1
		Runner up-1st	0.50
		Runner up-2nd	0.35

Award Distribution

As per the *Kayakalp* programme guidelines, 25% of the cash award will be given to the facility teams as a team incentive, while 75% of the cash award amount is to go to the Rogi Kalyan Samiti for investments in improving the amenities, upkeep and services (MoHFW, GoI, 2019). The award distribution at the facility level is depicted below:

Figure 1 Award distribution flow chart



Study Area

Rajasthan, one of the largest states of India, has shown considerable improvement in health care outcomes since the launch of the National Health Mission. Firstly, it is one of the states which falls under the category of high priority state as earlier it was known as one of the BIMARU states. Secondly, Banerjee, Duflo and Deaton (2004) examined the health care delivery in Rajasthan, India and observed that quality of public services is extremely low. Hence it becomes utmost important to administer the quality parameters of the program in the light of the cleanliness drive of the mission program. Total four districts that have been chosen for the study. The criteria for the district selection were the size as well as utilization of the respective programme fund where all the four districts have a remarkably higher fund utilization (more than 85%) in the year 2019-20; and the *Kayakalp* score where 2 districts i.e., Hanumangarh and Jhunjhunu had relatively higher score and two districts i.e., Ajmer and Churu had lower scores. Before visiting the different healthcare facilities, a state level meeting with *Kayakalp* nodal officers was held at Jaipur. The main motive of the interaction with same officials was to know the key challenges faced by the state in implementing the *Kayakalp* Programme and also take their opinion for the improvement of the programme. Interaction with the state level official provides an enriching insight about the government initiative in the entire state which can be help to generalize the findings for the entire Rajasthan. Since the programme was implemented across the districts of Rajasthan at the same point time and the state quality assurance cell also confirmed that the gaps in *Kayakalp* implementation largely remain consistent across Rajasthan, therefore the results of this study may hold throw for *Kayakalp* implementation in other districts as well. The Table (2) illustrates the details of all the visited healthcare facilities.

Table 2 List of visited healthcare facilities

Name of the Facilities		
District Hospital	CHC	PHC
DH, B D K Jhunjhunu	CHC, Bagar	PHC, Islampur
DH, Hanumangarh	CHC, Dholipal	PHC, Khotawali
A K Hospital Beawar Ajmer	CHC, Rupangarh	PHC, Saradhna
DH, D B Churu	CHC, Sahawa	PHC, Daudsar

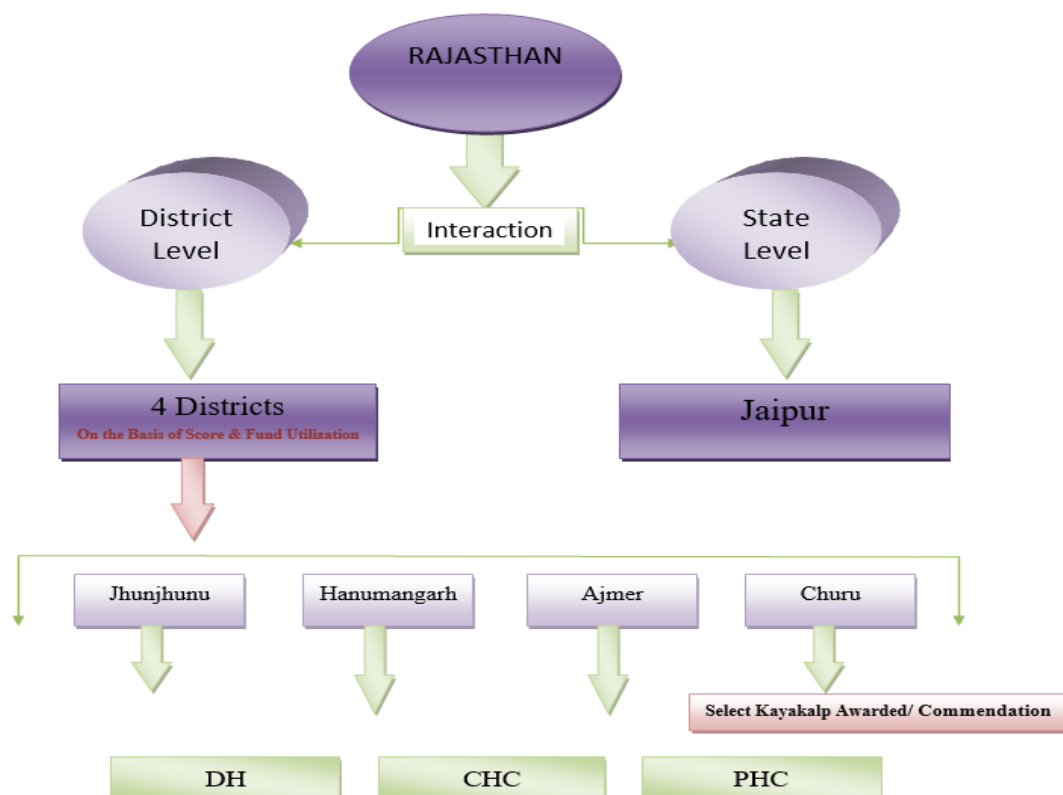
Objectives

- To examine the phase-wise growth in assessment process of public healthcare facilities in Rajasthan.
- To evaluate the impact of *Kayakalp* on cleanliness, sanitation, hygiene and infection control practices in public health care facilities.
- To examine the status of quality care training of the healthcare staff since the implementation of the programme.
- To analyze the allocation/ utilization of *Kayakalp* fund in public healthcare facilities.

Statistical Methods

The study is based on mainly primary data collected from 3 tiers of health facilities, i.e., DH, CHC and PHC in four districts of Rajasthan. The purposive random sampling method employed towards the said study. One healthcare facility at each health tier who was *Kayakalp* awardees/ commendation in 2019-20 has been included in the study. Pre-tested structured checklists were used for assessing the implementation adherence in healthcare facilities. Personal interviews by were conducted with facility in-charge, quality assurance nodal officers as well as the other associated staff. Wilcoxon Signed Rank Test, CAGR and AGR statistics measure has been used to calculate the test statistics. The Wilcoxon signed-rank test is a non-parametric statistical hypothesis used to determine whether the two dependent samples that we have selected from population are having the same distribution or mean ranks differ. The secondary data, collected through various reports and website i.e., NHSRC, State *Kayakalp* Portal, States Health System Resource Centre, States PIP Report, Financial Management Report, and *Kayakalp* Coffeetable Book.

Figure 2 Sample design of the study



Rationale

Quantifying quality in public health facilities has never been an easy task in India, despite quality framework/ standards being available (Agrawal, A., Srivastava, J. N., & Priyadarshi, M.

2019; Chaudhary, A. et. al 2019). In India reportedly 16 lakhs die every year owing to poor quality of healthcare facilities. However, India can save 3 out of 5 lives by providing good & quality health care (The Lancet Global Health Commission, 2018). Cleanliness & hygiene practices in any healthcare facility are regarded as important factors to determine the quality of services delivery. These promotion practices have been regarded among the most neglected areas with respect to public health care facilities. Banerjee, Duflo and Deaton (2004) examined the health care delivery in Rajasthan, India and observed that quality of public services is extremely low. Rajasthan, one of the largest states of India, is one of the states which falls under the category of high priority state as earlier it was known as one of the BIMARU states. Hence it becomes utmost important to administer the quality parameters of the program in the light of the cleanliness drive of the mission program in the said state. In this context, this study will try to assess the impact of the *Kayakalp* programme in maintaining & promoting cleanliness, hygiene and infection control practices in the public healthcare facilities. Furthermore, after the gets rolled out this programme, healthcare facilities are enthusiastically participating under the said scheme every year, which shows the tremendous interest towards the programme. The study will be beneficial in revealing the achievement of this program, showing a tremendous increase in sanitation standards facilities. Thus, in turn, it will depict the picture of high level of quality services in public healthcare premises against the general fact, particularly in high priority state. The present study will also help to understand the performance pattern of different districts in terms of funds utilization processes; therefore, helps the government in making future policy direction.

Contemporary Status of *Kayakalp* Programme in Rajasthan State

After the implementation of the programme, the number of *Kayakalp* awarded facilities has tremendously increased as clearly depicted in figure 3. In the year 2019-20, Around 400 facilities were *Kayakalp* awarded (score more than 70%), while in the year 2016-17 it was only 47.

Table 3 Status of *Kayakalp* programme in Rajasthan State, 2019-20

Health Facilities	Total Public Healthcare facilities	Awarded Facilities	Percent Awarded
DH	36	16	44%
SDH/CHC	748	124	17%
PHC	2886	257	9%
SC	14583	NA	-
Total	18253	397	2.2%

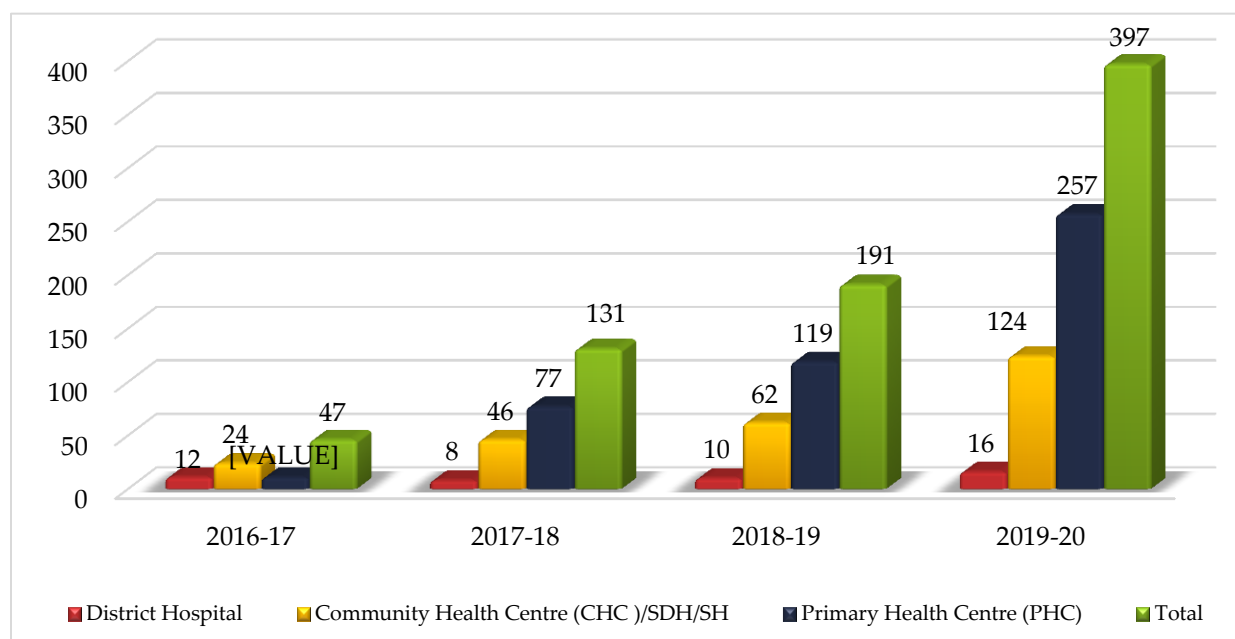
Source: HMIS, 2019-20 & *Kayakalp* Data obtain from State level Meeting as on January, 2021

Note: Earlier SCs were not covered under *Kayakalp* programme.

Overall, in Rajasthan, there are a total 18253 public healthcare facilities in the year 2019-20. As seen in table 3 out of the total district hospital, 44 percent DH has been *Kayakalp* certified. While in case of CHC & PHCs only 17 % and 9 % has been awarded respectively. Despite almost five years of implementation, low percolation of the program on the downstream can be observed

due to the fact that most of the district hospitals have dedicated quality assurance staff i.e., hospital quality manager to maintain and ensure the quality care standards. While at CHC/PHC level, there is no particular post assigned for the same. When need arises, the District Quality Consultant visits at CHC/PHC level. Furthermore, there is no District Quality Assurance Consultant (DQAC) assigned at district/block level in the Rajasthan state. In the absence of the DQAC, *Kayakalp* programme is often looked over by the CMHO/ DPMU Office. This has possibly created a performance gap that could be averted by ensuring the quality managing staffs at the said levels.

Figure 3 Trends in *Kayakalp* awarded facilities: since the program implementation



District-wise Status of *Kayakalp* programme in Rajasthan, 2019-20

For the purpose of improving quality care services in the state, all the public healthcare facilities in the state are classified into four categories based on achieved *Kayakalp* percentage score in external assessment for the year 2019-20. Table 4 further throws light on the categorization of external assessment scores from higher to lower ranges. The definitions of the four categories are as follows:

Category A: >90% score of the facility in state in the year of 2019-20.

Category B: *Kayakalp* score range 80.1-90 % of the facility in state in the year of 2019-20.

Category C: *Kayakalp* score range 70.1-80 % of the facility in state in the year of 2019-20.

Category D: ≤ 70% score of the facility in the state in the year of 2019-20.

In the First category A i.e., > 90% score, the total number of *Kayakalp* facilities are 55, out of which there are 4 DH, 18 CHCs, 21 PHCs, 11 UPHCs and 1 Clinic Dispensary. In the same way

Second category B i.e., score range 80.1-90 %, the total number of *Kayakalp* awarded facilities are 127, out of which there are 3 DH, 28 CHC, 65 PHC and 22 UPHC.

In the third Category C i.e., 70.1 to 80 % score, the total number of *Kayakalp* awarded facilities are 305, out of which there are 9 DH, 67 CHC and 164 PHC. Only five SDH have scored more than 70% score. Finally, in the Fourth category D ($\leq 70\%$ score), total 478 facilities fall in this category, of which around 170 are from CHC and 4 are from DH.

Table 4 District-wise status of *Kayakalp* programme in Rajasthan, 2019-20

	Total Public Facilities (1)	SC (2)	Eligible for <i>Kayakalp</i> (1-2)	Participated in external	No. of Facilities (as per External Assessment)					District Ranking	
					Cat A	Cat B	Cat C	Cat D	Score >70% (A +B+C)	Maximum no. of facility scoring>70 %	Maximum no. of facility scoring $\leq$ 70 %
Sikar	857	692	165	97	4	14	34	45	52	1	1
Jaipur	1009	736	273	71	8	12	16	35	36	2	2
Jhunjhunu	848	654	194	62	10	9	8	35	27	3	2
Pali	742	500	242	54	1	9	17	27	27	3	4
Jodhpur	893	708	185	58	-	7	17	34	24	4	3
Ganganagar	529	440	89	34	3	2	18	11	23	5	12
Kota	305	217	88	35	1	7	14	12	23	5	11
Ajmer	548	416	132	40	4	8	11	18	22	6	7
Jhalawar	402	337	65	41	-	5	17	19	22	6	6
Udaipur	827	666	161	29	3	1	14	11	18	7	12
Bikaner	550	448	102	28	-	5	13	10	18	7	13
Sirohi	279	233	46	33	-	5	12	16	17	8	8
Churu	603	468	135	34	2	6	8	18	16	9	7
Tonk	387	306	81	38	2	2	12	22	16	9	5
Rajsamand	338	273	65	19	-	4	10	5	14	10	17
Banswara	563	473	90	22	1	1	11	9	13	11	14
Hanumangarh	469	387	82	18	3	1	7	7	11	12	16
Madhopur	365	294	71	12	2	3	6	1	11	12	20
Barmer	953	810	143	29	-	3	7	19	10	13	6
Jalore	513	426	87	18	-	4	6	8	10	13	15
Chittorgarh	497	403	94	17	3	1	6	7	10	13	16
Baran	354	280	74	18	1	3	6	8	10	13	15
Pratapgarh	258	211	47	10	1	2	7	0	10	13	21
Bhilwara	677	554	123	27	1	4	4	18	9	14	7
Bundi	268	215	53	18	1	2	5	10	8	15	13
Dholpur	300	251	49	11		3	4	4	7	16	18
Nagaur	1055	861	194	18	1	1	3	13	5	17	10
Dungarpur	455	372	83	5	2	1	1	1	4	18	20
Dausa	406	335	71	13	-	-	4	9	4	19	14
Jaisalmer	211	169	42	6	-	-	4	2	4	20	19
Alwar	932	746	186	17	1	1	1	14	3	21	9
Bharatpur	506	407	99	18	-	-	2	16	2	22	8
Karauli	354	295	59	15	-	1	-	14	1	23	9
Rajasthan	18253	14583	3670	965	55	127	305	478	487	-	-

Source: HMIS Standard report 2019-20 & *Kayakalp* Portal Rajasthan State

In category A, out of 55 facilities 33% facilities fall in the two districts of the state, namely, Jhunjhunu & Jaipur. Among the Category B facilities, 44 facilities fall in four districts i.e., Sikar, Jaipur, Jhunjhunu, and Pali. In the same way third Category C, maximum number of facilities fall in the district of Sikar, Ganganagar & Jodhpur respectively. Finally, in the fourth Category D, maximum number of facilities fall in Sikar, Jaipur & Jhunjhunu district. In the state, Total 478 number of facilities have scored $\leq 70\%$ in external assessment. The implementation of scheme is not efficiently working throughout the state. The main reason behind the same that, out of the total eligible facilities for *Kayakalp* around 15 % facilities do not even meet the minimum said criterion i.e., 70% score in assessment. The maximum number of such type facilities (Category D) fall in the Sikar, Jaipur and Jodhpur districts. The implementation of the scheme, which works equally in all health facilities in all districts, is extremely crucial for improving the quality care services in the state.

Assessment process-wise growth in *Kayakalp* facilities in Rajasthan state

Facilities go through internal, peer & external assessment process against predetermined criteria. All the healthcare facilities scored decline in peer assessment from the internal assessment process, except three facilities i.e., DH Hanumangarh, CHC Sahawa & PHC Saradhana. Therefore, in almost all the facilities a negative growth rate has been observed in their peer assessment score.

Table 5 Overall growth rate in assessment process of *Kayakalp* programme, 2019-20

<i>Kayakalp</i> Facilities	<i>Kayakalp</i> assessment score (%)			Growth Rate		Thematic Areas: Increase/ Decrease/ Constant performance	
	Internal	Peer	External	Internal to Peer	Peer to External	Internal to Peer	Peer to External
DH Level							
Ajmer	89	83.83	78.17	-5.8	-6.8	A, B, C, D, E, F, G	A, B, C, D, E, F, G
Churu	76.83	76.67	74.5	-0.2	-2.8	A, B, C, D, E, F, G	A, B, C, D, E, F, G
Hanumangarh	89.5	89.67	90.17	0.2	0.6	A, B, C, D, E, F, G	A, B, C, D, E, F, G
Jhunjhunu	88.83	82.66	91.66	-6.9	10.9	A, B, C, D, E, F, G	A, B, C, D, E, F, G
CHC Level							
Rupangarh (A)	83	74.67	72.5	-10.03	-2.90	A, B, C, D, E, F, G	A, B, C, D, E, F, G
Sahawa (C)	99.17	99.67	88.17	0.5	-11.5	A, B, C, D, E, F, G	A, B, C, D, E, F, G
Dholipal (H)	95.67	87.67	91.33	-8.4	4.2	A, B, C, D, E, F, G	A, B, C, D, E, F, G
Bagar (J)	91.33	85.33	95	-6.6	11.3	A, B, C, D, E, F, G	A, B, C, D, E, F, G
PHC Level							
Saradhana (A)	98.61	98.89	98.89	0.3	0	A, B, C, D, E, F, G	A, B, C, D, E, F, G
Daudsar (C)	94.44	80.83	80.28	-14.4	-0.7	A, B, C, D, E, F, G	A, B, C, D, E, F, G
Khotawali (H)	98.33	95.83	72.22	-2.5	-24.6	A, B, C, D, E, F, G	A, B, C, D, E, F, G
Islampur (J)	100	84.17	97.5	-15.8	15.8	A, B, C, D, E, F, G	A, B, C, D, E, F, G

Note: A- Facility Upkeep, B- Sanitation & Hygiene, C- Waste Management, D- Infection Control, E- Support Services, F- Hygiene Promotion, G- Beyond Facility Boundary

With regards to peer to external assessment process, around 50% facilities score fall for the same. Overall, in Jhunjhunu district, all the facilities had surpassed the score in external

assessment vis a vis peer assessment score while it was opposite trend in case of internal to external assessment. It can be clearly seen from the table 5 that a difference or a gap existed between internal, peer assessment and external assessment scores. There are two main reasons behind this.

Firstly, facility team is more liberal/ lenient while carrying out the internal assessment process at its own facility. On the other hand, external assessment refers to the evaluation of performance by outside team from the state. The team of external assessor has a better training as well as quite rich experience in the field of assessment, and they also have knowledgeable standards of the other healthcare facilities i.e., what standards exists in other healthcare facilities by which they can easily identify differences.

Secondly, due to gap/lack in training; the peer reviewed and internal processing, who might not understand the process of evaluation of *Kayakalp*, hence on reaching the external level, many facilities are found to be underscored or over-scored. Such err can be resolved by complete utilization of the training fund, which according to data is found to be underutilized.

Impact Assessment of Quality of Care: Pre & Post *Kayakalp* Implementation

Kayakalp programme has a significantly positive impact on the public health sector as observed in the selected health facilities. Facilities are taking an initiative towards clean healthcare provisioning. Better awareness cope for quality and good practices in a hospital/ facility environment has evolved. A series of training i.e., Hand washing, BMW segregation, Infection control practices, use of ETP in the lab, proper implementation of 5's done & use of PPE made the staff more conscious about the cleanliness because of which healthcare facilities have indeed reached a higher standard of cleanliness.

Facility Performance: Baseline assessment and Year of Award

The table 6 depicts comparison of external assessment score with baseline assessment score in all visited healthcare facilities. There have been significant differences observed in baseline score level and final assessment score at the time of awarding except PHC Khotawali, Hanumangarh. The maximum positive difference/change has been observed in PHC Islampur (Jhunjhunu) and PHC Saradhana (Ajmer) while the minimum positive difference was observed in PHC Sahawa, Churu. In Jhunjhunu and Ajmer district baseline scores at PHC level facilities were quite low due to poor record maintenance, lack of cleanliness, no proper mechanism of disposal of BMW & no infection control committee. Post *Kayakalp* implementation, in a short span of time both healthcare facility i.e., PHC Islampur and Saradhana have seen drastic changes in facility upkeep and cleanliness. A committee had been constituted for periodic monitoring of the cleanliness activities. Furthermore, to improve the performance in the healthcare facilities, the facility team started to work on fulfilling the gap.

Table 6 Comparison the external assessment score with baseline assessment score

BASELINE Vs AWARDED YEAR (%)						
Facility/ District		Year of Baseline assessment	Baseline <i>Kayakalp</i> score (%)	Awarded Year's score (2019-20)	% Change	Thematic Areas: Increase/ Decrease/ Constant performance
DH Level	Ajmer	2016-17	70	78.17	8.17	A, B, C, D, E, F
	Churu	2016-17	46.8	74.5	27.7	A, B, C, D, E, F
	Hanumangarh	2015-16	74.8	90	15.2	A, B, C, D, E, F
	Jhunjhunu	2015-16	70.6	91.17	20.57	A, B, C, D, E, F
CHC Level	Rupangarh (A)	2018-19	53.50	72.5	19	A, B, C, D, E, F
	Sahawa (C)	2017-18	86	88.17	2.17	Record not maintained
	Dholipal (H)	2016-17	82	91.33	9.33	A, B, C, D, E, F
	Bagar (J)	2016-17	81.40	95	13.6	A, B, C, D, E, F
PHC Level	Saradhana (A)	2016-17	50	98.89	48.89	A, B, C, D, E, F
	Daudsar (C)	2016-17	72	80.28	8.28	A, B, C, D, E, F
	Khotawali (H)	2016-17	87.67	77.22	-10.45	A, B, C, D, E, F
	Islampur (J)	2016-17	41.33	97.5	56.17	A, B, C, D, E, F

Note: A- Facility Upkeep, B- Sanitation & Hygiene, C- Waste Management, D- Infection Control, E- Support Services, F- Hygiene Promotion

The contrary result was observed in the case of PHC Khotawali, Hanumangarh. The total score obtained in 2016-17 was 87.67% and for the year 2019-20 it was 77.22%. To elaborate furthermore, significant negative differences are observed in almost all thematic areas except the infection Control. Thus, there has been a decline in total percent score (around 10%) due to the training gap. It was reported that facility staff didn't receive any training from the state level from the implementation period. Furthermore, only one training was received from the district level from 2015-16 to 2019-20. Another reason cited was that in the first assessment there are chances to biasness with regards to *Kayakalp* score as the facility considered this assessment process as informal instead of formal. Overall, there are variations across the different facilities but CHC Sahawa stands undisputedly different from the other districts due to its somewhat consistent performance in facility upkeep, cleanliness, hygiene, waste management, and infection control even pre and post implementation of the programme in the facility. As the building of the said CHC was newly constructed in the year 2016, which helps to adhere the cleanliness and sanitation standards in the premises.

In the case of Ajmer district, the lower tier of healthcare had shown good performance compared to the higher healthcare tier. Reason being, the DH has very high OPD load as compared to CHC/ PHC which puts DH at a limiting side; it indicates that somewhat modification can be done in the assessment methodology by taking into account the OPD load/

service delivery. Furthermore, effective leadership or supervision has also played a significant role in better functioning and as well as improving the performance of the facility. The staff of the said facility was observed to highly motivated towards the same instead of upper tier.

Theme-wise Pre and Post Assessment

Since the main aim of the study was to capture the performance standards of different healthcare facilities with respect to the guidelines of the flagship programme i.e., *Kayakalp*, this section will attempt to discuss in detail the cumulative quality standards along with the temporal comparison of pre and post *Kayakalp* implementation. A Wilcoxon Signed Rank Test for two dependent samples has been used to differentiate between the *Kayakalp* assessment score of pre and post implementation period. Results of the said Test are as follows:

Table 7 Wilcoxon signed rank test: pre and post implementation period

Test Statistics								
	Facility Upkeep	Sanitation& Hygiene	Waste Management	Infection Control	Support Services	Hygiene Promotion	Overall	%
Z	-2.395 ^b	-2.046 ^b	-2.580 ^b	-2.937 ^b	-2.762 ^b	-2.540 ^b	-2.981 ^b	-2.511 ^b
Asymp.Sig. (2-tailed)	.017	.041	.010	.003	.006	.011	.003	.012
p <0.05	True	True	True	True	True	True	True	True

The Wilcoxon signed-rank test clearly showed that *Kayakalp* Implementation did elicit a statistically significant change for the following subset domains: Facility Upkeep (Z= -2.395, p = .017), Sanitation and Hygiene (Z= -2.046, p = .041), Waste Management (Z= -2.580, p = .010), Infection Control (Z= -2.937, p = .003), Support Services (Z= -2.762, p = .006), Hygiene Promotion (Z= -2.540, p = .011), and overall (Z= -2.981, p= .003). Indeed, median Score rating was statistically different both pre- and post-*Kayakalp* Implementation for all the above-mentioned domains.

Status of Quality Care Training in Rajasthan State

Quality was a new concept for public health facilities which were struggling with infrastructure, funds and HR issues. The state conducted intensive training for familiarizing the staff with the concept and how it can be achieved and creating a desire among the staff to achieve the same. A total 95 number of trainings with regards to quality care were imparted by the state to the districts. Total of 6060 participants took part in the training. It was asserted that the state quality cell tried to involve officials till block level.

Table 8 Training status in Rajasthan state since the programme implementation

Year	Total no. of training	Approx. Beneficiaries
2016-17	36	2160
2017-18	7	700
2018-19	8	560
2019-20	44	2640
Total	95	6060

In Hanumangarh DH, 82% staff of received training by the state teams since the programme implementation. Jhunjhunu district is an exception where no training was observed at the CHC level by the district team. Likewise, another exception was Hanumangarh DH; however, almost 100 % of staff have received training at facility level. Churu, Jhunjhunu, & Ajmer are three such districts where facility level training is observed very less at DH level.

Table 9 Quality care training status of the health staff, 2015-16 to 2019-20

	Facilities	Total staff	% Staff received Training by:		
			State	District	Facility level
DH level	Ajmer	150	4%	2%	67%
	Churu	350	1%	3%	14%
	Hanumangarh	181	82%	-	100%
	Jhunjhunu	350	4%	30%	66%
CHC Level	Rupangarh (A)	22	-	5%	82%
	Sahawa (C)	32	13%	19%	100%
	Dholipal (H)	25	-	8%	88%
	Bagar (J)	47	4%	-	100%
PHC level	Saradhana (A)	15	-	20%	80%
	Daudsar (C)	14	14%	15%	100%
	Khotawali (H)	13	0%	8%	100%
	Islampur (J)	13	8%	8%	100%

Kayakalp Budget Allocation Vs Utilization: State Level Analysis

The utilization ratio in the Quality assurance training programme (69%) was higher than *Kayakalp* programme (67%) in the state. At the state level, it was asserted that the unutilized fund from a specific year is treated as savings, and these savings, after obtaining due permissions from the concerned officers, are used towards activities/areas which may not necessarily pertain to the *Kayakalp* programme.

Table 10 Rajasthan state fund details (As per FMR Report), 2019-20

Particulars	Total amount received (lakh)	Total Expenditure (lakh)	% Utilization
Under <i>Kayakalp</i> programme	1479.35	985.36	67%
Quality Assurance training prog.	2523.76	1746.30	69%

***Kayakalp* Budget Allocation Vs Utilization: District Level Analysis**

Using the case of the *Kayakalp* fund in Rajasthan state, it highlights how the rules and procedures that govern release of public funds affect utilization of budgeted resources. It analyzes the utilization of funds in 33 Districts of Rajasthan, and finds out that on average; only about 66-67 per cent of funds allocated for the programme were utilized in the year 2019-20.

Table 11 District-wise fund utilization in Rajasthan as on 30th March 2020 (In lakh)

Districts	Ranking (based on maximum no. of facility scoring >70 %)	<i>Kayakalp</i> Budget	Total Expenditure	Percent Utilization	Percent Share in State Budget
Sikar	1	51.15	51.13	100	3.5
Hanumangarh	12	42.66	42.11	99	2.9
Jhunjhunu	3	38.99	38.04	98	2.7
Jhalawar	6	38.32	36.85	96	2.6
Bundi	15	22.26	21.34	96	1.5
Pratapgarh	13	28.81	26.85	93	2.0
Churu	9	14.90	13.89	93	1.0
Pali	3	13.85	12.78	92	1.0
Bikaner	7	10.87	9.79	90	0.7
Udaipur	7	73.80	64.84	88	5.1
Jalore	13	41.65	36.62	88	2.9
Dausa	19	11.47	10.03	87	0.8
Dholpur	16	11.97	10.18	85	0.8
Ajmer	6	8.40	7.07	84	0.6
Tonk	9	24.21	20.06	83	1.7
Sawai Madhopur	12	5.05	4.06	80	0.3
Baran	13	28.00	21.90	78	1.9
Jaisalmer	20	23.95	18.74	78	1.7
Jaipur	2	13.3	9.99	75	0.9
Rajsamand	10	20.85	15.45	74	1.4
Alwar	21	91.25	63.78	70	6.3
Bhilwara	14	66.37	46.13	70	4.6
Jodhpur	4	101.47	70.41	69	7.0
Chittorgarh	13	56.12	38.84	69	3.9
Sirohi	8	44.11	28.85	65	3.0
Banswara	11	67.70	42.78	63	4.7
Barmer	13	122.98	66.07	54	8.5
Bharatpur	22	43.25	22.32	52	3.0
Nagaur	17	31.65	16.49	52	2.2
Karauli	23	68.14	32.64	48	4.7
Sri Ganganagar	5	108.65	45.04	41	7.5
Kota	5	82.07	26.27	32	5.7
Dungarpur	18	41.67	10.30	25	2.9
Rajasthan	-	1449.89	981.64	67.7	100

Note: *Kayakalp* budget includes budget for following activities as per GOI PIP sheet: Conducting *Kayakalp* assessments, Trainings, Award money, pest control activities for DH only & for ODF CHCs of the District.

Utilization of NHM funds was higher in some of the districts like Sikar, Bundi, Jhunjhunu, Jhalawar, Pratapgarh, Churu, Pali, than Dungarpur, Kota, Sri Ganga Nagar, and Karauli district. This can potentially accentuate the inequality in fund spending across districts. In contrast, Bhilwara, a relatively better-off District, stood at the average in fund utilization in the said year.

The table 11 shows that almost all districts showed noticeable levels of under-utilization of budget outlays except Sikar district. This district presents a relatively better picture of utilization of funds, followed by Hanumangarh & Jhunjhunu district respectively. However, Durgapur and Kota showed the highest levels of underutilization of funds with regards to the *Kayakalp* programme. The analysis clearly depicts that in the last financial year the main problem in Rajasthan is not of inadequate budget outlays but rather poor utilization of *Kayakalp* fund. It is true that unspent balance is high in Rajasthan State, particularly in Dungarpur and Kota. However, factors affecting the extent and quality of fund utilization in different districts differ to some extent. Some common factors that are the biggest hurdles in utilization of funds were cited as delayed flow of funds and poor planning.

***Kayakalp* Budget Allocation Vs Utilization: Facility Level Analysis**

The below table 12 clearly depicts that fund utilization varies across facility level. Overall, utilization seems satisfactory at the secondary level facilities (DH and CHC), while not at the PHC level.

Table 12 Facility-wise fund utilization, 2019-20

Name of the Facilities	Result Announce	Fund Received	No. of days	Amount Received	Total expenditure	Percent Utilization
DH Level						
Ajmer	20 Jan, 2019	22 Jan, 2019	2	20,00,000	16, 96, 235	84.8
Churu	20 Jan, 2020	4 th Feb, 2020	14	3,00,000	3,00,000	100
Hanumangarh	20 Jan, 2020	26 th Jan,2020	6	6,00,000	6,00,000	100
Jhunjhunu	26 Jan, 2018	2 nd Feb, 2018	7	50,00,000	50,00,000	100
CHC Level						
Rupangarh (A)	20 Jan, 2020	-	-	Not Received	-	-
Sahawa (C)	20 Jan, 2020	27 th Apr, 2020	97	1,00,000	99,806	99.8
Dholipal (H)	20 Jan, 2020	15 th Mar,2020	54	1,00,000	99,949	99.95
Bagar (J)	20 Jan, 2020	13 ^h Mar, 2020	52	1,00,000	1,00,000	100
PHC Level						
Saradhana (A)*	20 Jan, 2020	-	-	Not Received	-	-
Daudsar (C)	20 Jan, 2020	1 st Apr, 2020	71	50,000	22,300	46.60
Khotawali (H)	20Jan, 2020	6 th Mar, 2020	45	50,000	37,977	75.95
Islampur (J)	20 Jan, 2020	-	-	Not Received	-	-

At DH level, the time taken for release of funds from district health society (DHS) to facility account in Ajmer & Hanumangarh district was much lower as compared to other districts. In 2019-20, about 100% of all funds received by DH were credited in less than two days.

Furthermore, all the district hospitals received their funds on an average within one or two weeks. At CHC Level, all the funds were credited in more than two months (average no. of days 68). Similarly, at PHC level, time taken for release of funds from BHS to facility level was observed around two months. The procedures for fund release are unduly lengthy at lower tier system of healthcare. The substantial delay in release of funds from the DHS to BHS and from BHS to the facility account has adversely affected the utilization of funds in Rajasthan state, particularly at PHC level. For instance, PHC Saradhana and Islampur were awarded in the year 2019-20 but did not receive their fund till February 2021. In Rupangarh CHC too, the delays had adverse effects on utilization of funds. Delayed in funds affects the proper utilization of funds at facility, asserted by the PHC Daudsar. Therefore, delay in funds observed only at CHC and PHC Level. However, average time taken for release of funds at PHC level was much lower than the CHC level. It was asserted that funds which were not utilized carried forward over to next year. According to the district accountant/ facility in-charge the health facility carried forward the *Kayakalp* fund in Ledger Portfolio under MRS account.

Table 13 Facility- wise total fund utilization: 2015-16 to 2019-20

Name of Facilities	Total Amount Received	Total expenditure	Percent Utilization
DH Level			
Ajmer	29,00,000	29,00,000	100 %
Churu	6,00,000	6,00,000	100%
Hanumangarh	79,00, 000	79,00,000	100%
Jhunjhunu	62,00,000	62,00,000	100%
CHC Level			
Rupangarh (A)	0	0	0%
Sahawa (C)	17,00,000	13,51,834	80%
Dholipal (H)	4,00,000	3,97,348	99%
Bagar (J)	13,00,000	13,00,000	100%
PHC Level			
Saradhana (A)*	3,00,000	1,35,000	45%
Daudsar (C)	3,00,000	2,72,300	91%
Khotawali (H)	1,50,000	1,37,979	92%
Islampur (J)	3,00,000	3,00,000	100%

At DH level 100 % of funds utilization practices were seen. With regards to PHC & CHC level underutilization of funds practices were observed at all the district except Jhunjhunu district. The contrary result was observed in case of PHC Saradhana, Ajmer. The total fund received was around 3 lakhs, of which around 45 % utilization was observed. The table 14 shows the disbursement of staff incentive under *Kayakalp* at facility tier in 2019-20. In DH Hanumangarh and CHC Bagar, Jhunjhunu staff incentive was not distributed among staff with the staff consent showing by yellow color in the table. The facilities shown with green color indicates the fund disbursement as per Guidelines, while in red color highlighted the facility which non-adherence the guidelines. It was observed that PHC Daudsar, Churu were not aware about the

fund utilization guidelines with respect to staff incentive. The disbursement of the *Kayakalp* staff fund was not being done in the ratio as recommended by the programme guidelines.

Table 14 Disbursement of staff incentive under *Kayakalp* programme at facility tier, 2019-20

Particulars	To tal sta ff	Per perso n incen tive	Total Expend iture	To tal sta ff	Per perso n incen tive	Total Expend iture	To tal sta ff	Per perso n incen tive	Total Expend iture	To tal sta ff	Per perso n incen tive	Total Expend iture
Nursing & Paramedical staff	11 7	1083	126800	7	2678	18746	No expenditure on staff			12 5	2500	312500
Administration & Doctors	43	4558	196000	5	6750	33750				65	5000	325000
4 th Class employees	42	878	36875	5	2250	11250				47	4000	188000
Office Staff	6	3833	23000	3	2500	7500				32	4000	128000
Others, including cleaning staff (Outsourced through agency)	14 6	774	113000	3	1250	3750				25	2500	62500
Award Function if any	-	-	-	-	-	-				-	-	250000
Total expenditure	35 4	11126	495675	23	15428	74996				29 4	18000	1266000
Nursing & Paramedical staff	N A	NA	NA	15	750	11250	7	893	6250	No expenditure on staff		
Administration & Doctors	N A	NA	NA	8	781.2 5	6250	3	3750	11250			
4 th Class employees	N A	NA	NA	12	312.5	3750	2	625	1250			
Office Staff	N A	NA	NA	6	416	2500	4	625	2500			
Others, including cleaning staff (Outsourced through agency)	N A	NA	NA	4	312.5	1250	1	3750	3750			
Award Function if any	N A	NA	NA	-	-	-	-	-	-			
Total expenditure		Fund Delay		45	2572. 25	25000	17	9643	25000			
Nursing & Paramedical staff	N A	NA	NA	8	892	7136	5	625	3125	N A	NA	NA
Administration & Doctors	N A	NA	NA	2	892	1784	1	5625	5625	N A	NA	NA
4 th Class employees	N A	NA	NA	1	904	904	2	937	1875	N A	NA	NA
Office Staff	N A	NA	NA	3	892	2676	2	312	625	N A	NA	NA
Others, including cleaning staff (Outsourced through agency)	N A	NA	NA	0	0	0	0	0	0	N A	NA	NA
Award Function if any	N A	NA	NA	-	-	-	-	-	-	N A	NA	NA
Total expenditure		Fund Delay		14	3580	12500	10	7499	11250	Fund Delay		

Figure 4 Theme-wise expenditure in Rajasthan under *Kayakalp* Programme, 2019-20



Utilization of Funds

More than 40% of the funds received are utilized for Infrastructure repair and maintenance and facility appearance, while less funds goes for Workplace Management, Crowd management,

Patient privacy and Security Services. There is no expenditure observed under the segment of waiting area & kitchen services.

Table 15 Theme-wise percent share of total *Kayakalp* fund utilization at different facility tier

Particular	Percent share in Total Funds at:		
	DH Level	PHC Level	CHC Level
Infrastructure repair & maintenance	26.9	16.4	8.5
Facility Appearance	16.7	4.0	0.0
Staff Duty & Rest Room	7.0	5.2	15.1
Others	6.6	4.0	0.0
Maintenance of furniture and fixture/ removal of junk material	5.4	4.8	26.8
Maintenance of open area	5.0	5.9	4.1
Laundry services & linen management	4.7	3.3	0.0
Water conservation & Sanitation	4.0	4.5	25.5
Lab/ Radiology services	3.8	0.4	0.0
Functions	3.5	0.8	0.0
Landscape & Gardening	3.5	1.6	0.0
Illumination	3.1	0.8	0.0
Pest & Animal Control	2.5	8.5	0.0
Strengthening of infection control/ Quality Improvement	2.5	8.4	10.1
Outsourced service management	1.7	1.2	0.0
Work place management	1.0	0.8	9.9
Queue/ crowd management	1.0	0.8	0.0
Patient Privacy	1.0	2.0	0.0
Security Services	0.1	24.5	0.0
Library	0.0	0.0	0.0
Kitchen Services	0.0	0.0	0.0
Waiting area	0.0	2.0	0.0
Total	100	100	100

The above table 15 depicts the *Kayakalp* fund utilization pattern at different facility tier. The fund utilization pattern of *Kayakalp* award money reveals that at the District Hospital level, *Kayakalp* award money is accorded to infrastructure upkeep and maintenance. However, at the PHC and the CHC level, the majority of the award money is used towards strengthening the sanitation and/or security services.

Discussion

This section will highlight the major bottlenecks and impact of the *Kayakalp* Programme, in light of the observations made at the studied facilities:

Major Bottlenecks of the Programme

The major bottlenecks specific to the programme implementation have been categorized below:

Disbursement of *Kayakalp* Funds

- The study highlights that there is a huge problem related to timely disbursement of the *Kayakalp* funds.

- The process of Fund flow was observed to be poor in Rajasthan State. Fund could not reach timely at the PHC level. It was asserted that the low utilization was associated with a delay in release of funds from the block level.
- It was observed that some facilities were not aware about the fund utilization guidelines with respect to staff incentive. The disbursement of the *Kayakalp* staff fund was not being done in the ratio that is compliant with the programme guidelines. It was observed that there was an unfair distribution/allocation of budget among healthcare staff.

Funds Flow

There are several reasons for delays in funds flow in the Treasury Route, some of which have been enumerated below:

- Fund crunch at the block level leads to backlog of payments/funds at facility level.
- Delayed in funds disbursement was observed in Rajasthan state due to prioritization of other things i.e., COVID management.
- Delayed release of funds from districts to Block (Delay to be calculated from the date of sanction of annual action plan to the date of release of funds).

There is backlog of payments in many blocks. These backlogs create further problems in efficient utilization of funds.

Trainings

- There is significant gap/lack with regards to routine re-orientation training regarding *Kayakalp* assessment and scoring procedures. The difference between peer-reviewed/internal scores is wide in relation to the external assessment score, reason being that facility team is more liberal/ lenient while carrying out the internal assessment process at its own facility. On the other hand, external assessment refers to the evaluation of performance by outside team from the state. The team of external assessor has a better training as well as quite rich experience in the field of assessment, and they also have knowledgeable standards of the other healthcare facilities i.e., what standards exists in other health facilities by which they can easily identify differences.
- Frequency of trainings from the state level up-till the facility level present a huge scope of further strengthening. Such err can be resolved by complete utilization of the training fund, which according to data is found to be underutilized.

Manpower

- As per the GoI recommendations for the implementation of the *Kayakalp* programme, the state does not comply with the mandate of deputing a District Quality Assurance Consultant (DQAC) at district level. In the absence of the DQAC, *Kayakalp* programme is

often looked over by the CMHO/ DPMU Office. This arrangement adds to the work-burden of programme managers and leads to considerable oversight.

- Most of the district hospitals have dedicated quality assurance staff i.e., hospital quality manager to maintain and ensure the quality standards. While at CHC/ PHC level, there is no particular post assigned for the same. There is no Quality Assurance staff assigned even at block level in the said state. This has possibly created a performance gap between upper and lower health tier.
- Hospital upkeep and service delivery efficiency varies strongly for public health hospitals where a Hospital Manager has been deputed. Various districts in the state still have vacant positions for a hospital manager in the respective district hospitals.
- Quality team faces the constant challenge of transfer and deputation of the staff from certified or selected facilities.

Major Impact of the Programme

- State has launched *Kayakalp* portal in 2016 for better and real time monitoring of the programme. The platform is being used by the state for implementation of the programme as all the health facilities of the state are mandated to upload the filled *Kayakalp* checklist through the portal only. On the basis of input, automatic ranking of institutes is generated which is available on the portal. The major impact of the *Kayakalp* programme is mentioned below:
- After the implementation of the scheme, the number of *Kayakalp* awarded facilities has increased significantly.
- Undoubtedly, public healthcare facilities have attained a higher standard of cleanliness as well as services after *Kayakalp* implementation. With the launch of the programme most of the facilities were found to be using three bucket system for cleanliness. Regular & appropriate training for cleaning/mopping have led to more awareness & knowledge generation among the staff members thereby ensuring a clean environment.
- There is a heightened reduction in hospital acquired infections due to adopting better infection control practices. A large number of healthcare service providers have been trained with respect to quality improvements at the facility level.
- Drastic improvement in the Bio Medical Waste management system of the facilities, as earlier there was very less awareness regarding waste segregation especially at the lower tier healthcare facilities. All the wastes were clubbed into one bin with little provision for management of bio degradable products.

- Documentation & record maintenance have improved considerably following the launch of the programme.
- The *Kayakalp* programme has helped to meet the minimum quality standard with further up gradation of services in a relatively short span of time.
- The programme has inculcated a culture of healthy competition among the healthcare facilities and providers.
- The awareness with respect to the scheme is being spread widely not among the staff members of the respective facility but also within the beneficiaries (as was observed in district Churu). Beneficiaries knew the word *Kayakalp* and associated its meaning to cleanliness which in itself is an achievement and commendable.

To round-up the discussion, it is important to mention the strength and limitations of the study. The study could be extended to a larger number of districts and health facilities. However, owing to the methodological needs of the study where fund utilization criterion was used to select districts across the state, the present study is limited to the findings of 4 districts only. The present study provides evidence-based estimates of the *Kayakalp* programme implementation across the districts of Rajasthan. The study includes not only a quantitative approach to study the improvements across the domains of *Kayakalp* programme, but also specifies qualitative inputs, as received from key stakeholders of the programme. Since the programme was implemented across the districts of Rajasthan at the same point time and the state quality assurance cell also confirmed that the gaps in *Kayakalp* implementation largely remain consistent across Rajasthan, therefore the results of this study may hold true for *Kayakalp* implementation in other districts as well.

Conclusion

The *Kayakalp* programme has significantly benefited the public healthcare facilities in the study districts of Rajasthan. The comparison of the baseline and the current *Kayakalp* score across all modalities of healthcare reveals a significant improvement in terms of overall facility strengthening. *Kayakalp* programme has ensured quality improvement action by mandating organizations to take purposeful action in response to observations, feedback or self-reflection resulting from the award-recognition process. The *Kayakalp* programme ensues self-assessment of facilities to validate the efforts of its staff members and demonstrate quality standards.

The fund utilization pattern of *Kayakalp* award money reveals that at the District Hospital level, *Kayakalp* award money is accorded to infrastructure upkeep and maintenance. However, at the PHC and the CHC level, the majority of the award money is used towards strengthening the sanitation and/or security services. Findings of the study reveal that the *Kayakalp* awarding process has acted as a mechanism to stimulate quality improvement actions. The provision of

knowledge through the certification process draws attention to best practices, at both, the micro and macro level.

The study has found some discrepancy with respect to the fund flow mechanism. CHC Rupangarh, and PHC Islampur & Saradhana, reported significant delay of award money disbursement. The study advocates a strong need for strengthening the fund flow mechanism at the lower tiers. Further, clear dissemination of guidelines with respect to staff share of the award money must be communicated to the healthcare staff at regular intervals. It could also be communicated via IEC material and display.

Recommendations

- There exists a dire need to depute a *Kayakalp* Quality Officer at the district level to look at the specificities of the said programme, in order to ensure effective implementation of the programme mandates.
- It is imperative for the effective implementation of quality-oriented programmes that a health manager must be assigned at the DH level.
- For the sustained success of the *Kayakalp* programme, micro-level vigilance is strongly recommended. Block-level officers designated specifically to the quality programmes can aid in ensuring the reach of the programme till the remotest facility.
- Improvement in the frequency of trainings and the number of beneficiaries trained is pivotal in ensuring the success of the programme. Such err can be resolved by complete utilization of the training fund, which according to data is found to be underutilized.
- For sustaining the programme for the long run, it is utmost important that the trained staff from the certified facility shouldn't be shuffled frequently.
- Delay in fund disbursement leads to *underutilization* of *Kayakalp* funds. The prompt transfer of funds can lead to efficient utilization at the facility level.
- The assessment methodology of *Kayakalp* scores must be revised to take into consideration some aspects of service delivery. Quality assessment in the backdrop of facility footfall must be studied and given due weight-age in the assessment process.

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11

Changes in Rural Health Infrastructure of India

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Abstract

Even after India's Independence, its population is still brewing under the scourge of degraded health system. There are nearly 716 million rural people who are constantly battling for basic healthcare services in their habitat. This condition has been aggravated by worsening living conditions of rural habitats. Nearly 75 per cent of health infrastructure and other health resources are concentrated in urban areas. The poor state of the health system in rural areas is not the outcome of a particular occurrence but a consolidated outgrowth of degraded system. In this context, the proposed study aimed to evaluate key indicator of rural health infrastructure in terms of physical and human during 2005-2019 in Indian states.

Introduction

Good health is a state of physical and mental wellbeing necessary to live a meaningful and productive life. Long healthy life is the basic aspiration of human development. Hence, health has become an important indicator of human development. Health infrastructures are “the basic services or social capital of a country” structures that support public health, having both tangible and intangible aspects and existing inside and outside the government sector. Health infrastructure is an important indicator for understanding the health care policy and welfare mechanism in a country. Primary Health Care in rural is a vital strategy which is a backbone of Health Service delivery for our country. India was one of the first few countries to recognize the importance of Primary Health Care Approach for rural segment.

India's National Health Policy, 2017 envisions the goal of attaining highest possible level of health and well-being for all and for all ages through a preventive and promotive health care orientation in all developmental policies and universal access to good quality health care services without financial hardship to the citizens. The gains of India in many health related indicators helped the country to make progress in achieving MDGs. More efforts, however, are required to reach the goals of Universal Health Coverage and those envisioned in SDG.

Rural Health Care System–The Structure and Current Scenario

The health care infrastructure in rural areas has been developed as a three-tier system and is based on the following population norms:

Table 1 Population norms for health infrastructure in India

Centre	Plain Area	Hilly/Tribal/Difficult Area
Sub-Centre	5000	3000
Primary Health Centre	30,000	20,000
Community Health Centre	1,20,000	80,000

Source: Rural Health Statistics; Rural Health Care System India final

Review of Literature

Ramani & Mavalankar (2006) made an attempt to see those building health systems that are responsive to community needs, particularly for the poor. This working paper identified the roles and responsibilities of various stakeholders for building health systems that are responsive to the community needs, particularly for the poor. Saikia (2014) focused on the progress in physical infrastructure, available health care facilities, status of existing manpower, etc. and highlights the challenges faced by the rural health care sector in the region. The results found that there has been significant improvement in the health care infrastructure after the implementation of NRHM in 2005 and suggested need for rigorous State policies to strengthen the rural health care infrastructure in the region. Lyngdoh (2015) studied is to understand the rural public healthcare system in North East India. The study concluded that there is an urgent need to address the shortcomings faced by the public healthcare system in the north eastern states. Barman & Roy (2018) examined healthcare facilities, development and problems of public health situation in Koch Bihar district using geographic information system (GIS). The results found that the health care system the service in rural areas is neglected. It is largely service based on urban areas. Although, there are number of RHs and PHCs in rural areas but infrastructure is less in the district. Sana & Archana (2019) were examined changes in health indicators against the backdrop of rural health infrastructure in India. There is definite improvement in the health indicators even as the situation in physical and human infrastructure has not improved as much as desired.

Statement of the Problem

Even after India's Independence, its population is still brewing under the scourge of degraded health system. There are nearly 716 million rural people who are constantly battling for basic healthcare services in their habitat. This condition has been aggravated by worsening living conditions of rural habitats. Nearly 75 per cent of health infrastructure and other health resources are concentrated in urban areas. The poor state of the health system in rural areas is not the outcome of a particular occurrence but a consolidated outgrowth of degraded system. It signifies not only lacunae in existing policy and infrastructure but blockage in potential development also. The rural public health facilities are battling with the problems of inadequate manpower. There exists shortfall across all cadres in rural health system. The deficiency of trained doctors and medical professionals has paralyzed the rural health facilities.

Need

Besides above focus, in a fast-growing economy, importance of healthcare could not be underestimated. There is an immense need to address to rural health challenges and work on solutions. There is a need to develop infrastructure, rural health workforce, and basic health resources to empower rural health growth. In this context, the proposed study aimed to evaluate key indicator of rural health infrastructure in terms of physical and human during 2005-2019 in Indian states.

Objectives

The main objective of the study is to analyse the growth scenario of health infrastructure in rural India and to study the key demographic indicators in rural areas. The specific objectives of the study are:

- To know the changing pattern of rural health infrastructure across the sample states.
- To suggest measures that may be useful to the policy makers both at the micro and macro levels for the improvement of health infrastructure and key demographic indicators in India.

Data Source and Methodology

The study is mainly based on secondary data obtained from annual reports of Rural Health Statistics published by Ministry of Health and Family Welfare, Government of India. The study covered all annual reports of Rural Health Statistics published by Ministry of Health and Family Welfare website from 2005 to 2019. Among all states and union territories of India 9 highly populated states and union territories and 9 least populated states and union territories based on census population of 2011 have been selected for the study. The time period of study is 15 years from 2005 to 2019 which is divided into 3 phases viz., phase -I (2005-2009), phase-II (2010-2014) and phase-III (2015-2019) for in depth analysis. The change between the years is calculated by subtracting base year value in actual numbers from the current year value in actual numbers.

Analysis

Sub-Centres

The Sub Centre is the most peripheral and first contact point between the primary health care system and the community. Sub Centres are assigned tasks relating to interpersonal communication in order to bring about behavioural change and provide services in relation to maternal and child health, family welfare, nutrition, immunization, diarrhoea control and control of communicable diseases programmes.

The below table shows the change in available sub centres by phase wise and for entire study period in the selected highly populated states of the study. It is observed from the table that in the phase-I Uttar Pradesh reports highest change in available sub centres with 18996 followed by Rajasthan, Maharashtra, Madhya Pradesh and Tamil Nadu with 1025, 852, 34 and 24 sub centres. Bihar reports a negative change in the available sub centres with a reduction of 1479. In the

phase-II there is more increase in available sub centres in Rajasthan with 2920 followed by Karnataka, Uttar Pradesh and Bihar with 1121, 270 and 33 respectively. The states like Maharashtra, West Bengal, Tamil Nadu and Gujarat reports no change. Madhya Pradesh is the only state which reports a reduction in the available sub centres with 105. Whereas in the third phase Gujarat, Madhya Pradesh, Karnataka, Uttar Pradesh, Bihar, Maharashtra and Tamil Nadu states shows an increase in the available sub centres with 1103, 1034, 494, 261, 220, 88 and 7 respectively. West Bengal reports no change and Rajasthan reports a negative change with a decrease of 895 in the available sub centres. For the entire study period Uttar Pradesh reports highest change in the available sub centres with 19257 followed Rajasthan, Gujarat, Karnataka, Madhya Pradesh, Maharashtra and Tamil Nadu with 3586, 1892, 1615, 1391, 941 and 31 respectively. West Bengal is the only state which reports no change and Bihar is the only state which reports a negative change with 388 in available sub centres for the entire study period.

Table 2 Phase wise and for total study period change in the available sub centres in the selected highly populated states and union territories

S. No.	State	Phase-I	Phase-II	Phase-III	Total Study Period
1	Uttar Pradesh	18996	270	261	19257
2	Maharashtra	852	0	88	941
3	Bihar	-1479	33	220	-388
4	West Bengal	0	0	0	0
5	Madhya Pradesh	34	-105	1034	1391
6	Tamil Nadu	24	0	7	31
7	Rajasthan	1025	2920	-895	3586
8	Karnataka	0	1121	494	1615
9	Gujarat	0	0	1103	1892

Source: Rural Health Statistics 2005-2019, Ministry of Health and Family Welfare, Government of India.

The change in the available sub centres in the selected least populated states and union territories by phase wise and for entire study period has been presented in the following table. It can be found from the table that in the phase-I Arunachal Pradesh reports highest increase with 216 followed by Mizoram with 19, Andaman & Nicobar Islands with 7, Daman & Diu with 5 and Chandigarh with 3. Sikkim, Dadra & Nagar Haveli and Lakshadweep are with no change and Puducherry is the only union territory which reports negative change in available sub centres with a reduction of 22 sub centres. It is also noticed from the table regarding phase-II all selected least populated states and union territories are reported no change in the available sub centres except Andaman & Nicobar Islands and Dadra & Nagar Haveli with 5 and 1 respectively. In the phase-III Chandigarh and Daman & Diu reports a negative change with a reduction of 16 and 3 sub centres from 2015 to 2019. Whereas Arunachal Pradesh reports highest positive change with 99 followed by Sikkim, Dadra & Nagar Haveli and Andaman & Nicobar Islands with 29, 15 and 2 sub centres respectively. Puducherry, Mizoram and Lakshadweep are reported no change. In the entire study period Puducherry and Chandigarh reports a negative change with a reduction of 21 and 13 sub centres correspondingly. Dadra & Nagar Haveli shows more increase in the available

sub centres with 33 followed by Sikkim, Mizoram, Andaman & Nicobar Islands, Arunachal Pradesh and Daman & Diu with 29, 19, 17, 9 and 2 accordingly. It is an alarming issue that in the entire study period Lakshadweep is in stationary position with regard to available sub centres.

Table 3 Phase wise and for total study period change in the available sub centres in the selected least populated states and union territories

S. No.	State	Phase-I	Phase-II	Phase-III	Total Study Period
1	Arunachal Pradesh	216	0	99	9
2	Puducherry	-22	0	0	-21
3	Mizoram	19	0	0	19
4	Chandigarh	3	0	-16	-13
5	Sikkim	0	0	29	29
6	Andaman & Nicobar Islands	7	5	2	17
7	Dadra & Nagar Haveli	0	1	15	33
8	Daman & Diu	5	0	-3	2
9	Lakshadweep	0	0	0	0

Source: Rural Health Statistics 2005-2019, Ministry of Health and Family Welfare, Government of India

Primary Health Centres

The PHCs were envisaged to provide an integrated curative and preventive health care to the rural population with emphasis on preventive and promotive aspects of health care. The PHCs are established and maintained by the State governments under the Minimum Needs Programme (MNP)/ Basic Minimum Services (BMS) Programme.

The below table shows the change in available primary health centres by phase wise and for entire study period in the selected highly populated states for the study. It is observed from the table that in the phase-I Maharashtra reports highest change in available primary health centres with 622 followed by Karnataka and Bihar with 514 and 128 respectively. There is a small increase in size of primary health centres in Uttar Pradesh and Gujarat with 50 and 14. Unfortunately in West Bengal, Rajasthan, Tamil Nadu and Madhya Pradesh states the size of primary health centres have been decreased with 251, 172, 103 and 39 accordingly. In the phase-II there is more increase in available primary health centres in Rajasthan with 578 followed by Tamil Nadu, Gujarat, Karnataka, Bihar and Madhya Pradesh with 86, 62, 40, 20 and 2 respectively. West Bengal reports no change. Uttar Pradesh and Maharashtra is the states shows a fall in the available primary health centres with 195 and 5. In the third phase of the study there is a rapid decline in the available primary health centres in Uttar Pradesh and Karnataka with 561 and 226. Gujarat reports maximum increase in the available primary health centres with 229 followed by Bihar, Tamil Nadu and Madhya Pradesh with 66, 50 and 28 respectively. Whereas in the total study period Maharashtra, Karnataka, Rajasthan, Gujarat, Bihar, Tamil Nadu and Madhya Pradesh shown an increase in the available primary health centres with 634, 448, 407, 406, 251, 42 and 5 respectively. Uttar Pradesh and West Bengal reports a damaging change with 704 and 265 in available primary health centres for the entire study period.

Table 4 Phase wise and for total study period change in the available primary health centres in the selected highly populated states and union territories

S. No.	State	Phase-I	Phase-II	Phase-III	Total Study Period
1	Uttar Pradesh	50	-195	-561	-704
2	Maharashtra	622	-5	17	634
3	Bihar	128	20	66	251
4	West Bengal	-251	0	-1	-265
5	Madhya Pradesh	-39	2	28	5
6	Tamil Nadu	-103	86	50	42
7	Rajasthan	-172	578	-1	407
8	Karnataka	514	40	-226	448
9	Gujarat	14	62	229	406

Source: Rural Health Statistics 2005-2019, Ministry of Health and Family Welfare, Government of India

The change in the available primary health centres in the selected least populated states and union territories by phase wise and for entire study period has been presented in the following table. It can be found from the table that in the phase-I Arunachal Pradesh reports highest increase with 38. Mizoram, Chandigarh, Sikkim, Dadra & Nagar Haveli and Lakshadweep reports with no change and Pondicherry, Andaman & Nicobar Islands and Daman & Diu reports a negative change in the available primary health centres with a reduction of 15, 1 and 1 respectively. It is also noticed from the table regarding phase-II all selected least populated states and union territories are reported no change in the available primary health centres except Arunachal Pradesh, Andaman & Nicobar Islands and Dadra & Nagar Haveli with 20, 3 and 1 respectively. In the phase-III Arunachal Pradesh reports highest positive change with 26 followed by Sikkim, Mizoram, Dadra & Nagar Haveli and Daman & Diu with 5, 2, 2 and 1 sub centres respectively. Puducherry, Chandigarh, Andaman & Nicobar Islands and Lakshadweep are reported no change. In the entire study period Puducherry reports a negative change with a reduction of 15 sub centres correspondingly. Arunachal Pradesh shows more increase in the available sub centres with 65 followed by Sikkim, Dadra & Nagar Haveli, Mizoram, Andaman & Nicobar Islands and Daman & Diu with 5, 3, 2, 2 and 1 accordingly. It is an alarming problem that in the entire study period Lakshadweep is in stationary position with regard to available primary health centres.

Table 5 Phase wise and for total study period change in the available primary health centres in the selected least populated states and union territories

S. No.	State	Phase-I 2005 to 2009	Phase-II 2010 to 2014	Phase-III 2015 to 2019	Study Period 2005 to 2019
1	Arunachal Pradesh	38	20	26	65
2	Puducherry	-15	0	0	-15
3	Mizoram	0	0	2	2
4	Chandigarh	0	0	0	0
5	Sikkim	0	0	5	5
6	Andaman & Nicobar Islands	-1	3	0	2
7	Dadra & Nagar Haveli	0	1	2	3
8	Daman & Diu	-1	0	1	1
9	Lakshadweep	0	0	0	0

Source: Rural Health Statistics 2005-2019, Ministry of Health and Family Welfare, Government of India

Community Health Centres

Community health centres are being established and maintained by the State government under MNP/BMS programme. As per minimum norms, a CHC is required to be manned by four medical specialists i.e. surgeon, physician, gynaecologist and paediatrician supported by 21 paramedical and other staff. It has 30 in-door beds with one OT, X-ray, labour room and laboratory facilities.

The below table shows the change in available community health centres by phase wise and for entire study period in the selected highly populated states of the study. It is observed from the table that in the phase-I there is rapid increase in the size community health centres with 239 in West Bengal followed by Uttar Pradesh, Tamil Nadu, Maharashtra, Madhya Pradesh, Karnataka, Rajasthan and Gujarat with 221, 221, 149, 106, 71, 69 and 10 respectively. In the phase-II there is an increase in available community health centres in Uttar Pradesh with 258 followed by Rajasthan, Tamil Nadu, Gujarat and Madhya Pradesh with 199, 129, 10 and 1 respectively. Bihar reports no change. Karnataka, Maharashtra and West Bengal are the states which report a reduction in the available community health centres with 132, 5 and 1. Whereas in the third phase Bihar, Gujarat, Maharashtra, Rajasthan and West Bengal states shows an increase in the available community health centres with 80, 42, 4, 3 and 1 respectively. Tamil Nadu reports no change and Uttar Pradesh, Madhya Pradesh and Karnataka reports a decrease of 94, 25 and 8 from 2015 to 2019. For the entire study period Uttar Pradesh reports highest change in the available sub centres with 385 followed by Tamil Nadu, Rajasthan, West Bengal, Maharashtra, Gujarat, Madhya Pradesh and Bihar with 350, 273, 253, 137, 91, 82 and 49 respectively and Karnataka is the only state which reports a negative change with 55 in available community health centres for the entire study period.

Table 6 Phase wise and for total study period change in the available community health centres in the selected highly populated states and union territories

S. No.	State	Phase-I	Phase-II	Phase-III	Total Study Period
1	Uttar Pradesh	221	258	-94	385
2	Maharashtra	149	-5	4	137
3	Bihar	-31	0	80	49
4	West Bengal	239	-1	1	253
5	Madhya Pradesh	106	1	-25	82
6	Tamil Nadu	221	129	0	350
7	Rajasthan	69	199	3	273
8	Karnataka	71	-132	-8	-55
9	Gujarat	10	10	42	91

Source: Rural Health Statistics 2005-2019, Ministry of Health and Family Welfare, Government of India

The change in the available community health centres in the selected least populated states and union territories by phase wise and for entire study period has been presented in the following table. It can be found from the table that in the phase-I Arunachal Pradesh reports highest increase with 13 followed by Chandigarh with 1 and Daman & Diu with 1. Andaman & Nicobar Islands, Dadra & Nagar Haveli and Lakshadweep are reporting with no change in the entire

phase. Unfortunately Sikkim reports maximum reduction with 4 followed by Mizoram and Puducherry with 3 and 1 community health centres. It is also noticed from the table regarding phase-II majority of the selected least populated states and union territories are reported no change excluding Arunachal Pradesh with a decline of 115 and Sikkim with advance of 2 in the available community health centres. In the phase-III Chandigarh, Daman & Diu and Puducherry reports a negative change with a reduction of 2, 2 and 1 community health centres from 2015 to 2019. Arunachal Pradesh reports highest positive change with 11 followed by Dadra & Nagar Haveli with 1 community health centres. Whereas in the case of Mizoram, Sikkim, Andaman & Nicobar Islands and Lakshadweep with no change in the entire phase. In the entire study period Mizoram reports highest decline with 3 followed by Puducherry, Sikkim, Chandigarh and Daman & Diu with a fall of 2, 2, 1 and 1 community health centres correspondingly. Arunachal Pradesh shows more increase in the available community healthcentres with 32 followed by Dadra & Nagar Haveli with 1 accordingly. It is also identified from the table that in the entire study period Andaman & Nicobar Islands and Lakshadweep is in stationary position with regard to available community health centres.

Table 7 Phase wise and for total study period change in the available community health centres in the selected least populated states and union territories

S. No.	State	Phase-I	Phase-II	Phase-III	Total Study Period
1	Arunachal Pradesh	13	-115	11	32
2	Puducherry	-1	0	-1	-2
3	Mizoram	-3	0	0	-3
4	Chandigarh	1	0	-2	-1
5	Sikkim	-4	2	0	-2
6	Andaman & Nicobar Islands	0	0	0	0
7	Dadra & Nagar Haveli	0	0	1	1
8	Daman & Diu	1	0	-2	-1
9	Lakshadweep	0	0	0	0

Source: Rural Health Statistics 2005-2019, Ministry of Health and Family Welfare, Government of India.

Conclusions

It is summarised that in the entire study period Uttar Pradesh reports highest change in the available sub-centres with an increase of 19257 in number followed Rajasthan, Gujarat, Karnataka, Madhya Pradesh, Maharashtra and Tamil Nadu with 3586, 1892, 1615, 1391, 941 and 31 respectively. West Bengal is the only state which reports no change and Bihar is the only state which reports a negative change with 388 in available sub-centres for the entire study period. Dadra and Nagar Haveli show more increase in the available sub-centres with 33 followed by Sikkim, Mizoram, Andaman and Nicobar Islands, Arunachal Pradesh and Daman and Diu with 29, 19, 17, 9 and 2 accordingly. In the total study period Maharashtra, Karnataka, Rajasthan, Gujarat, Bihar, Tamil Nadu and Madhya Pradesh showed an increase in the number of available primary health centres with 634, 448, 407, 406, 251, 42 and 5 respectively. Arunachal Pradesh shows more increase in the available primary health centres with 65 followed by Sikkim, Dadra and Nagar Haveli, Mizoram, Andaman and Nicobar Islands and Daman and Diu with 5, 3, 2, 2

and 1 accordingly. It is also observed that for the entire study period Uttar Pradesh reports highest change in the available primary health centres with 385 followed by Tamil Nadu, Rajasthan, West Bengal, Maharashtra, Gujarat, Madhya Pradesh and Bihar with 350, 273, 253, 137, 91, 82 and 49 respectively and Karnataka is the only state which reports a negative change with 55 in available community health centres for the entire study period. Arunachal Pradesh shows more increase in the available community healthcentres with 32 followed by Dadra and Nagar Haveli with 1 accordingly. It is also identified from the table that in the entire study period Andaman and Nicobar Islands and Lakshadweep is in stationary position with regard to available community health centres.

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12

Availability and Accessibility of Urban Primary Health Centres for Child Care in Lucknow City of Uttar Pradesh

Manoj Kumar Agarwal and Sunanda Mishra

Abstract

High infant and under-5 mortality rate is attributed to limited availability and accessibility to health interventions especially at the primary level. Around 41-72 per cent such deaths can be avoided through adequate health care interventions. Focusing upon child health the present paper attempts to gauge into the availability and accessibility factors governing the quality of care provided to children under-5 years of age at sample Urban Primary Health Centres. In this regard a field survey was conducted in Lucknow the capital city of the most populous state of Uttar Pradesh. The questionnaire included 120 caretakers who visited UPHC to seek treatment for infants below 5 years. Logistic regression shows that UPHCs are utilised more for vaccination than any other treatment. Further the dependency of the below poverty line and uneducated strata is more upon UPHCs as compared to other income groups.

Introduction

Seeking healthcare is a human right that has been recognised at national as well as international level and the right to treatment involves right to accessibility (Stellenberg, 2015). A high infant mortality is attributed to limited access to the available health interventions especially at the primary level (Kahabuka, 2011). Estimates reveal that around 41-72 per cent infant deaths can be avoided through adequate coverage of health care interventions (Rutherford, 2010). The importance of providing prompt and adequate healthcare facility to the infants has been highlighted in the guidelines of WHO Integrated Management of Childhood Illness as: "If families wait 'until the child is extremely sick, or take the child to an untrained provider, the child is more likely to die from the illness"(WHO 2005).

The primary healthcare package in India aims at providing services that are responsive to the needs of an individual and are provided at a cost that the society and community can afford. Owing to the fast pace of urbanisation the proportion of urban poor has also increased. As a consequence of this the urban poor become the sufferers of adverse health outcomes which generally do not get reflected in the public health statistics. Currently the urban poverty in India is 25 per cent which is lower than the rural poverty but gradually this gap is getting closer and by 2030 this figure is projected to reach 50 per cent. In order to cater to this the Government of India launched the historic National Urban Health Mission in 2014 to fill up the unmet needs of urban health care. Even the recent National Health Policy (2017) of India is working for strengthening of public health system in order to address the factors affecting child survival. This issue is mentioned as target 3.2 of Goal 3 of the Sustainable Development Goals of the United Nations for 2015-30.

Given the Indian set up for healthcare, the primary health centres play a crucial role. The National Urban Health Mission has laid provision for setting up of Urban Primary health Centre to cater to 50,000-60,000 urban population. It is providing the basic health care since it serves as a first point of access to a qualified doctor and improved access to essential healthcare holds a great potential for reducing the infant mortality (Rutherford, 2010). Services that form a part of the minimum requirement of a primary healthcare centre with regard to infant and child care involve regular and full immunisation, examination by a qualified doctor, lesser waiting time, potable water and adequate supply of essential medicines that may be provided in a suitable manner (Aminu U. Kaoje, 2018).

Now we talk about Uttar Pradesh. This is a state in India which has shown progressive improvement towards better health and nutrition for its population of around 200 million. Still, it lags not only behind India but also the three other Empowered Action Group (EAG) states viz. Bihar, Madhya Pradesh and Rajasthan in terms of child immunisation and nutritional outcomes. A major factor that impacts the healthcare of a child is the quality of care that is provided to him at the primary level. The beneficiary's perception about getting the complete healthcare is governed by two major determinants; first of which is the availability of the treatment required and second is the accessibility to the available treatment in an appropriate manner. Given this backdrop, the present study attempts to gauge into the availability of services and accessibility factors that govern the quality of care provided to the children at Urban Primary Health Centres (UPHC) in Uttar Pradesh, the biggest state of India residing its one-sixth population.

Methods and Materials

Study area

An observational study was conducted between October 2018 and March 2019 which was spread over 12 Urban Primary Health Centres that were located within the Lucknow district of Uttar Pradesh. Our presumption while selecting the city of Lucknow is that being the state capital, it has a possibility to serve as a benchmark for the health centres located in other cities of Uttar Pradesh. The study population consisted of 120 children below 5 years of age who visited the UPHC to seek any sort of treatment. A well-structured interview schedule was developed based upon the quality standards for urban primary health centres as prepared by the Ministry of Health and Family Welfare.

Study design

The study revolves around availability and accessibility factors that play a crucial role in providing quality healthcare to children below 5 years of age. The dimensions considered for generating availability variable involve waiting time, availability of medicines, examination by a qualified doctor, communication regarding method of consuming medicines and treatment provided. Further we have generated the accessibility variable by considering dimensions like distance travelled to reach the health centre, UPHC easy to locate and signage and directions easy to follow inside the UPHC building. Hence by recoding all these dimensions we shall arrive at availability and accessibility variables.

The respondents in our case are the accompanists who brought the beneficiary to the health centre for seeking treatment. Here, in most cases the respondents were the parents of the beneficiaries.

Outcome variable

The availability of services and accessibility to the Urban Primary Health Centres are the two outcome variables used in our analysis. However, the outcome variables were only analysed for the children below 5 years of age who visited Urban Primary Health Centres (UPHCs) to seek any sort of treatment during our study period.

Independent variables

A number of independent variables are used in our analysis, which reflects the socio-economic as well as demographic status of the households. These variables include gender, educational qualification (uneducated, primary, upper primary, secondary, higher secondary, college and above), monthly income (less than Rs 5000, Rs 5000-8000, Rs 8000-10,000, more than Rs 10,000) and source of income (labour, domestic worker, government job, private job, small scale business/others). Apart from this, two other independent variables that relate to the Urban Primary Health Centres chosen were frequency of visiting UPHC (once in two months, once in a month, twice a month, more than twice, first visit) and reason for visiting (vaccination, cold and cough, fever, others).

Statistical Methods

Descriptive statistics and logistic regression were used to predict the probability of respondents finding the services available as well as accessible at the concerned urban primary health centres. The outcome variable has been coded as 1 if the services at the UPHC were available and coded as 0 otherwise. Similarly, if the UPHC was found to be accessible by the respondent, it has been coded as 1 and 0 if not accessible. The general form of regression model for both the outcome variables has been given as:

$$\text{logit}(\pi_i) = \alpha + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5 + \beta_6 X_6 + e_i$$

Outcome variable: π_i : probability of respondent finding the services available at the UPHC

Independent variable:

X_1 = gender

X_2 = educational attainment

X_3 = monthly income

X_4 = source of income

X_5 = frequency of visit

X_6 = reason for visiting

Where α is the intercept and β 's are the slope parameters. A similar form of regression model has been used for gauging into the accessibility of the health centre. Further, the predictive probabilities are used to analyse the regression results with respect to children visiting UPHC.

Results

Socio-economic and demographic profile of the sample population

Table 1 presents the socio-economic and demographic characteristics of the study participants. A total of 120 people were interviewed where our target population were individuals who brought sick children of age less than 5 years to seek treatment at UPHC. It was found that majority of the target population were biological fathers (45.8 per cent) followed by biological mothers (41.7 per cent). However, the children who sought treatment were majorly male (60 per cent). The background characteristics of the respondents reveal that majority of them were illiterate (30 per cent) followed by people who attended college and above (18.3 per cent), upper primary (16.7 per cent) and primary education (15 per cent). The monthly income was largely found to be less than Rs 5000 (51.7 per cent) followed by Rs 5000-8000 (21.7 per cent) and Rs 8000-10,000 (14.2 per cent). However, the source of income of the people who preferred UPHC for treatment of children were basically labour class (38.3 per cent) followed by domestic workers and people involved in private job the frequency of which was found to be 24.2 per cent each.

Table 1 Socio-demographic characteristics of the respondents

Variables	Frequency (n=120)	Percentage
Child sex		
Male	72	60
Female	48	40
Relationship with the respondent		
Biological Mother	50	41.7
Biological Father	55	45.8
Grandmother	07	5.8
Grandfather	05	4.2
Others	03	2.5
Respondents Educational Attainment		
Uneducated	36	30.0
Primary	18	15.0
Upper Primary	20	16.7
Secondary	16	13.3
Higher Secondary	8	6.7
College and above	22	18.3
Respondents Monthly Income		
Less than Rs 5000	62	51.7
Rs 5000-Rs 8000	26	21.7
Rs8000-Rs 10,000	17	14.2
More than Rs 10,000	15	12.5
Respondents Source of Income		
Labour	46	38.3
Domestic Worker	29	24.2
Government Job	8	6.7
Private Job	29	24.2
Small Scale Business / Others	8	6.67

Source: Survey Data

To gauge into the accessibility and availability of services at UPHC, questions were also asked regarding the frequency and reasons for visiting UPHC results of which are shown in Table 2. It was found that majority of the study population visited the urban health centre once a month

(36.7 per cent) followed by visiting once in two months (24.2 per cent) and more than twice in a month (19.2 per cent). However, the reasons for visiting health centre with regard to child care was in most cases vaccination (55.8 per cent) followed by others (16.7 per cent) where treatment was to be sought regarding minor injuries, skin problem and general disorders. The proportion of beneficiaries who visited health centres to seek treatment regarding fever and cold & cough was found to be 15 per cent and 12.5 per cent respectively.

Table 2 Frequency and reason for visiting UPHC

Variables	Frequency (n=120)	Percentage
Frequency of Visiting UPHC		
Once in two months	29	24.2
Once in a month	44	36.7
Twice a month	13	10.8
More than Twice	23	19.2
First Visit	11	9.2
Reason for Visiting		
Vaccination	67	55.8
Cold and Cough	15	12.5
Fever	18	15.0
Others	20	16.7

Source: Survey Data

Accessibility of services at the Urban Primary Health Centres

Accessibility is governed by both the demand as well as supply side factors. Demand side focuses upon the ability to use the health service. The supply side involves the factors that hinder the individual to uptake the service (Bart Jacobs, 2011). Inaccessibility of services leads to delay in health seeking behaviour which is directly linked with mortality and morbidity (Stellenberg, 2015). Considering this aspect, the accessibility variable was generated the results of which are shown in Table 3. It was found that majority (77.5 per cent) of the respondents found the UPHC accessible.

Table 3 Accessibility of UPHC

Accessibility	Frequency	Percentage
UPHC Accessible	93	77.5
UPHC not Accessible	27	22.5
Total	120	100

Source: Survey Data

Table 4 presents the predicted probability of accessibility of UPHC adjusted for socio-economic and demographic characteristics of study population. The results suggest that the predicted probability of finding the urban health centre accessible was higher for male child (0.27) as compared to females (0.17). The predictive probability of accessibility to UPHC was 0.28 for individuals with monthly income between INR 5000-8000 whereas it was 0.13 for individuals with monthly income more than INR 10000 and 0.25 for individuals with monthly income of

less than INR 5000. Amongst the category of workers who found the UPHC accessible, it was mainly the government servants who had a higher predictive probability (0.38) of finding the UPHC and the services provided in it to be more accessible. Among other categories, the people involved in private jobs had a predictive probability of 0.28 followed by the domestic workers (0.23) and small scale businessmen (0.14). For illiterate individuals, the predictive probability of accessibility to UPHC was 0.26 as compared to 0.19 for sample possessing primary education and 0.17 for secondary and higher education.

Table 4 Predicted probabilities of accessibility of UPHC-socio-economic and demographic characteristics

Variables	Accessibility of UPHC	
	Predictive Probability	[95 per cent Conf. Interval]
Gender		
Female	0.170*	(0.078, 0.263)
male	0.266*	(0.126, 0.406)
Monthly Income		
Less than 5000	0.250*	(0.081, 0.418)
Rs5000-8000	0.283*	(0.092, 0.474)
Rs 8000-10000	0.078	(-0.060, 0.216)
More than 10000	0.131	(-0.101, 0.363)
Source of Income		
Labour	0.139**	(0.010, 0.268)
Domestic Worker	0.231*	(0.061, 0.402)
Government Job	0.381	(-0.149, 0.912)
Private Job	0.285**	(0.021, 0.549)
Small scale business	0.144	(-0.172, 0.460)
Educational Attainment		
Uneducated	0.267*	(0.093, 0.441)
Primary Education	0.196*	(0.057, 0.334)
Secondary and higher	0.171*	(0.049, 0.293)
Frequency of Visit		
once in two months	0.197**	(0.035, 0.359)
Once in a month	0.222*	(0.083, 0.362)
twice a month	0.131	(-0.052, 0.315)
more than twice	0.164**	(0.001, 0.327)
first visit	0.406**	(0.018, 0.794)
Reason for Visiting		
vaccination	0.186*	(0.078, 0.293)
cold and cough	0.129	(-0.045, 0.304)
fever	0.240**	(0.013, 0.467)
others	0.323*	(0.064, 0.582)

Source: Estimated from Survey Data

Note: * & ** shows significance at 1 per cent and 5 per cent respectively

Further the children who visited the urban primary health centre for the first time had a higher predictive probability of accessibility to UPHC were 0.40 for first visit to UPHC. However, it was 0.16 for those who visited more than twice and 0.13 for those who visited twice a month. In

terms of reasons for visiting the UPHC the predictive probability for accessibility for the category of other reasons was found to be 0.32 which involved several diseases for which the children sought medication such as injury, skin problem, boils, etc. Moreover, the predictive probability of accessibility to UPHC for the children seeking treatment for fever was found to be 0.24 followed by children visiting UPHC for vaccination (0.18).

Availability of Services at the Urban Primary Health Centres

A major issue linked with the quality of health care is the unavailability of services at the UPHC which leads to a delay in treatment and gradually pose a burden upon the health system. In order to gauge into the service availability at urban primary health centres, the availability variable was generated by incorporating several dimensions. Considering the results shown in Table 5 it was found that out of our total study population, 69.75 per cent of the respondents found the service available at the health centre whereas 30.83 per cent found the services unavailable.

Table 5 Availability of services at UPHC

Availability of Services	Frequency	Percentage
Services Available	83	69.75
Services Unavailable	37	30.83
Total	120	100

Source: Survey Data

Now just as in case of accessibility if we consider the results of predicted probability of availability of services at UPHC adjusted for socio-economic and demographic characteristics of study population (Table 6), firstly it was found that the predicted probability of male child was 0.29 and female child was 0.26 in terms of finding the services available at UPHC. When the monthly income of the study population was considered, the category belonging to INR 8000-10,000 had a predicted probability of 0.43 in finding the services available which was followed by the category of more than INR 10,000 (0.39) and for less than INR 5000 income the predictive probability was 0.28. A pattern similar to that of accessibility was observed in case of predictive probability of availability of services when the educational qualification was taken into account. Under this variable too the uneducated class had a predictive probability of 0.33 in terms of finding the services available at UPHC whereas the class which possessed primary education had a predictive probability of 0.21 followed by secondary and higher education (0.30).

Table 6 Predicted probabilities of availability of services at UPHCs-socio-economic and demographic characteristics

Variables	Availability of Services	
	Predictive Probability	[95 per cent Conf. Interval]
Gender		
Female	0.268*	(0.157, 0.379)
Male	0.298*	(0.151, 0.445)
Monthly Income		
Less than 5000	0.284*	(0.091, 0.476)
Rs5000-8000	0.150**	(-0.002, 0.302)
Rs 8000-10000	0.434**	(-0.005, 0.873)
More than 10000	0.394*	(-0.043, 0.831)
Source of Income		
Labour	0.299*	(0.082, 0.515)
Domestic Worker	0.317*	(0.098, 0.536)
Government Job	0.461**	(-0.050, 0.971)
Private Job	0.191	(-0.052, 0.435)
Small scale business	0.243	(-0.149, 0.635)
Educational Attainment		
Uneducated	0.334*	(0.155, 0.512)
Primary Education	0.210*	(0.067, 0.354)
Secondary and higher	0.304*	(0.147, 0.461)
Frequency of Visit		
once in two months	0.230*	(0.063, 0.397)
Once in a month	0.253**	(0.110, 0.397)
twice a month	0.468*	(0.128, 0.809)
more than twice	0.363*	(0.141, 0.584)
first visit	0.177	(-0.085, 0.439)
Reason for Visiting		
vaccination	0.269*	(0.146, 0.391)
cold and cough	0.335*	(0.073, 0.596)
fever	0.437*	(0.177, 0.697)
others	0.171**	(-0.007, 0.349)

Source: Estimated from Survey Data

Note: * & ** shows significance at 1 per cent and 5 per cent respectively

Considering the frequency of visit, children who were brought to the UPHC twice a month had a predictive probability of 0.46 in finding the services available and the children who visited UPHC more than twice had a predictive probability of 0.36. However, the predictive probability of availability of services at UPHC was found low for children visiting the UPHC first time (0.17) or once in two months (0.23). In terms of reasons for visiting UPHC children who visited UPHC to seek treatment for fever had a predictive probability of 0.43 of getting the available treatment whereas the children who visited UPHC for vaccination had a predictive probability of 0.26 followed by children seeking treatment for cold & cough (0.33). Hence the pattern of predictive probability was found to be similar in many of the co-variates considered for accessibility and availability.

Discussion

Our study shows that availability and accessibility are two major variables that play a crucial role in providing healthcare especially to infants. Some of the reasons behind finding the UPHC services unavailable were medicines being partially available inside the UPHC building. In fact, our survey data reveals that around 32 per cent of the total respondents failed to find the medicines available at UPHC. Further, out of this proportion 49.2 per cent had to rely upon private players to purchase medicines. It is to be highlighted here that majority of the respondents represent people from below poverty line and unavailability of medicines in the public system leads to catastrophic expenditures thereby posing economic burden on the poor households. Other reasons behind unavailability of services at UPHC include the absence of examination by a qualified doctor which discourages the patients from trusting the treatment provided. Further, lack of information regarding the method of consumption of medicines and less or no information regarding the treatment shared were also some of the factors that led the beneficiary to find the services unavailable at the UPHC.

Accessibility contributes to the quality of care and takes into account not only the distance that is required to be covered to reach the UPHC but more so it has to do with the utilisation of available services (Stellenberg, 2015). Considering our sample, the factors associated with the inaccessibility of services at UPHC include difficulty in locating the UPHC along with problems associated with following the signage and directions inside the UPHC compound. Apart from this, the inaccessibility to the services inside UPHC is also linked with the overcrowding at the health centres especially on the days prescribed for vaccination. Overcrowding coupled with poor management leads to longer waiting hours and poor patient satisfaction thereby. It is suggested that the facility for vaccination may be spread over the entire week which may help in minimising the burden of children seeking vaccination on selected days.

Recommendations

Some key areas that require attention are that there is a need to motivate the doctors and staff to devote proper time and attention to the infants which can help in bringing about a change in the perception of the people in accepting the services of UPHC and will increase the utilisation of these urban health centres thereby. Apart from this, it is well-known that the medical officers in the UPHC are technically more equipped and resourceful in dealing with a particular health issue. Still it has been gauged that the infants were mostly dealt by paramedical and support staff which somehow gives them lesser satisfaction on the treatment and counselling provided. Hence making provision for a proper line of treatment can positively impact the availability and accessibility of the urban primary health centres especially for infants.

Conclusion

The urban primary health centres have a crucial role to play in catering to the child health services. Hence it is expected out of these health centres to make the masses aware of the services provided

free of charge in order to enhance its utilisation. National Family Health Survey-4 reveals that Uttar Pradesh has still a long way to go in terms of providing adequate cure for child health. Uttar Pradesh tops the list of infant mortality rate where it accounts for 64 deaths per 1000 live births under one year of age and 78 deaths per 1000 live births under 5 years of age. The reason behind this is higher dependency upon quacks for medical treatment and inability to access the health centres. It has been revealed in our study that people are majorly dependent upon the UPHCs for vaccination reason being that they are unaware of the other child health services provided here. Apart from this, our study highlights that the accessibility and availability of services at UPHC is more for the male child and it is the lower income group which is in a greater position to find the services accessible at the UPHC due to greater dependence upon UPHC for primary care. However, if the availability of services are to be considered then they are found to be more for the higher income class due to the literacy factor which makes them much aware of their rights that can be exercised inside the UPHC with regard to medicines and treatment.

Appendix I

Case Study 1

The NFHS-4 data reveals that India is towards progressive end when it come to the provision for health coverage for infants. However, the ground realities speak differently. Sarita a mother of 4 children who resides behind an Urban Primary Health Centre reveals that she often visits the UPHC to seek treatment for her children but goes back unsatisfied. In her words *“This centre doesn’t has treatment for any diseases you visit for. All my children are ill and I don’t have that much money to get them treated by a better doctor. My children are getting weak day by day. This is the reason why poor are always the sufferers”* Apart from this we also came across patients who travelled a long distance to reach the UPHC and spent substantial amount upon conveyance were also found unsatisfied since the entire staff of the UPHC had gone to perform outreach service leaving no one back at the centre to take care of out-patient duties. We can comment upon the existing urban primary health centres that they are required to be adequately equipped by way of both availability of service as well as staff. This would lead to a greater patient satisfaction and utilisation.

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Section VI
Sex Composition, Education,
Insurance and Data Quality

Trend Analysis of Sex-Ratio Among the Tribal of Andhra Pradesh

B. Muniswamy, Ch. Suribabu and R. Vijaya Krishna

Abstract

Tribal communities are less developed with social and economic backward with poor living conditions in our country. Generally, the tribal groups live in remote areas and high mountainous slopes, undisturbed by modern civilization. Most of the tribal groups live in villages which are situated on hill tops or the valleys of tribal areas. The tribal areas are near and in the midst of forests and mountains. Forests and tribal people are inseparable and existence and development of one depends on the other. Tribal people have got symbiotic and nature interrelationship with the forests and mountains around them. Further, this study explored the dynamics of population predictions through the trend analysis of the tribal communities.

Introduction

The tribal communities are enriched with diversity and culture across the country. These tribal communities are mostly living in the forest irrespective of geographical areas. In fact, our country is most populated tribal communities after Africa which shows the tribal culture and their identity. The majority of the tribal communities are occupied in our country such as Mizoram, Nagaland, Meghalaya and Arunachal Pradesh and Union Territories of Dadra and Nagar Haveli too. The total population of Scheduled Tribes in the country is about 707.29 Lakhs which constitutes 6.88 per cent of the total population as per 2001 census reports. There are thirty-four identified Scheduled Tribe Communities residing in both hilly and plains areas of the state. The tribal areas of Andhra Pradesh have major variety of communities such as Gonds, Koya, Hill Reddi, Bagata, Valmiki, Savara and Khond. The study attempted to provide the distribution of the tribal communities in Andhra Pradesh by bringing various concepts to comprehend the population dynamics. The following studies have been covered the trend analysis of the tribal communities by reporting variety of social mechanisms (Pati & Jagatdeb 1991; Saraswathy & Devi 2009; Saha & Paul 2017; Prasad 2020)

The current study provided based on the various Census of India data to provide changing trends in tribal sex ratio, the present study to analyze the trends in Overall tribal Sex Ratio (OSR) is proposed in all the thirteen districts of Andhra Pradesh. In this, an attempt made to understand the different contextual analysis of trends in tribal sex ratio based on different secondary sources. It is also made some of arguments and suggestions to analyse the overall

tribal sex ratio and child sex ratio. For that, the data has been collected from the different secondary sources which are obtained from Census of India, Census of India-Primary Census Abstracts, Internet Sources and Research articles. The collected data has been included from the period of 1991 to 2011 census and some of the new recent reports also included substantiating the arguments in this report. The secondary sources of data from census of India and Andhra Pradesh have been measured based on the formulas which are used by the Population Census of India to analyse the tribal sex ratio and its trends.

Methodology

This study has been started to understand the context of tribal sex ratio in different districts in the given state of Andhra Pradesh. In the keeping view on tribal sex ratio in Andhra Pradesh has been reported at different levels to promote attitude towards increase of tribal sex ratio. This report provides the information related tribal sex ratio which helps to the researchers, research institutions, organizations and stakeholders for the strengthening of policy regimes. To understand the issues related to tribal sex ratio, the thirteen districts have included for this report. However, it is not only strictly following the Andhra Pradesh, it also provides the general scenario of tribal sex ratio in India. This study provides holistic approach to understand the tribal sex ratio and its issues, challenges and strategies related to gender issues particularly to trend analysis of tribal sex ratio. This study is based on the secondary source data to understand the trend analysis of the tribal communities of Andhra Pradesh.

Objectives

The main objective of the study is to make a trend analysis in overall tribal sex ratio in the Andhra Pradesh States during the period 1991 to 2011.

- To know the sex ratio of working engagement and participation of tribal population in the districts of Andhra Pradesh.
- To know the overall sex ratio among the tribals of Andhra Pradesh.
- To assess the distribution of the trend analysis of work participation of tribal population in the districts of Andhra Pradesh.

Trend Analysis

Andhra Pradesh is the fourth largest and fifth most populous state of India that lies between latitudes of 00 00'12 40"N and 19 50'N and longitudes of 76 45'E and 84 40'E. It is surrounded by Chhattisgarh in the north-west and Maharashtra in the north direction, Tamil Nadu in the south, Karnataka in the west, Orissa in the north-east and coastal area of Bay of Bengal in the east. Total area of the state is 275,045 km of which 16.9% covered by different types of forest. Population density of Andhra Pradesh is 22308 per km which is lower than national average of 382 km.

Table 1 Trends in tribal sex ratio in India and Andhra Pradesh during 1961-2011

Year	India	Andhra Pradesh
1961	987	976
1971	982	973
1981	983	962
1991	972	960
2001	977	972
2011	990	1012

Source: Census of India

The above data from the census 1961 to 2011 shows that tribal sex ratio in has shown a declining trend 987 in 1961to 977 in 2001, from 2001 to 2011 there is an increasing trend with 990 in India. In Andhra Pradesh also shows the same in 1961 the sex ratio is 976 and it has declined to 960 in 1991, from there onwards it has increased to 972 in 2001 and 1012 in 2011. It is an evident fact that there are some fluctuations in tribal sex ratio by 2011 census the sex ratio in both India and Andhra Pradesh is good.

Table 2 District wise tribal sex ratio and variations in Andhra Pradesh during 1991-2011

District	Census Years			Variation		
	1991	2001	2011	2001-1991	2011 -2001	2011 – 1991
Srikakulam	994	1009	1041	15	32	47
Vizianagaram	1002	1025	1054	23	29	52
Visakhapatnam	987	1003	1042	16	39	55
East Godavari	995	1011	1042	16	31	47
West Godavari	1008	1018	1044	10	26	36
Krishna	948	949	985	1	36	40
Guntur	939	962	975	23	13	36
Prakasam	937	949	971	12	22	34
Nellore	946	961	970	15	9	24
Chittoor	940	969	996	29	27	56
Cuddapah	926	939	967	13	28	41
Anantapur	935	935	962	0	27	27
Kurnool	933	929	970	-4	41	37

Source: Census of India

Above the table shows that constant tribal sex ratio has been reported in all the districts significantly from 2001 to 2011. However, during 2001-1991 has been failed to maintain tribal sex ratio in districts of Kurnool and Anantapur. During 2001-1991, Kurnool reported decline of 933 (1991) to 929 and later it has been increased to 970; and Anantapur reported decline of 935 (1991) to 935 (2001) later it has been increased to 962. The districts of East Godavari and Cuddapah maintained variations in their respective sex ratios from 1991 to 2001, and 2001 to 2011 as follows: East Godavari (16 to 31) and Cuddapah (13 to 28). In fact, the sex ratio during 1991 was constant but during 1991-2001 it was declined and from 2001-2011 has been increased in all the districts in Andhra Pradesh significantly.

Table 3 Trends in overall sex ratio and variations of scheduled tribes in rural and urban areas of Andhra Pradesh during 1991-2011

Andhra Pradesh	Census Year			Variation		
	1991	2001	2011	2001-1991	2011-2001	2011-1991
Rural	973	986	1014	13	28	41
Urban	922	957	977	35	20	55

Source: Census of India

This study provided also on tribal communities of Andhra Pradesh based on comparative of 1991, 2001 and 2011 census. If the rural and urban sex ratios are considered in census data it shows overall improvement in sex ratio (adults and children combined) in all categories, Scheduled Tribes between 1991, 2001 and 2011. This sex ratio is more visible in rural areas than urban area is low. The sex ratio among rural area is better than that of all categories. The census data shows Tribal sex ratio has increased to 973, 986 from 1014 in rural areas. On the other hand, sex ratio of general population in Urban areas improved to 922, 957 from 977. The state sex ratio of urban population is improving slowly in comparison to rural population of tribals.

Table 4 District wise overall sex ratio and variations of scheduled tribes in rural areas of Andhra Pradesh during 1991-2011

District	Census Year			Variation		
	1991	2001	2011	2001-1991	2011-2001	2011-1991
Srikakulam	999	1010	1045	11	35	46
Vizianagaram	1004	1026	1050	22	25	46
Visakhapatnam	991	1008	1055	17	47	64
East Godavari	1003	1014	1049	11	35	46
West Godavari	1013	1020	1048	7	28	35
Krishna	945	949	986	4	37	41
Guntur	943	960	970	17	10	27
Prakasam	931	945	962	14	17	31
Nellore	942	956	969	14	10	27
Chittoor	941	966	985	25	19	44
Cuddapah	929	938	955	9	17	26
Anantapur	939	934	957	-5	23	18
Kurnool	954	936	987	-20	51	33

Source: Census of India

The districts of Andhra Pradesh have been reported with positive growth in sex ratio. The trend is observed in districts such as Prakasam and Cuddapah has been maintaining constant sex ratio from 1991 to 2011. Visakhapatnam and Srikakulam, Vizianagaram, East Godavari has been reported with highest sex ratio comparatively with other districts. It is reported that sex ratio has been decline at the district of Anantapur, Kurnool with minor variation. The trend has been observed during 2001-1991, the slight variations in decline of sex ratio during this period.

**Table 5 District wise overall sex ratio and variations of scheduled tribes
in urban areas of Andhra Pradesh during 1991-2011**

District Name	Census Year			Variation		
	1991	2001	2011	2001-1991	2011-2001	2011-1991
Srikakulam	782	932	929	150	-3	147
Vizianagaram	940	991	1149	51	158	209
Visakhapatnam	779	870	865	91	-5	86
East Godavari	833	962	950	129	-12	117
West Godavari	955	998	1000	43	2	45
Krishna	959	950	983	-9	33	24
Guntur	923	969	993	46	24	70
Prakasam	962	972	1013	10	41	51
Nellore	969	992	977	23	-15	8
Chittoor	928	993	1044	65	51	116
Cuddapah	907	951	1008	44	57	101
Anantapur	899	943	982	44	39	83
Kurnool	851	893	909	42	16	58

Source: Census of India

The urban sex ratio has been increased from 1991 to 2011 which indicates the positive aspect among urban sphere. During 1991 to 2011, the sex ratio enormously increased the following districts such as Vizianagaram 940 (1991) to 1149 (2011) with variation 209; Srikakulam 782 (1991) to 929 (2011) with variation 147; East Godavari 833 (1991) to 950 (2011) with variation 117; Chittoor 928 (1991) to 1044 (2011) with variation 116; Cuddapah 907 (1991) to 1008 (2011) with variation 101. However, the district of Srikakulam reported that the decline of sex ratio 932 (2001) to 929 (2011); Visakhapatnam 870 (2001) to 865 (2011); East Godavari 962 (2001) to 950 (2011); Nellore 992 (2001) to 977 (2011); Krishna 959 (1991) to 983 (2011) which is negative trend against other districts. Almost all the remaining districts have maintained the variation of sex ratio has been reported.

**Table 6 District wise rural and urban overall sex ratio of scheduled tribes in
Andhra Pradesh during 1991-2011**

District	Rural			Urban		
	1991	2001	2011	1991	2001	2011
Srikakulam	999	1010	1045	782	932	929
Vizianagaram	1004	1026	1050	940	991	1149
Visakhapatnam	991	1008	1055	779	870	865
East Godavari	1003	1014	1049	833	962	950
West Godavari	1013	1020	1048	955	998	1000
Krishna	945	949	986	959	950	983
Guntur	943	960	970	923	969	993
Prakasam	931	945	962	962	972	1013
Nellore	942	956	969	969	992	977
Chittoor	941	966	985	928	993	1044
Cuddapah	929	938	955	907	951	1008
Anantapur	939	934	957	899	943	982
Kurnool	954	936	987	851	893	909

Source: Census of India

As mentioned earlier tables regarding overall sex ratio reported that positive changes occurred in sex ratio gradually increased in both rural and urban population during 1991-2011. However, the

Visakhapatnam district at urban sex ratio declined 870 (2001) to 865 (2011) has been reported. Except Visakhapatnam districts, remaining district-maintained sex ratio among urban population.

Table 7 Trends in tribal literacy rates in India during 1961-2011

Year	Male	Female	Total
1961	13.83	3.16	8.53
1971	17.63	4.85	11.30
1981	24.52	8.04	16.35
1991	40.65	18.19	29.60
2001	59.17	34.76	47.10
2011	68.53	49.35	58.96

Source: Census of India

It is clear from table-3 there has an increasing trend in tribal literacy rates both in male literacy and female literacy as well as in total literacy. The total literacy in 1961 is 8.53 per cent and it has increased to 58.96 per cent in 2011 census that indicates that almost all there is 7 times increase in total tribal literacy from 1961 to 2011 census. As per as male literacy concerned in 1961 it is 13.83 per cent and increased to 68.53 per cent in 2011. Whereas in female literacy there is a drastic change in 1961 it is only 3.16 per cent and increased to 49.35 per cent in 2011. It is concluded that the trends in tribal literacy rates shows a favourable indication in overall, male and female literacy rates to their development.

Table 8 Trends and variation in sex ratio of literate population of scheduled tribes in rural and urban areas of Andhra Pradesh during 1991-2011

Andhra Pradesh	Census Year			Variation		
	1991	2001	2011	2001-1991	2011-2001	2011-1991
Rural	385	597	720	212	123	335
Urban	495	679	780	184	101	285

Source: Census of India

The data shows that the tribal literate sex ratio has been reported as increased trend between census 1991 to 2011 and the variation also increased between these censuses.

Table 9 District wise trends and variations in sex ratio in literate population of scheduled tribes in Andhra Pradesh during 1991-2011

District	Census Years			Variation		
	1991	2001	2011	2001-1991	2011 -2001	2011 – 1991
Srikakulam	359	579	724	220	145	365
Vizianagaram	388	603	738	215	135	350
Visakhapatnam	305	492	653	187	161	348
East Godavari	594	782	853	188	71	259
West Godavari	655	843	914	188	71	259
Krishna	435	662	751	227	89	316
Guntur	366	569	665	203	96	299
Prakasam	372	586	735	214	149	363
Nellore	528	753	833	225	80	305
Chittoor	433	657	776	224	119	343
Cuddapah	346	550	676	204	126	330
Anantapur	340	504	658	164	154	318
Kurnool	303	478	667	175	189	364

Source: Census of India

The data shows that the tribal literate sex ratio between the census 1991 to 2011 as per district wise. The trend also reported that the gradual increase both in the census years and variation too in all the districts.

Table 10 District wise trends in sex ratio in literate population of scheduled tribes in rural and urban areas of Andhra Pradesh during 1991-2011

District Name	Rural			Urban		
	1991	2001	2011	1991	2001	2011
Srikakulam	355	579	723	425	583	738
Vizianagaram	370	594	719	632	770	1055
Visakhapatnam	298	482	647	424	628	701
East Godavari	603	780	855	529	805	836
West Godavari	663	848	915	590	777	898
Krishna	374	652	721	553	699	810
Guntur	333	548	638	442	646	741
Prakasam	327	565	715	549	693	814
Nellore	511	752	838	579	761	815
Chittoor	405	637	745	555	759	876
Cuddapah	322	544	650	426	583	747
Anantapur	307	479	634	489	634	747
Kurnool	296	472	682	317	497	628

Source: Census of India

The data related to the tribal literates and sex ratio among the rural spheres has been reported from the years 1991 to 2011. From the 1991 census to 2011 census, there is gradual increase in the sex ratio and similarly the variation also increased in all the trends. The tribal literates and sex ratio among the urban spheres has been reported from the years 1991 to 2011 like rural spheres. From the 1991 census to 2011 census, there is gradual increase in the sex ratio and similarly the variation also increased in all the trends.

Table 11 Trends and variation in sex ratio of main worker of scheduled tribes in rural and urban areas of Andhra Pradesh during 1991-2011

Andhra Pradesh	Census Year			Variation		
	1991	2001	2011	2001-1991	2011-2001	2011-1991
Rural	750	776	804	26	28	54
Urban	508	430	487	-78	57	-21

Source: Census of India

The data shows that the main workers sex ratio of Andhra Pradesh from the census 1991 to 2011. The variation of this census has been drastically increased in the rural sphere and urban sphere it has been reported with decline trend.

Table 12 District wise trends in sex ratio of main workers of scheduled tribes in rural and urban areas of Andhra Pradesh during 1991-2011

District Name	Rural			Urban		
	1991	2001	2011	1991	2001	2011
Srikakulam	827	694	770	412	251	529
Vizianagaram	724	851	852	335	397	452
Visakhapatnam	695	854	848	196	231	369
East Godavari	761	703	773	354	296	435
West Godavari	858	778	864	483	439	452
Krishna	807	758	810	466	419	494
Guntur	832	830	881	594	510	587
Prakasam	776	737	801	604	487	560
Nellore	747	672	667	563	432	451
Chittoor	692	713	759	464	382	448
Cuddapah	637	616	732	456	434	493
Anantapur	749	741	759	390	384	422
Kurnool	838	840	890	534	619	570

Source: Census of India

The data related to main workers sex ratio among the rural and urban has been reported from the census 1991 to 2011. The trend also shows that the gradual increase in both rural and urban areas in all the districts of Andhra Pradesh.

Table 13 Trends and variations in sex ratio of cultivators of scheduled tribes in rural and urban areas of Andhra Pradesh during 1991-2011

Andhra Pradesh	Census Year			Variation		
	1991	2001	2011	2001-1991	2011-2001	2011-1991
Rural	564	718	628	154	-90	64
Urban	244	360	315	116	-45	71

Source: Census of India

The above table explains the trends and variations in sex ratio of cultivators of scheduled tribes in rural and urban areas of Andhra Pradesh during 1991-2011. It is clearly found that there is an increasing trend between 1991 and 2001 and declining trend between 2001 and 2011. It is also reported that the variation declined between 2001 to 2011 census and increased between census 2011 to 1991.

Table 14 District wise trends in sex ratio of cultivators of scheduled tribes in rural and urban areas of Andhra Pradesh during 1991-2011

District Name	Rural			Urban		
	1991	2001	2011	1991	2001	2011
Srikakulam	584	557	473	143	310	222
Vizianagaram	510	749	602	108	206	200
Visakhapatnam	652	832	690	61	674	517
East Godavari	587	633	652	239	682	133
West Godavari	484	569	499	111	667	300
Krishna	312	352	324	356	308	556
Guntur	271	517	509	610	621	208
Prakasam	257	475	498	342	583	240

District Name	Rural			Urban		
	1991	2001	2011	1991	2001	2011
Nellore	334	397	274	143	198	162
Chittoor	459	601	619	224	267	174
Cuddapah	210	395	397	0	800	263
Anantapur	366	489	392	205	143	266
Kurnool	256	611	622	112	71	553

Source: Census of India

The data shows that the trends of cultivator sex ratio among the districts of Andhra Pradesh. The trends of both rural and urban spheres have been reported with gradual decline from the census 1991 to 2011.

Table 15 Trends and variation in sex ratio of agricultural labourers of scheduled tribes in rural and urban areas of Andhra Pradesh during 1991-2011

Andhra Pradesh	Census Year			Variation		
	1991	2001	2011	2001-1991	2011-2001	2011-1991
Rural	999	916	936	-83	20	-63
Urban	841	733	790	-108	57	-51

Source: Census of India

The data shows that the agricultural labourers sex ratio of Andhra Pradesh 1991 to 2011. The data revealed that the decline between both 2001 to 1991 census and 2011 to 1991 census.

Table 16 District wise trends in sex ratio of agricultural labourers of scheduled tribes in rural and urban areas of Andhra Pradesh during 1991-2011

District Name	Rural			Urban		
	1991	2001	2011	1991	2001	2011
Srikakulam	955	812	873	667	359	863
Vizianagaram	983	1011	944	643	954	905
Visakhapatnam	1069	1033	1162	758	526	759
East Godavari	1116	848	871	504	433	611
West Godavari	1215	994	969	766	686	613
Krishna	1058	988	957	858	889	915
Guntur	1091	994	1025	976	773	952
Prakasam	988	915	947	899	665	824
Nellore	846	758	718	842	689	592
Chittoor	815	804	845	714	710	732
Cuddapah	850	736	889	531	912	676
Anantapur	1237	1136	1106	763	800	830
Kurnool	1186	1052	1058	774	1442	819

Source: Census of India

The data shows that agricultural labourers sex ratio among the rural and urban population between censuses 1991 to 2011. It is also reported that the increase trend in both rural and urban between these censuses.

Conclusion

The study indicate that tribal sex ratio has improved both in Andhra Pradesh and in India since, 1961 though the speed of increase has been much faster in the former during a period of fifty

years. At the district level a wide spread variations is recorded in the tribal sex ratio during in 2011. The fact that all districts in the State increased their relative share of females is a welcome sign. The study would recommend the need to create more awareness on the gender roles in the community development in the tribal areas to improve sex ratio among them. It also suggests to improve the educational facilities among the tribal groups to promote further gender equality as well as sex ratio.

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14

Awareness and Coverage of Health Insurance Schemes Among Urban Poor in Kerala

Suresh Kumar S and Sajini B Nair

Abstract

Health insurance has emerged as a key instrument in current health financing reforms in India to achieve the goal of Universal Health Coverage. Awareness about health insurance schemes and their coverage aspects among urban poor in Kerala is the subject of study here and data from 10240 urban respondents who availed services from all the 83 Urban Primary Health Centres in Kerala are used for analysis. The most popular health insurance schemes among the urban poor in Kerala are RSBY-CHIS, Karunya Benevolent Fund Scheme and Employment State Insurance Scheme. However, a substantial proportion of the urban poor are still outside the coverage of health insurance benefits. Among those enrolled, four in five respondents are in RSBY-CHIS scheme. The extent of coverage of claims under insurance schemes for treatment is only one-quarter (27.6 percent) of those who are enrolled in any scheme. During the two years preceding the survey, hospitalization for treatment of diseases was required among 36.2 percent of the respondents. Differentials by place of treatment are not significant as patients chose Government and Private hospital in almost similar proportion. However, out of those who required hospitalization only three-fifths had enrolled in any insurance scheme. The schemes have not helped the respondents to cover all their expenditure and OOOPE is evident in spite of being enrolled in the insurance scheme. Steps need to be taken to create awareness among the urban poor to increase their enrolment in the insurance schemes.

Introduction

Kerala is a rapidly urbanizing State. Notable instance of rapid urbanization has happened in Kerala where the urban population grew from 25.9 percent in 2001 to 47.7 per cent which is the highest growth for any State (Registrar General of India, 2011). The urban population is poised to increase further to 70 percent by 2021 (National Commission on Population, 2019). Consequent to this, the health care needs of urban people have also been increasing. The Sustainable Development Goals by 2030 aims at achieving Universal Health Coverage through various health programmes. It is true that programmes for their welfare are well placed as the primary health care needs of the urban poor are catered to a good extent by UPHCs established in the Corporations and Municipalities in the State. But a large proportion of the urban poor still remain unreached and they depend heavily on private hospitals, nursing homes, scanning centres, diagnostic centres thereby spending huge amount on health care. So, a large number of people, especially those below poverty line, borrow money or sell assets to pay for the

treatment in private hospitals. They are constantly exposed to environmental hazards due to over-urbanization and they struggle against multidimensional vulnerability. Studies echo the deprivation of health needs of the urban poor and show that out of pocket expenditures still exist in public health facilities. Increasing urbanization and widening inequalities, unmatched by the development of affordable services, could lead to restrictions in access to care and higher propensity to resort to self-treatment among the poor (Castro-Leal et al. 2000) or could even refrain from seeking medical help (Pillai et al. 2003). Health insurance has emerged as a key instrument in current health financing reforms in India. Only a health insurance could help the people to access good quality medical care and provides financial protection against high medical expenses.

In this context it is worthwhile to explore the health insurance schemes in place. The urban poor in Delhi have been shown to have limited access to health insurance with only 18 percent covered by ESIC/CGHS/RSBY, and only 9.4 percent of the eligible households with access to the RSBY (Kusuma et.al, 2018). National data sets reveal rural urban differentials in coverage of health insurance in India when the findings of LASI India wave 1 report (2020), showed that among older adults age 45 and above who have any type of health insurance, 86 percent in rural and 61 percent in urban areas are covered under the *Rashtriya Swasthya Bima Yojana*. Moreover, the proportion of older adults age 45 and above covered under private schemes (such as privately purchased commercial health insurance) is higher in urban areas (13%) than in rural areas (1%).

The basket of choice on health insurance schemes offered by Government and Private sector in India is rather wide. All the schemes have different eligibility conditions for enrolment. Among the Government health insurance schemes that have till date wide coverage, the Employees State Insurance Scheme (ESIC), the Central Government Health Scheme (CGHS) have long tradition of existence and the *Rashtriya Swasthya Bima Yojana* (RSBY) has the widest coverage in India.

The ESI Scheme has the credit of being the first health insurance programme in independent India established way back in 1952. The existing wage limit for coverage under the Act effective from 01/01/2017 is Rs.21,000/- per month [Rs. 25,000/- per month in the case of persons with disability] (Reddy et.al, 2011). Central Government Health Scheme (CGHS) launched in 1954 has been providing comprehensive medical care to the Central Government employees and pensioners enrolled under the scheme. This scheme covers OPD treatment including issue of medicines, Specialist Consultation at Polyclinic/Govt. Hospitals, indoor treatment at Government and Empanelled Hospitals and investigations at Government and Empanelled Diagnostic centres. Presently there are 436 Wellness Centres. As per the recent statistics the number of card holders (as on August 2020) is over 12.56 lakh and the number of beneficiaries is over 36.83 lakh (CGHS, Government of India).

Rashtriya Swasthya Bhima Yojana (RSBY), a national level public funded health insurance scheme, was launched in 2008 by Government of India aiming to provide health insurance to Below Poverty Line (BPL) households and people from unorganized economic sector so that they receive financial protection against CHE and OOPE and also have improved access to quality health care. RSBY was modified by different States to suit to their own health needs. Government extended the scheme by adding the Above Poverty Line (APL) households and relaunched it as Comprehensive Health Insurance scheme (CHIS). CHISPLUS scheme has been formulated by Kerala Government for providing treatment for cancer, heart diseases and kidney diseases. Free treatment is up to Rs 70,000/- for treatment of these diseases. This is in addition to the free treatment available under RSBY/CHIS and is available to all smartcard holders except APL. The scheme is implemented through all the Government Medical Colleges, Regional Cancer Centre, Sree Chitra Tirunal Institute for Medical Sciences & Technology, Malabar Cancer Centre and all District Headquarters Hospitals since January, 2010 (www.chiak.org). Senior citizens now get additional top up benefits to take care of additional geriatric diseases when Government of India enhanced coverage exclusively for senior citizens over and above the benefits being provided under RSBY. From 2017-18 onwards, the State Government extended this additional coverage of Rs. 30,000/- each to the Senior Citizens (beneficiaries of age 60 years & above) in the RSBY/CHIS family under Senior Citizens Health Insurance Scheme (SCHIS). This scheme is implemented alongside RSBY/CHIS through the same empanelled hospitals (<https://www.chiak.org>). Nearly 41 lakh families have been enrolled by the year 2018-19.

Ayushman Bharat - National Health Protection Mission subsumes the on-going centrally sponsored schemes RSBY and the SCHIS with well defined benefit cover of Rs. 5 lakh per family per year and enabling the beneficiaries to avail benefits in both public and empanelled private facilities. Registrations crossed the one crore mark during May 2020 (www.india.gov.in/spotlight/ayushman-bharat-national-health-protection-mission). As the scheme offers flexibility to the States, Kerala combined the already existing Karunya Benevolent Fund Scheme (KBF), the AB-PMJAY and formed the AB-PMJAY- Karunya Arogya Suraksha Padathi (KASP) which is now operational through CHIAK. Karunya Benevolent Fund provided financial aid to under-privileged people suffering from acute ailments like Cancer, Haemophilia, Kidney and Heart diseases and for Palliative Care by raising funds through Kerala lottery. KASP is the health care scheme which aims at providing a health cover of Rs. 5 lakhs per family per year for secondary and tertiary care hospitalization to over 42 Lakhs poor and vulnerable families (approximately 64 lakhs beneficiaries) that form the bottom 40% of the Kerala population" (<https://sha.kerala.gov.in/>).

Are the urban population who seek care in UPHCs aware of the health insurance schemes and are they adequately covered by the existing schemes is a question in itself? Studies resonate

evidences of low awareness and low coverage among urban poor in different States in India. Awareness and coverage of ESIC/CGHS/RSBY was low among urban poor in Delhi (Kusuma et.al, 2018). RSBY coverage was known to 29.7 percent of the 6000 HHs included and coverage was 21.6 percent in Maharashtra (Thakur. H, 2016). In Gujarat, out of the 2920 HHs covered in the study, Devadasan et. al, (2013) found that RSBY provided only partial financial coverage to the poor as nearly 60% of insured and admitted patients reporting OOOPE. A comparative study revealed that the mandatory health insurance schemes (ESIS and CGHS) better served the healthcare needs of the beneficiaries as compared with the RSBY (Kusuma et.al, 2018). In this context the authors delve upon the level of awareness about health insurance schemes and its coverage among urban poor in Kerala.

Objectives

The main objectives of the study are:

- To assess the awareness about Health Insurance among the urban poor in Kerala
- To examine the coverage of Health Insurance schemes among the urban poor in Kerala.

Data and Methodology

Data for the present study were taken from the beneficiary exit interviews of the study on “Utilization of Health Care Services in the UPHCs in Kerala”. The study was conducted during 2018-19 by PRC Kerala, with the financial aid of National Urban Health Mission, Arogyakeralam, Govt. of Kerala. The field survey was conducted during December 2018 - March 2019. Kerala has established 83 UPHCs spread over 14 districts. UPHCs were set up to cater to the needs of slum dwellers, unorganized workers including notified and non-notified slums. Beneficiaries of UPHC services, who form the majority, are the urban poor. So, exit interviews from the beneficiaries of services of the UPHC form the sample population. The number of beneficiaries in each UPHC was proportional to the average OPD per day during the study period. So, from the UPHCs the number of beneficiaries varied between 100 to 120. If the care seeker is a child or unmarried person, the parent who accompanied the child was the respondent. Thus, information was captured from a total of 10240 beneficiaries. The response rate was 99 percent as it was carried out as exit interviews at UPHCs after taking informed consent. Beneficiaries willing to take part in the study were only included.

The present study uses a part of the data collected on health insurance from the larger study on utilization of health care services of UPHCs. UPHCs are located mostly in Slums and Coastal areas in the State catering to the needs of poor and to some extent the middle class. The occupation category depicting the employment status showed that the majority of the respondents do not have any income or a steady income being either unemployed or working in unorganized sector or unskilled labourers and hence were included as ‘urban poor’ in the study. Kerala offers a wide basket of choice of health insurance schemes. But the urban poor

form a very vulnerable group exposed to diseases of all kinds but seldom being covered by any health insurance schemes that benefit the poor because of simply being an urban resident. So, such an analysis would definitely be of importance in understanding the coverage of health insurance schemes among the urban poor in order to formulate apt policies for their welfare. Univariate and bivariate classification derives the level of awareness on health insurance schemes and how it varies by background characteristics of the respondents.

Limitations

The study uses data collected as part of a larger study on 'Utilization of health care services of UPHCs in Kerala' which had administered the questionnaire to the mixed population including Urban poor and middle-income class who seek treatment in UPHCs in Kerala. UPHCs are established under NUHM to cater to the needs of slum population, unorganized workers etc. But since majority of the respondents in the study by their occupation did not have any income, or a steady income being either unemployed or working in unorganized sector or unskilled labourers, they were included as the urban poor. This may be considered as one of the limitations in the classification as urban poor.

Ethical Consideration

Institutional approval was taken prior to the survey. Informed consent in English/Malayalam was taken before the interview.

Findings

UPHCs in Kerala are placed so as to reach the urban poor and deliver health care serves. Although some of the UPHCs are located in areas where urban rich people reside, the services of the UPHC are availed mostly by the urban poor and middle-class residents of neighbouring wards. Every UPHC in the State caters to the needs of 4 or more wards depending on the geography of settlements.

Break up by the age of respondents reveal over one in three to be the aged population which is not surprising due to the higher incidence of NCDs among the adult and aged population. An equivalent proportion is in the 40–59-year age group when the onset of NCDs necessitates seeking primary health care. Representation of children and teenagers are relatively less. Gender differentials are evident as females tend to seek care more than males. Currently married group form more than half of the respondents (57.6 percent). Religious wise distribution of the sample population is more or less similar to that of the State with Hindus forming just over half the respondents, 30 percent are Muslims and 17.5 percent are Christians.

Educational background of the respondents as evident from Table 1 points to 12.6 percent of the respondents not having any formal schooling and two-thirds having 1-10 years of schooling, nearly one in five have more than 10 years of education inclusive of a good number of graduates and post graduates. As the UPHCs address the health care needs of the urban poor, it

is mostly the unemployed and who avail services and so we find that 44 percent of the respondents are unemployed. Only 7.8 percent are Government/Private/Retired employees and this group is from the urban rich area where the UPHC is located. One in five respondents are unskilled workers, students form 13.6 percent and 8.3 percent are housewives and only 5.6 percent of the health care seekers are children. Half the respondents are from households with 5 or more members and 13 percent have only up to 2 members.

Table 1 Percentage distribution of respondents by their background characteristics

Characteristics of respondents			Characteristics of respondents		
Percent			Percent		
Age of Respondent (in years)	0-14	15.1 (1544)	Religion	Hindu	51.9 (5316)
	15-29	10.9 (1117)		Muslim	30.6 (3125)
	30-39	10.0 (1024)		Christian	17.5 (1795)
	40-49	12.9 (1319)	Education of Respondent	No formal Schooling	12.6 (1288)
	50-59	18.3 (1878)		1-10 years	68.6 (7029)
	60+	32.8 (3358)		>10 years	18.8 (1923)
Gender of respondent	Male	37.2 (3809)	Employment status	Unemployed	44.0 (4504)
	Female	62.8 (6431)		Govt/Pvt Job/Retired	7.8 (799)
Marital Status	Never Married	7.0 (714)		Unskilled workers	20.7 (2124)
	Currently Married	57.6 (5895)		Student	13.6 (1389)
	Divorced\ Separated\ Widowed	18.1 (1856)		Housewife	8.3 (847)
	NA*	17.3 (1775)		Other	5.6 (577)
Household Size	<=2 members	13.0 (1333)			
	3 -4 members	36.9 (3782)			
	>= 5 members	50.0 (5125)			
Total No. of respondents		10240	Total No. of respondents		10240

* NA inclusion: Children

Awareness about health insurance schemes

Awareness of the respondents regarding health insurance is appreciable. More than 98 percent of the respondents under study have heard about Health Insurance Scheme which is a very promising note.

The study included a list of health insurance schemes that are accessible to the urban population including the poor and middle class as evident from the enrolment statistics available from either research studies or respective websites of the schemes. Those who had heard about the health insurance schemes were further asked whether they knew about different types of schemes. The result obtained shows that the most popularly known health insurance scheme among the respondents is RSBY-CHIS scheme. Karunya Benevolent Fund Scheme is the next popularly known one (96.6 percent). About 91 percent of the respondents knew about Employment State Insurance Scheme and 81 percent knew about Central Government Health Scheme. The new initiative of Govt. of India, 'Ayushman Bharath' is known to about 61 percent of the respondents. Other Insurance Schemes in the Public Sector are

known to 29.6 percent like the insurance schemes offered by the Labour Welfare Board and other insurance schemes of public sector insurance companies. Some Private Sector Insurance Schemes are also known to 12 percent of the respondents like the Star Insurance and Royal Sundaram Insurance.

Table 2 Percentage distribution of respondents by awareness about health insurance schemes

Heard about health insurance schemes	Percentage
Yes	98.2 (10052)
No	1.8 (186)
Total	10238
Awareness of different types of health insurance	
RSBY- CHIS	99.6 (10039)
Karunya Benevolent Fund Scheme	96.6 (9709)
Employment State Insurance Scheme	91.2 (9164)
Central Government Health Scheme	81.0 (8138)
Ayushman Bharath Health Insurance Scheme	60.9 (6039)
Other Insurance Scheme in the Public Sector	29.6 (2973)
Other Insurance Scheme in the Private Sector	12.0 (1211)
Other Schemes	3.1 (310)

* Multiple response as one person is enrolled in more than one health insurance scheme

Table 3 District wise percentage distribution of respondents by awareness about health insurance schemes

S. No.	District	Awareness about Health Insurance Schemes
1	Thiruvananthapuram	98.2
2	Kollam	99.2
3	Alappuzha	100
4	Pathanamthitta	100
5	Kottayam	100
6	Ernakulam	99.0
7	Idukki	99.2
8	Thrissur	99.3
9	Palakkad	99.8
10	Wayanad	100
11	Malappuram	91.8
12	Kozhikode	99.8
13	Kannur	99.8
14	Kasaragod	100
	KERALA	98.2 (10052)

Coverage of Health Insurance Schemes

Even though the knowledge of Health Insurance Schemes is very high among the respondents, it can be seen that only a little above half (53.4 percent) of the respondents have enrolled in any of the Health Insurance Schemes. So, a substantial proportion of the urban poor are still outside the coverage of health insurance benefits, which is really a matter of concern.

Table 4 Coverage of health insurance schemes

Enrolled in any Health Insurance Scheme	
Yes	53.4 (5365)
No	46.6 (4687)
Total	10052
Type of Health Insurance enrolled*	
RSBY- CHIS	86.6 (4648)
Karunya Benevolent Fund Scheme	0.9 (48)
Employment State Insurance Scheme	3.2 (174)
Central Government Health Scheme	0.6 (33)
Ayushman Bharath Health Insurance Scheme	1.0 (54)
Other Insurance Scheme in the Public Sector	1.2 (64)
Other Insurance Scheme in the Private Sector	4.9 (262)
Don't Know	1.4 (77)
Total	5365
Difficulty in enrollment	
Yes	1.0 (54)
No	99.0 (5311)
Total	5365
Renewal of Health Insurance Scheme	
Yes	99.1 (5316)
No	0.9 (49)
Total	5365

** Multiple response as one person is enrolled in more than one health insurance scheme*

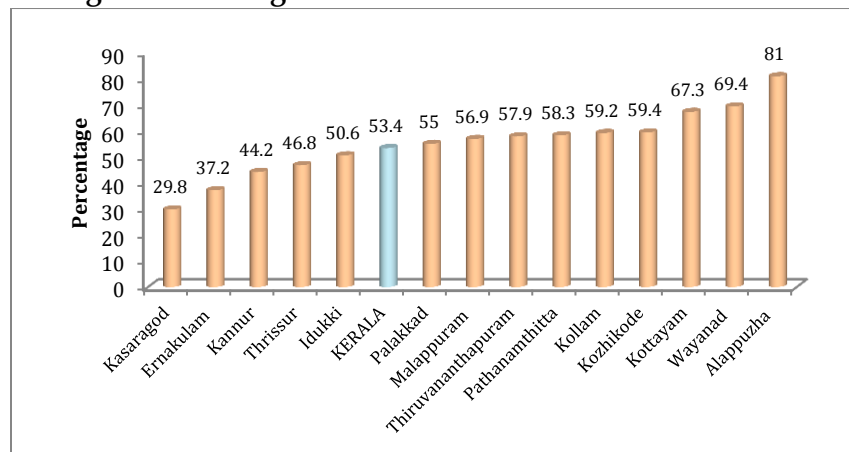
Analysis by the type of scheme in which the respondent is enrolled revealed that majority (86.6 percent) of the urban poor with a health insurance policy are enrolled in RSBY-CHIS scheme. Only about 3 percent of the respondents are enrolled in Employment State Insurance Scheme and only 0.6 percent in Central Government Health Scheme as majority of the respondents are from poor background and do not have the eligibility conditions to enrol in the schemes. Enrolment in Ayushman Bharath Health Insurance Scheme is only one percent and it may be due to the fact that the implementation of the new scheme was only started in Kerala at the time of our data collection. Enrolment in Karunya Benevolent Fund (KBF) Scheme is not like other health insurance schemes. It is a trust model implemented through Lottery department for helping the poor patients using the profit earned through 'Karunya Lottery'. Only the eligible patients under treatment can apply for the grant which keeps enrolment in KBF is very small. Share of 'Other Insurance Scheme in the Public Sector' is 1.2 percent. The share of Other Insurance Scheme in the Private Sector is only 4.9 percent. It may be due to the fact that the respondents, majority of whom belong to urban marginalised sector, may not be in a financial situation to afford a private health insurance scheme as the premium amount is quite high compared to Government schemes.

Table 5 District wise percentage distribution of respondents enrolled in any health insurance scheme

S. No.	District	Enrolled in any Health Insurance Scheme
1	Thiruvananthapuram	57.9
2	Kollam	59.2
3	Alappuzha	81.0
4	Pathanamthitta	58.3
5	Kottayam	67.3
6	Ernakulam	37.2
7	Idukki	50.6
8	Thrissur	46.8
9	Palakkad	55.0
10	Wayanad	69.4
11	Malappuram	56.9
12	Kozhikode	59.4
13	Kannur	44.2
14	Kasaragod	29.8
	KERALA	53.4 (5365)

District wise distribution point to enrolment being highest in Alappuzha (81.0 percent) followed by Wayanad (69.4 percent). Coverage is poor in Kasaragod (29.8 percent) and Ernakulam (37.2 percent) compared to other districts. Dearth of studies among urban poor on health insurance coverage limits the scope of comparison of these findings. Rural urban differentials in coverage of health insurance scheme in Kerala as pointed out by the findings of LASI (2017-18) revealed 46 percent of rural older adults 45 years and above and 34.5 percent urban counterparts to be covered, but still evidence on coverage among urban poor at district level are rare. So, the reasons for the wide inter district differentials in coverage of health insurance schemes among urban poor needs further extensive analysis.

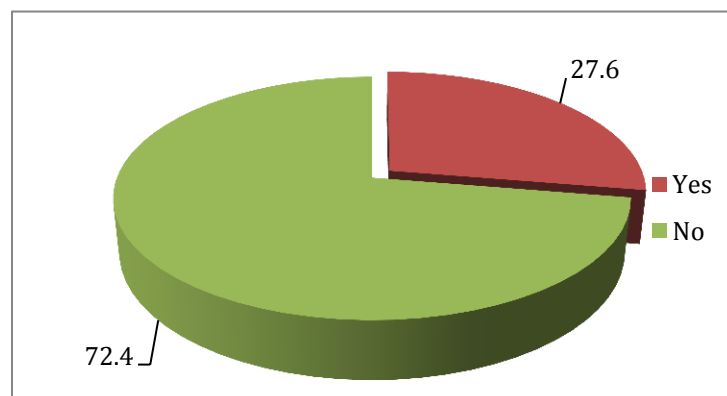
Figure 1 Coverage of health insurance schemes in Kerala



Utilization of benefits of health insurance schemes

An assessment of the extent of coverage of claims under insurance schemes for treatment revealed more than one quarter (27.6 percent) of the 5365 respondents who possess Health Insurance policy to have utilised the insurance scheme for treatment of diseases.

Figure 2 Utilization of health insurance scheme for treatment



Almost all health insurance schemes stipulate hospitalization as a mandatory condition for claims towards treatment. During the two years preceding the survey, hospitalization for treatment of diseases was required among 36.2 percent of the respondents out of the 10238 respondents who participated in the survey.

Table 6 Percentage distribution of respondents requiring hospitalization during the two years preceding the survey and coverage of health insurance scheme

Hospitalization	
Yes	36.2 (3704)
No	63.8 (6532)
	10236
Place of Treatment	
Government hospital	50.6 (1866)
Private hospital	49.4 (1836)
	3702
Enrolment in Insurance Scheme among those who required hospitalization	
In Govt. Hospital	59.3 (1106)
In Pvt. Hospital	40.7 (747)

Differentials by place of treatment are not significant as patients chose Government and Private hospital in almost similar proportion. However out of those who required hospitalization only 59.3 percent had enrolled in any insurance out of which 59 percent availed treatment in Govt. Hospitals and 41 percent in Private hospitals. Among the various diseases for which the respondents received insurance benefits for treatment during 2 years preceding the survey, Fever and Fits associated with fever (16.2 percent) was the most reported disease as the present-day situation of communicable diseases reported are mostly dengue, chikunguniya, H1N1, viral infections etc which require inpatient care in hospitals. 'Fits' reported by the respondents, are associated with high fever due to different types of communicable diseases.

Table 7 Insurance benefit received for treatment during two years preceding the survey by type of disease

Diseases	Percentage*
Thyroid treatment/surgery	1.8 (17)
Heart ailment	15.6 (149)
ENT surgery	8.9 (85)
Hip replacement	0.4 (4)
Urinary problems	4.1 (39)
Joint pain	3.1 (30)
Cancer	3.6 (34)
Stroke	1.9 (18)
Pains	0.6 (6)
Respiratory problems	8.2 (78)
Delivery	7.1 (68)
Fever and Fits	16.2 (155)
Stomach problem/Jaundice/Liver problems/	4.2 (40)
Hernia surgery	2.6 (25)
Accident-related treatments	9.0 (86)
BP/Diabetes related treatment	8.9 (85)
Tumour surgery	3.5 (33)
Total Number of Respondent who received Insurance benefits	955

* Multiple response

Heart ailments was the next common disease for which the benefit of enrolment in health insurance scheme helped 15.6 percent of the respondents. It was for ENT surgery that about 8.9 percent of the respondents utilized the schemes. Around 7 percent of the respondents utilised the scheme for delivery care and accident-related treatments formed 9 percent of all diseases. Treatment of problems related to BP/Diabetes (8.9 percent) which required hospitalization were covered under insurance schemes. Treatment/surgery for stomach problem/jaundice/liver problems, Cancer, tumour surgery, urinary problems, joint pain, hernia surgery and thyroid were the other diseases for which the health insurance policy was utilised. Respondents who received benefits from health insurance policy were further asked about the purposes for which the insurance amount has been utilised.

Table 8 Purpose for which Insurance scheme was utilized

Purpose for which Insurance scheme was utilized	Percentage*
Medicines	92.3 (881)
Lab Tests	81.7 (780)
Hospital inpatient bed charge	70.4 (672)
Doctor's Fee	54.5 (520)
Surgery charge	38.1 (364)
Food	8.9 (85)
Consumables	1.2 (11)
Other services	0.4 (4)
Total Number of Respondent who claimed Insurance benefits	955

* Multiple response

Majority of the respondents received the coverage amount for medicines (92.3 percent), followed by charges of Lab tests (81.7 percent) and Hospital inpatient bed charge (70.4 percent).

About 55 percent of the respondents could meet the Doctor's Fee and 38.1 percent met Surgery charges through the health insurance payments. District wise data of the respondents who received benefits from the health insurance schemes (Figure 4) shows utilisation of health insurance scheme is 40 percent or higher only in two districts namely Alappuzha and Pathanamthitta. Utilisation of health Insurance is below 20 percent in Kozhikode, Palakkad, Wayanad, Kannur and Kasaragod. Percentage of respondents who utilised health insurance for treatment is highest in Alappuzha (48.1 percent), followed by Pathanamthitta (40.0 percent), Idukki (38.2 percent) and Kottayam (38.0 percent). Utilisation of health Insurance is least in Kasaragod (1.4 percent), which is very low compared to other districts. Kannur (6.0 percent) and Wayanad (9.5 percent) also show lower utilization.

Figure 3 District wise distribution of utilization of health insurance schemes for treatment

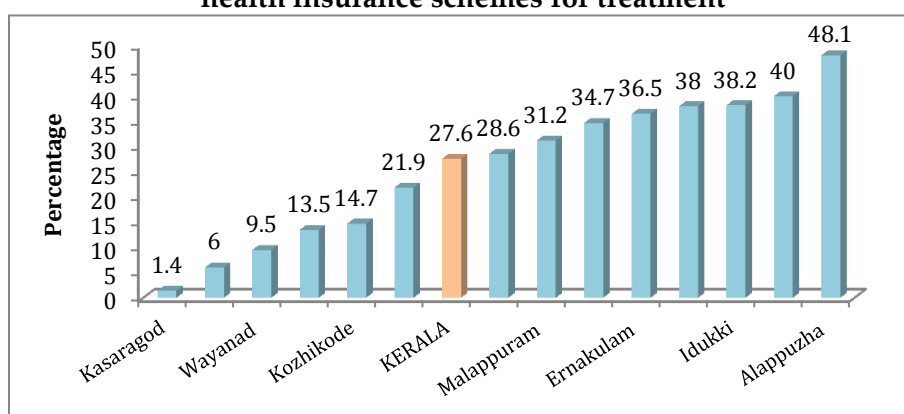


Table 9 District wise Percentage distribution of respondents who received benefits of health insurance schemes by heads cover

District	Insurance coverage received towards					
	Hospital inpatient bed charge	Surgery	Medicines	Lab tests	Doctor's Fee	Consumables
Thiruvananthapuram	82.9	44.5	96.7	95.0	79.3	1.3
Kollam	53.4	48.5	88.3	54.4	26.2	0.0
Alappuzha	40.4	19.7	86.7	69.7	33.0	0.5
Pathanamthitta	53.6	10.7	67.9	75.0	32.1	3.6
Kottayam	53.8	37.6	94.6	86.0	47.3	1.1
Ernakulam	71.3	34.7	92.0	77.3	66.1	1.6
Idukki	55.3	31.9	95.7	87.2	55.3	0.0
Thrissur	63.6	45.5	92.7	83.6	74.5	5.5
Palakkad	95.6	33.3	97.8	88.9	95.6	0.0
Wayanad	100.0	37.5	100.0	100.0	100.0	0.0
Malappuram	92.9	35.7	88.2	77.7	42.9	0.8
Kozhikode	64.8	55.6	98.1	97.2	13.9	0.0
Kannur	87.5	68.8	93.8	87.5	37.5	0.0
Kasaragod	100.0	0.0	100.0	100.0	100.0	0.0
Total Number	672	364	881	780	520	11

The study does not intend to make a district wise comparison on the insurance benefit extended on different heads. As claims and benefits received varies by disease and beneficiary needs. For instance, Kasaragod has the least coverage of insurance schemes and all those who utilized the insurance scheme for treatment received benefit on all the heads considered here and no one needed surgery as part of treatment. Alappuzha district had reported higher utilization but the type of diseases for which the insurance benefits were claimed and the type of insurance enrolled dictates the claims received for different heads. In general, variations are discernible with insurance benefit covering medicine cost, lab tests and hospital in-patient bed charge being more common. Most of the insurance schemes do not cover the cost of consumables, have only partial coverage of surgery and doctors' fee.

Expenditure on Hospitalization (Out of pocket expenditure)

The cost incurred on treatment of various diseases during hospitalization inspite of holding an insurance card points to partial coverage of the insurance schemes. Two in five patients who required hospitalization for treatment was not enrolled in any health insurance scheme. Among those with insurance membership, 41.5 percent did not incur any expenditure for treatment, one in three spent less than Rs. 10000/-, 10 percent incurred cost between Rs. 10000/- and Rs. 25000/- and 16.3 spent more than 25 thousand. Out of those who availed and received benefits from the health insurance scheme, two out of three patients were able to meet all treatment costs within the assured amount they were eligible. Since all items are not covered in all the schemes one in five patients spent less than Rs, 10000/- and 11.2 percent had to spent more than a quarter lakh rupee in spite of claiming insurance benefit.

Out of the total respondents who had membership in any health insurance schemes, only 44 percent (955) had used their insurance for treatment within the two-year preceding the survey. The diseases for which they claimed benefit was more for treatment of fever and also heart ailments. But for diseases that required major surgeries like hip replacement three out of the four patients who underwent surgery met expenses on their own.

Three patients who required treatment of liver problems incurred no cost towards treatment. For treatment of diseases like thyroid, jaundice, BP/diabetes that required hospitalization, major share of the expenses was covered under the insurance scheme or expenditure less than Rs. 10000/- were reported. As we found out of pocket expenditure in spite of holding an insurance card, an inquiry was made to all the respondents on their requirements on coverage of insurance schemes which are not available now.

Table 10 Expenditure on treatment of diseases

		Expenditure during Hospitalization for treatment (during 2 years preceding the survey)				
		0 (No expenditure)	Rs.100 to Rs.10000	Rs.10001 to Rs.25000	> Rs.25000	Total
Insurance Membership	Yes	41.5 (898)	32.1 (695)	10.0 (216)	16.3 (353)	59.3(2162)
	No	12.0 (178)	47.0 (696)	14.1 (209)	26.9 (399)	40.7(1482)
Out of pocket expenditure of Insurance Beneficiaries on hospitalization						
Disease Condition						
Thyroid		76.5	17.6	0	5.9	17
Heart ailment		67.8	14.1	5.4	12.8	149
ENT operation		60.0	17.6	7.1	15.3	85
Hip replacement		25.0	0	0	75.0	4
Urinary Problem		61.5	20.5	2.6	15.4	39
Joint Pain		60.0	16.7	10.0	13.3	30
Cancer		52.9	20.6	2.9	23.5	34
Stroke		50.0	16.7	11.1	22.2	18
Body pain		66.7	16.7	0	16.7	6
Respiratory problem		69.2	17.9	7.7	5.1	78
Fits		50.0	25.0	25.0	0	4
Delivery		42.6	44.1	7.4	5.9	68
Fever		62.3	24.5	6.0	7.3	151
Cough & Cold		66.7	33.3	0	0	3
Stomach problem		69.2	19.2	3.8	7.7	26
Jaundice		81.8	18.2	0	0	11
Hernia		72.0	16.0	0	12.0	25
Accident		51.2	24.4	7.0	17.4	86
BP/Diabetes		72.9	21.2	0	5.9	85
Liver problem		100	0	0	0	3
Tumour		54.5	21.2	12.1	12.1	33
No. of beneficiaries who claimed insurance benefits		62.0(592)	21.3(203)	5.5(53)	11.2(107)	100.0(955)

Table 11 Requirements of beneficiaries on coverage of health insurance schemes

Beneficiary requirements on coverage of Health Insurance Schemes	
More Coverage for children	92.0 (1206)
Include treatment in private Hospitals	81.5 (1206)
Include Lab tests done in other labs, Medicines purchased from outside Pharmacy	53.2 (787)
Include OP charges	37.8 (559)
Increase total claim amount	8.5 (126)
Cover all members of family	2.0 (201)
Total	10238

* Multiple response

The greatest demand among majority of the respondents was 'more coverage for children' (92.0 percent). About 82 percent were of the opinion that treatment taken from private hospitals also should be covered under the RSBY- CHIS scheme. More than half of the respondents were of the opinion that lab tests done at private labs should also be included under the coverage of the scheme and 37.8 percent demanded that OP treatment also should be covered under the scheme. Increase in the total claim amount was claimed by 8.5 percent of the beneficiaries. Some

of the beneficiaries (2.0 percent) also wanted the RSBY-CHIS scheme to cover all members of the family irrespective of number of members.

With regard to renewal of insurance schemes, only very few respondents (0.9 percent) failed to renew the insurance on time due to which they lost their chance to claim benefit for treatment.

Discussion and Conclusion

One of the top priorities of universal health coverage is minimizing the out-of-pocket expenditure on health care services. Health insurance schemes play an important role in this regard and hence Central and State Governments in India have rolled out many health insurances schemes for universal coverage. The awareness and coverage of health insurance scheme among urban poor is a challenge in itself in achieving this goal. The findings of the present study among the urban poor who seek treatment in UPHCs in Kerala which is their nearest point of health care shows awareness to be widespread but coverage to be less. Since RSBY entails enrolment of member of poor households, majority of the respondents who have a health insurance coverage are enrolled in RSBY. We found a coverage of any health insurance scheme only among 53 percent of the urban poor. Similar findings of higher awareness but lower coverage was observed by Swetha and Jose (2020). Coverage of 74 percent was found among the general population in a cross-sectional study focusing on those seeking care in a public sector hospital in Kerala (Harish et. al, 2020). Also, national level data sets reveal (LASI 2020) only 21 percent of older adults in age group 45 and above are covered under health insurance in India and the coverage in Kerala is 40.4 percent which increases the possibility of out-of-pocket expenditure for health care.

Among the urban poor enrolled in health insurance, 45 percent expressed dissatisfaction regarding available services. District wise differentials are evident in coverage with Alappuzha district showing highest enrolment and Kasaragod and Ernakulam the least. Further large sample studies on coverage are essential as districts like Ernakulam which is more urbanized and with a larger proportion of urban poor has lower coverage of health insurance schemes.

Utilization of the health insurance schemes is found to be less as only little over quarter proportion of respondents reported using the schemes for coverage of treatment cost. Only three in five respondents hospitalized had been enrolled in any scheme. The findings point to larger proportion insurance claims being made for heart ailments and treatment of other NCDs. But the schemes have not helped the respondents to cover all their expenditure. One reason for OOOPE among this group of urban poor is due to their dependence on private hospitals in their vicinity as nearly half the respondents report seeking treatment in private sector. From the response on various claims made for treatment, lack of coverage of all surgery expenditure, inpatient bed charges, Doctor's fee, lab tests and consumables are observed. These observations are validated from the suggestions of respondents on their further requirements for coverage. The greatest demand among majority of the respondents were 'more coverage for children',

‘treatment taken from private hospitals also should be covered under the RSBY- CHIS scheme’, ‘lab tests done at private labs should be included’ and ‘OP treatment also should be covered under the scheme’. Other studies too echo the difficulties in claiming the benefits. A study on CHIS scheme in Kerala pointed to the lack of knowledge regarding the list of empanelled hospitals/benefit package, referral from an empanelled hospital to a non-empanelled hospital and refusal from the empanelled hospital to accept the card, as the main reasons for not making use of the card (Neena et. al. 2017).

Kerala’s urban population is poised to increase to 71 percent of the total population of the State as per the latest projections (RGI, 2019) which is a jump from 47 percent within ten years. So, problems associated with increasing urbanization are also poised to increase and the health care needs have to be better taken care of. Majority of the respondents are enrolled in RSBY. Better coverage of RSBY is required in this context which calls for inclusion of more private hospitals in the empanelled list of hospitals. Similar findings are echoed in the study by Singh et. al (2017) in a study in three cities in India. As evident from the findings of our study, the respondents who seek care in UPHCs also need to be sensitized on the empanelled private hospitals so that they seek treatment at such hospitals to reduce the OOPe because it is usually the urban poor who lack detailed information on the conditions and process of insurance claims. The suggestions from the respondents in the present study point to this aspect of ignorance on assessing whether the scheme provides benefits in the private hospital they seek treatment or not. The study findings point to lower coverage in insurance scheme although awareness on health insurance coverage was high which means awareness does not lead to subscription to an insurance plan. Further in depth research on coverage of insurance schemes is necessary as Kerala has rolled out the KASP scheme in line with the AB PMJAY. As the study findings show there is still a large group of uninsured urban residents who seek care at UPHCs in the State who bear the treatment cost on their own, creating awareness on the benefits of the schemes for increasing coverage and steps to facilitate the poor in enrolment may be prioritized. It is expected that with the integration of AB PMJAY with the existing insurance schemes in the State, the coverage would increase in the coming years.

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15

Health Management Information System: A Journey Towards Public Health Data Ecosystem in India

Ugra Mohan Jha and Ritu Arora

Abstract

Health information systems serve multiple users and a wide array of purposes that can be summarized as the generation of information to enable decision-makers at all levels of the health system to identify problems and make evidence-based decisions on health policy and allocate resources optimally. Prior to 2008, there was no uniform MIS portal to cater the need for Ministry of Health and Family Welfare in India. There were undue delays in receipt of data, retrieval of old manual records was ineffective and time consuming, duplication of records was a setback to correct understanding, Monthly reports sent as hard copy was a challenge for data consolidation and analysis. Health Management Information System (HMIS) was thus developed in 2008 to facilitate better healthcare management system, as well as to integrate healthcare management at the various level such as village, Sub-district district and State levels. The Journey of HMIS started in 2008 with initial objective to upload district level consolidated figures on monthly basis, subsequently facility-based reporting was initiated in 2011-12. Due to old technology (Hardware and Software), inter-operability issues, non-availability of web API, programmatic need and lack of decision support system, revamping of HMIS was initiated in 2019 and HMIS 2.0 was launched in 2020.

Introduction

Sound and reliable information is the foundation of decision-making across all health system and is essential for health system policy development and implementation, governance and regulation, health research, human resources development, health education and training, service delivery and financing. A good Management Information System (MIS) has many roles to perform like the decision support role, the performance monitoring role and the functional support role. In today's world most agile and efficient data driven decision support system is needed by every organization be it Government or Private.

A good health information system brings together all relevant partners to ensure that users of health information have access to reliable, authoritative, useable, understandable, comparative data. Health information systems serve multiple users and a wide array of purposes that can be summarized as the generation of information to enable decision-makers at all levels of the health system to identify problems and needs, make evidence-based decisions on health policy and allocate scarce resources optimally (WHO, 2008). Data from different sources are used for multiple purposes at different levels of the health care system.

Prior to 2008, there was no uniform MIS portal to cater the need for Ministry of Health and Family Welfare. There was undue delays in receipt of data, Retrieval of old manual records was ineffective and time consuming, duplication of records was a set-back to correct understanding, Monthly reports sent as hard copy was a challenge for data consolidation and analysis.

In such a scenario, there was clearly a pressing need to provide tools and solutions that would improve healthcare management in the country. The only solution was to introduce a digital platform. World Health Organisation (WHO) identified Health Management Information System (HMIS) as a key building block of health system (WHO, 2007). In India, the emphasis has been laid on creation and utilization of an effective HMIS system under the National Rural Health Mission (NRHM), flagship healthcare program under the Ministry of Health and Family Welfare, Government of India.

National Health Mission (NHM) is a flagship programme of the Government of India to address the health needs of under-served rural areas and health concerns of the urban poor population. The National Health Mission (NHM) encompasses its two Sub-Missions, the National Rural Health Mission (NRHM) and the newly launched National Urban Health Mission (NUHM). The main programmatic components include Health System Strengthening in rural and urban areas- Reproductive-Maternal- Neonatal-Child and Adolescent Health (RMNCH+A), and Communicable and Non-Communicable Diseases. The NHM envisages achievement of universal access to equitable, affordable & quality health care services that are accountable and responsive to people's needs (NHM, 2005).

Health Management Information System (HMIS) was thus launched in 2008 to facilitate better healthcare management system as well as to integrate healthcare management at the various level such as village, Sub-district district and State levels.

Objective

Following are two primary objectives of the study:

- To document the journey of Health Management Information System (HMIS) as a monitoring tool for health programmes in India.
- To understand the challenges faced by HMIS towards ensuring data quality.

Methods

The study is descriptive research in which secondary sources of information available for design, development and implementation of HMIS by the Ministry of Health and Family Welfare in India available in Annual Reports 2020-21 and HMIS 2019-20- An Analytical Reports were used. Also, information observed during the field visits undertaken by the authors in different States in India during 2019 to 2021 were used. The HMIS information was critically analyzed and based on the data synthesis, results were provided.

Results

The Health Management Information System (HMIS) portal facilitates the flow of physical performance from the facility level to the Sub-district, District, State and National level using a web-based Health Management Information System (HMIS) interface. The portal provides periodic reports on the status of the health services performances and Human Resources and Infrastructure services facilities available.

Health facility level data enable health care managers to determine resource needs, guide purchasing decisions for drugs, equipment and supplies, and develop community outreach. The health facilities mapped on HMIS are Sub-Centres (SCs), Primary Health Centres (PHCs), Community Health Centres (CHCs), Sub-District Hospital (SDH) and District Hospitals (DHs). Medical Colleges are mapped at SDH and DH level in respective States/UTs. As on 31st August, 2021, there were 726 districts and 2.17 lakh health facilities mapped in HMIS Portal including 341 Medical Colleges (Mapped as SDH & DH in HMIS) and around 75 thousand Health and Wellness Centres (Mapped as SC & PHC in HMIS) reporting through HMIS (HMIS, 2020).

HMIS captures over 300 plus data items under Service Delivery and over 400 plus data items under Infrastructures and Human Resources on monthly basis. Service Delivery data items includes Reproductive, Maternal and Child Health, Immunization, Family Planning, Vector borne disease, Tuberculosis, Morbidity and Mortality, OPD, IPD Services, Surgeries etc. and Infrastructure and Human Resources includes Human Resources, Equipment, Cleanliness, Building, Availability of Medical Services such as Surgery etc., Super Specialties services such as Cardiology etc., Diagnostics, Para Medical and Clinical Services etc.

The Journey

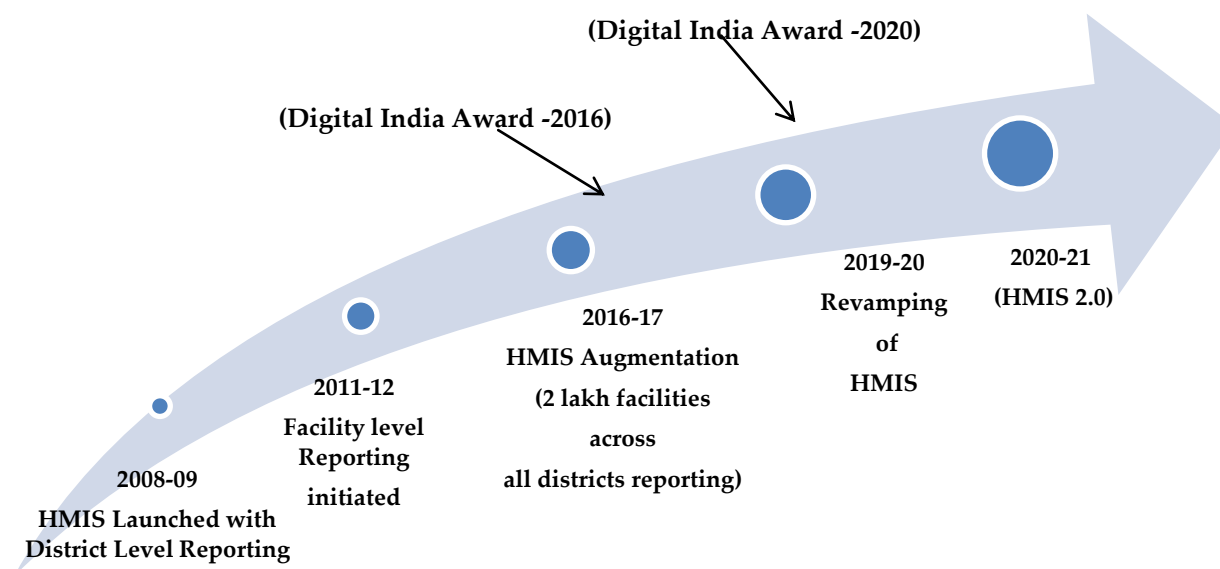
In 2008, HMIS was developed with initial objective to upload district level consolidated figures on monthly basis for NHM programme monitoring. Subsequently from 2011-12, the facility-based reporting was initiated and in 2016-17. The revamping and development of new HMIS (HMIS.20) was initiated during 2019-20 and finally in December 2020, HMIS 2.0 was thus launched. The details of journey of HMIS in India is provided in Figure 1.

Due to proactive, timely and regular release of datasets/resources through Web Services/APIs on the Open Government Data (OGD) Platform (<https://data.gov.in>) in compliance with the National Data Sharing and Accessibility Policy (NDSAP), Ministry of Health and Family Welfare received Digital India award in 2016 and 2020 under Open Data Champion category (DIA, 2020). This award was received by the ministry primarily due to Health Management Information System (HMIS), timely and regular release of datasets in Open Government Data (OGD) Platform.

The development of a new HMIS (HMIS 2.0) was felt need as old HMIS (HMIS 1.0) due to its old technology (Hardware and Software), inter-operability issues, non-availability of web API,

programmatic need and lack of decision support system. Therefore, keeping in view the need to revamp the old HMIS (HMIS 1.0) application and also ministry's view of bringing of all health information system at a single Platform; it was decided to get a new HMIS (HMIS 2.0) portal on Integrated Health Information Platform (IHIP).

Figure 1 Journey of HMIS in India



The primary objective of IHIP was to enable the creation of standards compliant Electronic Health Records (EHRs) of the citizens on all India basis along with the integration and interoperability of the EHRs through a comprehensive Health Information Exchange (HIE) as part of this centralized accessible platform. Integrated Health Information Platform (IHIP) of MoHFW has provision for capturing data at facility and beneficiary level (IHIP, 2017). The integration of HMIS online portal with IHIP ensure full use of all applicable government standards, including standardized facility listing, geographic and entity names, all geographic boundaries (up to the level of villages), population data and other relevant information.

The new HMIS (HMIS 2.0) was developed with technical support from the World Health Organization (WHO). This was developed after series of discussion/deliberation with the software developers of the WHO.

The key features of new HMIS (HMIS 2.0) portal are following:

- Person Specific User Credentials (mapped to an Individual) for fixing responsibility to user for entering complete and correct data in the system.
- Compatibility with multiple devices such as Mobile, Tab etc.
- Flexible: Multiple facilities to one credentials, one facility to one credential

- National Identification Number (NIN) and Local Government Directory (LGD) have been used for unique identification of health facilities at village/block levels.
- State specific administrative hierarchy as per LGD has been implemented and States can tag their health facilities and generate report accordingly.
- Provision for real time data entry and real time monitoring, alerts, Analytics GIS Integration with layers up to road, village boundary etc.
- Data Visualization through GIS and Interactive Dashboard
- API (Application Program Interface) developed to ensure direct linkage of HMIS online portal with other programme/software application of the ministry to enable automated extraction of aggregate data to HMIS.
- State specific generic API for fetching data from State MIS portal and
- Interactive Dashboards and GIS

The new HMIS (HMIS 2.0) system developed is an excellent example of Re-engineering of Digital Transformation under Government of India which will be able to reduce the work-load on the frontline workers at facility level by integrating multiple health applications/software under single platform. HMIS 2.0 architecture is based on e-Governance Data and Information Standards (UID, NIN, ICD-10/11, LG Code etc.). It has ability to connect with My Health Record, e-Hospital Systems, AB-PMJAY etc. It has ability to connect with public and private hospitals, laboratories, and to exchange health data. There are 16 cloud servers procured from BSNL for HMIS 2.0. The Cloud servers are distributed among States as per requirements for efficient data management.

Quality Assurance

Data from health facilities can provide immediate and ongoing information relevant to public health decision-making. The data must be of high quality, relate to all facilities (public and private), and be representative of the services available to the population as a whole. Out of a total 2.17 lakh health facilities reporting in the HMIS, public health facilities are predominant though around 12 thousand private health facilities also participate in the process.

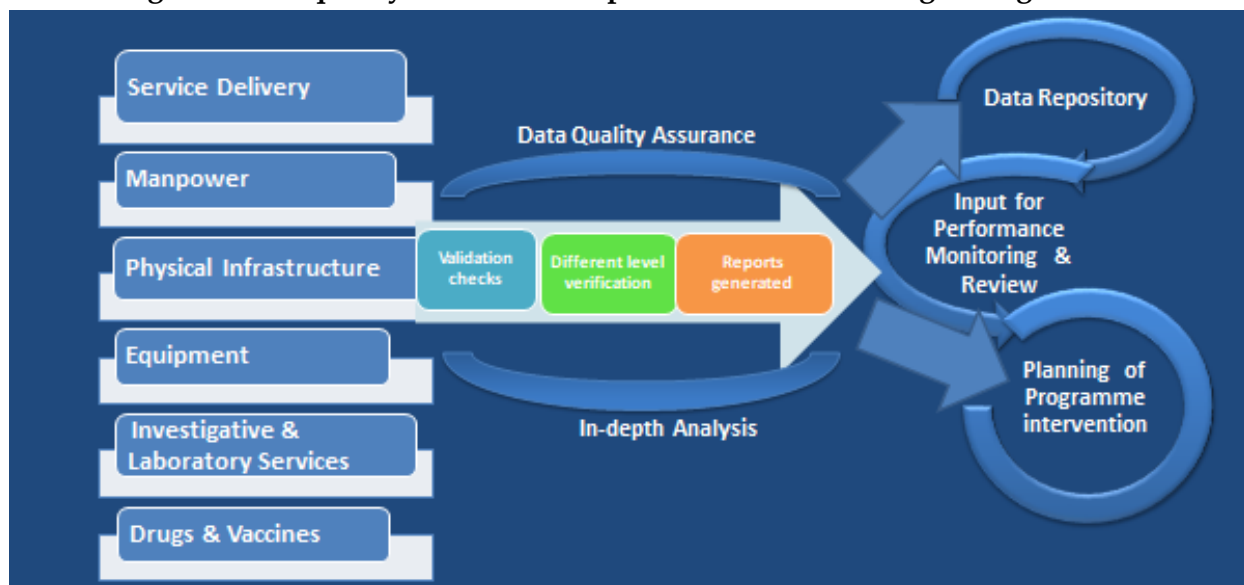
Data quality of statistics is measured in the following dimensions: (i) relevance, (ii) accuracy (iii) timeliness and punctuality, (iv) accessibility and clarity, (v) comparability and (vi) Coherence. However, all these criteria do not apply to the administrative data.

Administrative data quality dimensions can be described by (i) Technical Checks: Technical usability of the file and the data in the file (iii) Accuracy: The extent to which data are correct reliable and certified (iv) Completeness: Degree to which a data source includes data describing

the corresponding set of real-world objects and variables (v) Time related Dimension: Indicators that are time or stability related.

HMIS data is extensively validated at different levels before it is used for performance monitoring of National Health Mission and other National Programmes. The details of Data quality assurance mechanism adopted and in-depth analysis done for performance Monitoring is provided at Figure 2.

Figure 2 Data quality assurance and performance monitoring through HMIS



Accuracy is defined as the extent to which data are correct, reliable and certified. In HMIS all these checks are built-in to see the accuracy of data.

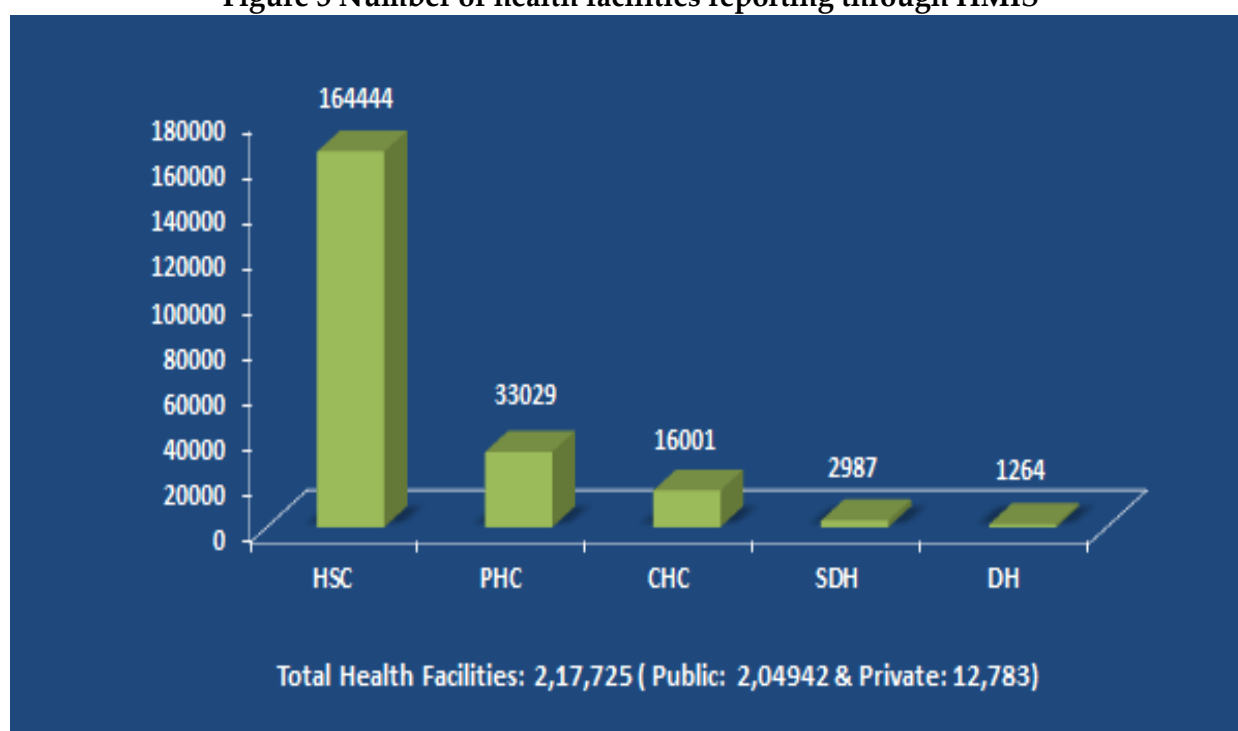
- a. Compare option
 - b. Inter-data validation checks
 - c. Random Checks of Registers
- a. **Compare option:** Compare option compare data with previous month compared. Data fields are highlighted as per comparison status. This gives instant view to data entry operator for any data quality/data entry related issues.
 - b. **Inter-data validation checks:** While saving data, as per prior inbuilt validation rules. There are two types of error shown on windows, which include warning and validation error. This informs the data entry operator for data quality issues (if any) and accordingly correction can be made. e.g., Number of registered women within 1st trimester (within 12 weeks) cannot be greater than Total number of pregnant women registered for ANC during the reporting month is example of validation error and

- c. **Random Check of Registers:** Random checks of HMIS data in the registers at Facility Level is undertaken during field visits. Supportive supervision visits undertaken by the Staff at the various levels to verify HMIS data. During Common Review Mission of National Health Mission also, HMIS data is extensively used and verified. Population Research Centre (PRCs) are also involved in the data verification exercise of the HMIS data.

Scale

There are 36 States/UT's, 726 Districts, 6858 Sub-districts, over 2 lakh health facilities, more than 3 Terabytes of database size (approx.) and around 1500-2000 concurrent users. Number of health facilities reporting through HMIS (HMIS, 2020) is provided at Figure 3.

Figure 3 Number of health facilities reporting through HMIS



Challenges of Data Quality

During field visits undertaken by the authors to different health facilities in India in 2019-21, it has been observed that there are challenges in ensuring data qualities in HMIS. The error could mainly happen due to misreporting from field or data entry error at the time of punching of data into the system. The most important people in the data flow chain at health facilities level and Data entry level for HMIS are the data entry operators and ANMs, At the Supervisory level, Block Data Manager, District Data Manager and State Data Manager. But due to frequent changes of the data entry operators and lack of understanding of data definition, adequate supervisory checks at the field's level and lack on digitalisation of registers are some of the challenges of the HMIS.

Discussion

The journey of HMIS from 2008 to 2021 has been quite remarkable with district level consolidated data reporting in 2008 to Facilities based reporting in 2016-17 and new revamped HMIS implementation in 2020-21. Health Management Information System (HMIS) is one of the important components of National Rural Health Mission (NRHM)(Dehury, 2016). Apart from Monitoring and Evaluation of National Health mission, HMIS data is extensively used for performance evaluation of States/Union Territories (UTs) for Annual Plan of States/UTs. It is used in monitoring and supervision visits by officer of Central, State and district to health facilities. It is also used in annual Common Review Mission of States/UTs by Ministry of Health & Family Welfare.

Ranking of District Hospitals has created awareness among district hospitals regarding the HMIS and the importance of proper record-keeping (Aayog, 2021).State Health Index of NITI Aayog extensively used HMIS data. Due to extensive uses of HMIS data for NITI's index etc., the further emphasis on quality of data has increased. The data validation and discussions among state and central level programme managers during State Health Index has helped reinforce good practices related to data scrutiny and validation of HMIS data, and thus improving quality of HMIS data (NITI Aayog, 2021).

According to a study conducted during 2008–2011 by the National Health Systems Resource Centre (NHSRC) on a sample of 35 districts of India, gaps were found in data collection. The quality of data remains a major challenge for efficient policy-making and improving maternal health (Mishra A, 2012). New HMIS system (HMIS 2.0) developed and implemented during 2020-21 is tries to ensure data quality on technical issues (with inbuilt validation checks and reports) however administrative issues (frequent changes of staff and lack of understanding of data items) are addressed through periodic training and motivational measures.

Conclusion

Health Programmes monitoring has seen many changes over the year from paper based reporting to computer based reporting. When National Health Mission (NHM) was launched in 2005, development of a HMIS with the aim to monitor the flagship programmes of NHM was conceived. With this vision, Health Management Information System (HMIS) was launched in 2008. The journey of HMIS has been outstanding from 2008 with District level consolidated reporting followed by Facility level reporting 2011-12 onwards. 2020-21 saw a revamped, updated version of HMIS functioning with latest technology. HMIS has faced many challenges in ensuring data qualities but the system always tried to ensure data quality with its inbuilt validation checks and reports.

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